ROYAL

SEALS FOR HYDRAULICS & PNEUMATICS



ROYAL COMPOSITES PVT LTD
INDIA

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Certificate of Registration

QUALITY MANAGEMENT SYSTEM - ISO 9001:2008

This is to certify that

Royal Composites Private Limited
Howrah Polypark Near Sankrail Toll Plaza
Plot-5, Unit PPII - 4
Village & PO: Kandua, P.S.: Sankrail
Howrah 711 302
West Bengal
India

Holds Certificate No: FM 00046

and operates a Quality Management System which complies with the requirements of ISO 9001:2008 for the following scope:

The Design, Development, Manufacture, Marketing and Export of Seals for Hydraulic and Pneumatic Applications.

For and on behalf of BSI:

Managing Director, Mid India, Operational Assets

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Page: 1 of 1

BSI



Details contained in this catalogue have been prepared through years of exhaustive research at our facility. The material data and values is not to be construed as guaranteeing specific properties and can not be used to deduce the suitability for a particular application.

No warranty or guarantee is expressed in our standard terms and condition of sale, since the condition of use are beyond our control.

All the material data contained are for reference purpose only.

Royal is continuously improving its range and reserves right to withdraw or modify any item shown in this catalogue.

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SEALS FOR HYDRAULICS & PNEUMATICS

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Royal is a leading group of India, pioneer in designing, developing, manufacturing and marketing Hydraulic & Pneumatic seals with a reliability of performance. ROYAL'S headquarters is located in Kolkata, India. Besides its wide range of standard seal profiles, ROYAL also manufactures customized sealing solutions as per design requirement and other high-tech machinery parts made of various synthetic polymeric materials like Polyurethane, PTFE, NBR, FPM, Polyamide, Polyester Resin and various other engineering plastics, their blends, alloys and composites.





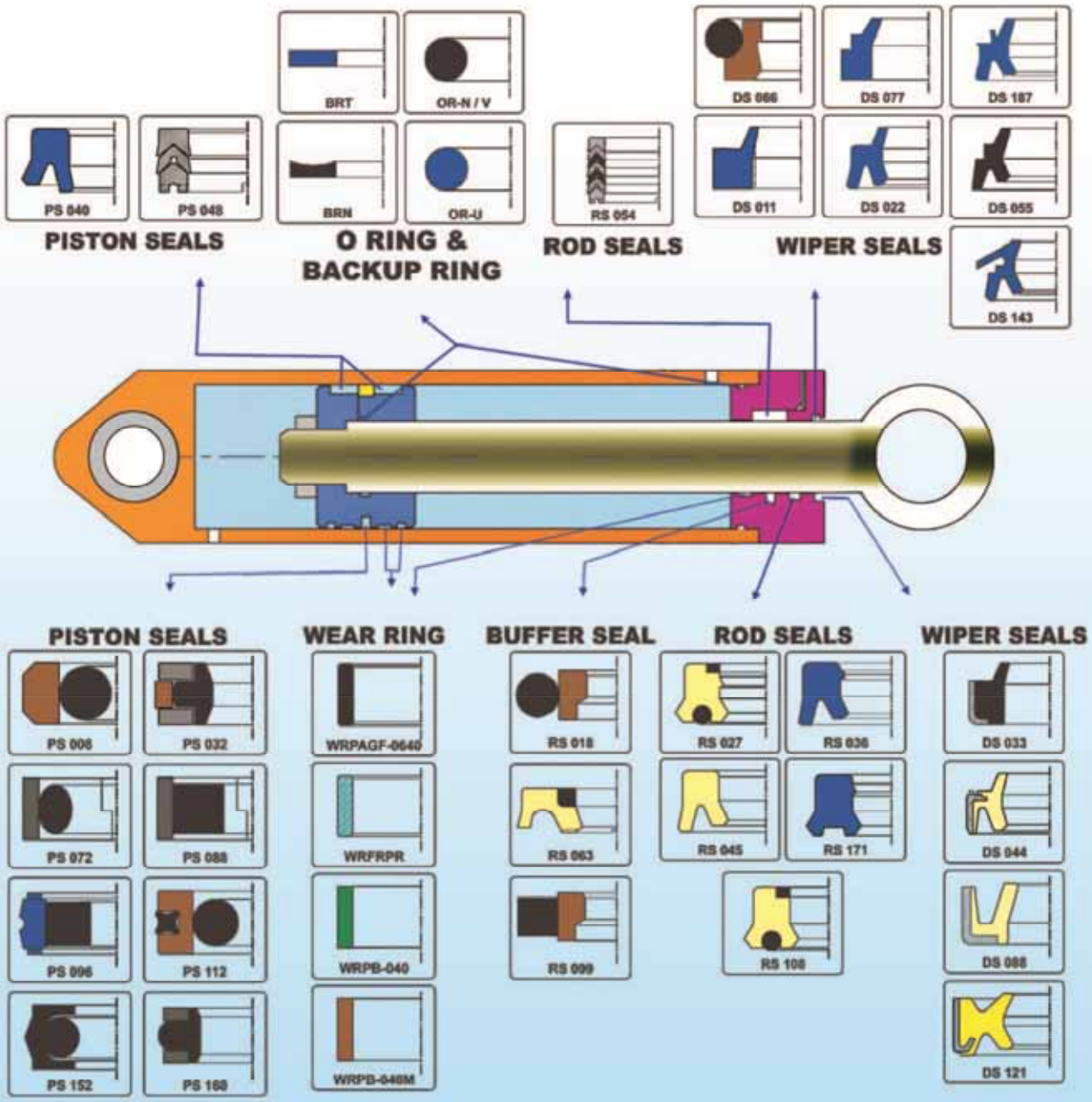
Inspired by the developed world, we took up on our own, mission of providing with perfection the most delicate part of the cylinder – its sealing system. Hydraulically or pneumatically controlled system has become an integral part of this machine age. Good & perfect sealing system is the back bone for the success of any hydraulic application. Our design engineers have developed a unique system to overcome failure modes common to small, medium and heavy-duty hydraulic cylinders. We have done it by combining advanced seal materials with the most progressive seal profiles. Our constant efforts for improvement and commitment to quality has led us to adopt latest manufacturing technology and advanced quality control system.

Considering the critical application of sealing elements under high hydraulic pressures, we with our masterly knowledge of advanced polymer science and years of experience of manufacturing are confident of providing configurations of complete combination of seals and even new seal designs for zero leakage, smooth running and a very long lasting sealing solution to our customers.

With its modern manufacturing facilities, very strong quality control department and a team of highly motivated group of professionals, Royal is in a position to understand, design, manufacture and service all your sealing requirement.

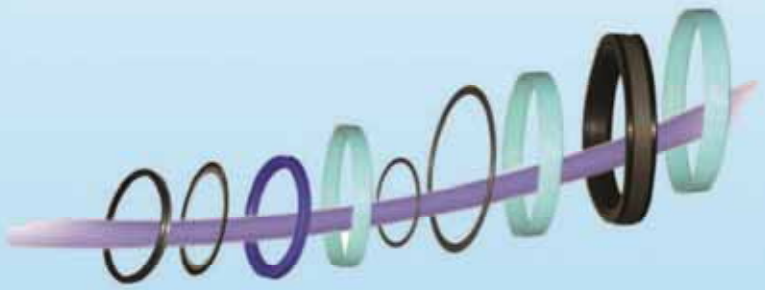


SEAL APPLICATIONS



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Section - I Seal Material

Appropriate choice of seal material is very important in sealing systems to operate leak free and to maintain a long service life. Material performance and seal design are the two most crucial factors in terms of having the best sealing solution. Elastomers are one of the most widely used seal material. They possess properties like elasticity, resilience and chemical resistance which are ideal for sealing applications. Below is a brief description of the most widely used seal materials.

Polyurethane – (PUR)

Polyurethane has emerged as one of the most versatile seal material. Apart from its high wear resistance characteristics compared to other materials it possesses elastic qualities of rubber. The flexibility in chemically engineering TPU's to get a wide range of properties including fluid and moisture resistance has led to a wide acceptance of this material. Polyurethane seals offer outstanding performance in high pressure hydraulic systems with abrasive contamination, high shock loads, and related adverse conditions provided temperature is below 120°C.

Polyester and Polyether grade of PUR are generally used in the seal industry. Polyester grades generally have superior mechanical properties whereas Polyether grades have better chemical and hydrolytic stability. Special polyurethane materials having good low temperature behavior are being designed and used for specific applications.

The oil resistance and swell characteristics can be controlled to a desired extent by changing the type and order of the constituent hard and soft segment of PUR. These characteristics along with the processing ease have made PUR one of the most widely accepted seal material.

Fluid / Chemical Resistance:

- Resistant to Mineral oil and Greases.
- Not resistant to non-mineral oil, automotive brake fluids etc.

Nitrile Rubber – (NBR)

Nitrile is the most widely used elastomer in the seal Industry. Excellent resistance to petroleum products and its ability to be compounded for service over a wide temperature range has made Nitrile a very popular material.

Nitrile is a copolymer of butadiene and acrylonitrile. Specific properties can be achieved by varying the proportions of these individual polymers. Acrylonitrile content in Nitrile can be varied from 18% to 50% by weight. Nominal designations are 'low' (18 to 24%), 'medium' (28% to 33%) and 'high' (38% and higher) Nitrile. An increase in acrylonitrile content increases heat as well as chemical resistance to petroleum base oils and fuels but decreases low temperature flexibility. Nitrite provides very low compression set, good tear resistance, cold flow and abrasion resistance. The major limiting properties of nitrile are its poor ozone and weather resistance and moderate heat resistance. Hydrogenated – NBR are used in applications requiring high wear and chemical resistance.

Fluid / Chemical Resistance:

- Resistant to Mineral oil and Greases, water and many other chemicals, hydrocarbon fuels etc.
- Not resistant to non-mineral automotive brake fluids

Fluorocarbon Rubber - (FPM)

Fluorocarbon combines high temperature and chemical resistance to a broader range of fluids than any of the other elastomer. Low compression set even at high temperatures makes it an appropriate choice of material in most conditions. Its working temperature range varies from 230°C for high to -20°C for low temperature applications. Three main factors contribute to the remarkable heat and fluid resistance of fluoroelastomers. First, there are extremely strong bonds between the carbon atoms comprising the polymer backbone and the attached

Seal Material

(Pendant) fluorine atoms.Second, fluoroelastomers features a high fluorine-to-hydrogen ratio. Third, the carbon backbone is fully saturated. Fluorocarbons are a good choice of material for both static and dynamic applications.

Fluid/Chemical Resistance:

- Resistant to Mineral oil and Greases, hydrocarbon fuels etc.
- Non resistant to certain ketones and acids.

Polyoxymethylene - (POM)

POM is a high strength engineering plastic suitable for bearings, bushings and back-up ring applications. The main molecular chain of acetal homopolymer is the backbone of the acetal copolymer chain, with ethylene oxide being randomly added throughout the acetal homopolymer. The simple molecular nature of the main chain allows a very stable crystalline structure. This crystallinity leads to high heat resistance, high modulus, excellent wear resistance, and resistance to solvents. The added ethylene linkages of acetal copolymer increase the thermal stability, elongation and abrasion resistance. It also displays good low temperature impact resistance, high dimensional stability, favourable frictional properties and low water absorption. It offers excellent gliding characteristics when blended with PTFE and good temperature resistance. Owing to its chemical structure, POM has low moisture absorption as well as good chemical resistance.

Fluid/Chemical Resistance:

- Resistant to Mineral oil and Greases, Water etc.
- Non resistant to certain solvents and acids.

GF-Polyamide

Glass filled polyamide has proved to be an excellent choice for bearing and bushing application. Its high compression modulus owing to the glass fiber reinforcement and excellent dimensional stability is very important in its functioning. GF-polyamides have high load bearing capacity which makes it a good choice for back-up rings and guide ring applications. It has good temperature and wear resistance as well.

Fluid/Chemical Resistance:

- Resistant to Mineral oil and Greases, Water etc.
- Non resistant to certain solvents and acids.

Polytetrafluoroethylene - (PTFE)

PTFE is an elastomer with excellent chemical, tribological and thermal properties. Reinforced grades of PTFE offers good wear and extrusion resistance. Bronze filler provides slightly superior properties than glass filler as far as tribological properties are concerned. PTFE is practically resistant to all hydraulic fluids. It is excellent for dynamic applications and pressures upto 6000 psi or 400 bars.

Fluid/Chemical Resistance:

- Excellent all round resistance to acids, alkalis, ketones, ester, amines, water, solvents etc.

For ROYAL compound detail please see the raw material data sheet see section III

Section - III Royal Seal Material

Shore hardness

Hardness may be defined as the resistance to indentation under specific conditions. When judging the potential effectiveness of molded seal (processed), hardness is one of the most common criteria in the rubber industry. Though the elastomeric lip of most standard seals fall in the 70 to 95 Shore A range, the application in question will always govern the necessary hardness. Softer compounds offering less resistance may be perfectly fine for low-pressure seals, but high-pressure seals will likely require a harder, tougher lip material.

Resilience

Resilience (also known as rebound) refers to a compound's ability to regain its original size and shape following temporary deformation. Though compounding may improve an elastomer in this area, it is largely an inherent property. As a general rule, resilience is most critical in dynamic seals. This can be seen, for example, when a shaft seal's elastomeric lip flexes (is distorted) to follow a shaft imperfection, and then returns to its original ("resting") position. In many cases resilience is closely linked to flexibility.

Tear Strength

Tear resistance (or tear strength) is the ability of a material to resist the growth of a cut or nick when tension is applied. Tear resistance is an important consideration, both as the finished article is being removed from the mold *and* as it performs in actual service. Compounds with low tear resistance are at risk for installation damage, especially in designs featuring non-smooth areas (as with burrs, slots, threads) and/or sharp, non-radiused (non-rounded) corners.

Abrasion Resistance

Measured as a loss percentage based on original weight, abrasion resistance is the resistance of a material to wearing away by contact with a moving abrasive surface. On one hand where the cutting or nicking of a sealing lip is an instantaneous event, abrasive rubbing or scraping is much more progressive phenomenon that develops over time. Seals in motion are most susceptible to abrasion. Hard compounds tend to exhibit less abrasive wear than soft compounds, but use of a harder compound can also increase friction in dynamic seals, and increased friction generates seal-degrading heat.

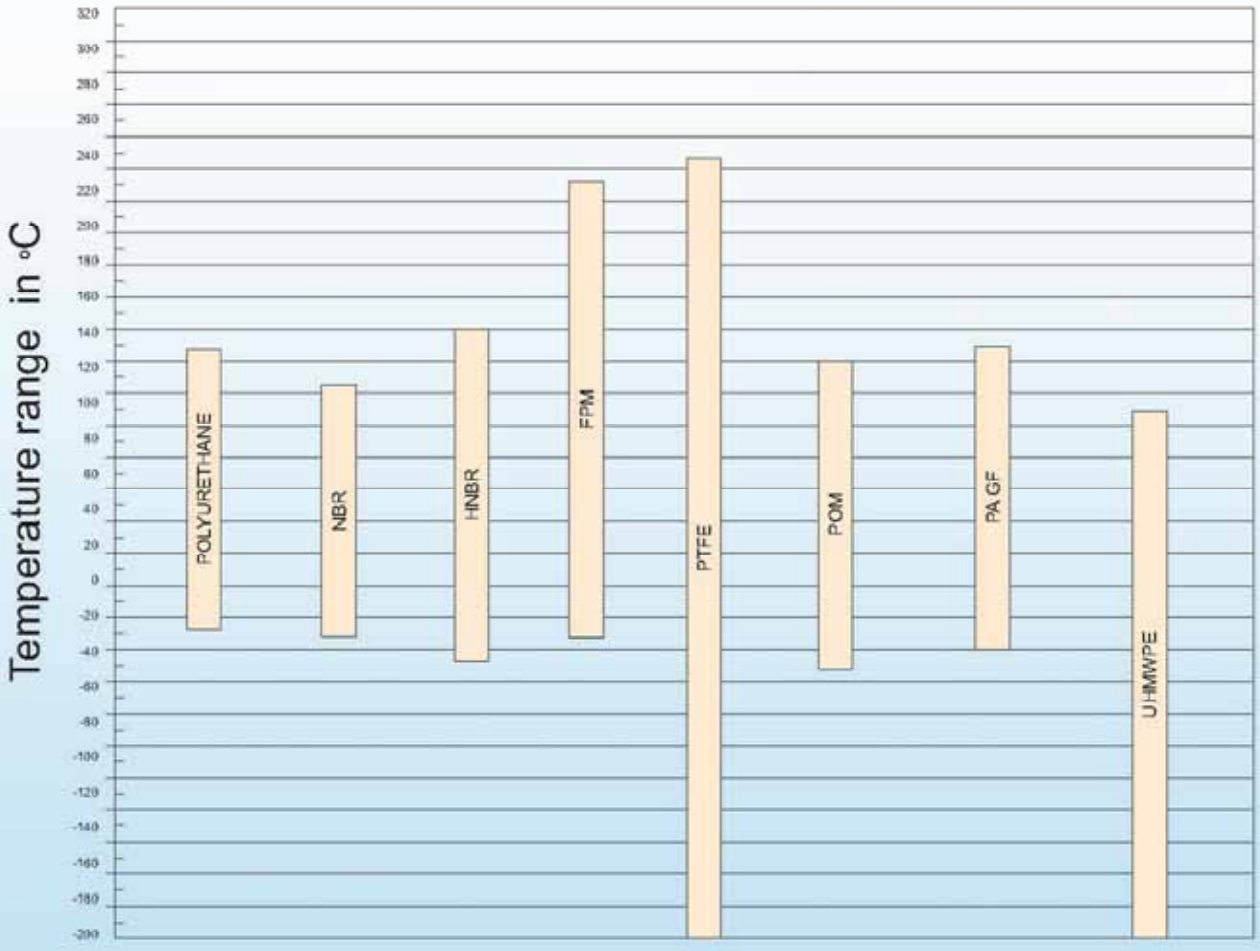
Compression Set

Compression set is the end result of progressive stress relaxation, which is the steady decline in sealing force that results when an elastomer is compressed over a period of time. In terms of the life of a seal, stress relaxation is like dying, whereas permanent compression set is like death. Compression set tests find their widest use in selecting materials for use as *O-rings, U cup seals and other seals* that rely on an optimal degree of compression.

Important Seal Properties

Operating temperature range

Working temperature range is one of the most important properties with regard to selection of seal material. Chart below shows the operating temprature range of various polymers used in the seal industry.



For special applications please consult with ROYAL before selection of raw material.

Section - III Royal Seal Material

TYPES AND CHARACTERISTICS OF ROYAL COMPOUNDS

ROYAL MATERIAL CODE	MATERIAL	COLOUR	FEATURES	MAJOR APPLICATION
RUBBER COMPOUNDS				
NBR70	NBR	Black	Very low compression set	Energizer rings for piston seal and rod seals, O rings"
NBR70L	NBR	Black	Very good low temperature Low compression set	Energizer rings for piston seal and rod seals, O rings"
NBR75	NBR	Black	Low compression set, self lubricating, wear resistant.	Sealing element for PS024
NBR78	NBR	Black	Very low compression set, High modulus.	Energizer rings for piston seal (Eg. PS096)
NBR80	NBR	Black	Very low compression set, High modulus.	Energizer rings for piston seal (Eg. PS032)
NBR90	NBR	Black	High modulus, high strength, low compression set	'O' ring fo static sealing.
PVCNBR85	PVC Blended	Black	High wear resistance, hydrolysis resistance	Seals for water hydraulics. friction.
XNBR80	X-NBR	Black	Very high wear resistance comparable to polyurethane	Seals for heavy duty and high speed application
HNBR80	H-NBR	Black	High temperature and chemical resistance.	Special purpose seal
FPM80	FPM	Ash Black	Very high heat resistance with excellent mechanical properties.	Seals for high temperature applications
FPM90	FPM Ash	Black/Brown	Very high heat resistance with excellent mechanical properties.	O rings for high temperature applications
SILICON70	Silicon rubber	Red	Very good low temperature flexibility, good thermal resistance.	Seals for very low temperature applications.
NBRCF	NBR + Fabric	Ash Black	Fabric reinforced, high wear resistance resistance, high modulus	'V' Packing
FPMCF	FPM + Fabric	Brown	Fabric reinforced, high temperature resistance, high wear resistance, high modulus	'V' Packing for high temperature application
POLYURETHANES & POLYESTERS				
PUAB85	Polyurethane	Yellow / Blue	Low friction, Good flexibility	Pneumatic Seals
PUAB92	Polyurethane	Ocean Blue	Low friction, Low wear loss.	Pneumatic Seals, Wipers
PUAB94	Polyurethane	Yellow / Blue	Excellent wear resistance	Wiper Seal, Dust Seal
PUAB98	Polyurethane	Blue / Green	Excellent wear resistance	Piston Seal, Back-ups
PUDB65	Polyurethane	Deep Yellow	High Hardness, High Modulus	Support Rings.
PUET93/ PUHP93	High Performance Polyether Polyurethane	Translucent Blue	Hydrolysis resistance, Low compression set at elevated temperature	Rod Seals, Piston Seals
PUHP94	High Performance Polyurethane	Translucent Blue	Very low compression set at elevated temperature, Excellent mechanical properties	Rod Seals, Piston Seals
PUHP96	High Performance Polyurethane	Opaque Yellow	High modulus, low compression set	Centre swivel rings.
HY55	Thermoplastic Polyester	Red / Black	Excellent wear resistance. High Mechanical strength.	Anti Extrusion rings, Back-Ups, Piston Seals

Section - III Royal Seal Material

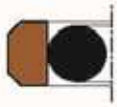
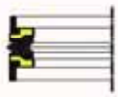
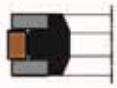
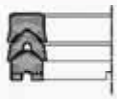
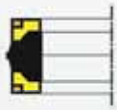
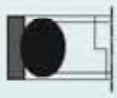
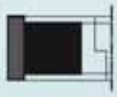
TYPES AND CHARACTERISTICS OF ROYAL COMPOUNDS

ROYAL MATERIAL CODE	MATERIAL	COLOUR	FEATURES	MAJOR APPLICATION
PTFE & FILLED COMPOUNDS				
TFPR	Pure PTFE	White	General purpose	Back-up rings
TB40	40% Bronze filled PTFE	Green	Cold extrusion resistant	Guide tape, soft wear rings
TB40M	40% Bronze Filled PTFE With MoS2	Brown Black	Low coefficient of friction, Cold extrusion resistant	Piston seal PS008, Rod seal Rs018
TB60M	60% Bronze Filled PTFE With MoS2	Brown Black	Extrusion resistant, higher hardness with self lubrication	Speial purpose seal
TG20B20M	20% Bronze/20% glass filled PTFE with MoS2	Brown Black	Heat dissipating, low friction, extrusion resistant	Piston seal PS032
TG25	25% glass filled PTFE	Blue/Yellow	Extrusion resistant, High Pressure, Good abrasion resistance	Special application, PS160
TG20M	20% glass filled PTFE with MoS2	Grey	Lower friction, Extrusion resistant, High Pressure, Good abrasion resistance	Special application, PS120
TC23G2	23% Carbon 2% Graphite	Black	Excellent wear resistant at high speed friction.	Dry pneumatic applications
TCF10	10% Carbon Fibre filled PTFE	Grey	Excellent wear resistant, Good mechanical properties	Piston Seals, Special Application
TG15	15% graphite filled PTFE	Grey	Very low friction, Good abrasion resistance	Piston seals, V Packings
POLYAMIDE, COMPOSITES & FILLED COMPOUNDS				
PA6000	Heat Stabilized Polyamide	Black	General purpose	Back-ups, Anti extrusion ring
PA6600	Heat Stabilized Polyamide 6'6	Black / Yellow	General purpose	Back-ups, Anti extrusion ring
PG6030	30% Glass Fibre Filled Polyamide	Black/Brown	Good mechanical properties Good abrasion resistance	Wear Rings
PG6040	40% Glass Fibre Filled Polyamide	Black	Good mechanical properties Good abrasion resistance	Medium & Heavy duty wear rings
PG6040T	40% Glass Fibre Filled Polyamide with PTFE micro powder	Black	Low friction, Good mechanical properties Good abrasion resistance	Medium & Heavy duty wear rings
PG6040M	40% Glass Fibre Filled Polyamide with MoS2 powder	Black	Low friction, Good mechanical properties Good abrasion resistance	Piston seals, Hydraulic and pneumatic wear rings.
PG6650	Heat stabilized 50% Glass Fibre Filled Polyamide 6'6	Brown / Black	Very high Compressive strength, High dimensional stability,	Special Applications
FRPR	Polyester PTFE Resin + Fabric	Light Blue	Very high Compressive strength, Good Abrasion Strength, No water swell	Medium to very Heavy duty wear rings
POM	Polyoxymethylene	White	High compressive strength, Low friction	Wear Rings & Anti extrusion ring

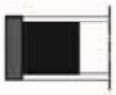
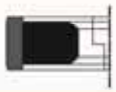
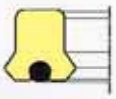
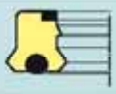
NOTE:

NBR : Nitrile Butadiene Rubber | X-NBR : Carboxilated Nitrile Butadiene Rubber | H-NBR : Hydrogenated Nitrile Butadiene Rubber
FPM : Fluoro-carbon Rubber | For other compound details, contact Royal at : info@royalseals.co.in

Section - IV Royal Seal Profile

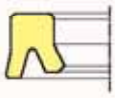
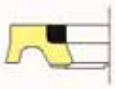
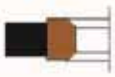
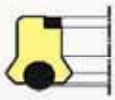
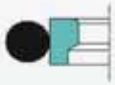
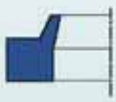
Royal Profiles	Standard Profiles	Working Parameters			Description	PG No.
		Max Pressure (Bar)	Speed (m/s)	Temp Range (°C)		
PISTON SEALS						
	PS008N	400	4	-20 to +100	PTFE seal energized with O ring - double Acting	17
	PS008F	400	4	-20 to +200		
	PS016N	400	4	-20 to +100	PTFE seal energized with O ring - Single Acting	★
	PS016F	400	4	-20 to +200		
	PS024	315	0.5	-20 to +100	5 Pc Piston Seal - Double Acting	25
	PS032	500	1.5	-20 to +100	4 Pc Capped T Seal - Robust seal design for high pressure application - Double Acting	28
	PS040U	400	0.5	-30 to + 110	U type Asymmetric Seal - For Low and medium duty Application - Single Acting	35
	PS040F	160	0.5	-20 to + 200		
	PS048N	400	0.5	-20 to +100	Chevron Packing for Piston application - Robust design for heavy duty single Acting Application	42
	PS048F	400	0.5	-15 to +140		
	PS056	315	0.5	-20 to +100	5 pc Piston Seal - Double Acting Application	45
	PS064	400	4	-20 to +100	PTFE seal energized with square / rectangular Rubber energizer	47
	PS072	500	1.5	-20 to +100	Step cut Polyamide Ring as main seal with Nitrile Oval ring as energizer	51
	PS088	500	1.5	-20 to +100	Step cut Glass filled Polyamide Ring as main seal with Nitrile square / rectangular ring as energizer	57
	PS096	315	0.5	-20 to +100	Polyurethane ring as main seal with Nitrile square / rectangular ring as energizer	61
	PS112	400	1	-20 to +100	PTFE seal with X Ring energized with O ring	67

Royal Seal Profile

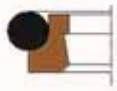
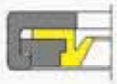
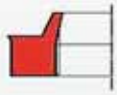
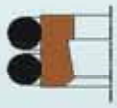
Royal Profiles	Standard Profiles	Working Parameters			Description	PG No.
		Max Pressure (Bar)	Speed (m/s)	Temp Range (°C)		
	PS120	400	4	-20 to +100	Glass filled PTFE with MoS2 as main seal energized with square / rectangular ring as energizer	71
	PS136	400	0.5	-20 to +100	U type Assymetric Seal with AE ring	★
	PS152	315	1	-20 to +100	Polyurethane / Polyester ring as main Seal with Nitrile O Ring as energizer	74
	PS160	315	4	-20 to +100	Square PTFE seal energized with O Ring as Energizer	77
	PS168	315	1.5	-20 to +100	3 pc Piston Seal - NBR T Seal with polyamide back up	★
	PS184	400	0.5	-20 to +100	Step cut Pure Polyamide Ring as main seal with nitrile square / rectangular ring as energizer	★
SYMMETRIC SEALS						
	SS001	400	0.5	-20 to +100	Symmetric Seal loaded with O Ring	81
	SS002	400	1	-20 to +100	U Type Symmetric Seal	★
	SS003	400	0.5	-20 to +100	U Type Symmetric Seal energized with NBR Loader	★
	SS004	400	0.5	-20 to +100	Non-bevelled Symmetric Seal Loaded with O Ring. Also used as wiper.	★
ROD SEALS						
	RS009N	400	4	-20 to +100	PTFE Seal energized O Ring	★
	RS009F	400	4	-20 to +200		
	RS018N	400	4	-20 to +100	PTFE Seal energized O Ring	87
	RS018F	400	4	-20 to +200		
	RS027	500	0.5	-20 to +100	Rod Seal with secondary lip and Anti-Extrusion ring loaded with NBR O Ring	96

★ Contact Royal for size list at : info@royalseals.co.in

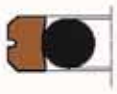
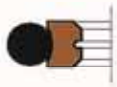
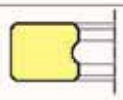
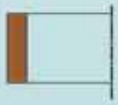
Royal Seal Profile

Royal Profiles	Standard Profiles	Working Parameters			Description	PG No.
		Max Pressure (Bar)	Speed (m/s)	Temp Range (°C)		
	RS036U	400	0.5	-30 to +110	U Type Asymmetric Seal with secondary lip	99
	RS045F	160	0.5	-20 to +200	U Type Asymmetric Rod Seal	109
	RS045U	400	0.5	-20 to +120		
	RS054N	400	0.5	-20 to +100	Chevron Packing for Rod Application Robust Design for Heavy Duty Application	116
	RS054F	400	0.5	-15 to +140		
	RS063	400	0.5	-20 to +100	U type Asymmetric buffer seal with Anti Extrusion ring	120
	RS090	400	4	-20 to +100	PTFE seal energized with ring square ring - Double Acting	★
	RS099	400	4	-20 to +100	PTFE Seal with NBR square ring Energize	124
	RS108	400	0.5	-20 to +100	Rod Seal with Anti - Extrusion ring loaded with NBR O Ring	127
	RS117	315	1	-20 to +100	Polyurethne buffer seal with NBR O Ring Energizer	129
	RS171	400	0.5	-20 to +100	U Type Symmetric Seal with Secondary lip	★
WIPER SEALS						
	DS011U	-	0.5	-20 to +100	Single lip Wiper for low and medium duty application	131
	DS011N	-	0.5	-20 to +100		
	DS011F	-	0.5	-20 to +200		
	DS022U	-	0.5	-20 to +100	Double lip Wiper for medium to heavy duty application	136
	DS033U	-	0.5	-20 to +100	Single lip Metallic Wiper for medium and heavy duty application	140
	DS033N	-	0.5	-20 to +100		
	DS033F	-	0.5	-20 to +200		

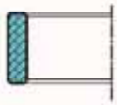
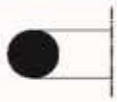
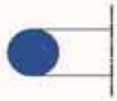
Royal Seal Profile

Royal Profiles	Standard Profiles	Working Parameters			Description	PG No.
		Max Pressure (Bar)	Speed (m/s)	Temp Range (°C)		
	DS044U	-	0.5	-20 to +100	Double lip Metallic Wiper for Medium and heavy duty application	144
	DS055N	-	0.5	-20 to +100	Double lip Step Wiper for Pneumatic and light to medium duty application	147
	DS055F	-	0.5	-20 to +200		
	DS055U	-	0.5	-20 to +100		
	DS066N	-	4	-20 to +100	Filled PTFE wiper energized with O ring for medium to heavy duty hydraulic application	151
	DS066F	-	4	-20 to +200		
	DS077U	-	1	-20 to +100	Single lip Step Wiper for low to medium duty application	156
	DS088	-	1	-20 to +100	Single lip U type Metallic Wiper for pin dust and medium duty application	160
	DS099	-	1	-20 to +100	Single lip Metallic Wiper for pin dust and medium duty application	163
	DS110	-	0.5	-20 to +100	Single lip Wiper with crush lip and ribs for heavy duty application	165
	DS121	-	0.5	-20 to +100	Double lip Metallic Wiper for medium and Heavy duty application	167
	DS143	-	0.5	-20 to +100	Double lip wiper with dust protecting cap for medium to heavy duty application	★
	DS154	-	0.5	-20 to +100	Filled with PTFE wiper energized with two O Rings for medium to heavy duty application	★
	DS165	-	0.5	-20 to +100	Single lip wiper with metal inside for medium and heavy duty Application	★
	DS187	-	0.5	-20 to +100	Double lip wiper with dust protecting lip	★
	DS198	-	0.5	-20 to +100	Single lip Wiper with pressure release slots for low to medium duty application	★

Royal Seal Profile

Royal Profiles	Standard Profiles	Working Parameters			Description	PG No.
		Max Pressure (Bar)	Speed (m/s)	Temp Range (°C)		
ROTARY SEALS						
	ROP011N	400	4	-20 to +100	Rotary Seal for Piston Application	170
	ROP001F	400	4	-20 to +200		
	ROR002N	400	4	-20 to +100	Rotary Seal for Rod Application	173
	ROR002F	400	4	-20 to +200		
	ROS004U	400	0.5	-20 to +100	Rotary Seal for application like Swivel Joints	177
PNEUMATIC SEALS						
	PNPS007	16	1	-20 to +100	Polyurethane based Piston Seal	180
	PNRS010	16	1	-20 to +100	Polyurethane based rod cum wiper seal	182
	PNCS012	16	1	-20 to +100	Polyurethane based Cushion Seal	184
WEAR RINGS						
	WRPAGF	-	1.5	-55 to +140	Glass filled polyamide Wear Ring - used in medium and heavy duty hydraulic application	193
	WRPAGF - L TYPE	-	1.5	-55 to +140	Glass Filled Polyamide T Wear ring - used in telescopic Rod Application	★
	WRPAGF -T TYPE	-	1.5	-55 to +140	Glass filled Polyamide T type wear ring - used in special application	★
	WRPBRG	-	4	-55 to +200	Bronze Filled PTFE Wear Ring - used in light duty hydraulic and pneumatic application	186
	WRPBRM	-	4	-55 to +220	Bronze filled PTFE with Moly Wear Ring - Used in light duty hydraulic and pneumatic application	★

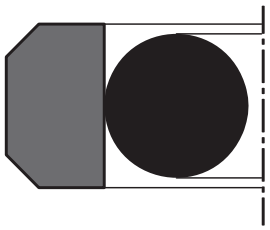
Royal Seal Profile

Royal Profiles	Standard Profiles	Working Parameters			Description	PG No.
		Max Pressure (Bar)	Speed (m/s)	Temp Range (°C)		
	WRFRPR	-	5	-20 to +100	Polyester Composite based Wear Ring - used in medium and heavy hydraulic application	189
OTHER PROFILE						
	OR-N	-	0.5	-20 to +100	Standard NBR O ring - used in static and dynamic applications	197
	OR-F	-	0.5	-20 to +200	Standard FPM O ring - used in static and dynamic applications	197
	OR-PU	-	1	-20 to +100	Standard TPU O ring - used in static, dynamic and certain high pressure applications	★
	BRN	-	0.5	-20 to +100	Standard NBR back-up used in static applications	★
	BRT	-	1	-20 to +100	Standard PTFE back-up used in static and dynamic applications	★

★ Contact Royal for size list at : info@royalseals.co.in

All drawings and dimensions in this catalogue are in millimeters (mm)

* Apart from the Seal Profiles mentioned herein, Royal does manufacture many other Seal Profiles. Please contact **ROYAL** at : info@royalseals.co.in for details

PISTON SEAL	PS008	

Features:

Piston seal profile PS008 consists of two piece seal set for sealing pistons with one seal ring and one energizer O-ring.

Composition:

PS008 have two different type of combination, PS008-N & PS008-F for different temperature range as per following table :-

SEAL COMBINATION		PS008-N	PS008-F
Seal ring	Material	PTFE+ Bronze + MoS2	PTFE+ Bronze + MoS2
	Royal code	TB40M	TB40M
O Ring	Material	Nitrile Butadiene rubber (NBR)	Fluoro Carbon Rubber (FPM)
	Royal code	NBR70	FPM80
Working Temperature Range Medium : Hydraulic Oil		-20°C to +100°C	-20°C to +200°C

For compound detail please refer Royal seal materials. (Section - III)

Properties:

- Low friction, free of stick slip
- High abrasion resistance
- Wide temperature Range
- Good thermal conductivity
- Low wear and high extrusion resistance
- Wide fluid application range
- Resistance against cold flow
- Minimum installation space required

Application:

- Mobile Hydraulics
- Injection moulding machines
- Machine tools
- Material Handling Equipments
- Control and regulating apparatus

General Technical Data:

Seal Profile	PS008-N	PS008-F
Working Pressure(max.)	400 bar	400 bar
Surface Speed(max.)	4 m/s	4 m/s

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Extrusion gap dimension :

The gap dimension on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the minimum clearance Ø and max bore Ø.

Profile dimension	Maximum permissible gap (S)			
H	160 bar	260 bar	320 bar	400 bar
2,2	0,35	0,3	-	-
3,2	0,4	0,35	-	-
4,2	0,5	0,4	0,3	-
6,3	0,6	0,5	0,45	0,40
8,1	0,7	0,6	0,55	0,5

Installation :

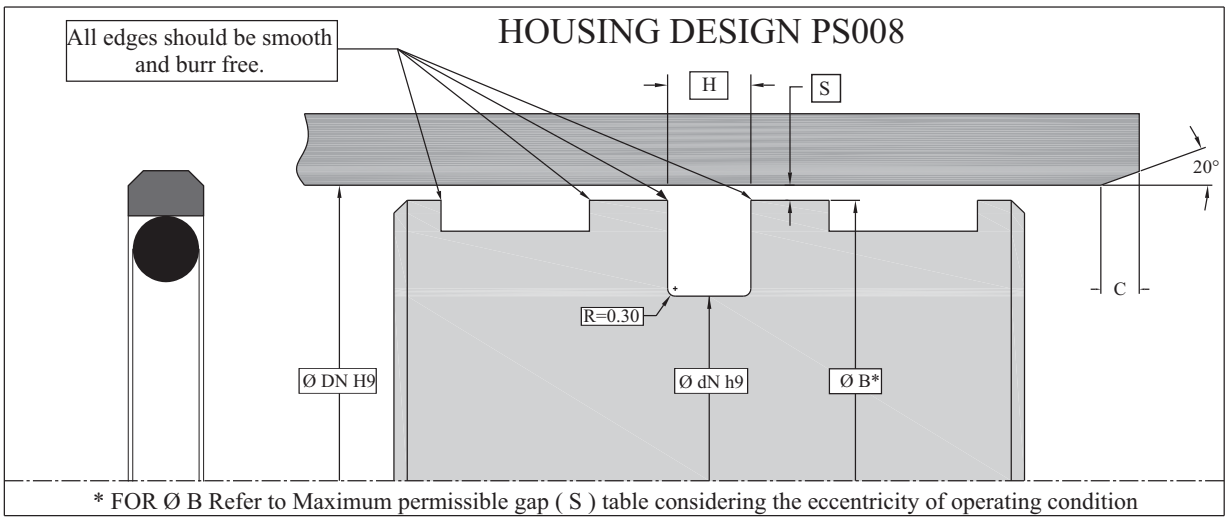
Ps008 piston seals can be fitted on single-piece pistons. First insert the rubber 'O' Ring into the housing carefully to avoid twisting of O Ring Chord. The PTFE ring is then carefully pushed over the piston into place above the O Ring. An assembly tool eases installation of small, PTFE rings. The PTFE seal normally returns to its original size after installation. It is advisable to calibrate the PTFE ring by mounting it on a steel tube (duly machined and polished), having Inner diameter equal to cylinder bore diameter. Immediately after calibration, the piston should be installed in the cylinder.

Ordering format :

Please mention ROYAL order code in your order.
Example: Size 10 X 5.1 X 2.2
For Temp range: -20°C to +100°C (O ring material is NBR70) Order code is PS008-N-01000-00510-0220
For Temp range: -20°C to +200°C (O ring material is FPM80) Order code is PS008-F-01000-00510-0220

We have given all metric(mm) sizes here. For Imperial (inch) sizes please contact ROYAL at
info@royalcomposites.co.in / info@royalseals.co.in

HOUSING DESIGN PS008

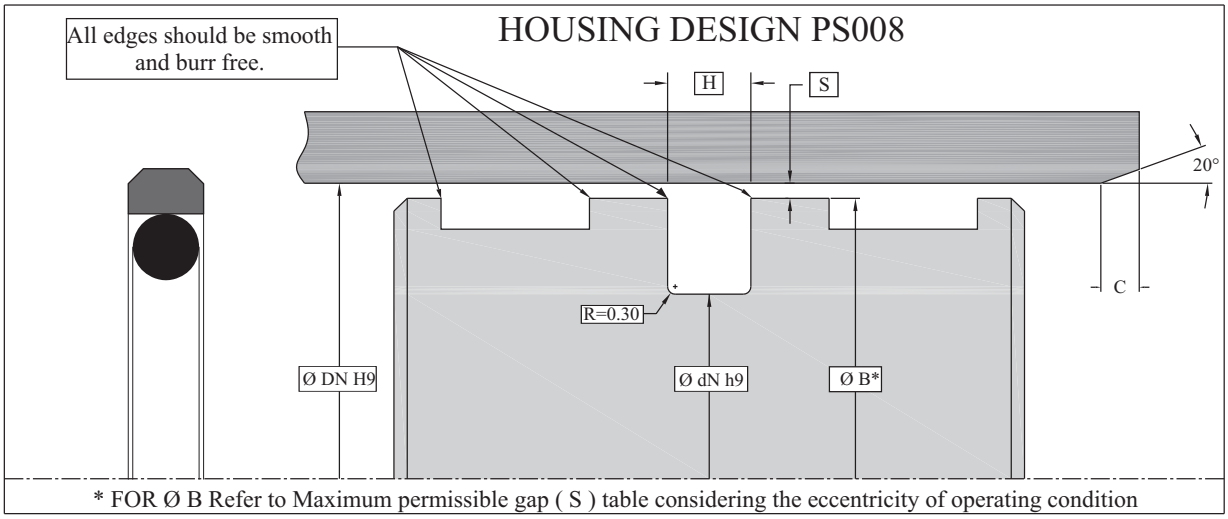


* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

ØDN (H9)	ØdN (h9)	H (+0.20)	C	O-RING SIZE	ORDER CODE For Temp range: -20°C to +100°C use PS008-N -20°C to +200°C use PS008-F
8.00	3.10	2.20	3.00	3.10 X 1.78	PS008-N/F-00800-00310-0220
10.00	5.10	2.20	3.00	4.80 X 1.78	PS008-N/F-01000-00510-0220
12.00	7.10	2.20	3.00	6.74 X 1.78	PS008-N/F-01200-00710-0220
12.70	7.80	2.20	3.00	7.65 X 1.78	PS008-N/F-01270-00780-0220
14.00	9.10	2.20	3.00	8.74 X 1.78	PS008-N/F-01400-00910-0220
15.00	7.50	3.20	4.50	7.59 X 2.62	PS008-N/F-01500-00750-0320
16.00	8.50	3.20	4.50	7.59 X 2.62	PS008-N/F-01600-00850-0320
18.00	10.50	3.20	4.50	9.19 X 2.62	PS008-N/F-01800-01050-0320
19.05	11.55	3.20	4.50	10.77 X 2.62	PS008-N/F-01905-01155-0320
20.00	12.50	3.20	4.50	11.35 X 2.62	PS008-N/F-02000-01250-0320
22.00	14.50	3.20	4.50	13.94 X 2.62	PS008-N/F-02200-01450-0320
25.00	17.50	3.20	4.50	17.12 X 2.62	PS008-N/F-02500-01750-0320
25.00	14.00	4.20	6.00	13.87 X 3.53	PS008-N/F-02500-01400-0420
25.40	17.90	3.20	4.50	17.12 X 2.62	PS008-N/F-02540-01790-0320
28.00	20.50	3.20	4.50	20.29 X 2.62	PS008-N/F-02800-02050-0320
28.00	17.00	4.20	6.00	17.04 X 3.53	PS008-N/F-02800-01700-0420
30.00	22.50	3.20	4.50	21.89 X 2.62	PS008-N/F-03000-02250-0320
31.70	24.20	3.20	4.50	23.47 X 2.62	PS008-N/F-03170-02420-0320
32.00	24.50	3.20	4.50	23.47 X 2.62	PS008-N/F-03200-02450-0320
32.00	21.00	4.20	6.00	20.22 X 3.53	PS008-N/F-03200-02100-0420
35.00	27.50	3.20	4.50	26.64 X 2.62	PS008-N/F-03500-02750-0320
35.00	24.00	4.20	6.00	23.40 X 3.53	PS008-N/F-03500-02400-0420

HOUSING DESIGN PS008



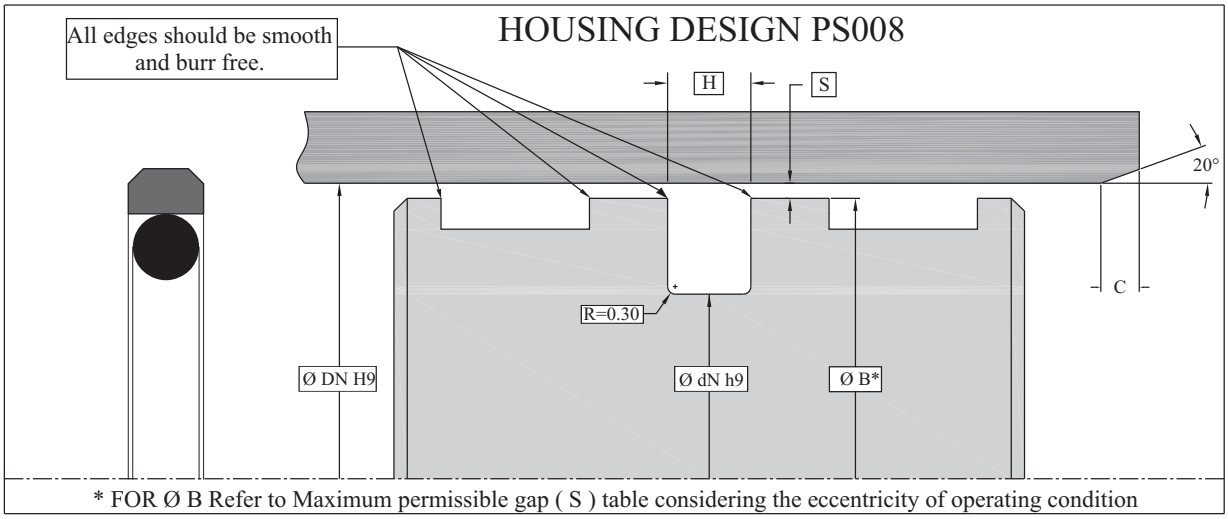
* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

ØDN (H9)	ØdN (h9)	H (+0.20)	C	O-RING SIZE	ORDER CODE For Temp range: -20°C to +100°C use PS008-N -20°C to +200°C use PS008-F
36.00	28.50	3.20	4.50	28.24 X 2.62	PS008-N/F-03600-02850-0320
36.00	25.00	4.20	6.00	24.99 X 3.53	PS008-N/F-03600-02500-0420
38.00	30.50	3.20	4.50	29.82 X 2.62	PS008-N/F-03800-03050-0320
40.00	29.00	4.20	6.00	28.17 X 3.53	PS008-N/F-04000-02900-0420
42.00	31.00	4.20	6.00	29.75 X 3.53	PS008-N/F-04200-03100-0420
44.45	36.95	3.20	6.00	36.17 X 2.62	PS008-N/F-04445-03695-0320
45.00	34.00	4.20	6.00	32.92 X 3.53	PS008-N/F-04500-03400-0420
48.00	37.00	4.20	6.00	36.09 X 3.53	PS008-N/F-04800-03700-0420
50.00	39.00	4.20	6.00	37.69 X 3.53	PS008-N/F-05000-03900-0420
50.80	39.80	4.20	6.00	38.70 X 3.53	PS008-N/F-05080-03980-0420
55.00	44.00	4.20	6.00	44.04 X 3.53	PS008-N/F-05500-04400-0420
56.00	45.00	4.20	6.00	44.04 X 3.53	PS008-N/F-05600-04500-0420
57.10	46.10	4.20	6.00	44.04 X 3.53	PS008-N/F-05710-04610-0420
60.00	49.00	4.20	6.00	47.22 X 3.53	PS008-N/F-06000-04900-0420
63.00	52.00	4.20	6.00	50.39 X 3.53	PS008-N/F-06300-05200-0420
63.00	47.50	6.30	8.00	46.99 X 5.33	PS008-N/F-06300-04750-0630
63.50	52.50	4.20	6.00	50.39 X 3.53	PS008-N/F-06350-05250-0420
65.00	54.00	4.20	6.00	53.57 X 3.53	PS008-N/F-06500-05400-0420
69.85	58.85	4.20	6.00	56.74 X 3.53	PS008-N/F-06985-05885-0420
70.00	59.00	4.20	6.00	58.33 X 3.53	PS008-N/F-07000-05900-0420
70.00	54.50	6.30	8.00	53.34 X 5.33	PS008-N/F-07000-05450-0630
75.00	64.00	4.20	6.00	63.09 X 3.53	PS008-N/F-07500-06400-0420

ROYAL STANDARD SIZE LIST					
ØDN (H9)	**ØdN (h9)**	**H (+0.20)**	**C**	**O-RING SIZE**	**ORDER CODE** For Temp range: -20°C to +100°C use PS008-N -20°C to +200°C use PS008-F
75.00	59.50	6.30	8.00	58.10 X 5.33	PS008-N/F-07500-05950-0630
76.20	65.20	4.20	6.00	63.09 X 3.53	PS008-N/F-07620-06520-0420
80.00	69.00	4.20	6.00	66.27 X 3.53	PS008-N/F-08000-06900-0420
80.00	64.50	6.30	8.00	62.87 X 5.33	PS008-N/F-08000-06450-0630
85.00	69.50	6.30	8.00	69.22 X 5.33	PS008-N/F-08500-06950-0630
88.90	73.40	6.30	8.00	72.39 X 5.33	PS008-N/F-08890-07340-0630
90.00	79.00	4.20	6.00	78.97 X 3.53	PS008-N/F-09000-07900-0420
90.00	74.50	6.30	8.00	72.39 X 5.33	PS008-N/F-09000-07450-0630
90.00	69.00	8.10	10.50	68.00 X 6.99	PS008-N/F-09000-06900-0810
95.00	84.00	4.20	6.00	82.14 X 3.53	PS008-N/F-09500-08400-0420
95.00	79.50	6.30	8.00	78.74 X 5.33	PS008-N/F-09500-07950-0630
100.00	89.00	4.20	6.00	88.49 X 3.53	PS008-N/F-10000-08900-0420
100.00	84.50	6.30	8.00	81.92 X 5.33	PS008-N/F-10000-08450-0630
100.00	79.00	8.10	10.50	78.00 X 6.99	PS008-N/F-10000-07900-0810
101.60	86.10	6.30	8.00	85.09 X 5.33	PS008-N/F-10160-08610-0630
105.00	89.50	6.30	8.00	88.27 X 5.33	PS008-N/F-10500-08950-0630
110.00	94.50	6.30	8.00	94.62 X 5.33	PS008-N/F-11000-09450-0630
110.00	89.00	8.10	10.50	88.00 X 6.99	PS008-N/F-11000-08900-0810
115.00	99.50	6.30	8.00	97.79 X 5.33	PS008-N/F-11500-09950-0630
115.00	94.00	8.10	10.50	93.00 X 6.99	PS008-N/F-11500-09400-0810
120.00	104.50	6.30	8.00	104.40 X 5.33	PS008-N/F-12000-10450-0630
120.00	99.00	8.10	10.50	98.00 X 6.99	PS008-N/F-12000-09900-0810
125.00	109.50	6.30	8.00	107.32 X 5.33	PS008-N/F-12500-10950-0630
125.00	104.00	8.10	10.50	103.00 X 6.99	PS008-N/F-12500-10400-0810
127.00	111.50	6.30	8.00	110.49 X 5.33	PS008-N/F-12700-11150-0630
130.00	114.50	6.30	8.00	113.67 X 5.33	PS008-N/F-13000-11450-0630
130.00	109.00	8.10	10.50	107.31 X 6.99	PS008-N/F-13000-10900-0810
135.00	119.50	6.30	8.00	116.84 X 5.33	PS008-N/F-13500-11950-0630
135.00	114.00	8.10	10.50	113.67 X 6.99	PS008-N/F-13500-11400-0810
140.00	124.50	6.30	8.00	123.19 X 5.33	PS008-N/F-14000-12450-0630
140.00	119.00	8.10	10.50	116.84 X 6.99	PS008-N/F-14000-11900-0810
145.00	129.50	6.30	8.00	126.37 X 5.33	PS008-N/F-14500-12950-0630
145.00	124.00	8.10	10.50	123.19 X 6.99	PS008-N/F-14500-12400-0810
150.00	134.50	6.30	8.00	132.72 X 5.33	PS008-N/F-15000-13450-0630

HOUSING DESIGN PS008



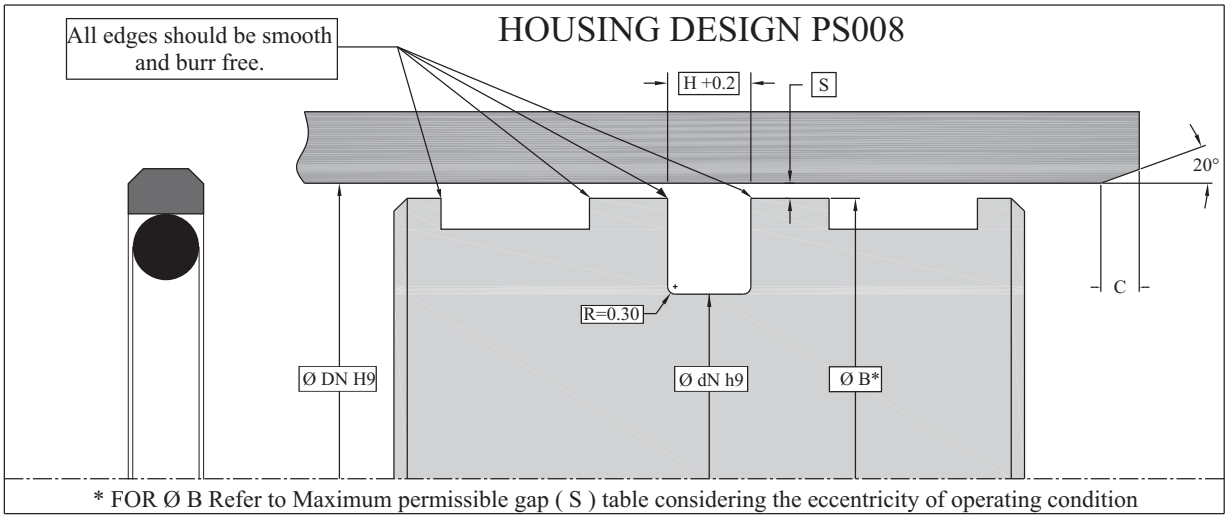
* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

ØDN (H9)	ØdN (h9)	H (+0.20)	C	O-RING SIZE	ORDER CODE For Temp range: -20°C to +100°C use PS008-N -20°C to +200°C use PS008-F
150.00	129.00	8.10	10.50	126.37 X 6.99	PS008-N/F-15000-12900-0810
152.40	131.40	8.10	10.50	129.54 X 6.99	PS008-N/F-15240-13140-0810
155.00	139.50	6.30	8.00	139.07 X 5.33	PS008-N/F-15500-13950-0630
155.00	134.00	8.10	10.50	132.72 X 6.99	PS008-N/F-15500-13400-0810
160.00	144.50	6.30	8.00	142.24 X 5.33	PS008-N/F-16000-14450-0630
160.00	139.00	8.10	10.50	139.07 X 6.99	PS008-N/F-16000-13900-0810
165.00	144.00	8.10	10.50	142.24 X 6.99	PS008-N/F-16500-14400-0810
170.00	154.50	6.30	8.00	154.48 X 5.33	PS008-N/F-17000-15450-0630
170.00	149.00	8.10	10.50	148.59 X 6.99	PS008-N/F-17000-14900-0810
175.00	159.50	6.30	8.00	158.12 X 5.33	PS008-N/F-17500-15950-0630
175.00	154.00	8.10	10.50	151.77 X 6.99	PS008-N/F-17500-15400-0810
180.00	159.00	8.10	10.50	158.12 X 6.99	PS008-N/F-18000-15900-0810
185.00	164.00	8.10	10.50	158.12 X 6.99	PS008-N/F-18500-16400-0810
190.00	169.00	8.10	10.50	164.47 X 6.99	PS008-N/F-19000-16900-0810
190.00	174.50	6.30	8.00	170.82 X 5.33	PS008-N/F-19000-17450-0630
200.00	184.50	6.30	8.00	183.52 X 5.33	PS008-N/F-20000-18450-0630
200.00	179.00	8.10	10.50	177.17 X 6.99	PS008-N/F-20000-17900-0810
210.00	189.00	8.10	10.50	183.52 X 6.99	PS008-N/F-21000-18900-0810
220.00	199.00	8.10	10.50	196.22 X 6.99	PS008-N/F-22000-19900-0810
225.00	204.00	8.10	10.50	202.57 X 6.99	PS008-N/F-22500-20400-0810
230.00	209.00	8.10	10.50	208.92 X 6.99	PS008-N/F-23000-20900-0810
240.00	219.00	8.10	10.50	215.27 X 6.99	PS008-N/F-24000-21900-0810

ROYAL STANDARD SIZE LIST					
ØDN (H9)	**ØdN (h9)**	**H (+0.20)**	**C**	**O-RING SIZE**	**ORDER CODE** For Temp range: -20°C to +100°C use PS008-N -20°C to +200°C use PS008-F
250.00	229.00	8.10	10.50	227.97 X 6.99	PS008-N/F-25000-22900-0810
260.00	239.00	8.10	10.50	236.00 X 6.99	PS008-N/F-26000-23900-0810
265.00	244.00	8.10	10.50	240.67 X 6.99	PS008-N/F-26500-24400-0810
270.00	249.00	8.10	10.50	240.67 X 6.99	PS008-N/F-27000-24900-0810
275.00	254.00	8.10	10.50	253.37 X 6.99	PS008-N/F-27500-25400-0810
280.00	259.00	8.10	10.50	253.37 X 6.99	PS008-N/F-28000-25900-0810
290.00	269.00	8.10	10.50	266.07 X 6.99	PS008-N/F-29000-26900-0810
300.00	279.00	8.10	10.50	278.77 X 6.99	PS008-N/F-30000-27900-0810
310.00	289.00	8.10	10.50	278.77 X 6.99	PS008-N/F-31000-28900-0810
320.00	299.00	8.10	10.50	291.47 X 6.99	PS008-N/F-32000-29900-0810
325.00	304.00	8.10	10.50	304.17 X 6.99	PS008-N/F-32500-30400-0810
330.00	305.50	8.10	10.50	304.17 X 6.99	PS008-N/F-33000-30550-0810
340.00	315.50	8.10	10.50	316.87 X 6.99	PS008-N/F-34000-31550-0810
350.00	325.50	8.10	10.50	316.87 X 6.99	PS008-N/F-35000-32550-0810
360.00	335.50	8.10	10.50	329.57 X 6.99	PS008-N/F-36000-33550-0810
370.00	345.50	8.10	10.50	342.27 X 6.99	PS008-N/F-37000-34550-0810
380.00	355.50	8.10	10.50	354.97 X 6.99	PS008-N/F-38000-35550-0810
390.00	365.50	8.10	10.50	354.97 X 6.99	PS008-N/F-39000-36550-0810
400.00	375.50	8.10	10.50	367.67 X 6.99	PS008-N/F-40000-37550-0810
410.00	385.50	8.10	10.50	380.37 X 6.99	PS008-N/F-41000-38550-0810
415.00	390.50	8.10	10.50	393.07 X 6.99	PS008-N/F-41500-39050-0810
420.00	395.50	8.10	10.50	393.07 X 6.99	PS008-N/F-42000-39550-0810
430.00	405.50	8.10	10.50	405.26 X 6.99	PS008-N/F-43000-40550-0810
440.00	415.50	8.10	10.50	405.26 X 6.99	PS008-N/F-44000-41550-0810
450.00	425.50	8.10	10.50	417.96 X 6.99	PS008-N/F-45000-42550-0810
460.00	435.50	8.10	10.50	430.66 X 6.99	PS008-N/F-46000-43550-0810
465.00	440.50	8.10	10.50	430.66 X 6.99	PS008-N/F-46500-44050-0810
470.00	445.50	8.10	10.50	443.36 X 6.99	PS008-N/F-47000-44550-0810
480.00	455.50	8.10	10.50	456.07 X 6.99	PS008-N/F-48000-45550-0810
500.00	475.50	8.10	10.50	468.76 X 6.99	PS008-N/F-50000-47550-0810
520.00	495.50	8.10	10.50	494.16 X 6.99	PS008-N/F-52000-49550-0810
530.00	505.50	8.10	10.50	494.16 X 6.99	PS008-N/F-53000-50550-0810
545.00	520.50	8.10	10.50	506.86 X 6.99	PS008-N/F-54500-52050-0810
550.00	525.50	8.10	10.50	506.86 X 6.99	PS008-N/F-55000-52550-0810

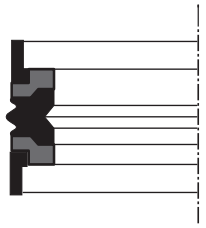
HOUSING DESIGN PS008



* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

ØDN (H9)	ØdN (h9)	H (+0.20)	C	O-RING SIZE	ORDER CODE For Temp range: -20°C to +100°C use PS008-N -20°C to +200°C use PS008-F
570.00	545.50	8.10	10.50	540.00 X 6.99	PS008-N/F-57000-54550-0810
600.00	575.50	8.10	10.50	556.00 X 6.99	PS008-N/F-60000-57550-0810
620.00	595.50	8.10	10.50	582.68 X 6.99	PS008-N/F-62000-59550-0810
630.00	605.50	8.10	10.50	600.00 X 6.99	PS008-N/F-63000-60550-0810
650.00	625.50	8.10	10.50	620.00 X 6.99	PS008-N/F-65000-62550-0810
660.00	635.50	8.10	10.50	633.48 X 6.99	PS008-N/F-66000-63550-0810
680.00	655.50	8.10	10.50	650.00 X 6.99	PS008-N/F-68000-65550-0810

PISTON SEAL	PS024	

Features:

Piston seal profile PS024 consists of five piece seal set for sealing pistons with one elastomer sealing component, two anti-extrusion support rings and two L-shape guide rings.

Composition:

Component	Sealing Ring	Backup rings	L rings
Material	Nitrile Butadiene Rubber	Polyurethane	PA6 + 40% GLASS
Royal Code	NBR78	PUDB65	PG6040

For further compound detail please refer Royal seal materials.

Properties:

- Less space required for seal and guide elements.
- Prevents distortion inside the groove.
- Single Piece Piston with one groove.
- No extrusion into the radial sealing gap
- Suitable for rapid energisation.
- Sealing component protected against twisting.
- Zero leakage.

Application:

- Mobile Hydraulics
- Injection moulding machines
- Machine tools
- Piston Accumulator.
- Agricultural equipment
- Standard cylinders

General Technical Data:

Working Pressure (max.)	315 Bar
Temperature range	-20 to 100°C
Surface Speed (max.)	0.5 m/s.

Fitting:

PS024 is very easy to fit. Careful fitting of the seal is a prerequisite for it's perfect functioning.

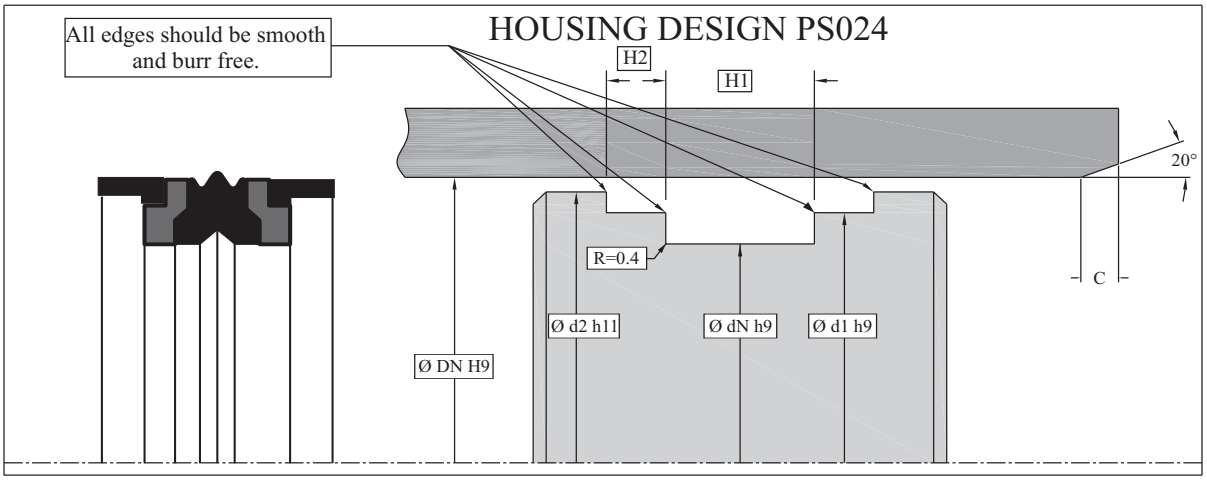
Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Ordering format:

Please mention ROYAL Order Code in your order.
Example: Size 25 X 15 X 12 Order Code is PS024-N-02500-01500-01200

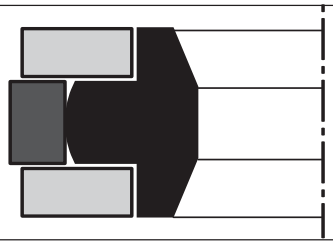
Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list



ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	d1	d2	H1 (+0.20)	H2 (+0.10)	C	ORDER CODE
25.00	15.00	21.00	23.00	12.00	4.00	4.00	PS024-N-02500-01500-1200
32.00	22.00	28.50	30.50	16.40	6.40	4.00	PS024-N-03200-02200-1640
40.00	30.00	35.40	38.50	16.40	6.40	4.00	PS024-N-04000-03000-1640
40.00	24.00	35.40	38.70	18.40	6.40	4.00	PS024-N-04000-02400-1840
50.00	34.00	46.00	49.00	20.50	3.10	4.00	PS024-N-05000-03400-2050
50.00	34.00	45.40	49.00	18.40	6.40	4.00	PS024-N-05000-03400-1840
55.00	39.00	50.40	54.00	18.40	6.40	4.00	PS024-N-05500-03900-1840
60.00	44.00	55.40	58.70	18.40	6.40	4.00	PS024-N-06000-04400-1840
60.00	44.00	56.00	59.00	20.50	3.10	4.00	PS024-N-06000-04400-2050
63.00	47.00	58.40	61.70	18.40	6.40	4.00	PS024-N-06300-04700-1840
65.00	49.00	61.00	64.00	20.50	3.10	4.00	PS024-N-06500-04900-2050
70.00	50.00	64.20	68.30	22.40	6.40	4.00	PS024-N-07000-05000-2240
70.00	58.00	66.00	69.40	20.50	4.20	4.00	PS024-N-07000-05800-2050
75.00	55.00	69.20	73.30	22.40	6.40	4.00	PS024-N-07500-05500-2240
80.00	60.00	74.20	78.30	22.40	6.40	5.00	PS024-N-08000-06000-2240
80.00	62.00	76.00	79.00	22.50	3.60	5.00	PS024-N-08000-06200-2250
85.00	65.00	79.20	83.30	22.40	6.40	5.00	PS024-N-08500-06500-2240
90.00	70.00	84.20	88.30	22.40	6.40	5.00	PS024-N-09000-07000-2240
90.00	72.00	86.00	89.00	22.50	3.60	5.00	PS024-N-09000-07200-2250
95.00	75.00	89.20	93.30	22.40	6.40	5.00	PS024-N-09500-07500-2240
100.00	75.00	93.20	98.00	22.40	6.40	5.00	PS024-N-10000-07500-2240
100.00	82.00	96.00	99.00	22.50	3.60	5.00	PS024-N-10000-08200-2250
100.00	86.00	96.00	99.40	22.50	5.20	5.00	PS024-N-10000-08600-2250
105.00	80.00	98.10	103.00	22.40	6.40	5.00	PS024-N-10500-08000-2240
110.00	85.00	103.10	108.00	22.40	6.40	5.00	PS024-N-11000-08500-2240

DN (H9)	dN (h9)	d1	d2	H1 (+0.20)	H2 (+0.10)	C	ORDER CODE
115.00	90.00	108.10	113.00	22.40	6.40	6.00	PS024-N-11500-09000-2240
120.00	95.00	113.10	118.00	22.40	6.40	6.00	PS024-N-12000-09500-2240
125.00	100.00	118.10	123.00	25.40	6.40	6.00	PS024-N-12500-10000-2540
125.00	103.00	121.00	124.00	26.50	5.10	6.00	PS024-N-12500-10300-2650
130.00	105.00	123.10	128.00	25.40	6.40	6.00	PS024-N-13000-10500-2540
130.00	105.00	122.70	127.50	25.40	9.50	6.00	PS024-N-1300-1050-0254-4450
135.00	110.00	128.10	133.00	25.40	6.40	6.00	PS024-N-13500-11000-2540
135.00	110.00	127.60	132.50	25.40	9.50	6.00	PS024-N-1350-1100-0254-4450
140.00	115.00	133.00	138.00	25.40	6.40	6.00	PS024-N-14000-11500-2540
140.00	115.00	132.60	137.50	25.40	9.50	6.00	PS024-N-1400-1150-0254-4450
140.00	118.00	136.00	139.00	26.50	5.10	6.00	PS024-N-14000-11800-2650
145.00	120.00	138.30	143.00	25.40	6.40	6.00	PS024-N-14500-12000-2540
145.00	120.00	137.90	142.50	25.40	9.50	6.00	PS024-N-1450-1200-0254-4450
150.00	125.00	143.00	148.00	25.40	6.40	6.00	PS024-N-15000-12500-2540
150.00	125.00	142.60	147.50	25.40	9.50	6.00	PS024-N-1500-1250-0254-4450
160.00	130.00	153.00	157.50	25.40	6.40	6.00	PS024-N-16000-13000-2540
160.00	135.00	152.60	157.50	25.40	9.50	6.00	PS024-N-1600-1350-0254-4450
175.00	150.00	166.70	172.10	25.40	12.70	6.00	PS024-N-17500-15000-2540
180.00	155.00	171.70	177.10	25.40	12.70	6.00	PS024-N-18000-15500-2540
200.00	175.00	191.60	197.00	25.40	12.70	6.00	PS024-N-20000-17500-2540
225.00	200.00	216.60	222.00	25.40	12.70	6.00	PS024-N-22500-20000-2540
250.00	225.00	241.60	247.00	25.40	12.70	6.00	PS024-N-25000-22500-2540
IMPERIAL SIZES							
50.80	34.92	46.23	49.53	19.05	6.35	4.00	PS024-IN-02000-01375-0750
63.50	47.63	58.95	62.23	19.05	6.35	4.00	PS024-IN-02500-01875-0750
76.20	57.15	70.46	74.68	23.80	6.35	5.00	PS024-IN-03000-02250-0937
88.90	69.85	83.16	87.38	23.80	6.35	5.00	PS024-IN-03500-02750-0937
101.60	82.55	95.83	100.08	23.80	6.35	5.00	PS024-IN-04000-03250-0937

PISTON SEAL	PS032	

Features:

Piston seal profile PS032 consists of four piece seal set with one main sealing component, one T shaped energizer ring and two anti-extrusion ring which support the seal on both sides and prevent contamination to the energizer and main seal ring.

Composition:

Component	Sealing Ring	Backup Rings	Energizer T Ring
Material	Modified PTFE+Bronze+Glass+MoS2	Polyamide	Nitrile Butadiene Rubber(NBR)
Royal Code	TG20B20M	PA6600	NBR80

For further compound detail please refer Royal seal materials.

Properties:

- Double acting
- Allow large radial gaps due to backup rings
- Very good extrusion reliability
- High resistance to abrasion
- Low friction, free of stick-slip
- High contact pressure through rubber profile ring
- No extrusion into the radial sealing gap
-

Application:

- Mobile Hydraulics
- Machine tools
- Agricultural equipment
- Standard cylinders

General technical data:

Working Pressure (max.)	500 Bar
Temperature range	-20°C to 100°C
Surface Speed (max.)	1.5 m/s

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Gap dimension:

The largest gap dimension appearing in operation on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the minimum clearance Ø and max bore Ø.

Profile dimension	Maximum permissible gap (S)				
H	160 bar	260 bar	320 bar	400 bar	500 bar
8.00	0.60	0.50	0.40	0.36	0.30
9.00	0.80	0.70	0.50	0.40	0.32
10.77 / 11.00	1.20	1.00	0.65	0.50	0.40
12.50 / 12.70	1.20	1.00	0.65	0.50	0.40
14.71 / 16.00 / 16.26	1.80	1.40	0.90	0.70	0.50
17.50 / 19.05	1.80	1.40	0.90	0.70	0.50

Fitting :

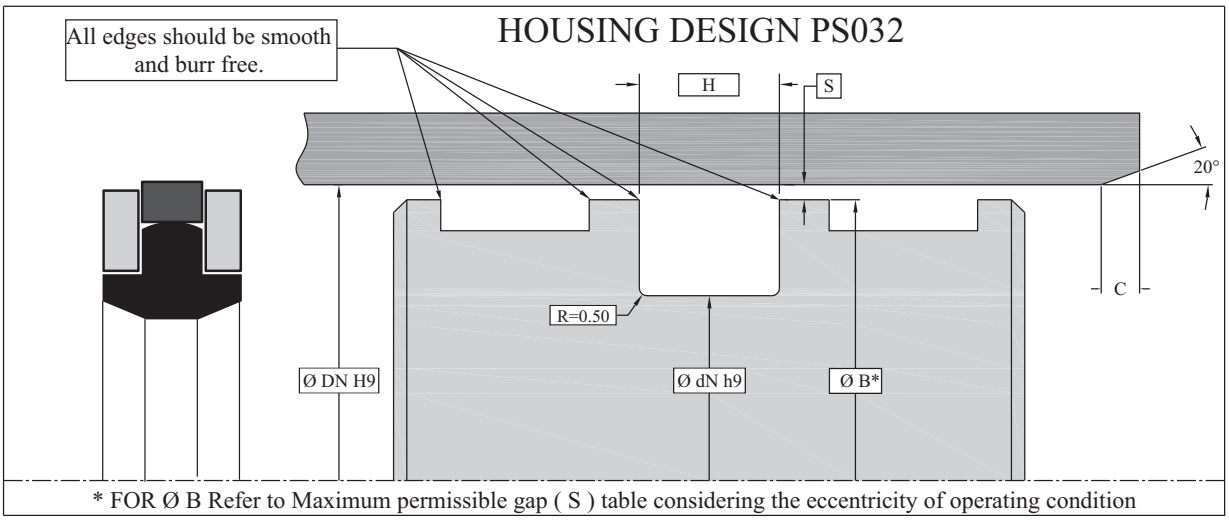
PS032 is very easy to fit. Careful fitting of the seal is a prerequisite for it's perfect functioning.

Ordering format :

Follow Royal Order Code No in Dimension chart.
for example 50 X 36 X 9 Royal Code is PS032-N-05000-03600-0900

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN PS032

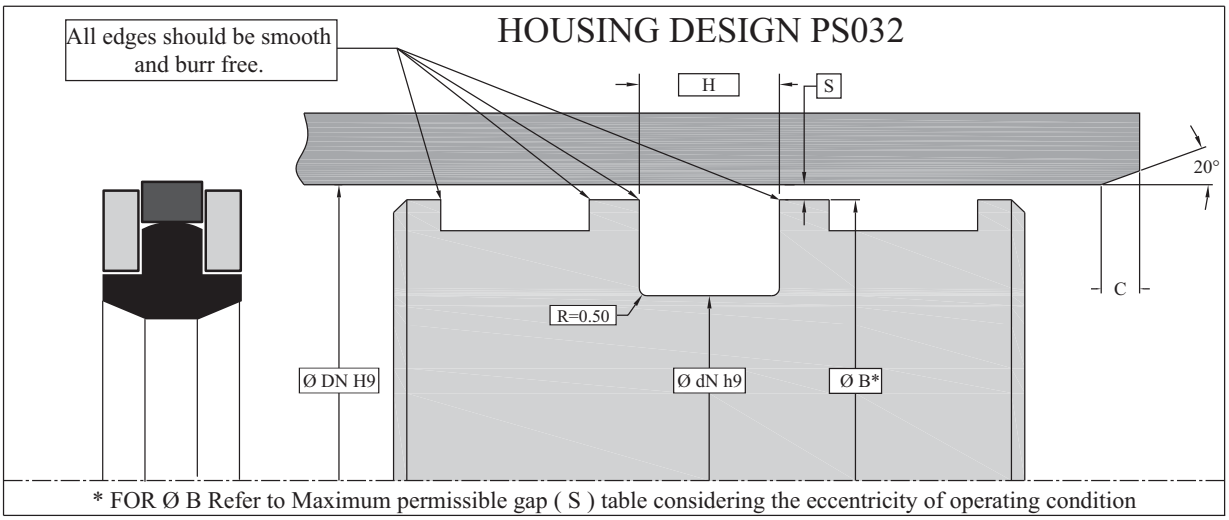


* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)	C	ORDER CODE
50.00	36.00	9.00	4.00	PS032-N-05000-03600-0900
50.00	40.00	8.00	4.00	PS032-N-05000-04000-0800
55.00	41.00	9.00	4.00	PS032-N-05500-04100-0900
60.00	46.00	9.00	4.00	PS032-N-06000-04600-0900
63.00	48.00	11.00	5.00	PS032-N-06300-04800-1100
63.00	53.00	8.00	5.00	PS032-N-06300-05300-0800
65.00	50.00	11.00	5.00	PS032-N-06500-05000-1100
70.00	55.00	11.00	5.00	PS032-N-07000-05500-1100
70.00	60.00	8.00	5.00	PS032-N-07000-06000-0800
75.00	60.00	11.00	5.00	PS032-N-07500-06000-1100
80.00	65.00	11.00	5.00	PS032-N-08000-06500-1100
80.00	65.00	12.00	5.00	PS032-N-08000-06500-1200
80.00	65.00	12.50	5.00	PS032-N-08000-06500-1250
85.00	70.00	11.00	5.00	PS032-N-08500-07000-1100
90.00	75.00	11.00	5.00	PS032-N-09000-07500-1100
90.00	75.00	12.00	5.00	PS032-N-09000-07500-1200
90.00	75.00	12.50	5.00	PS032-N-09000-07500-1250
95.00	80.00	11.00	5.00	PS032-N-09500-08000-1100
95.00	80.00	12.50	5.00	PS032-N-09500-08000-1250
100.00	85.00	12.50	6.50	PS032-N-10000-08500-1250
105.00	90.00	12.50	6.50	PS032-N-10500-09000-1250
110.00	95.00	12.50	6.50	PS032-N-11000-09500-1250
115.00	100.00	12.50	6.50	PS032-N-11500-10000-1250
120.00	100.00	16.00	6.50	PS032-N-12000-10000-1600

ROYAL STANDARD SIZE LIST				
DN (H9)	**dN (h9)**	**H (+0.20)**	**C**	**ORDER CODE**
120.00	105.00	12.50	6.50	PS032-N-12000-10500-1250
125.00	102.00	16.00	6.50	PS032-N-12500-10200-1600
125.00	105.00	16.00	6.50	PS032-N-12500-10500-1600
125.00	108.00	16.00	6.50	PS032-N-12500-10800-1600
130.00	105.00	25.00	6.50	PS032-N-13000-10500-2500
130.00	107.00	16.00	6.50	PS032-N-13000-10700-1600
130.00	110.00	16.00	6.50	PS032-N-13000-11000-1600
135.00	112.00	16.00	6.50	PS032-N-13500-11200-1600
140.00	117.00	16.00	6.50	PS032-N-14000-11700-1600
140.00	120.00	16.00	6.50	PS032-N-14000-12000-1600
145.00	122.00	16.00	6.50	PS032-N-14500-12200-1600
150.00	127.00	16.00	6.50	PS032-N-15000-12700-1600
155.00	132.00	16.00	6.50	PS032-N-15500-13200-1600
160.00	137.00	16.00	6.50	PS032-N-16000-13700-1600
160.00	140.00	16.00	6.50	PS032-N-16000-14000-1600
165.00	142.00	16.00	6.50	PS032-N-16500-14200-1600
170.00	147.00	16.00	6.50	PS032-N-17000-14700-1600
175.00	152.00	16.00	6.50	PS032-N-17500-15200-1600
180.00	157.00	16.00	6.50	PS032-N-18000-15700-1600
180.00	160.00	16.00	6.50	PS032-N-18000-16000-1600
180.00	165.00	12.50	6.50	PS032-N-18000-16500-1250
185.00	162.00	16.00	6.50	PS032-N-18500-16200-1600
190.00	167.00	16.00	6.50	PS032-N-19000-16700-1600
195.00	172.00	16.00	6.50	PS032-N-19500-17200-1600
195.00	182.00	16.00	6.50	PS032-N-19500-18200-1600
200.00	177.00	16.00	6.50	PS032-N-20000-17700-1600
210.00	187.00	16.00	6.50	PS032-N-21000-18700-1600
215.00	192.00	16.00	6.50	PS032-N-21500-19200-1600
220.00	197.00	16.00	6.50	PS032-N-22000-19700-1600
225.00	202.00	16.00	6.50	PS032-N-22500-20200-1600
230.00	207.00	16.00	6.50	PS032-N-23000-20700-1600
240.00	217.00	16.00	6.50	PS032-N-24000-21700-1600
250.00	222.00	17.50	7.50	PS032-N-25000-22200-1750
260.00	232.00	17.50	7.50	PS032-N-26000-23200-1750
270.00	242.00	17.50	7.50	PS032-N-27000-24200-1750
280.00	252.00	17.50	7.50	PS032-N-28000-25200-1750

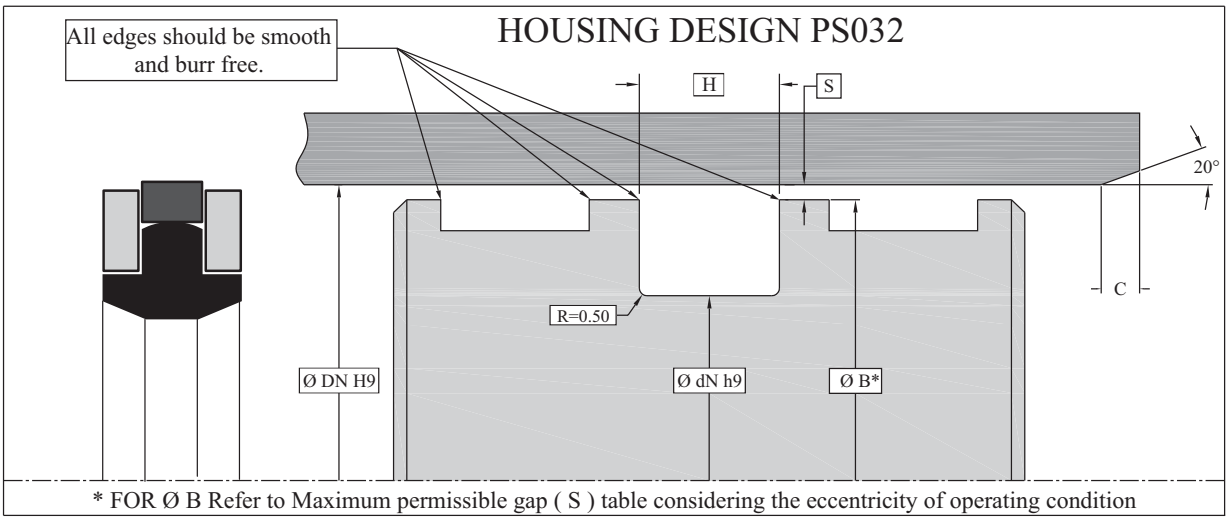


ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)	C	ORDER CODE
300.00	272.00	17.50	7.50	PS032-N-30000-27200-1750
300.00	275.00	19.00	7.50	PS032-N-30000-27500-1900
320.00	292.00	17.50	7.50	PS032-N-32000-29200-1750
350.00	322.00	17.50	7.50	PS032-N-35000-32200-1750
360.00	332.00	17.50	7.50	PS032-N-36000-33200-1750
380.00	352.00	17.50	7.50	PS032-N-38000-35200-1750
IMPERIAL SIZES				
25.40	15.88	10.77	4.00	PS032-IN-N-01000-00625-0424
31.75	22.22	10.77	4.00	PS032-IN-N-01250-00875-0424
38.10	28.58	10.77	4.00	PS032-IN-N-01500-01125-0424
44.45	34.92	10.77	4.00	PS032-IN-N-01750-01375-0424
50.80	41.28	10.77	4.00	PS032-IN-N-02000-01625-0424
57.15	47.63	10.77	4.00	PS032-IN-N-02250-01875-0424
63.50	53.97	10.77	5.00	PS032-IN-N-02500-02125-0424
69.85	60.32	10.77	5.00	PS032-IN-N-02750-02375-0424
76.20	62.03	10.77	5.00	PS032-IN-N-03000-02442-0424
76.20	64.01	14.71	5.00	PS032-IN-N-03000-02520-0579
82.55	70.36	14.71	5.00	PS032-IN-N-03250-02770-0579
88.90	76.71	14.71	5.00	PS032-IN-N-03500-03020-0579
95.25	83.06	14.71	5.00	PS032-IN-N-03750-03270-0579
101.60	87.43	10.77	6.50	PS032-IN-N-04000-03442-0424
101.60	89.41	14.71	6.50	PS032-IN-N-04000-03520-0579
107.95	95.76	14.71	6.50	PS032-IN-N-04250-03770-0579

ROYAL STANDARD SIZE LIST				
DN (H9)	**dN (h9)**	**H (+0.20)**	**C**	**ORDER CODE**
114.30	97.28	12.70	6.50	PS032-IN-N-04500-03830-0500
114.30	102.11	14.71	6.50	PS032-IN-N-04500-04020-0579
120.65	108.46	14.71	6.50	PS032-IN-N-04750-04270-0579
127.00	107.19	16.26	6.50	PS032-IN-N-05000-04220-0640
127.00	108.46	19.05	6.50	PS032-IN-N-05000-04270-0750
127.00	112.88	10.77	6.50	PS032-IN-N-05000-04444-0424
133.35	114.81	19.05	6.50	PS032-IN-N-05250-04520-0750
139.70	119.89	16.26	6.50	PS032-IN-N-05500-04720-0640
139.70	121.16	19.05	6.50	PS032-IN-N-05500-04770-0750
146.05	127.56	19.05	6.50	PS032-IN-N-05750-05022-0750
152.40	129.29	16.26	6.50	PS032-IN-N-06000-05090-0640
152.40	133.86	19.05	6.50	PS032-IN-N-06000-05270-0750
158.75	135.64	16.26	6.50	PS032-IN-N-06250-05340-0640
158.75	140.21	19.05	6.50	PS032-IN-N-06250-05520-0750
165.10	141.99	16.26	6.50	PS032-IN-N-06500-05590-0640
165.10	146.56	19.05	6.50	PS032-IN-N-06500-05770-0750
171.45	152.96	19.05	6.50	PS032-IN-N-06750-06022-0750
177.80	154.69	16.26	6.50	PS032-IN-N-07000-06090-0640
177.80	159.26	19.05	6.50	PS032-IN-N-07000-06270-0750
184.15	161.04	16.26	6.50	PS032-IN-N-07250-06340-0640
184.15	165.61	19.05	6.50	PS032-IN-N-07250-06520-0750
190.50	167.39	16.26	6.50	PS032-IN-N-07500-06590-0640
190.50	171.96	19.05	6.50	PS032-IN-N-07500-06770-0750
196.85	173.74	16.26	6.50	PS032-IN-N-07750-06840-0640
196.85	178.36	19.05	6.50	PS032-IN-N-07750-07022-0750
203.20	184.66	19.05	6.50	PS032-IN-N-08000-07270-0750
209.55	184.15	16.26	6.50	PS032-IN-N-08250-07250-0640
209.55	191.01	19.05	6.50	PS032-IN-N-08250-07520-0750
215.90	190.50	16.26	6.50	PS032-IN-N-08500-07500-0640
215.90	197.36	19.05	6.50	PS032-IN-N-08500-07770-0750
222.25	203.76	19.05	6.50	PS032-IN-N-08750-08022-0750
228.60	210.06	19.05	6.50	PS032-IN-N-09000-08270-0750
254.00	225.55	17.53	7.50	PS032-IN-N-10000-08880-0690
254.00	235.46	19.05	7.50	PS032-IN-N-10000-09270-0750
260.35	231.90	17.53	7.50	PS032-IN-N-10250-09130-0690
266.70	238.25	17.53	7.50	PS032-IN-N-10500-09380-0690

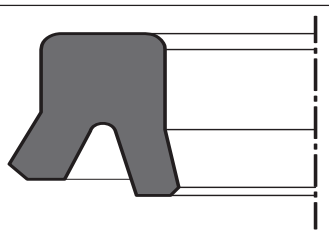
HOUSING DESIGN PS032



* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)	C	ORDER CODE
279.40	250.95	17.53	7.50	PS032-IN-N-11000-09880-0690
279.40	260.86	19.05	7.50	PS032-IN-N-11000-10270-0750
292.10	263.65	17.53	7.50	PS032-IN-N-11500-10380-0690
304.80	286.26	19.05	7.50	PS032-IN-N-12000-11270-0750
330.20	301.75	17.53	7.50	PS032-IN-N-13000-11880-0690
355.60	337.06	19.05	7.50	PS032-IN-N-14000-13270-0750

PISTON SEAL	PS040	

Features:

Piston seal profile PS040 is a single acting U cup type seal which has a very good static and dynamic tightness. The seal can be used for double acting applications by fitting back to back in separate grooves. The outer dynamic lip is more robust to provide improved sealing and compression set characteristics. The seal is manufactured in two different material, PUET93 Polyurethane and FPM80 Fluorocarbon Rubber for different application temperature range.

Composition:

Seal Profile type	PS040-U	PS040-F
Material	Polyurethane	Fluorocarbon Rubber
Royal Code	PUET93	FPM80

For compound detail please refer Royal seal materials.

Properties:

- High resistance to abrasion
- Easy installation

Application:

- Mobile Hydraulics
- Injection molding machines
- Machine tools
- Agricultural equipment
- Standard cylinders

General Technical Data:

Seal Profile	PS040-U	PS040-F
Temperature range	-30°C to +110°C	-20°C to +200°C
Working Pressure(max.)	400 bar *	160 bar
Surface Speed(max.)	0.5 m/s	0.5 m/s

* Maximum working pressure can be extended for PS040-U up to 400 bar and for PS040-F up to 250 bar with the use of a back up ring. If this option is required ask ROYAL.

Surface finish:

Roughness depth	R(max.)	Ra
Sliding surface	≤ 2,5 µm	0,05-0,3 µm
Bottom of groove	≤ 6,3 µm	≤1,6 µm
Sides of groove	≤ 15 µm	≤ 3 µm

Gap dimension:

The largest gap dimension appearing in operation on the non-pressurized side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the minimum clearance Ø and max bore Ø.

Profile Type	Maximum permissible gap (S)				
	160 bar	260 bar	320 bar	400 bar	
PS040-U	0.5	0.4	Ask ROYAL	Ask ROYAL	
PS040-F	0.4	Ask ROYAL	-	-	

Fitting:

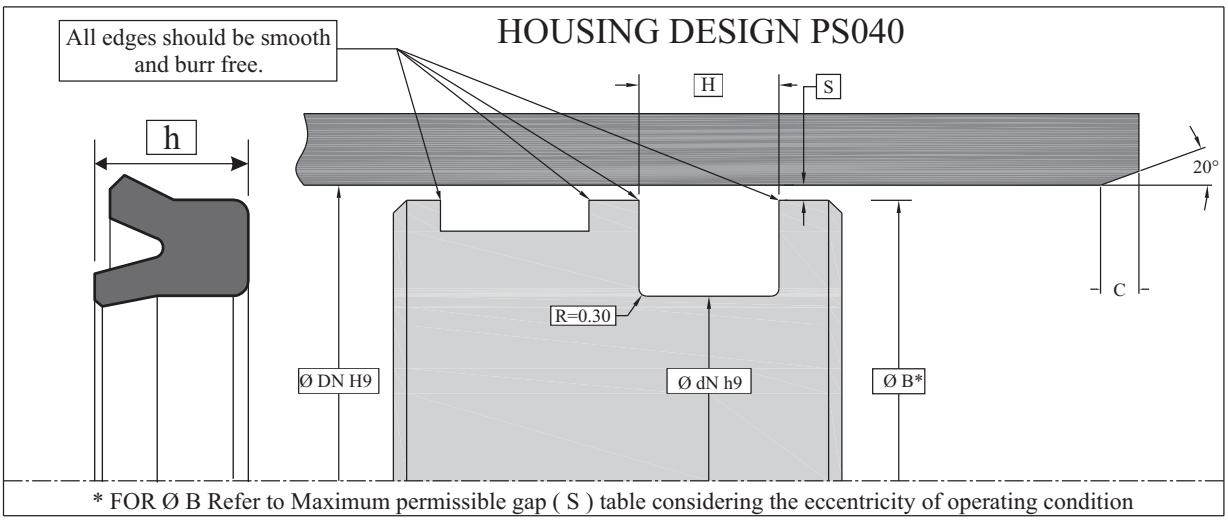
PS040 is very easy to fit. Careful fitting of the seal is a prerequisite for it's perfect functioning.

Ordering format:

Please mention ROYAL order code in your order.
Example: Size 16 X 8 X 6.3
For Temp range: -30°C to +110°C Order code is PS040-U-01600-00800-0630
For Temp range: -20°C to +200°C Order code is PS040-F-01600-00800-0630

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN PS040



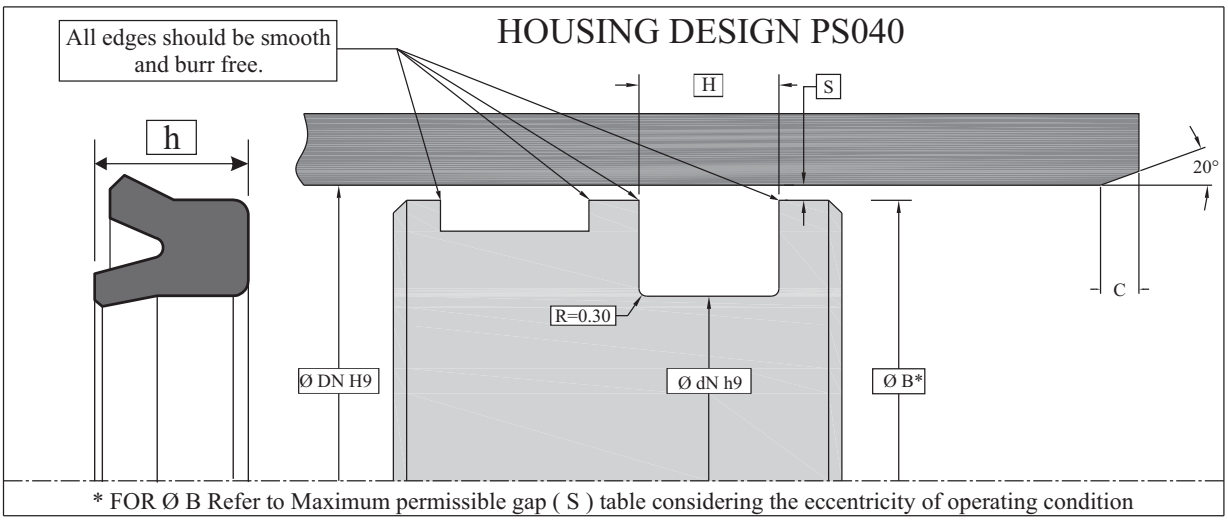
* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	h	H (+0.20)	**ORDER CODE** For temp. range -30 to +110°C use PS040-U -20 to +200°C use PS040-F*
16.00	8.00	5.70	6.30	PS040-U-01600-00800-0630
18.00	8.00	7.30	8.00	PS040-U-01800-00800-0800
20.00	12.00	5.70	6.30	PS040-U-02000-01200-0630
20.00	12.00	7.90	8.70	PS040-U-02000-01200-0870
22.00	14.00	5.70	6.30	PS040-U-02200-01400-0630
22.00	14.00	9.10	10.00	PS040-U-02200-01400-1000
25.00	17.00	5.70	6.30	PS040-U-02500-01700-0630
26.00	18.00	5.70	6.30	PS040-U-02600-01800-0630
28.00	20.00	5.70	6.30	PS040-U-02800-02000-0630
28.00	22.00	3.60	4.00	PS040-U-02800-02200-0400
32.00	24.00	5.70	6.30	PS040-U-03200-02400-0630
35.00	20.00	6.80	7.50	PS040-U-03500-02000-0750
35.00	25.00	7.30	8.00	PS040-U-03500-02500-0800
36.00	26.00	7.30	8.00	PS040-U-03600-02600-0800
40.00	25.00	10.00	11.00	PS040-U-04000-02500-1100
40.00	28.00	10.00	11.00	PS040-U-04000-02800-1100
40.00	30.00	7.30	8.00	PS040-U-04000-03000-0800
40.00	32.00	7.30	8.00	PS040-U-04000-03200-0800
40.00	30.00	10.00	11.00	PS040-U-04000-03000-1100
40.00	32.00	5.70	6.30	PS040-U-04000-03200-0630
45.00	30.00	10.00	11.00	PS040-U-04500-03000-1100
45.00	35.00	7.30	8.00	PS040-U-04500-03500-0800

* Please contact ROYAL for available size list of PS040 - F at info@royalcomposites.co.in

HOUSING DESIGN PS040



* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

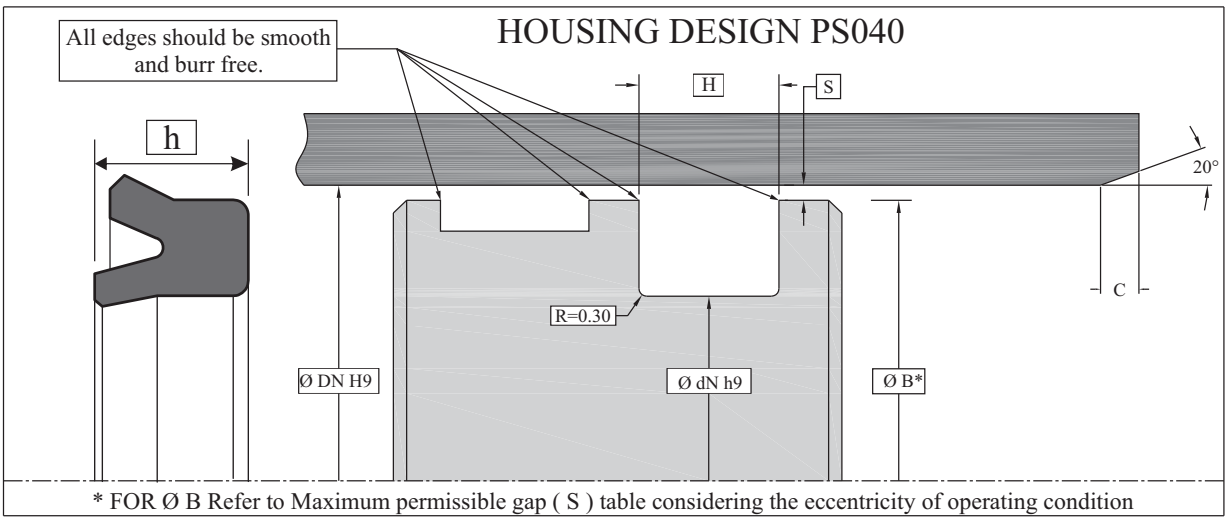
DN (H9)	dN (h9)	h	H (+0.20)	**ORDER CODE** For temp. range -30 to +110°C use PS040-U -20 to +200°C use PS040-F*
50.00	35.00	9.10	10.00	PS040-U-05000-03500-1000
50.00	36.00	10.00	11.00	PS040-U-05000-03600-1100
50.00	38.00	7.30	8.00	PS040-U-05000-03800-0800
50.00	40.00	7.30	8.00	PS040-U-05000-04000-0800
50.00	35.00	7.30	8.00	PS040-U-05000-03500-0800
50.00	35.00	11.80	13.00	PS040-U-05000-03500-1300
50.80	28.57	11.60	12.70	PS040-U-05080-02857-1270
60.00	40.00	9.10	10.00	PS040-U-06000-04000-1000
60.00	45.00	10.00	11.00	PS040-U-06000-04500-1100
60.00	50.00	7.30	8.00	PS040-U-06000-05000-0800
63.00	43.00	11.80	13.00	PS040-U-06300-04300-1300
63.00	45.00	10.00	11.00	PS040-U-06300-04500-1100
63.00	48.00	11.40	12.50	PS040-U-06300-04800-1250
63.00	50.00	7.30	8.00	PS040-U-06300-05000-0800
63.00	53.00	7.30	8.00	PS040-U-06300-05300-0800
63.00	53.00	9.10	10.00	PS040-U-06300-05300-1000
63.00	48.00	10.00	11.00	PS040-U-06300-04800-1100
65.00	45.00	10.00	11.00	PS040-U-06500-04500-1100
70.00	50.00	11.80	13.00	PS040-U-07000-05000-1300
70.00	55.00	10.00	11.00	PS040-U-07000-05500-1100
70.00	55.00	13.20	14.50	PS040-U-07000-05500-1450
70.00	58.00	13.20	14.50	PS040-U-07000-05800-1450

* Please contact ROYAL for available size list of PS040 - F at info@royalcomposites.co.in

ROYAL STANDARD SIZE LIST				
DN (H9)	**dN (h9)**	**h**	**H (+0.20)**	**ORDER CODE** For temp. range -30 to +110°C use PS040-U -20 to +200°C use PS040-F*
75.00	55.00	11.40	12.50	PS040-U-07500-05500-1250
75.00	63.00	8.20	9.00	PS040-U-07500-06300-0900
75.00	45.00	13.70	15.00	PS040-U-07500-04500-1500
75.00	63.00	8.60	9.50	PS040-U-07500-06300-0950
76.20	60.53	12.30	13.50	PS040-U-07620-06053-1350
78.00	54.30	11.40	12.50	PS040-U-07800-05430-1250
80.00	60.00	11.80	13.00	PS040-U-08000-06000-1300
80.00	65.00	8.20	9.00	PS040-U-08000-06500-0900
80.00	65.00	11.80	13.00	PS040-U-08000-06500-1300
80.00	68.00	8.60	9.50	PS040-U-08000-06800-0950
80.00	70.00	7.30	8.00	PS040-U-08000-07000-0800
80.00	72.00	10.90	12.00	PS040-U-08000-07200-1200
80.00	65.00	10.00	11.00	PS040-U-08000-06500-1100
80.00	68.00	13.30	14.60	PS040-U-08000-06800-1460
85.00	65.00	9.10	10.00	PS040-U-08500-06500-1000
90.00	70.00	11.80	13.00	PS040-U-09000-07000-1300
90.00	75.00	10.00	11.00	PS040-U-09000-07500-1100
90.00	78.00	8.60	9.50	PS040-U-09000-07800-0950
90.00	80.00	12.70	14.00	PS040-U-09000-08000-1400
90.00	78.00	7.70	8.50	PS040-U-09000-07800-0850
90.00	80.00	7.30	8.00	PS040-U-09000-08000-0800
100.00	75.00	18.20	20.00	PS040-U-10000-07500-2000
100.00	80.00	11.80	13.00	PS040-U-10000-08000-1300
100.00	80.00	12.70	14.00	PS040-U-10000-08000-1400
100.00	88.00	8.60	9.50	PS040-U-10000-08800-0950
100.00	90.00	9.60	10.50	PS040-U-10000-09000-1050
100.00	85.00	9.10	10.00	PS040-U-10000-08500-1000
100.00	90.00	6.80	7.50	PS040-U-10000-09000-0750
101.60	83.60	12.70	14.00	PS040-U-10160-08360-1400
105.00	95.00	11.80	13.00	PS040-U-10500-09500-1300
107.95	92.46	12.30	13.50	PS040-U-10795-09246-1350
110.00	85.00	18.20	20.00	PS040-U-11000-08500-2000
110.00	75.00	14.60	16.00	PS040-U-11000-07500-1600
110.00	90.00	11.80	13.00	PS040-U-11000-09000-1300

* Please contact ROYAL for available size list of PS040 - F at info@royalcomposites.co.in

HOUSING DESIGN PS040



* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

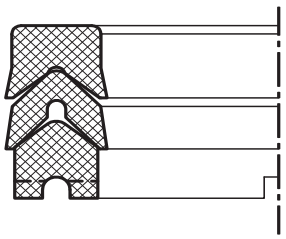
ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	h	H (+0.20)	**ORDER CODE** For temp. range -30 to +110°C use PS040-U -20 to +200°C use PS040-F*
110.00	94.00	18.20	20.00	PS040-U-11000-09400-2000
110.00	95.00	11.80	13.00	PS040-U-11000-09500-1300
110.00	98.00	13.70	15.00	PS040-U-11000-09800-1500
115.00	95.00	11.80	13.00	PS040-U-11500-09500-1300
120.00	100.00	11.80	13.00	PS040-U-12000-10000-1300
125.00	100.00	11.80	13.00	PS040-U-12500-10000-1300
125.00	100.00	14.60	16.00	PS040-U-12500-10000-1600
125.00	105.00	14.60	16.00	PS040-U-12500-10500-1600
125.00	103.00	14.60	16.00	PS040-U-12500-10300-1600
125.00	110.00	9.10	10.00	PS040-U-12500-11000-1000
130.00	110.00	11.80	13.00	PS040-U-13000-11000-1300
140.00	115.00	14.60	16.00	PS040-U-14000-11500-1600
140.00	120.00	14.60	16.00	PS040-U-14000-12000-1600
140.00	125.00	9.10	10.00	PS040-U-14000-12500-1000
140.00	128.00	14.60	16.00	PS040-U-14000-12800-1600
150.00	125.00	11.80	13.00	PS040-U-15000-12500-1300
150.00	125.00	14.60	16.00	PS040-U-15000-12500-1600
150.00	130.00	14.60	16.00	PS040-U-15000-13000-1600
150.00	135.00	10.90	12.00	PS040-U-15000-13500-1200
160.00	140.00	14.60	16.00	PS040-U-16000-14000-1600
160.00	140.00	11.40	12.50	PS040-U-16000-14000-1250
170.00	150.00	14.60	16.00	PS040-U-17000-15000-1600

* Please contact ROYAL for available size list of PS040 - F at info@royalcomposites.co.in

ROYAL STANDARD SIZE LIST				
DN (H9)	**dN (h9)**	**h**	**H (+0.20)**	**ORDER CODE** For temp. range -30 to +110°C use PS040-U -20 to +200°C use PS040-F*
180.00	160.00	14.60	16.00	PS040-U-18000-16000-1600
180.00	165.00	17.30	19.00	PS040-U-18000-16500-1900
200.00	170.00	18.20	20.00	PS040-U-20000-17000-2000
200.00	180.00	14.60	16.00	PS040-U-20000-18000-1600
250.00	234.00	17.30	19.00	PS040-U-25000-23400-1900
300.00	284.00	17.30	19.00	PS040-U-30000-28400-1900
IMPERIAL SIZE				
38.10	25.40	9.50	10.49	PS040-U-01500-01000-0413
44.45	31.75	9.50	10.49	PS040-U-01750-01250-0413
50.80	31.75	10.10	11.10	PS040-U-02000-01250-0437
50.80	38.10	9.50	10.49	PS040-U-02000-01500-0413
57.15	44.45	9.50	10.49	PS040-U-02250-01750-0413
63.50	50.80	9.50	10.49	PS040-U-02500-02000-0413
69.85	57.15	9.50	10.49	PS040-U-02750-02250-0413
76.20	63.50	9.50	10.49	PS040-U-03000-02500-0413
82.55	69.85	9.50	10.49	PS040-U-03250-02750-0413
88.90	76.20	9.50	10.49	PS040-U-03500-03000-0413
95.25	82.55	9.50	10.49	PS040-U-03750-03250-0413
101.60	88.90	9.50	10.49	PS040-U-04000-03500-0413
107.95	95.25	9.50	10.49	PS040-U-04250-03750-0413
114.30	101.60	9.50	10.49	PS040-U-04500-04000-0413
120.65	107.95	9.50	10.49	PS040-U-04750-04250-0413
127.00	114.30	9.50	10.49	PS040-U-05000-04500-0413
133.35	120.65	9.50	10.49	PS040-U-05250-04750-0413
139.70	127.00	9.50	10.49	PS040-U-05500-05000-0413

* Please contact ROYAL for available size list of PS040 - F at info@royalcomposites.co.in

PISTON SEAL	PS048	

Features:

Pistons seal profile PS048 is a seal set consist of one or more chevron ring with a header and pressure ring.

Composition:

Seal Profile	PS048-N		PS048-F	
	Material	Royal Code	Material	Royal Code
Header ring	Fabric+ NBR	NBRCF	Fabric + FPM	FPMCF
Chevron ring(fabric)	Fabric+ NBR	NBRCF	Fabric + FPM	FPMCF
Pressure ring type-1	Fabric + NBR	NBRCF	Fabric + FPM	FPMCF
Pressure ring type-2	Polyamide	PA6000	-	-

For further compound detail please refer Royal seal materials.(Section III)

Properties:

- High resistance to extrusion due to fabric reinforced header ring
- Easy installation
- Excellent sealing function
- Long working life in extreme conditions

Application:

- Mobile Hydraulics
- Machine tools
- Agricultural equipment
- standard cylinders
- hydraulic presses
- iron and steel industry
- marine hydraulics

General Technical Data:

Seal Profile	PS048-U	PS048-F
Temperature range	-20°C to +100°C	-15°C to +140 °C
Working Pressure(max.)	400 bar	400 bar
Surface Speed(max.)	0.5 m/s.	0.5 m/s.

Surface finish:

Roughness depth	R(max.)	Ra
Sliding surface	≤ 2,5 µm	0,05-0,3 µm
Bottom of groove	≤ 6,3 µm	≤ 1,6 µm
Sides of groove	≤ 15 µm	≤ 3 µm

Gap dimension:

The largest gap dimension appearing in operation on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the minimum clearance Ø and max bore Ø.

Maximum permissible gap dimension (s)			
160 bar	260 bar	320 bar	400 bar
0.65	0.45	0.40	0.35

Fitting:

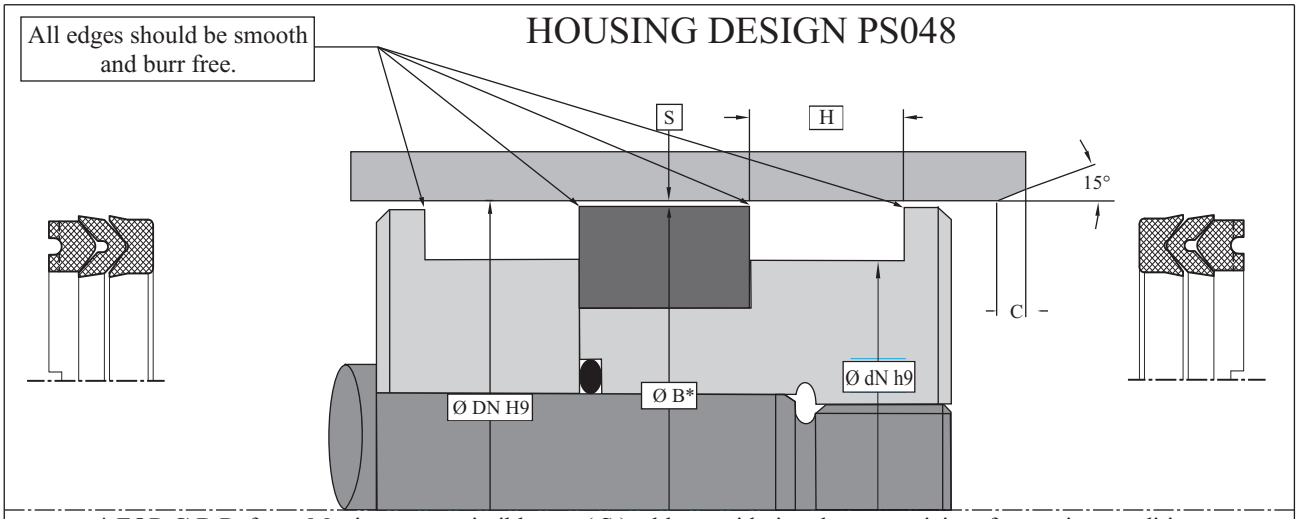
PS048 is very easy to fit. Careful fitting of the seal is a prerequisite for it's perfect functioning.

Ordering format:

Please mention ROYAL order code in your order.
Example: Size 40 X 25 X 11.5
For Temp range: -20°C to +100°C Order code is PS048-N-04000-02500-1150
For Temp range: -15°C to +140°C Order code is PS048-F-04000-02500-1150

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN PS048

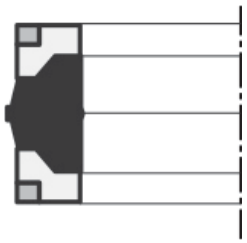


* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

Ø DN (H9)	Ø dN (h9)	Ø B	H (+0.20)	C	ORDER CODE For Temp range: -20°C to +100°C use PS048-N -15°C to +140°C use PS048-F
40.00	25.00	39.00	11.50	5.00	PS048-N/F-04000-02500-1150
45.00	30.00	44.00	11.50	5.00	PS048-N/F-04500-03000-1150
50.00	35.00	49.00	11.50	5.00	PS048-N/F-05000-03500-1150
56.00	41.00	55.00	11.50	5.00	PS048-N/F-05600-04100-1150
60.00	45.00	59.00	11.50	5.00	PS048-N/F-06000-04500-1150
63.00	48.00	62.00	13.00	5.00	PS048-N/F-06300-04800-1300
70.00	50.00	69.00	15.20	7.50	PS048-N/F-07000-05000-1520
80.00	60.00	79.00	15.20	7.50	PS048-N/F-08000-06000-1520
90.00	70.00	89.00	21.20	7.50	PS048-N/F-09000-07000-2120
100.00	80.00	99.00	21.20	7.50	PS048-N/F-10000-08000-2120
110.00	90.00	109.00	21.20	7.50	PS048-N/F-11000-09000-2120
115.00	95.00	114.00	21.20	7.50	PS048-N/F-11500-09500-2120
125.00	100.00	124.00	25.80	10.00	PS048-N/F-12500-10000-2580
140.00	115.00	139.00	25.80	10.00	PS048-N/F-14000-11500-2580
150.00	120.00	148.50	29.00	10.00	PS048-N/F-15000-12000-2900
160.00	130.00	158.50	29.00	10.00	PS048-N/F-16000-13000-2900
180.00	150.00	178.50	31.50	10.00	PS048-N/F-18000-15000-3150
200.00	170.00	198.50	33.50	10.00	PS048-N/F-20000-17000-3350
225.00	195.00	223.50	33.50	10.00	PS048-N/F-22500-19500-3350
250.00	220.00	248.50	33.50	10.00	PS048-N/F-25000-22000-3350
280.00	250.00	278.50	33.50	10.00	PS048-N/F-28000-25000-3350
300.00	270.00	298.00	33.50	10.00	PS048-N/F-30000-27000-3350

Please contact ROYAL at info@royalcomposites.co.in if your required size is not mentioned in Royal standard size list.

PISTON SEAL	PS056	

Features:

Piston seal profile PS056 consists of five piece seal set for sealing pistons with one elastomer sealing component, two support rings and two anti-extrusion rings.

Composition:

Component	Sealing Ring	Backup rings	AE rings
Material	Nitrile Butadiene Rubber	Polyurethane	PA6,6
Royal Code	NBR78	PUDB65	PA6600

For further compound detail please refer Royal seal materials.

Properties:

- Less space required for seal and guide elements.
- Prevents distortion inside the groove.
- Single Piece Piston with one groove.
- No extrusion into the radial sealing gap
- Suitable for rapid energisation.
- Sealing component protected against twisting.
- Zero leakage.

Application:

- Mobile Hydraulics
- Injection moulding machines
- Machine tools
- Piston Accumulator.
- Agricultural equipment
- Standard cylinders

General Technical Data:

Working Pressure (max.)	315 Bar
Temperature range	-20 to 100°C
Surface Speed (max.)	0.5 m/s.

Fitting:

PS056 is very easy to fit. Careful fitting of the seal is a prerequisite for it's perfect functioning.

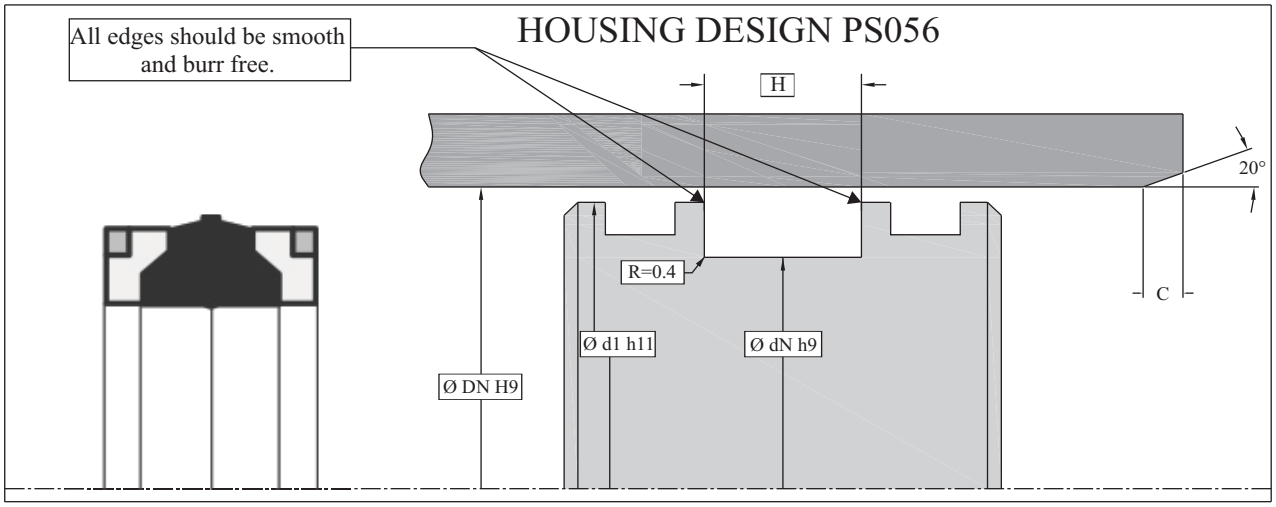
Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Ordering format:

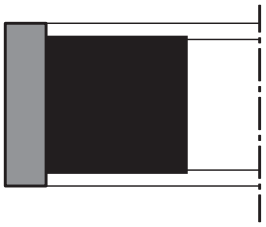
Please mention ROYAL Order Code in your order.
Example: Size 70 X 55 X 22.5 Order Code is PS056-N-07000-05500-02250

We have given all metric(mm) sizes here. For Imperial (inch) sizes please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in**



ROYAL STANDARD SIZE LIST

ØDN (H9)	ØdN (h9)	H (+0.20)	d1 (h11)	C	ORDER CODE
70.00	55.00	22.40	69.00	4.00	PS056-07000-05500-2240
80.00	65.00	22.40	79.00	5.00	PS056-08000-06500-2240
90.00	70.00	22.40	89.00	5.00	PS056-09000-07000-2240
100.00	80.00	30.00	99.00	5.00	PS056-10000-08000-3000
110.00	90.00	30.00	109.00	5.00	PS056-11000-09000-3000
120.00	100.00	30.00	119.00	6.00	PS056-12000-10000-3000
130.00	110.00	30.00	129.00	6.00	PS056-13000-11000-3000

PISTON SEAL	PS064	

Features:

Piston seal profile PS064 consists of two piece seal set for sealing pistons with one seal ring and one energizer Square / Rectangular ring.

Composition:

SEAL		PS064-
Seal ring	Material	PTFE+ Bronze + MoS2
	Royal code	TB40M
Square Ring	Material	Nitrile Butadiene rubber (NBR)
	Royal code	NBR70
Working Temperature Range Medium : Hydraulic Oil		-20°C to +100°C

For compound detail please refer Royal seal materials. (Section - III)

Properties:

- Low friction, free of stick slip
- High abrasion resistance
- Wide temperature Range
- Good thermal conductivity
- Low wear and high extrusion resistance
- Wide fluid application range
- Resistance against cold flow
- Minimum installation space required

Application:

- Mobile Hydraulics
- Injection moulding machines
- Machine tools
- Material Handling Equipments
- Control and regulating apparatus

General Technical Data:

Seal Profile	PS064-N
Working Pressure(max.)	400 bar
Surface Speed(max.)	4 m/s

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Extrusion gap dimension :

The gap dimension on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the minimum clearance Ø and max bore Ø.

Profile dimension	Maximum permissible gap (S)			
H	160 bar	260 bar	320 bar	400 bar
4,50	0,50	0,40	0,30	ASK ROYAL
7,50	0,60	0,50	0,40	ASK ROYAL
11,00	0,70	0,60	0,45	ASK ROYAL

Installation :

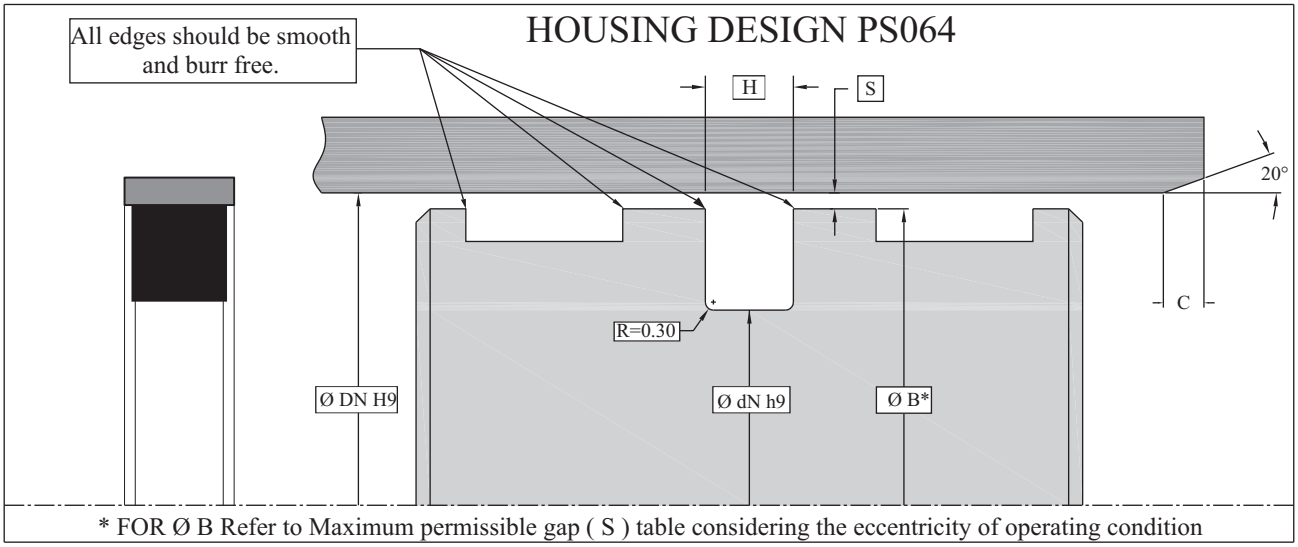
Ps064 piston seals can be fitted on single-piece pistons. First insert the rubber Square / Rectangular ring into the housing carefully to avoid twisting The PTFE ring is then carefully pushed over the piston into place above the Square / Rectangular ring. An assembly tool eases installation of small, PTFE rings. The PTFE seal normally returns to its original size after installation. It is advisable to calibrate the PTFE ring by mounting it on a steel tube (duly machined and polished), having inner diameter equal to cylinders bore diameter. Immediately after calibration, the piston should be installed in the cylinder.

Ordering format :

Please mention ROYAL order code in your order.
Example: Size 30 X 20.5 X 4.5
For Temp range: -20°C to +100°C (Square Ring material is NBR70) Order code is PS064-N-03000-02050-0450

We have given all metric(mm) sizes here. For Imperial (inch) sizes please contact ROYAL at
info@royalcomposites.co.in / info@royalseals.co.in

HOUSING DESIGN PS064

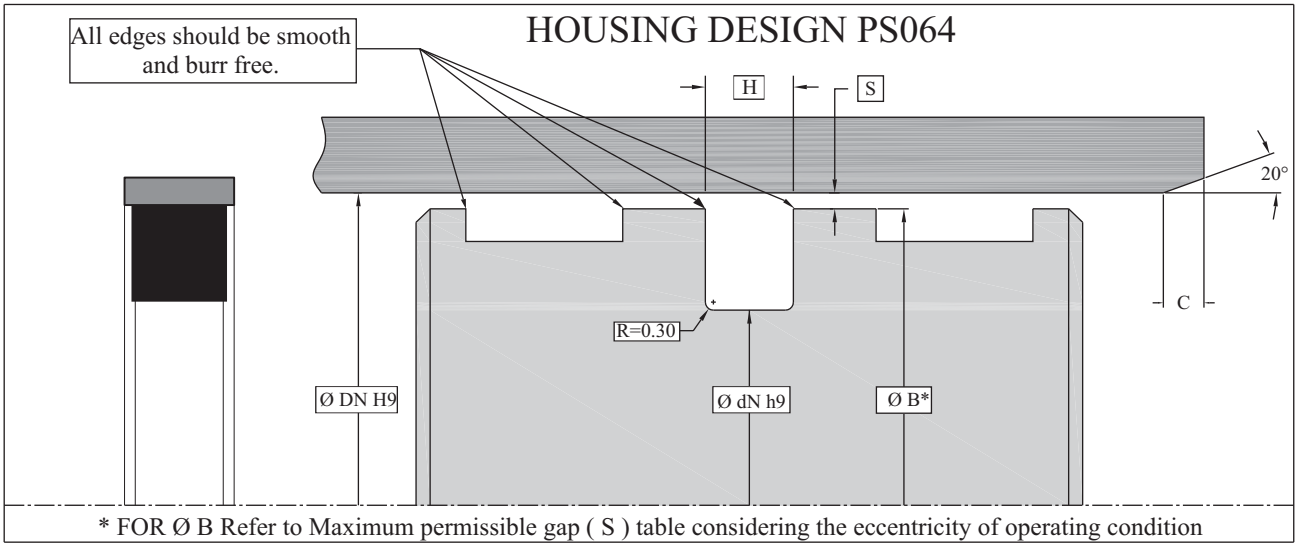


* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)	C	ORDER CODE
30.00	20.50	4.50	2.00	PS064-N-03000-02050-0450
31.50	22.00	4.50	3.50	PS064-N-03150-02200-0450
32.00	22.50	4.50	3.50	PS064-N-03200-02250-0450
35.00	25.50	4.50	3.50	PS064-N-03500-02550-0450
35.50	26.00	4.50	3.50	PS064-N-03550-02600-0450
40.00	30.00	4.50	3.50	PS064-N-04000-03000-0450
45.00	35.00	4.50	3.50	PS064-N-04500-03500-0450
50.00	40.00	4.50	4.00	PS064-N-05000-04000-0450
55.00	45.00	4.50	4.00	PS064-N-05500-04500-0450
56.00	46.00	4.50	4.00	PS064-N-05600-04600-0450
60.00	50.00	4.50	4.00	PS064-N-06000-05000-0450
63.00	48.00	7.50	4.00	PS064-N-06300-04800-0750
65.00	50.00	7.50	4.00	PS064-N-06500-05000-0750
70.00	55.00	7.50	5.00	PS064-N-07000-05500-0750
75.00	60.00	7.50	5.00	PS064-N-07500-06000-0750
80.00	65.00	7.50	5.00	PS064-N-08000-06500-0750
85.00	70.00	7.50	5.00	PS064-N-08500-07000-0750
90.00	75.00	7.50	5.00	PS064-N-09000-07500-0750
95.00	80.00	7.50	5.00	PS064-N-09500-08000-0750
100.00	85.00	7.50	5.00	PS064-N-10000-08500-0750
105.00	90.00	7.50	5.00	PS064-N-10500-09000-0750
108.00	92.00	7.50	5.00	PS064-N-10800-09200-0750

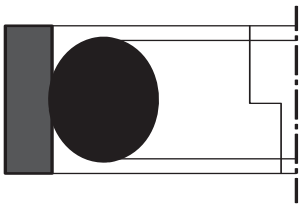
HOUSING DESIGN PS064



* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)	C	ORDER CODE
110.00	94.00	7.50	5.00	PS064-N-11000-09400-0750
112.00	96.00	7.50	6.50	PS064-N-11200-09600-0750
115.00	99.00	7.50	6.50	PS064-N-11500-09900-0750
120.00	104.00	7.50	6.50	PS064N-12000-10400-0750
125.00	109.00	7.50	6.50	PS064-N-12500-10900-0750
130.00	114.00	7.50	6.50	PS064-N-13000-11400-0750
135.00	119.00	7.50	6.50	PS064-N-13500-11900-0750
140.00	124.00	7.50	6.50	PS064-N-14000-12400-0750
145.00	129.00	7.50	6.50	PS064-N-14500-12900-0750
150.00	134.00	7.50	6.50	PS064-N-15000-13400-0750
155.00	139.00	7.50	6.50	PS064-N-15500-13900-0750
160.00	144.00	7.50	6.50	PS064-N-16000-14400-0750
170.00	148.00	11.00	6.50	PS064-N-17000-14800-1100
180.00	158.00	11.00	6.50	PS064-N-18000-15800-1100
200.00	178.00	11.00	6.50	PS064-N-20000-17800-1100

PISTON SEAL	PS072	

Features:

Piston seal profile PS072 consists of two piece seal set for sealing pistons with one seal ring and one energizer Oval-ring.

Composition:

PS 072 is a combition of split glass filled polyamide seal ring along with NBR Oval Energizer.

SEAL		PS072
Seal ring	Material	Glass Filled Polyamide
	Royal code	PG6040 / PG 6040M*
Oval Ring	Material	Nitrile Butadiene rubber (NBR)
	Royal code	NBR80
Working Temperature Range Medium : Hydraulic Oil		-20°C to +100°C

For compound detail please refer Royal seal materials. (Section - III)

Properties:

- High abrasion resistance
- Low wear and high extrusion resistance
- Wide fluid application range
- Resistance against cold flow
- Minimum installation space required

Application:

- Mobile Hydraulics
- Machine tools
- Material Handling Equipments
- Where Piston is Passing above an oil port

General Technical Data:

Seal Profile	PS072
Working Pressure(max.)	500 bar
Surface Speed(max.)	1.5 m/s

* On request

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Installation :

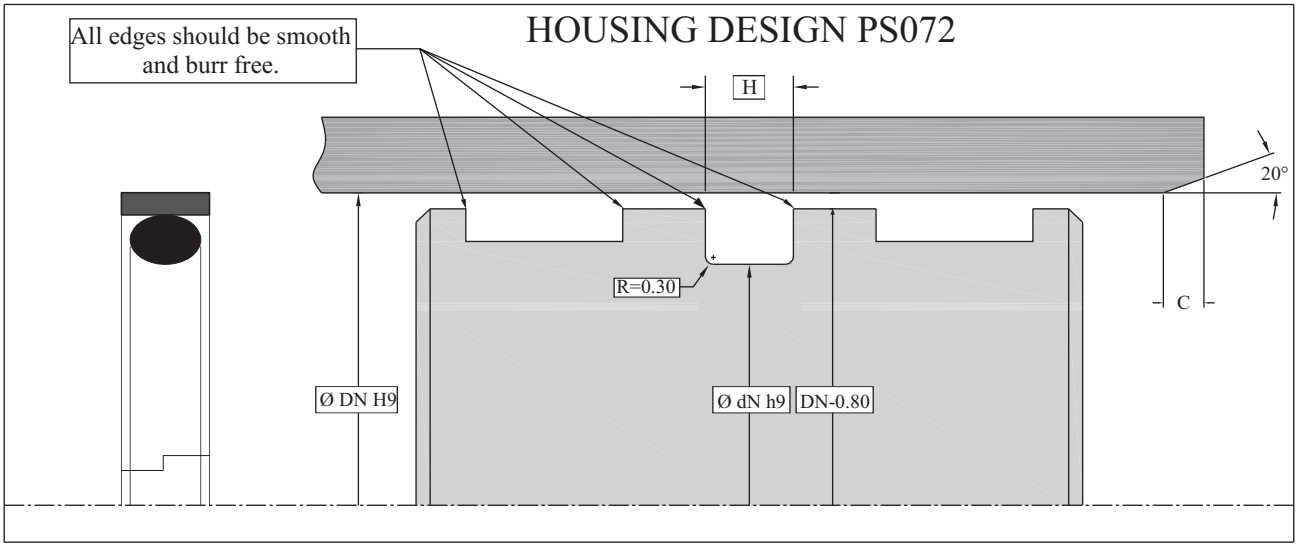
Ps072 piston seals can be fitted on single-piece pistons. First insert the rubber Oval Ring into the housing carefully to avoid twisting of Ring Chord. The filled polyamide ring is then carefully plased on the piston above the Oval Ring

Ordering format :

Please Follow ROYAL order code to place your order.
Like - for 100 X 84.84 X 12.70 - order cord is PS072-10000-08484-1270

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

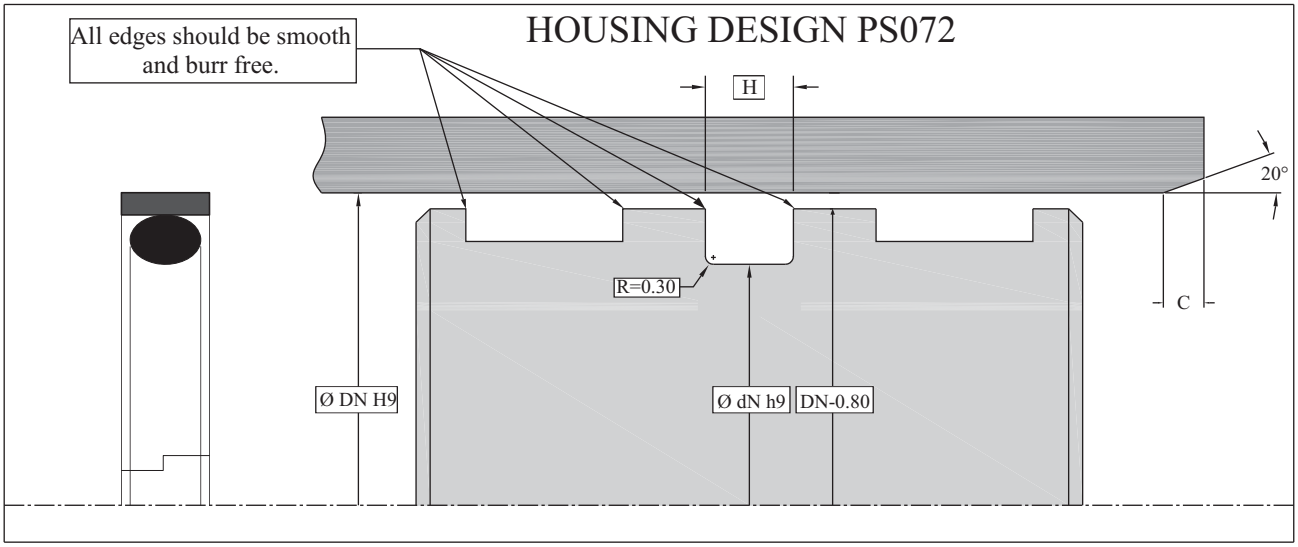
HOUSING DESIGN PS072



ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)	C	ORDER CODE
100.00	84.84	12.70		PS072-10000-08484-1270
100.00	85.75	7.24		PS072-10000-08575-0724
105.00	90.75	7.24		PS072-10500-09075-0724
110.00	95.00	12.70		PS072-11000-09500-1270
110.00	95.75	7.24		PS072-11000-09575-0724
115.00	100.75	7.24		PS072-11500-10075-0724
120.00	105.74	7.24		PS072-12000-10574-0724
130.00	115.74	7.24		PS072-13000-11574-0724
140.00	119.13	8.13		PS072-14000-11913-0813
140.00	120.62	9.63		PS072-14000-12062-0963
150.00	130.63	9.63		PS072-15000-13063-0963
160.00	140.61	9.63		PS072-16000-14061-0963
170.00	150.62	9.63		PS072-17000-15062-0963
180.00	160.62	9.63		PS072-18000-16062-0963
190.00	170.68	9.63		PS072-19000-17068-0963
200.00	174.73	9.63		PS072-20000-17473-0963
220.00	194.74	9.63		PS072-22000-19474-0963

HOUSING DESIGN PS072

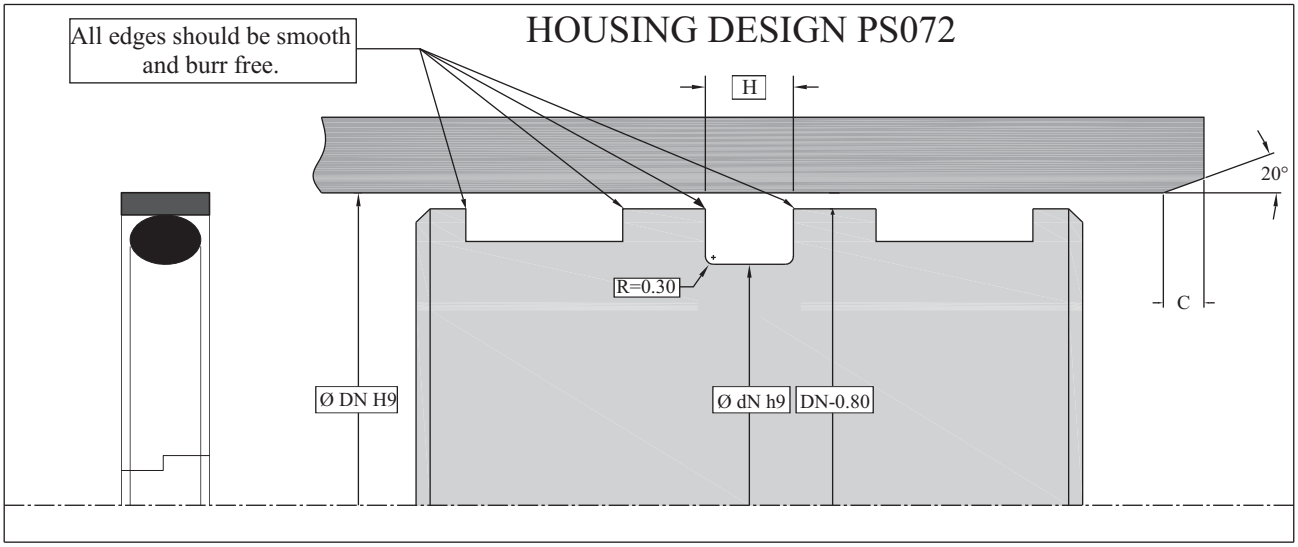


ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)		ORDER CODE
IMPERIAL SIZES				
50.80	35.76	7.92		PS072-IN-02000-01408-0312
50.80	37.13	7.16		PS072-IN-02000-01462-0282
57.15	43.48	7.16		PS072-IN-02250-01712-0282
63.50	48.46	7.92		PS072-IN-02500-01908-0312
63.50	49.83	7.16		PS072-IN-02500-01962-0282
63.50	54.10	4.83		PS072-IN-02500-02130-0190
69.85	56.18	7.16		PS072-IN-02750-02212-0282
76.20	61.16	7.92		PS072-IN-03000-02408-0312
76.20	62.03	7.16		PS072-IN-03000-02442-0282
76.20	62.48	4.83		PS072-IN-03000-02460-0190
82.55	68.38	7.14		PS072-IN-03250-02692-0281
88.90	73.86	7.92		PS072-IN-03500-02908-0312
88.90	74.73	7.16		PS072-IN-03500-02942-0282
88.90	75.18	4.83		PS072-IN-03500-02960-0190
89.54	74.73	7.14		PS072-IN-03525-02942-0281
95.25	81.08	7.16		PS072-IN-03750-03192-0282
95.25	81.53	4.83		PS072-IN-03750-03210-0190
101.60	81.53	4.83		PS072-IN-04000-03210-0190
101.60	86.56	7.92		PS072-IN-04000-03408-0312
101.60	87.43	7.16		PS072-IN-04000-03442-0282
107.95	91.19	6.35		PS072-IN-04250-03590-0250

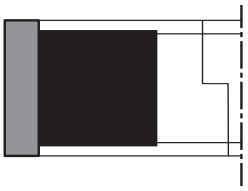
ROYAL STANDARD SIZE LIST				
DN (H9)	**dN (h9)**	**H (+0.20)**		**ORDER CODE**
114.30	97.54	6.35		PS072-IN-04500-03840-0250
114.30	97.54	12.70		PS072-IN-04500-03840-0500
114.30	99.26	7.92		PS072-IN-04500-03908-0312
114.30	100.13	7.16		PS072-IN-04500-03942-0282
120.65	103.89	6.35		PS072-IN-04750-04090-0250
120.65	103.89	16.00		PS072-IN-04750-04090-0630
120.65	106.48	7.16		PS072-IN-04750-04192-0282
127.00	107.44	9.65		PS072-IN-05000-04230-0380
127.00	107.44	16.00		PS072-IN-05000-04230-0630
127.00	107.70	9.58		PS072-IN-05000-04240-0377
127.00	112.83	7.16		PS072-IN-05000-04442-0282
127.00	112.83	7.16		PS072-IN-05000-04442-0282
133.35	113.28	9.65		PS072-IN-05250-04460-0380
133.35	114.05	9.58		PS072-IN-05250-04490-0377
139.70	120.14	9.65		PS072-IN-05500-04730-0380
139.70	120.14	16.00		PS072-IN-05500-04730-0630
139.70	120.40	9.58		PS072-IN-05500-04740-0377
152.40	129.29	9.65		PS072-IN-06000-05090-0380
152.40	129.29	16.00		PS072-IN-06000-05090-0630
152.40	133.10	9.58		PS072-IN-06000-05240-0377
158.75	135.64	9.65		PS072-IN-06250-05340-0380
158.75	135.64	16.00		PS072-IN-06250-05340-0630
165.10	141.99	16.00		PS072-IN-06500-05590-0630
165.10	145.80	9.58		PS072-IN-06500-05740-0377
177.80	154.69	16.00		PS072-IN-07000-06090-0630
177.80	158.50	9.58		PS072-IN-07000-06240-0377
184.15	161.04	16.00		PS072-IN-07250-06340-0630
190.50	167.39	16.00		PS072-IN-07500-06590-0630
190.50	171.20	9.58		PS072-IN-07500-06740-0377
196.85	173.74	16.00		PS072-IN-07750-06840-0630
203.20	183.90	9.58		PS072-IN-08000-07240-0377
209.55	184.40	16.00		PS072-IN-08250-07260-0630
215.90	190.75	16.00		PS072-IN-08500-07510-0630
215.90	193.65	9.58		PS072-IN-08500-07624-0377

HOUSING DESIGN PS072



ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)		ORDER CODE
215.90	196.60	9.58		PS072-IN-08500-07740-0377
228.60	206.35	9.58		PS072-IN-09000-08124-0377
234.95	209.80	9.65		PS072-IN-09250-08260-0380
235.71	209.80	9.65		PS072-IN-09280-08260-0380
241.30	216.15	9.65		PS072-IN-09500-08510-0380
241.30	219.05	9.58		PS072-IN-09500-08624-0377
254.00	225.55	17.53		PS072-IN-10000-08880-0690
254.00	231.75	9.58		PS072-IN-10000-09124-0377
260.35	231.90	17.53		PS072-IN-10250-09130-0690
266.70	238.25	17.53		PS072-IN-10500-09380-0690
266.70	244.45	9.58		PS072-IN-10500-09624-0377
279.40	250.95	17.53		PS072-IN-11000-09880-0690
279.40	257.15	9.58		PS072-IN-11000-10124-0377
292.10	263.65	17.53		PS072-IN-11500-10380-0690
298.45	276.20	9.58		PS072-IN-11750-10874-0377
304.80	282.55	9.58		PS072-IN-12000-11124-0377
317.50	289.05	17.53		PS072-IN-12500-11380-0690
330.20	301.75	17.53		PS072-IN-13000-11880-0690
342.90	314.96	17.53		PS072-IN-13500-12400-0690

PISTON SEAL	PS088	

Features:

Piston seal profile PS088 consists of two piece seal set for sealing pistons with one seal ring and one NBR square/rectangular energizer ring.

Composition:

PS 088 is a combination of split glass filled polyamide seal ring along with NBR square energizer ring.

SEAL		PS088
Seal ring	Material	Glass Filled Polyamide
	Royal code	PG6040 / PG6040M*
Oval Ring	Material	Nitrile Butadiene rubber (NBR)
	Royal code	NBR80
Working Temperature Range Medium : Hydraulic Oil		-20°C to +100°C

For compound detail please refer Royal seal materials. (Section - III)

Properties:

- High abrasion resistance
- Low wear and high extrusion resistance
- Wide fluid application range
- Resistance against cold flow
- Minimum installation space required

Application:

- Mobile Hydraulics
- Machine tools
- Material Handling Equipments

General Technical Data:

Seal Profile	PS088
Working Pressure(max.)	500 bar
Surface Speed(max.)	1.5 m/s

* On request

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

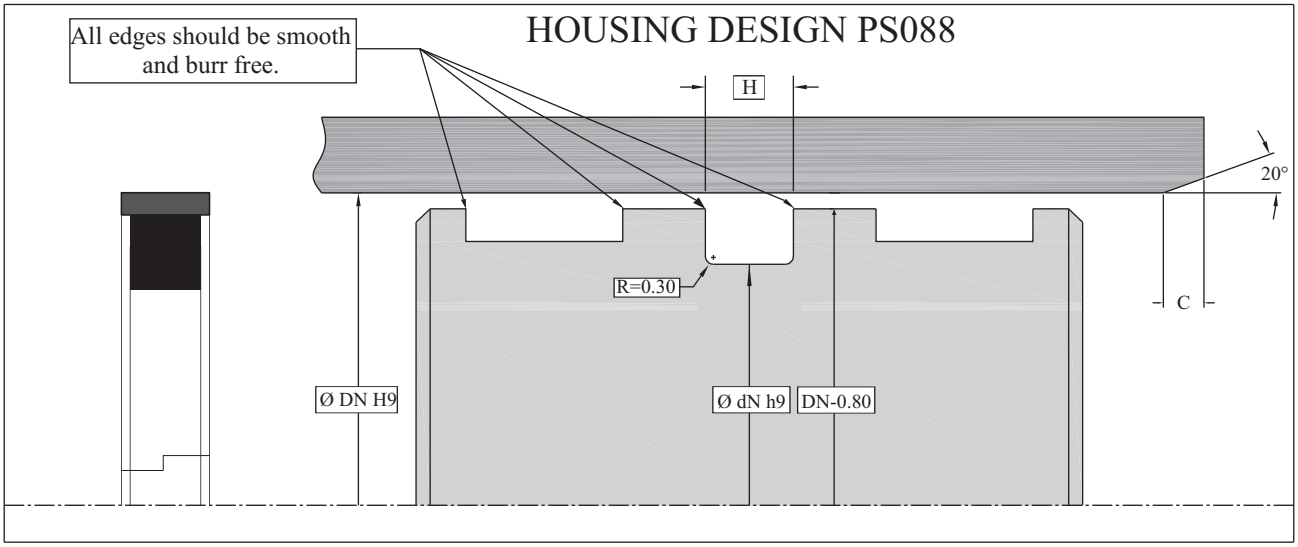
Installation :

Ps088 piston seals can be fitted on single-piece pistons. First insert the rubber Square Ring into the housing carefully to avoid twisting of Ring. The polyamide ring is then carefully placed on the piston above the Square Ring

Ordering format :

Please Follow ROYAL order code to place your order.
Like - for 50.80 X 37.13 X 7.16 - order cord is PS088-IN-02000-01462-0282

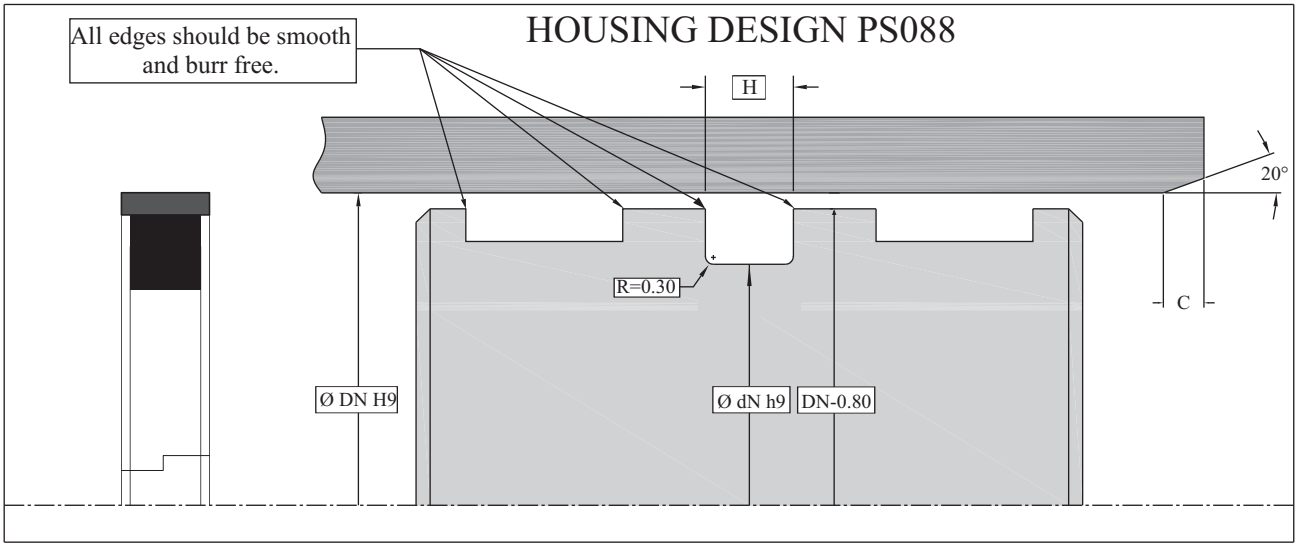
Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list



ROYAL STANDARD SIZE LIST

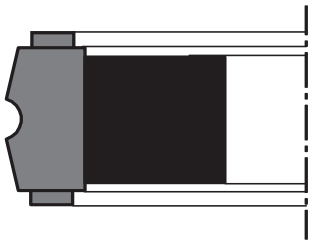
DN (H9)	dN (h9)	H (+0.20)		ORDER CODE
IMPERIAL SIZES				
50.80	37.13	7.16		PS088-IN-02000-01462-0282
50.80	35.76	7.92		PS088-IN-02000-01408-0312
57.15	43.48	7.16		PS088-IN-02250-01712-0282
63.50	49.83	7.16		PS088-IN-02500-01962-0282
63.50	48.46	7.92		PS088-IN-02500-01908-0312
69.85	56.18	7.16		PS088-IN-02750-02212-0282
76.20	62.03	7.16		PS088-IN-03000-02442-0282
76.20	61.16	7.92		PS088-IN-03000-02408-0312
82.55	68.38	7.14		PS088-IN-03250-02692-0281
88.90	74.73	7.16		PS088-IN-03500-02942-0282
88.90	73.86	7.92		PS088-IN-03500-02908-0312
89.54	74.73	7.14		PS088-IN-03525-02942-0281
95.25	81.08	7.16		PS088-IN-03750-03192-0282
101.60	87.43	7.16		PS088-IN-04000-03442-0282
101.60	86.56	7.92		PS088-IN-04000-03408-0312
114.30	100.13	7.16		PS088-IN-04500-03942-0282
114.30	99.26	7.92		PS088-IN-04500-03908-0312
120.65	106.48	7.16		PS088-IN-04750-04192-0282
127.00	112.83	7.16		PS088-IN-05000-04442-0282
127.00	107.70	9.58		PS088-IN-05000-04240-0377
133.35	114.05	9.58		PS088-IN-05250-04490-0377

HOUSING DESIGN PS088



ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)		ORDER CODE
IMPERIAL SIZES				
139.70	120.40	9.58		PS088-IN-05500-04740-0377
152.40	133.10	9.58		PS088-IN-06000-05240-0377
165.10	145.80	9.58		PS088-IN-06500-05740-0377
177.80	158.50	9.58		PS088-IN-07000-06240-0377
190.50	171.20	9.58		PS088-IN-07500-06740-0377
203.20	183.90	9.58		PS088-IN-08000-07240-0377
215.90	196.60	9.58		PS088-IN-08500-07740-0377
215.90	193.65	9.58		PS088-IN-08500-07624-0377
228.60	206.35	9.58		PS088-IN-09000-08124-0377
241.30	219.05	9.58		PS088-IN-09500-08624-0377
254.00	231.75	9.58		PS088-IN-10000-09124-0377
266.70	244.45	9.58		PS088-IN-10500-09624-0377
279.40	257.15	9.58		PS088-IN-11000-10124-0377
298.45	276.20	9.58		PS088-IN-11750-10874-0377
304.80	282.55	9.58		PS088-IN-12000-11124-0377

PISTON SEAL	PS096	

Features:

Piston seal profile PS096 consists of two piece seal set for sealing pistons with one polyurethane seal ring and one energizer square ring.

Composition:

Sealing ring	Material	Polyurethane
	Royal code	PUAB98
Square Ring	Material	Nitrile Butadiene rubber (NBR)
	Royal code	NBR75
Working Temperature Range Medium : Hydraulic Oil		-20°C. to +100°C.

For compound detail please refer Royal seal materials.(Section III)

Properties:

- High abrasion resistance.
- Low wear and high extrusion resistance.
- Easy installation.

Application:

- Mobile Hydraulics
- Injection moulding machines
- Machine tools
- Material Handling Equipments.
- Agricultural equipments

General Technical Data:

Working Pressure (max.)	315 Bar
Temperature range (max.)	-20°C to 100°C
Surface Speed (max.)	0.5 m/s.

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Gap dimension:

The gap dimension on the non-pressurised side of the seal is decisive for sealing function.

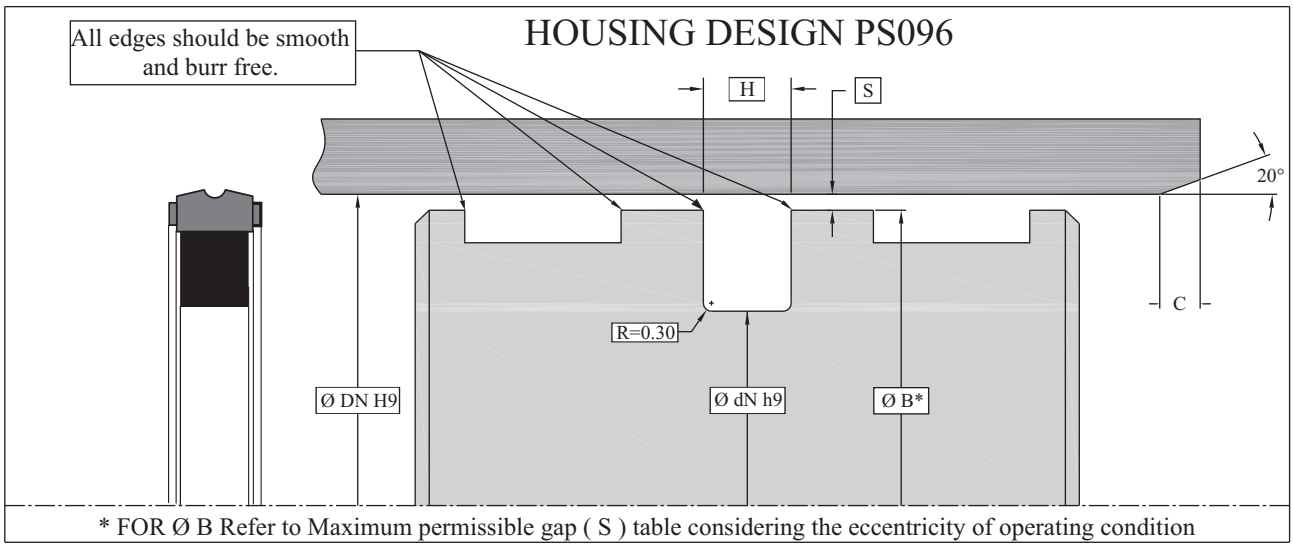
Profile dimension	Maximum permissible gap (S)			
H	16 Mpa	26 Mpa	32 Mpa	
3.20 / 3.28	0.30	0.20	-	
4.20	0.40	0.30	-	
6.30 / 7.19	0.50	0.40	0.25	
8.10 / 9.60	0.60	0.50	0.30	
10.50	0.65	0.55	0.40	

Installation:

PS096 piston seals can be fitted on single-piece pistons. First insert the rubber Square / Rectangular ring. into the housing carefully to avoid twisting of Square / Rectangular ring. . The polyurethane ring is then carefully pushed over the piston into place above the Square / Rectangular ring. The polyurethane seal normally returns to its original size after installation.

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN PS096

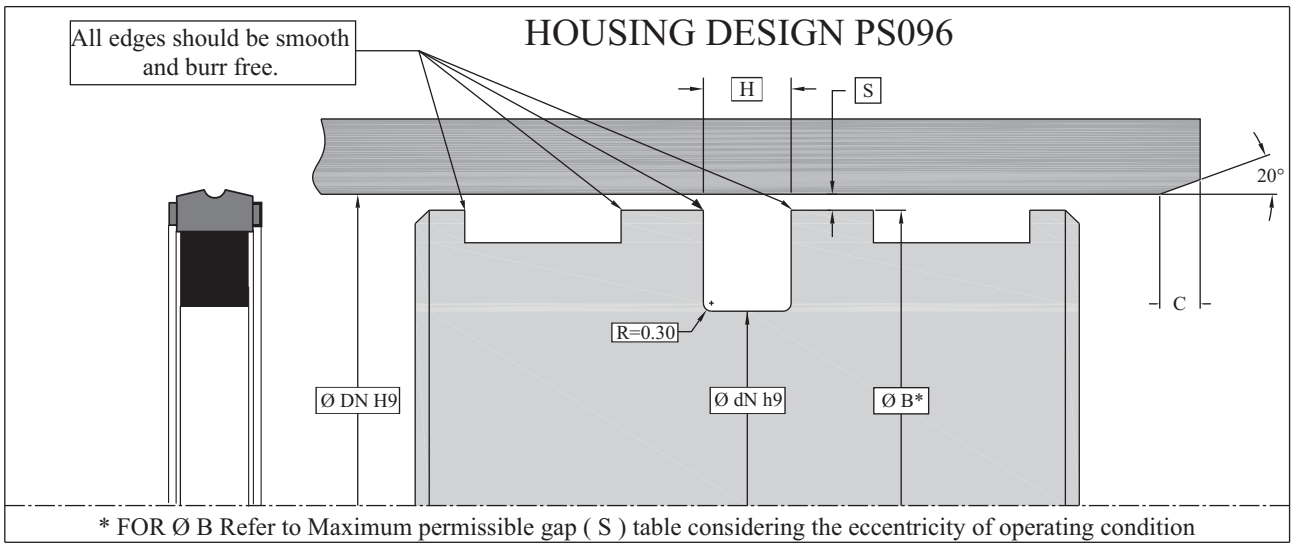


* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)	C	ORDER CODE
25.00	17.50	3.20	2.00	PS096-02500-01750-0320
30.00	22.50	3.20	2.00	PS096-03000-02250-0320
32.00	24.50	3.20	2.00	PS096-03200-02450-0320
40.00	29.00	4.20	2.50	PS096-04000-02900-0420
45.00	34.00	4.20	2.50	PS096-04500-03400-0420
50.00	39.00	4.20	2.50	PS096-05000-03900-0420
50.00	37.00	6.30	2.50	PS096-05000-03700-0630
55.00	39.50	6.30	3.00	PS096-05500-03950-0630
60.00	44.50	6.30	3.00	PS096-06000-04450-0630
60.00	49.00	6.30	3.00	PS096-06000-04900-0630
63.00	47.50	6.30	3.00	PS096-06300-04750-0630
63.00	48.00	7.50	5.00	PS096-06300-04800-0750
65.00	49.50	6.30	3.00	PS096-06500-04950-0630
65.00	52.00	6.30	3.00	PS096-06500-05200-0630
70.00	54.50	6.30	4.00	PS096-07000-05450-0630
75.00	62.00	6.30	3.00	PS096-07500-06200-0630
75.00	59.50	6.30	4.00	PS096-07500-05950-0630
80.00	64.50	6.30	4.00	PS096-08000-06450-0630
80.00	66.50	6.30	4.00	PS096-08000-06650-0630
90.00	74.50	6.30	4.00	PS096-09000-07450-0630
100.00	79.00	8.10	5.00	PS096-10000-07900-0810
100.00	84.50	6.30	4.00	PS096-10000-08450-0630

HOUSING DESIGN PS096

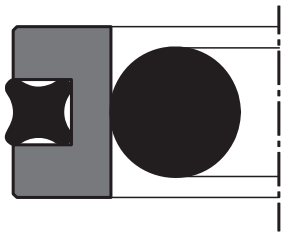


* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)	C	ORDER CODE
110.00	89.00	8.10	5.00	PS096-11000-08900-0810
120.00	99.00	8.10	5.00	PS096-12000-09900-0810
125.00	104.00	8.10	5.00	PS096-12500-10400-0810
130.00	109.00	8.10	5.00	PS096-13000-10900-0810
140.00	119.00	8.10	5.00	PS096-14000-11900-0810
140.00	119.00	10.50	7.00	PS096-14000-11900-1050
150.00	129.00	8.10	5.00	PS096-15000-12900-0810
150.00	129.00	10.50	7.00	PS096-15000-12900-1050
160.00	139.00	8.10	5.00	PS096-16000-13900-0810
165.00	144.00	8.10	5.00	PS096-16500-14400-0810
170.00	149.00	8.10	5.00	PS096-17000-14900-0810
180.00	159.00	8.10	5.00	PS096-18000-15900-0810
200.00	179.00	8.10	5.00	PS096-20000-17900-0810
IMPERIAL SIZES				
25.40	17.55	3.28	3.00	PS096-IN-N-01000-00691-0129
31.75	23.90	3.28	3.00	PS096-IN-N-01250-00941-0129
38.10	30.25	3.28	3.00	PS096-IN-N-01500-01191-0129
44.45	36.60	3.28	3.00	PS096-IN-N-01750-01441-0129
50.80	42.95	3.28	3.00	PS096-IN-N-02000-01691-0129
57.15	49.30	3.28	3.00	PS096-IN-N-02250-01941-0129
63.50	55.65	3.28	3.00	PS096-IN-N-02500-02191-0129

ROYAL STANDARD SIZE LIST				
DN (H9)	**dN (h9)**	**H (+0.20)**	**C**	**ORDER CODE**
69.85	62.00	3.28	3.00	PS096-IN-N-02750-02441-0129
76.20	62.53	7.19	5.00	PS096-IN-N-03000-02462-0283
82.55	68.88	7.19	5.00	PS096-IN-N-03250-02712-0283
88.90	75.23	7.19	5.00	PS096-IN-N-03500-02962-0283
95.25	81.58	7.19	5.00	PS096-IN-N-03750-03212-0283
101.60	87.93	7.19	5.00	PS096-IN-N-04000-03462-0283
107.95	94.28	7.19	5.00	PS096-IN-N-04250-03712-0283
114.30	100.63	7.19	5.00	PS096-IN-N-04500-03962-0283
120.65	106.98	7.19	5.00	PS096-IN-N-04750-04212-0283
127.00	113.33	7.19	5.00	PS096-IN-N-05000-04462-0283
133.35	114.40	9.60	7.50	PS096-IN-N-05250-04504-0378
139.70	120.75	9.60	7.50	PS096-IN-N-05500-04754-0378
146.05	127.10	9.60	7.50	PS096-IN-N-05750-05004-0378
152.40	133.45	9.60	7.50	PS096-IN-N-06000-05254-0378
158.75	139.80	9.60	7.50	PS096-IN-N-06250-05504-0378

PISTON SEAL	PS112	

Features:

Piston seal profile PS112 consists of three piece seal set for sealing pistons with one seal ring having a NBR quad ring installad on face groove and one energizer O-ring.

Composition:

SEAL COMBINATION		PS112-N
Seal ring	Material	PTFE+ Bronze + MoS2
	Royal code	TB40M
O Ring / Quad ring	Material	Nitrile Butadiene rubber (NBR)
	Royal code	NBR70
Working Temperature Range Medium : Hydraulic Oil	-20°C to +100°C	

For compound detail please refer Royal seal materials. (Section - III)

Properties:

- Low friction, free of stick slip
- Low wear and high extrusion resistance
- Wide fluid application range
- Suitable for dual media like nitrogen on one side and hydraulic oil on another side

Application:

- Piston Accumulators
- Dampers
- Control and regulating apparatus
- For Hydro pneumatic cylinders

General Technical Data:

Seal Profile	PS112-N
Working Pressure(max.)	400 bar
Surface Speed(max.)	1 m/s

Extrusion gap dimension (S) :

The gap dimension on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the minimum clearance Ø and max bore Ø.

Profile dimension	Maximum permissible gap (S)			
H	160 bar	260 bar	320 bar	400 bar
5.33	0.50	0.40	0.30	ASK ROYAL
7.32	0.60	0.50	0.40	ASK ROYAL
9.52	0.60	0.50	0.40	ASK ROYAL

Installation :

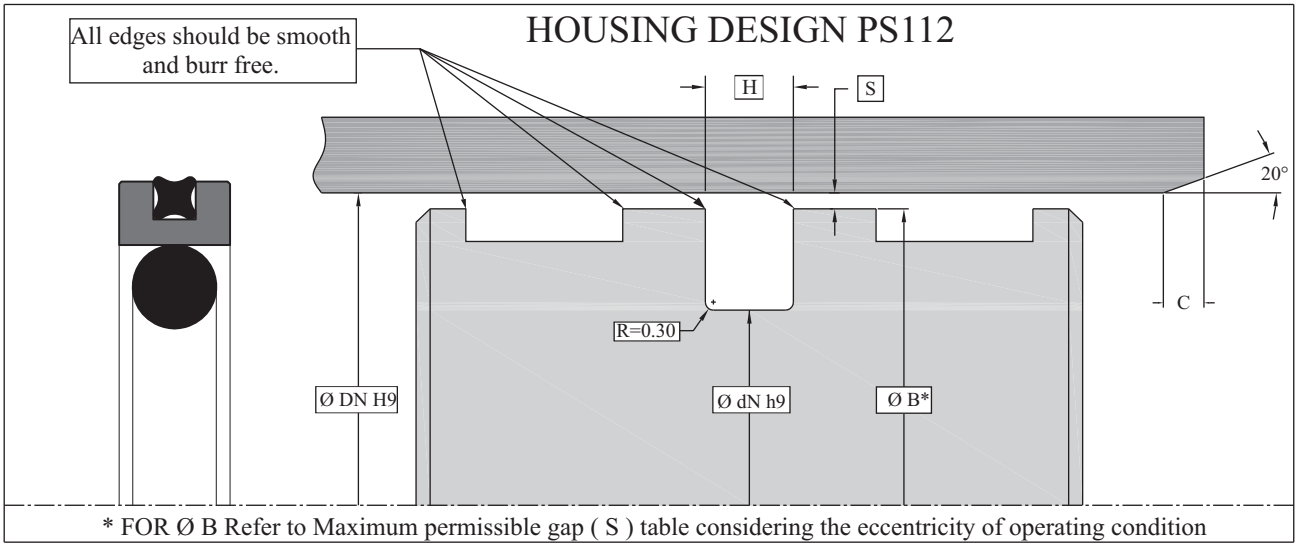
PS112 piston seals can be fitted on single-piece pistons. First insert the rubber O / Square Ring into the housing carefully to avoid twisting of Ring. Install the PTFE part carefully with fixture and then place the quad ring on the groove over PTFE part.

Ordering format :

Please Follow ROYAL order code to place your order.
Like - for 25.40 X 14.63 X 5.33 - order cord is PS112-IN-01000-00576-0210

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN PS112

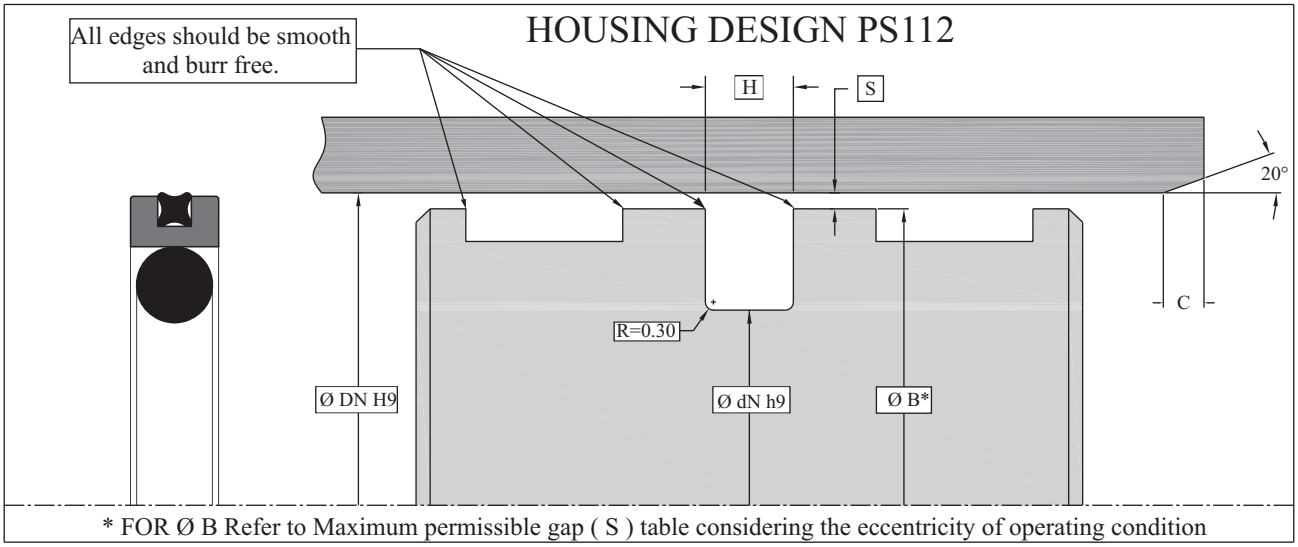


* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)	C	ORDER CODE
IMPERIAL SIZES				
25.40	14.63	5.33		PS112-IN-01000-00576-0210
28.58	17.81	5.33		PS112-IN-01125-00701-0210
31.75	20.98	5.33		PS112-IN-01250-00826-0210
34.92	24.16	5.33		PS112-IN-01375-00951-0210
38.10	27.33	5.33		PS112-IN-01500-01076-0210
41.28	30.51	5.33		PS112-IN-01625-01201-0210
44.45	33.68	5.33		PS112-IN-01750-01326-0210
47.63	36.86	5.33		PS112-IN-01875-01451-0210
50.80	35.15	7.32		PS112-IN-02000-01384-0288
53.97	38.33	7.32		PS112-IN-02125-01509-0288
57.15	41.50	7.32		PS112-IN-02250-01634-0288
60.32	44.68	7.32		PS112-IN-02375-01759-0288
63.50	47.85	7.32		PS112-IN-02500-01884-0288
66.68	51.03	7.32		PS112-IN-02625-02009-0288
69.85	54.20	7.32		PS112-IN-02750-02134-0288
73.02	57.38	7.32		PS112-IN-02875-02259-0288
76.20	60.55	7.32		PS112-IN-03000-02384-0288
79.38	63.73	7.32		PS112-IN-03125-02509-0288
82.55	66.90	7.32		PS112-IN-03250-02634-0288
85.73	70.08	7.32		PS112-IN-03375-02759-0288
88.90	73.25	7.32		PS112-IN-03500-02884-0288

HOUSING DESIGN PS112



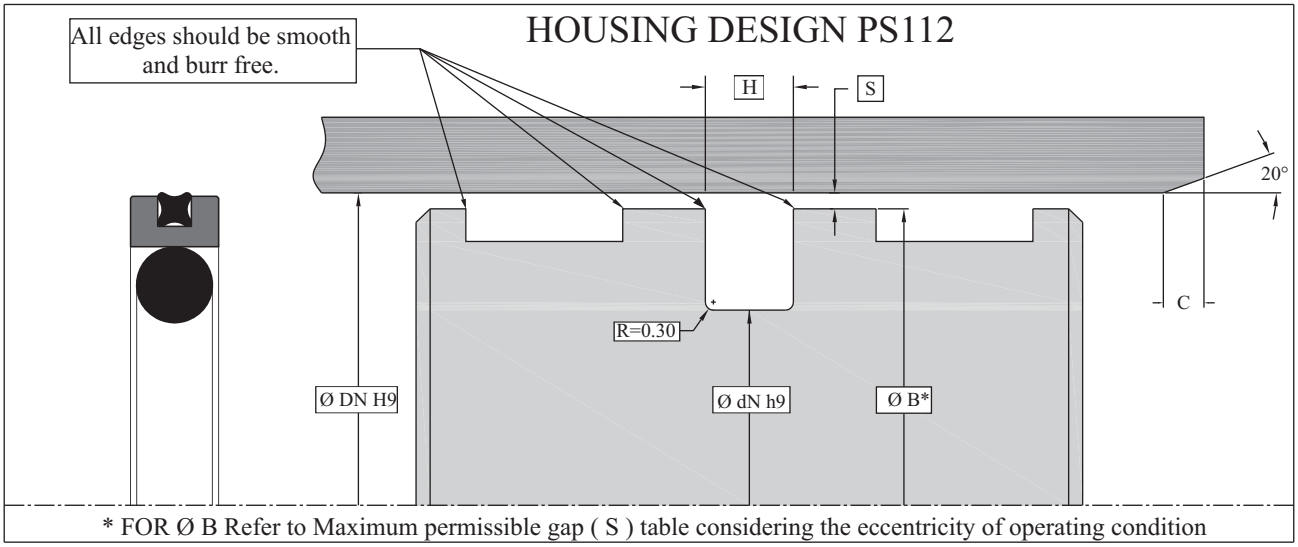
* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)		ORDER CODE
IMPERIAL SIZES				
92.07	76.43	7.32		PS112-IN-03625-03009-0288
95.25	79.60	7.32		PS112-IN-03750-03134-0288
98.43	82.78	7.32		PS112-IN-03875-03259-0288
101.60	85.95	7.32		PS112-IN-04000-03384-0288
104.77	89.13	7.32		PS112-IN-04125-03509-0288
107.95	92.30	7.32		PS112-IN-04250-03634-0288
111.13	95.48	7.32		PS112-IN-04375-03759-0288
114.30	98.65	7.32		PS112-IN-04500-03884-0288
117.48	101.83	7.32		PS112-IN-04625-04009-0288
120.65	105.00	7.32		PS112-IN-04750-04134-0288
123.82	108.18	7.32		PS112-IN-04875-04259-0288
127.00	111.35	7.32		PS112-IN-05000-04384-0288
130.17	114.53	7.32		PS112-IN-05125-04509-0288
133.35	117.70	7.32		PS112-IN-05250-04634-0288
136.53	115.19	9.52		PS112-IN-05375-04535-0375
139.70	118.36	9.52		PS112-IN-05500-04660-0375
142.88	121.54	9.52		PS112-IN-05625-04785-0375
146.05	124.71	9.52		PS112-IN-05750-04910-0375
149.23	127.89	9.52		PS112-IN-05875-05035-0375
152.40	131.06	9.52		PS112-IN-06000-05160-0375
155.57	134.24	9.52		PS112-IN-06125-05285-0375

ROYAL STANDARD SIZE LIST				
DN (H9)	**dN (h9)**	**H (+0.20)**		**ORDER CODE**
IMPERIAL SIZES				
158.75	137.54	9.52		PS112-IN-06250-05415-0375
161.92	140.72	9.52		PS112-IN-06375-05540-0375
165.10	143.89	9.52		PS112-IN-06500-05665-0375
168.27	147.19	9.52		PS112-IN-06625-05795-0375
171.45	150.37	9.52		PS112-IN-06750-05920-0375
174.63	153.54	9.52		PS112-IN-06875-06045-0375
177.80	156.72	9.52		PS112-IN-07000-06170-0375
184.15	163.07	9.52		PS112-IN-07250-06420-0375
190.50	169.42	9.52		PS112-IN-07500-06670-0375
196.85	175.64	9.52		PS112-IN-07750-06915-0375
203.20	182.12	9.52		PS112-IN-08000-07170-0375
209.55	188.34	9.52		PS112-IN-08250-07415-0375
215.90	194.69	9.52		PS112-IN-08500-07665-0375
222.25	201.04	9.52		PS112-IN-08750-07915-0375
228.60	207.39	9.52		PS112-IN-09000-08165-0375
234.95	213.87	9.52		PS112-IN-09250-08420-0375
241.30	220.09	9.52		PS112-IN-09500-08665-0375
247.65	226.57	9.52		PS112-IN-09750-08920-0375
254.00	232.79	9.52		PS112-IN-10000-09165-0375
260.35	237.24	9.52		PS112-IN-10250-09340-0375
266.70	243.33	9.52		PS112-IN-10500-09580-0375
273.05	249.94	9.52		PS112-IN-10750-09840-0375
279.40	256.03	9.52		PS112-IN-11000-10080-0375
285.75	262.64	9.52		PS112-IN-11250-10340-0375
292.10	268.73	9.52		PS112-IN-11500-10580-0375
298.45	275.34	9.52		PS112-IN-11750-10840-0375
304.80	281.43	9.52		PS112-IN-12000-11080-0375
311.15	288.04	9.52		PS112-IN-12250-11340-0375
317.50	294.13	9.52		PS112-IN-12500-11580-0375
323.85	300.74	9.52		PS112-IN-12750-11840-0375
330.20	306.83	9.52		PS112-IN-13000-12080-0375
336.55	313.31	9.52		PS112-IN-13250-12335-0375
342.90	319.53	9.52		PS112-IN-13500-12580-0375

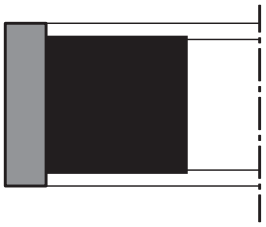
HOUSING DESIGN PS112



* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)		ORDER CODE
IMPERIAL SIZES				
349.25	326.14	9.52		PS112-IN-13750-12840-0375
355.60	332.23	9.52		PS112-IN-14000-13080-0375

PISTON SEAL	PS120	

Features:

Piston seal profile PS 120 consists of two piece seal set for sealing pistons with one seal ring and one square/ rectangular energizer ring.

Composition:

SEAL		PS120
Seal ring	Material	PTFE+ Glass + MoS2
	Royal code	TG20M
Square Ring	Material	Nitrile Butadiene rubber (NBR)
	Royal code	NBR75
Working Temperature Range Medium : Hydraulic Oil		-20°C to +100°C

For compound detail please refer Royal seal materials. (Section - III)

Properties:

- Low friction, free of stick slip
- High abrasion resistance
- Wide temperature Range
- Good thermal conductivity
- Wide fluid application range

Application:

- Mobile Hydraulics
- Injection moulding machines
- Machine tools
- Material Handling Equipments
- Control and regulating apparatus

General Technical Data:

Seal Profile	PS120
Working Pressure(max.)	400 bar
Surface Speed(max.)	4 m/s

Extrusion gap dimension (S) :

The gap dimension on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the minimum clearance Ø and max bore Ø.

Profile dimension	Maximum permissible gap (S)			
H	160 bar	260 bar	320 bar	400 bar
3.00 / 3.28 / 4.88	0.50	0.40	0.30	ASK ROYAL
6.40	0.60	0.50	0.40	ASK ROYAL
9.63 / 11.18	0.70	0.60	0.50	ASK ROYAL

Installation :

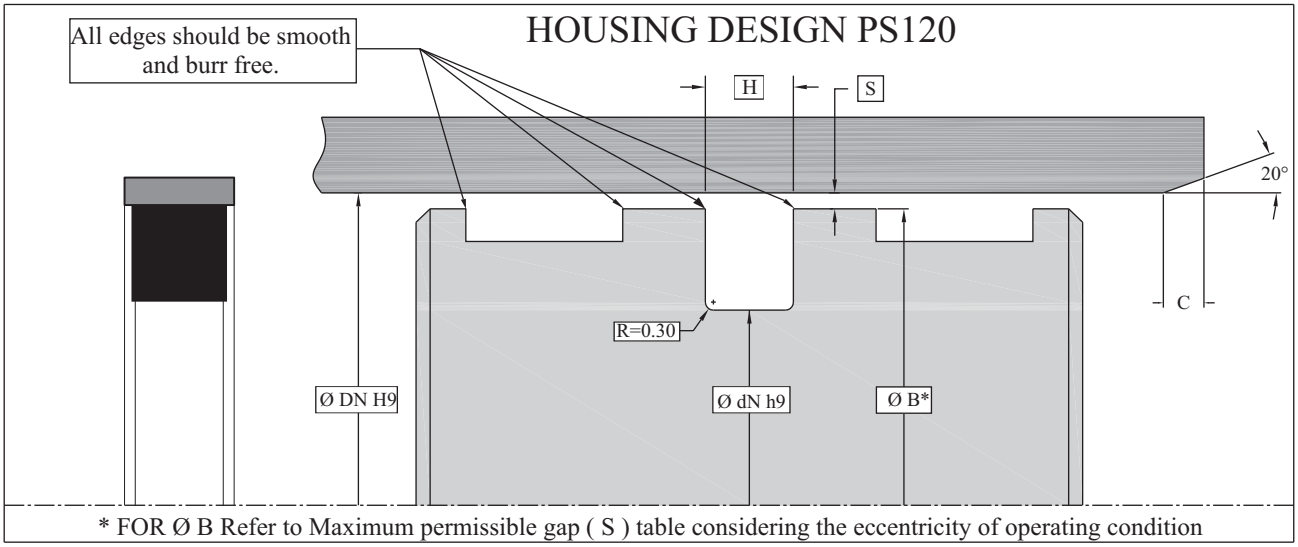
Ps120 piston seals can be fitted on single-piece pistons. First insert the rubber Square/ rectangular Ring into the housing carefully to avoid twisting of Ring. Install the PTFE part carefully with fixture

Ordering format :

Please Follow ROYAL order code to place your order.
Like - for 25.40 X 17.55 X 3.28 - order cord is PS120-IN-01000-00691-0129

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN PS120

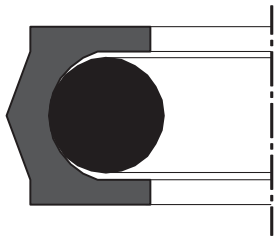


* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)		ORDER CODE
IMPERIAL SIZES				
25.40	17.55	3.28		PS120-IN-01000-00691-0129
31.75	23.90	3.28		PS120-IN-01250-00941-0129
38.10	28.09	3.00		PS120-IN-01500-01106-0118
44.45	36.60	3.28		PS120-IN-01750-01441-0129
50.80	41.28	4.88		PS120-IN-02000-01625-0192
57.15	47.63	4.88		PS120-IN-02250-01875-0192
63.50	53.97	4.88		PS120-IN-02500-02125-0192
69.85	60.32	4.88		PS120-IN-02750-02375-0192
76.20	62.48	4.88		PS120-IN-03000-02460-0192
82.55	72.64	3.28		PS120-N-03250-02860-0129
82.55	68.83	4.88		PS120-IN-03250-02710-0192
95.25	81.53	4.88		PS120-IN-03750-03210-0192
101.60	84.71	6.40		PS120-IN-04000-03335-0252
107.95	91.06	6.40		PS120-IN-04250-03585-0252
114.30	97.41	6.40		PS120-IN-04500-03835-0252
127.00	107.31	9.63		PS120-IN-05000-04225-0379
139.70	120.01	9.63		PS120-IN-05500-04725-0379
146.05	126.36	9.63		PS120-IN-05750-04975-0379
152.40	129.41	9.63		PS120-IN-06000-05095-0379
158.75	135.76	9.63		PS120-IN-06250-05345-0379
165.10	142.11	9.63		PS120-IN-06500-05595-0379

ROYAL STANDARD SIZE LIST				
DN (H9)	**dN (h9)**	**H (+0.20)**		**ORDER CODE**
IMPERIAL SIZES				
177.80	154.81	9.63		PS120-N-07000-06095-0379
184.15	161.16	9.63		PS120-N-07250-06345-0379
190.50	167.51	9.63		PS120-N-07500-06595-0379
196.85	173.86	9.63		PS120-N-07750-06845-0379
203.20	184.25	9.63		PS120-N-08000-07254-0379
209.55	184.28	9.63		PS120-N-08250-07255-0379
215.90	190.63	9.63		PS120-N-08500-07505-0379
228.60	206.68	9.63		PS120-N-09000-08137-0379
241.30	219.38	9.63		PS120-N-09500-08637-0379
266.70	238.25	11.18		PS120-N-10500-09380-0440
279.40	250.95	11.18		PS120-N-11000-09880-0440

PISTON SEAL	PS152	

Features:
Piston seal profile PS152 consists of one polyurethane / polyester seal ring energized with one NBR 'O' ring.

Composition:

SEAL COMBINATION		PS152	
Seal ring	Material	High Performance polyurethane	Thermoplastic polyester
	Royal code	PUHP 94	HY 55
O Ring	Material	Nitrile Butadiene rubber (NBR)	Nitrile Butadiene rubber (NBR)
	Royal code	NBR70	NBR70
Working Temperature Range Medium : Hydraulic Oil		-20°C to +100°C	-20°C to +100°C

For compound detail please refer Royal seal materials. (Section - III)

Properties:
- Low friction, free of stick slip
- High abrasion resistance
- Low wear
- Minimum installation space required

Application:
- Mobile Hydraulics
- Agricultural equipments
- Material Handling Equipments
- Control and regulating apparatus

General Technical Data:

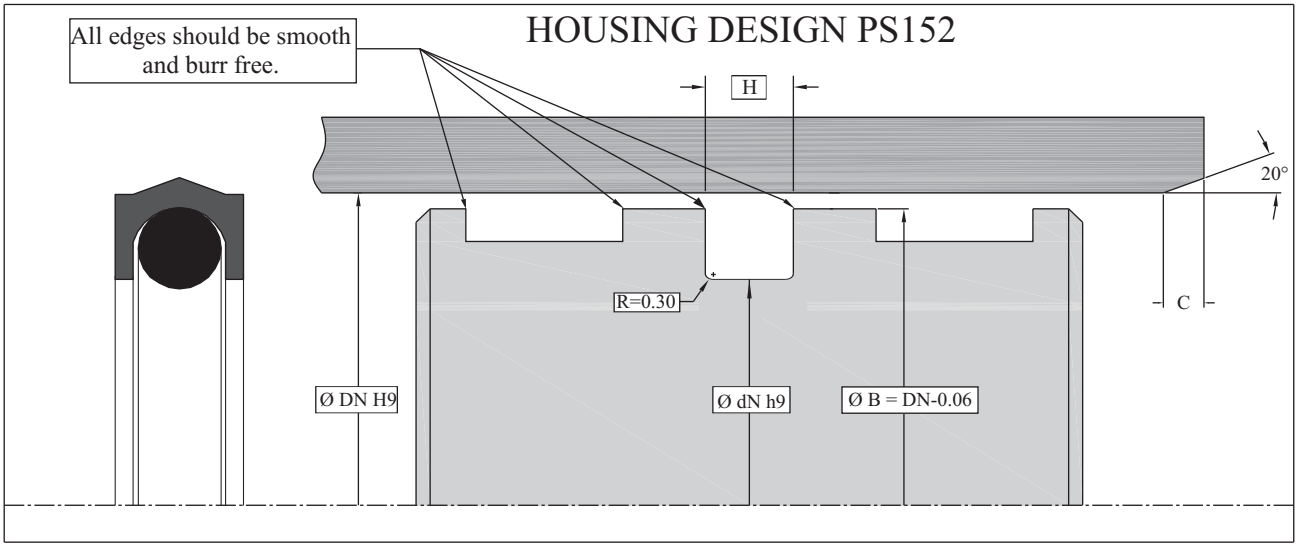
Seal Profile	PS152
Working Pressure(max.)	315 bar
Surface Speed(max.)	1 m/s

Installation :
PS152 piston seals can be fitted on single-piece pistons. First insert the rubber O Ring into the housing carefully to avoid twisting of Ring. Install the Polyurethane / polyester part carefully over the O ring.

Ordering format :
Please Follow ROYAL order code to place your order.
Like - for 25.40 X 19.25 X 4.75 - order cord is PS152-IN-01000-00758-0187

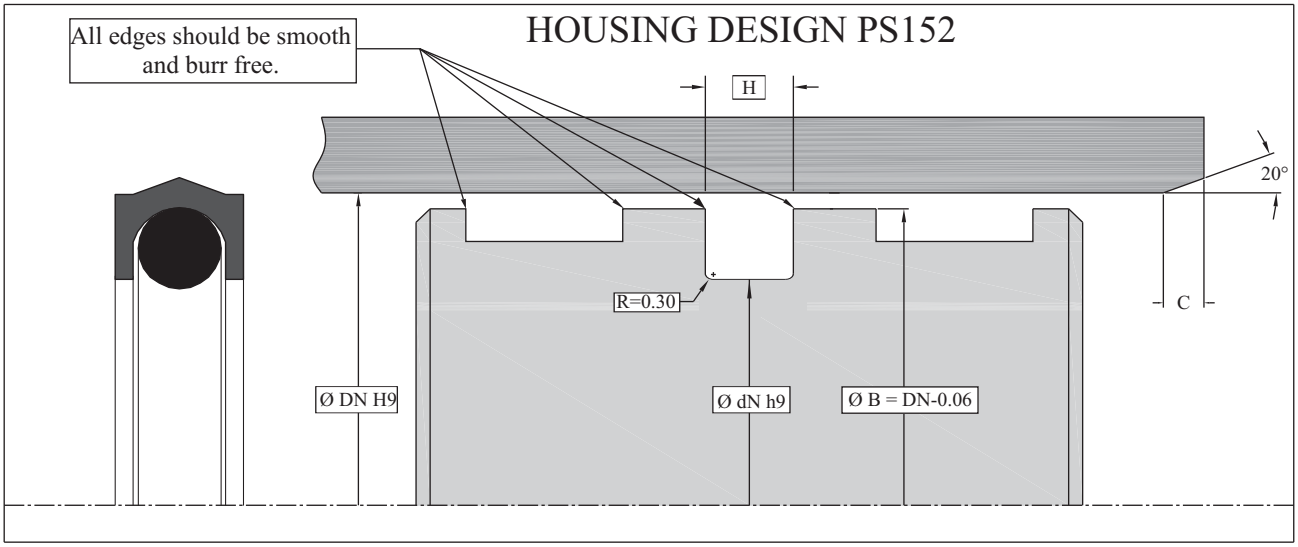
Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN PS152



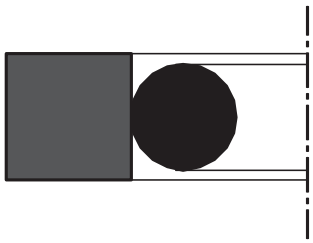
ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)	O RING REF. DASH NO.	ORDER CODE
IMPERIAL SIZES				
25.40	19.25	4.75	210	PS152-IN-01000-00758-0187
28.58	21.16	4.75	212	PS152-IN-01125-00833-0187
31.75	25.60	4.75	214	PS152-IN-01250-01008-0187
34.92	28.78	4.75	216	PS152-IN-01375-01133-0187
38.10	31.95	4.75	218	PS152-IN-01500-01258-0187
38.10	31.95	6.99	218-2	PS152-IN-01500-01258-0275
41.28	35.13	4.75	220	PS152-IN-01625-01383-0187
44.45	38.30	4.75	222	PS152-IN-01750-01508-0187
44.45	38.30	6.99	222-2	PS152-IN-01750-01508-0275
50.80	41.40	7.14	326	PS152-IN-02000-01630-0281
50.80	41.40	10.41	326-2	PS152-IN-02000-01630-0410
57.15	47.75	7.14	328	PS152-IN-02250-01880-0281
63.50	54.10	7.14	330	PS152-IN-02500-02130-0281
63.50	54.10	10.41	330-2	PS152-IN-02500-02130-0410
69.85	60.45	7.14	332	PS152-IN-02750-02380-0281
69.85	60.45	10.41	332-2	PS152-IN-02750-02380-0410
76.20	66.80	7.14	334	PS152-IN-03000-02630-0281
76.20	66.80	7.14	334-2	PS152-IN-03000-02630-0281
82.55	73.15	10.41	336	PS152-IN-03250-02880-0410
88.90	79.50	7.14	338	PS152-IN-03500-03130-0281
88.90	79.50	10.41	338-2	PS152-IN-03500-03130-0410



ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)	O RING REF. DASH NO.	ORDER CODE
95.25	85.85	7.14	340	PS152-IN-03750-03380-0281
101.60	92.20	7.14	342	PS152-IN-04000-03630-0281
101.60	92.20	10.41	342-2	PS152-IN-04000-03630-0410
107.95	98.55	7.14	344	PS152-IN-04250-03880-0281
107.95	98.55	10.41	344-2	PS152-IN-04250-03880-0410
114.30	104.90	7.14	346	PS152-IN-04500-04130-0281
114.30	104.90	10.41	346-2	PS152-IN-04500-04130-0410
120.65	111.25	7.14	348	PS152-IN-04750-04380-0281
120.65	111.25	10.41	348-2	PS152-IN-04750-04380-0410
127.00	117.60	7.14	350	PS152-IN-05000-04630-0281
127.00	115.01	9.52	425	PS152-IN-05000-04528-0375
127.00	115.01	13.67	425-2	PS152-IN-05000-04528-0538
139.70	127.71	9.52	429	PS152-IN-05500-05028-0375
152.40	140.41	9.52	433	PS152-IN-06000-05528-0375
152.40	140.41	13.67	433-2	PS152-IN-06000-05528-0538
165.10	153.11	9.52	437	PS152-IN-06500-06028-0375
177.80	165.81	9.52	439	PS152-IN-07000-06528-0375
184.15	172.16	9.52	440	PS152-IN-07250-06778-0375
190.50	178.51	9.52	441	PS152-IN-07500-07028-0375
196.85	184.86	9.52	441	PS152-IN-07750-07278-0375
203.20	191.21	9.52	443	PS152-IN-08000-07528-0375
228.60	216.61	9.52	446	PS152-IN-09000-08528-0375
254.00	242.01	9.52	448	PS152-IN-10000-09528-0375

PISTON SEAL	PS160	

Features:

Piston seal profile PS160 consists of two piece seal set for sealing pistons with one modified PTFE seal ring and one NBR 'O' ring as energizer.

Composition:

SEAL		PS160
Seal ring	Material	PTFE+ Glass Fiber
	Royal code	TG25
Square Ring	Material	Nitrile Butadiene rubber (NBR)
	Royal code	NBR70
Working Temperature Range Medium : Hydraulic Oil		-20°C to +100°C

For compound detail please refer Royal seal materials. (Section - III)

Properties:

- Low friction, free of stick slip
- High abrasion resistance
- Wide temperature Range
- Wide fluid application range
- Minimum installation space required

Application:

- Mobile Hydraulics
- Machine tools
- Control and regulating apparatus

General Technical Data:

Seal Profile	PS160
Working Pressure(max.)	315 bar
Surface Speed(max.)	4 m/s

Installation :

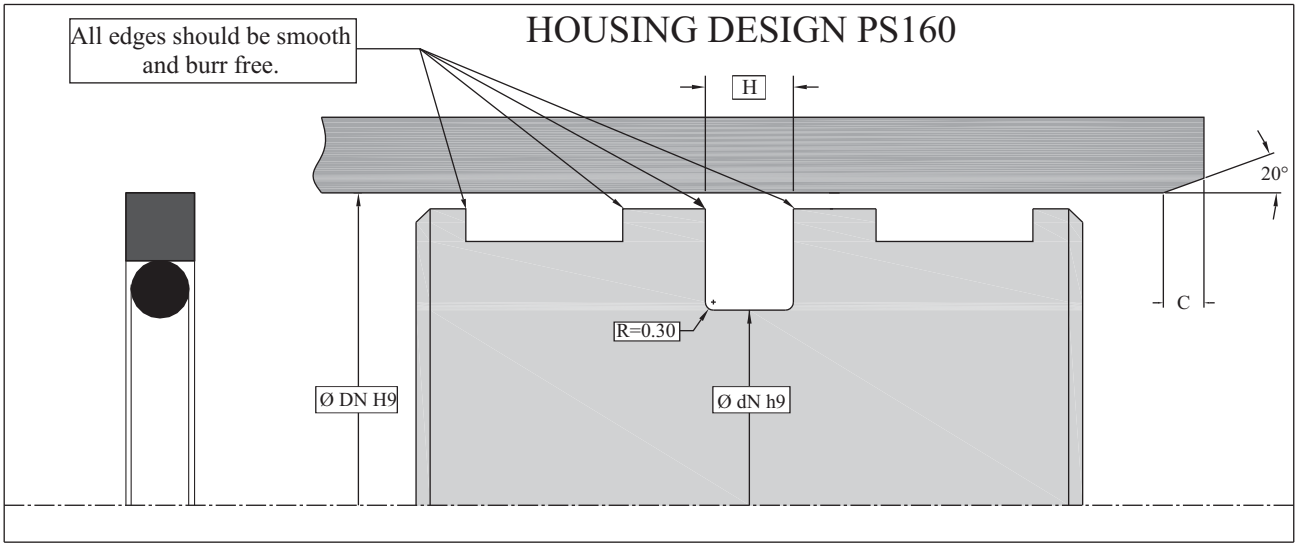
Ps160 piston seals can be fitted on single-piece pistons. First insert the rubber O Ring into the housing carefully to avoid twisting of Ring. Install the PTFE part carefully over the O ring.

Ordering format :

Please Follow ROYAL order code to place your order.
Like - for 12.70 X 6.12 X 2.11 - order cord is PS160-IN-00500-00241-0083

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN PS160

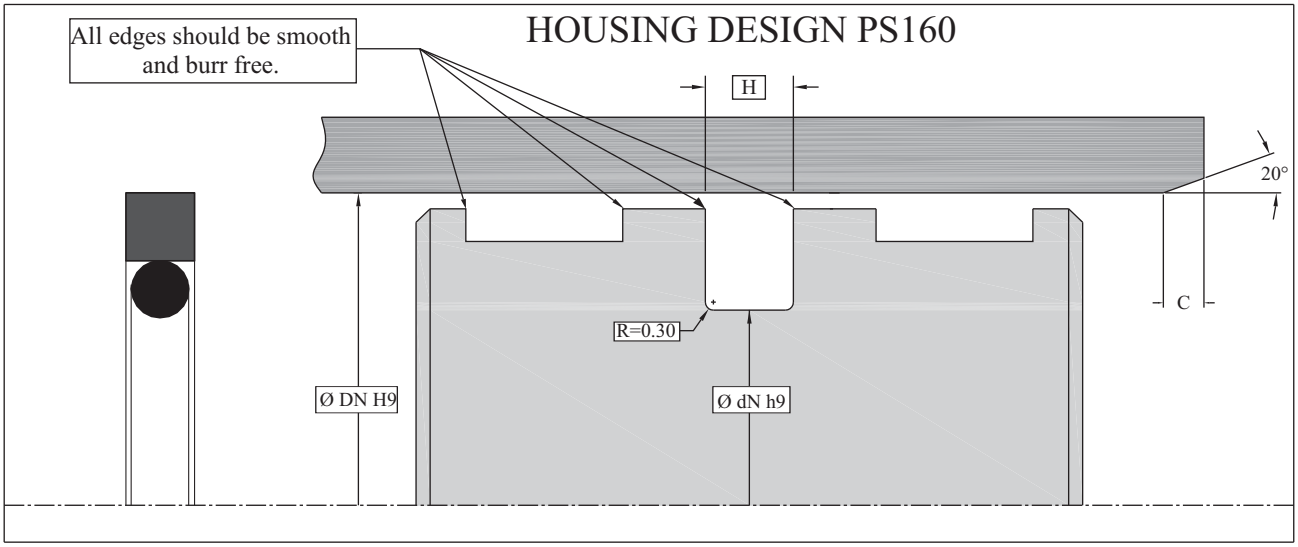


ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)		ORDER CODE
IMPERIAL SIZES				
12.70	6.12	2.11		PS160-IN-00500-00241-0083
15.88	9.30	2.11		PS160-IN-00625-00366-0083
19.05	12.47	2.11		PS160-IN-00750-00491-0083
25.40	18.82	2.11		PS160-IN-01000-00741-0083
28.58	22.00	2.11		PS160-IN-01125-00866-0083
31.75	25.17	2.11		PS160-IN-01250-00991-0083
34.92	28.35	2.11		PS160-IN-01375-01116-0083
38.10	31.52	2.11		PS160-IN-01500-01241-0083
41.28	31.37	3.10		PS160-IN-01625-01235-0122
44.45	34.54	3.10		PS160-IN-01750-01360-0122
47.63	37.67	3.10		PS160-IN-01875-01483-0122
50.80	40.79	3.30		PS160-IN-02000-01606-0130
53.97	44.02	3.30		PS160-IN-02125-01733-0130
57.15	47.14	3.30		PS160-IN-02250-01856-0130
60.32	50.32	3.30		PS160-IN-02375-01981-0130
63.50	53.49	3.30		PS160-IN-02500-02106-0130
69.85	59.84	3.30		PS160-IN-02750-02356-0130
76.20	66.19	3.30		PS160-IN-03000-02606-0130
79.38	69.42	3.30		PS160-IN-03125-02733-0130
82.55	72.54	3.30		PS160-IN-03250-02856-0130
85.73	75.77	3.30		PS160-IN-03375-02983-0130

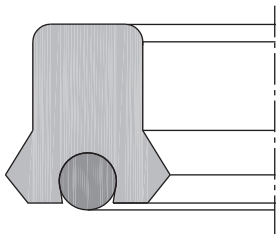
ROYAL STANDARD SIZE LIST				
DN (H9)	**dN (h9)**	**H (+0.20)**		**ORDER CODE**
IMPERIAL SIZES				
88.90	78.89	3.30		PS160-IN-03500-03106-0130
92.07	82.12	3.30		PS160-IN-03625-03233-0130
95.25	85.24	3.30		PS160-IN-03750-03356-0130
101.60	91.59	3.30		PS160-IN-04000-03606-0130
107.95	97.94	3.30		PS160-IN-04250-03856-0130
111.76	101.80	3.30		PS160-IN-04400-04008-0130
114.30	104.29	3.30		PS160-IN-04500-04106-0130
120.65	110.64	3.30		PS160-IN-04750-04356-0130
123.82	113.87	3.30		PS160-IN-04875-04483-0130
127.00	116.99	3.30		PS160-IN-05000-04606-0130
133.35	123.34	3.30		PS160-IN-05250-04856-0130
139.70	129.69	3.30		PS160-IN-05500-05106-0130
146.05	132.89	4.04		PS160-IN-05750-05232-0159
149.23	136.07	4.04		PS160-IN-05875-05357-0159
152.40	139.24	4.04		PS160-IN-06000-05482-0159
155.57	142.42	4.04		PS160-IN-06125-05607-0159
158.75	145.59	4.04		PS160-IN-06250-05732-0159
165.10	151.94	4.04		PS160-IN-06500-05982-0159
171.45	158.29	4.04		PS160-IN-06750-06232-0159
177.80	164.64	4.04		PS160-IN-07000-06482-0159
184.15	170.99	4.04		PS160-IN-07250-06732-0159
190.50	177.34	4.04		PS160-IN-07500-06982-0159
196.85	183.69	4.04		PS160-IN-07750-07232-0159
203.20	190.04	4.04		PS160-IN-08000-07482-0159
206.38	117.02	4.04		PS160-IN-08125-04607-0159
209.55	196.39	4.04		PS160-IN-08250-07732-0159
215.90	202.74	4.04		PS160-IN-08500-07982-0159
222.25	209.09	4.04		PS160-IN-08750-08232-0159
228.60	215.44	4.04		PS160-IN-09000-08482-0159
234.95	221.79	4.04		PS160-IN-09250-08732-0159
241.30	228.14	4.04		PS160-IN-09500-08982-0159
247.65	234.49	4.04		PS160-IN-09750-09232-0159
254.00	240.84	4.04		PS160-IN-10000-09482-0159

HOUSING DESIGN PS160



ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	H (+0.20)		ORDER CODE
IMPERIAL SIZES				
260.35	247.19	4.04		PS160-IN-10250-09732-0159
266.70	253.54	4.04		PS160-IN-10500-09982-0159
273.05	259.89	4.04		PS160-IN-10750-10232-0159
279.40	266.24	4.04		PS160-IN-11000-10482-0159
292.10	278.94	4.04		PS160-IN-11500-10982-0159
304.80	291.64	4.04		PS160-IN-12000-11482-0159
317.50	304.34	4.04		PS160-IN-12500-11982-0159
330.20	317.04	4.04		PS160-IN-13000-12482-0159
355.60	342.44	4.04		PS160-IN-14000-13482-0159

SYMMETRIC SEAL	SS001	

Features :

SS001 is a single acting U cup type seal with O ring energizer used for both piston and rod seal.

Composition:

Components	Material:	Royal Code
Main seal	Polyurethane	PUHP93
Energizer O ring	NBR	NBR70

For material detail please refer Royal seal materials.(Section III)

Properties:

- High resistance to abrasion
- Easy installation
- Second seal ring take load in high pressure conditions

Application:

- Mobile Hydraulics
- Injection molding machines
- Machine tools
- Agricultural equipment
- Standard cylinders

General Technical Data:

Temperature range	-20°C to +100°C
Working Pressure(max.)	400 bar*
Surface Speed(max.)	0.5 m/s

* Maximum working pressure can be extended up to 400 bar with the use of a back up ring. If this option is required ask ROYAL.

Extrusion Gap dimension:

The gap dimension on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the maximum clearance Ø and min rod Ø.

Maximum permissible gap dimension (s)			
160 bar	260 bar	320 bar	400 bar
0,4	0,4	0,35	0,3

Minimum Chamfer C:

DN - dN	Up to 8	8.1 to10	10.1 to 15	15.1 to 20	20.1 to 25	25.1 and above
C	4.0	4.5	6.0	7.5	8.0	10.0

Fitting:

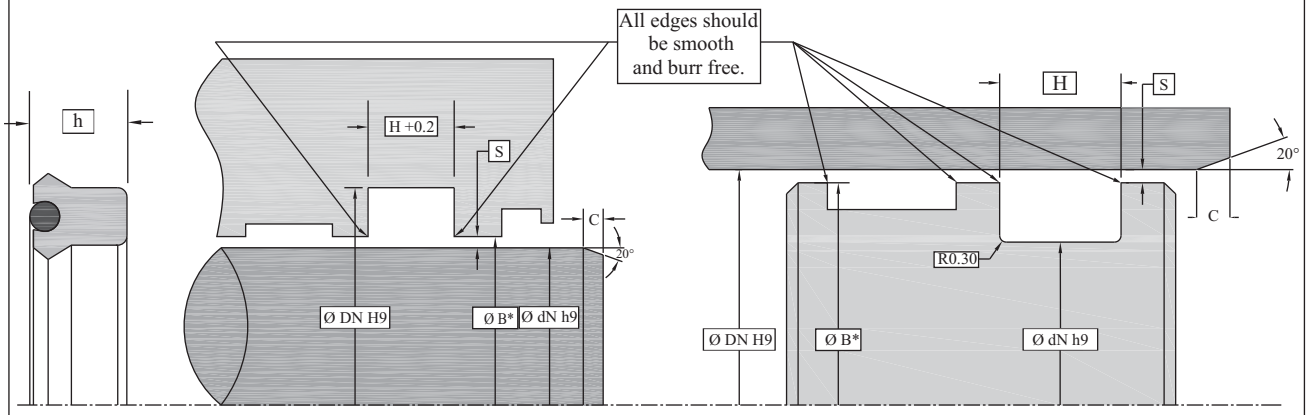
SS001 is very easy to fit. Careful fitting of the seal is a prerequisite for it's perfect functioning.

Ordering format:

Please mention ROYAL order code in your order.
Example: Size 40 X 55 X 12.5
Order code is SS001-U-04000-05500-1250

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

SS001 FOR ROD AND PISTON APPLICATION



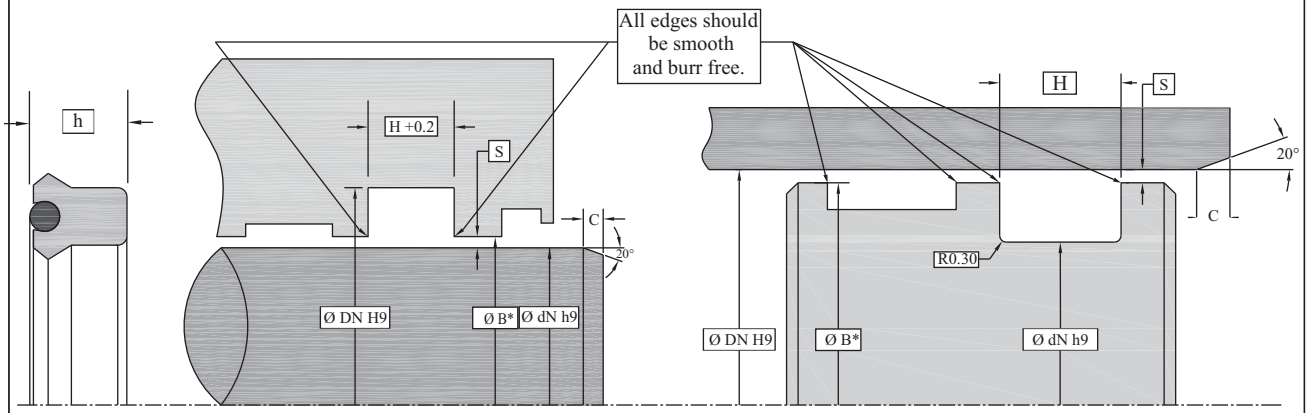
* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

dN (f8)	DN (H9)	h	H (+0.20)	ORDER CODE
5.00	12.00	5.90	6.50	SS001-U-00500-01200-0650
10.00	15.00	3.60	4.00	SS001-U-01000-01500-0400
12.00	18.00	5.50	6.00	SS001-U-01200-01800-0600
12.00	22.00	7.30	8.00	SS001-U-01200-02200-0800
18.00	26.00	5.30	5.80	SS001-U-01800-02600-0580
25.00	33.00	5.50	6.00	SS001-U-02500-03300-0600
25.00	33.00	5.50	6.00	SS001-U-02500-03300-0600
25.00	33.00	6.40	7.00	SS001-U-02500-03300-0700
25.00	38.00	9.10	10.00	SS001-U-02500-03800-1000
30.00	40.00	7.30	8.00	SS001-U-03000-04000-0800
30.00	40.00	9.10	10.00	SS001-U-03000-04000-1000
31.50	41.50	6.20	6.80	SS001-U-03150-04150-0680
35.00	45.00	7.30	8.00	SS001-U-03500-04500-0800
40.00	50.00	7.30	8.00	SS001-U-04000-05000-0800
40.00	55.00	11.40	12.50	SS001-U-04000-05500-1250
45.00	55.00	7.30	8.00	SS001-U-04500-05500-0800
45.00	55.00	9.10	10.00	SS001-U-04500-05500-1000
45.00	65.00	11.40	12.50	SS001-U-04500-06500-1250
50.00	60.00	7.30	8.00	SS001-U-05000-06000-0800
50.00	65.00	11.40	12.50	SS001-U-05000-06500-1250
50.00	70.00	14.50	16.00	SS001-U-05000-07000-1600
55.00	65.00	11.80	13.00	SS001-U-05500-06500-1300
55.00	70.00	9.10	10.00	SS001-U-05500-07000-1000

ROYAL STANDARD SIZE LIST				
dN (f8)	**DN (H9)**	**h**	**H (+0.20)**	**ORDER CODE**
55.00	70.00	10.90	12.00	SS001-U-05500-07000-1200
55.00	75.00	14.50	16.00	SS001-U-05500-07500-1600
56.00	66.00	10.00	11.00	SS001-U-05600-06600-1100
56.00	71.00	9.10	10.00	SS001-U-05600-07100-1000
60.00	75.00	11.40	12.50	SS001-U-06000-07500-1250
60.00	75.00	11.40	12.50	SS001-U-06000-07500-1250
60.00	80.00	11.80	13.00	SS001-U-06000-08000-1300
62.00	72.00	7.30	8.00	SS001-U-06200-07200-0800
62.00	72.00	10.00	11.00	SS001-U-06200-07200-1100
63.00	73.00	9.10	10.00	SS001-U-06300-07300-1000
65.00	80.00	11.40	12.50	SS001-U-06500-08000-1250
68.00	76.00	9.10	10.00	SS001-U-06800-07600-1000
70.00	80.00	11.80	13.00	SS001-U-07000-08000-1300
70.00	80.00	11.80	13.00	SS001-U-07000-08000-1300
70.00	85.00	11.40	12.50	SS001-U-07000-08500-1250
72.00	82.00	10.00	11.00	SS001-U-07200-08200-1100
72.00	87.00	9.10	10.00	SS001-U-07200-08700-1000
75.00	85.00	11.80	13.00	SS001-U-07500-08500-1300
75.00	95.00	14.50	16.00	SS001-U-07500-09500-1600
80.00	90.00	10.00	11.00	SS001-U-08000-09000-1100
80.00	100.00	14.50	16.00	SS001-U-08000-10000-1600
82.00	92.00	10.00	11.00	SS001-U-08200-09200-1100
85.00	100.00	9.10	10.00	SS001-U-08500-10000-1000
85.70	100.00	16.40	18.00	SS001-U-08570-10000-1800
86.00	94.00	9.10	10.00	SS001-U-08600-09400-1000
88.00	98.00	7.30	8.00	SS001-U-08800-09800-0800
90.00	110.00	11.80	13.00	SS001-U-09000-11000-1300
90.00	110.00	14.50	16.00	SS001-U-09000-11000-1600
94.00	104.00	11.80	13.00	SS001-U-09400-10400-1300
100.00	110.00	9.10	10.00	SS001-U-10000-11000-1000
100.00	115.00	11.40	12.50	SS001-U-10000-11500-1250
105.00	120.00	10.00	11.00	SS001-U-10500-12000-1100
112.00	122.00	7.30	8.00	SS001-U-11200-12200-0800
114.00	124.00	11.80	13.00	SS001-U-11400-12400-1300
130.00	160.00	17.30	19.00	SS001-U-13000-16000-1900

SS001 FOR ROD AND PISTON APPLICATION



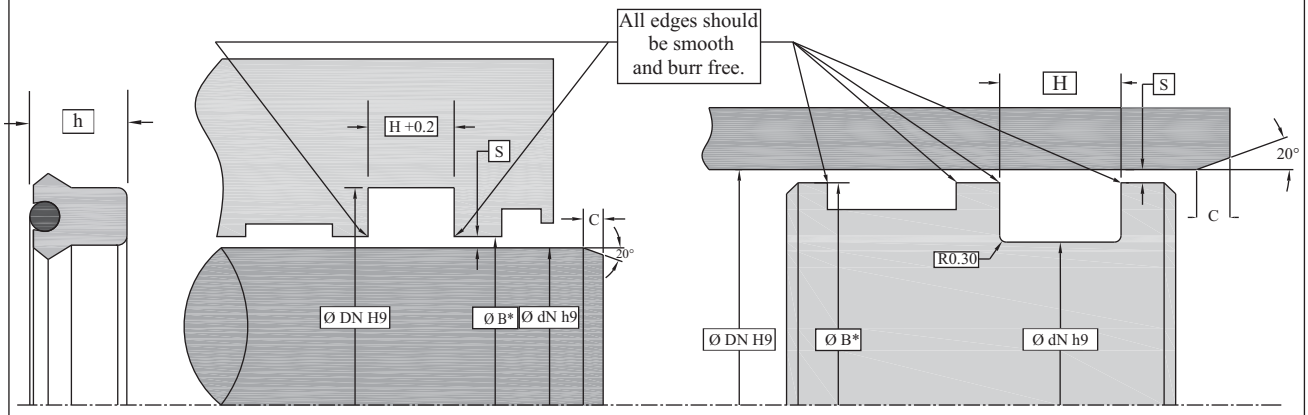
* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

dN (f8)	DN (H9)	h	H (+0.20)	ORDER CODE
210.00	230.00	14.50	16.00	SS001-U-21000-23000-1600
230.00	250.00	14.50	16.00	SS001-U-23000-25000-1600
IMPERIAL SIZES				
19.05	25.40	3.12	3.43	SS001-U-IN-00750-01000-0135
19.05	25.40	6.35	6.99	SS001-U-IN-00750-01000-0275
25.40	31.75	3.12	3.43	SS001-U-IN-01000-01250-0135
28.58	41.28	8.66	9.53	SS001-U-IN-01125-01625-0375
28.58	44.45	9.93	10.92	SS001-U-IN-01125-01750-0430
31.75	38.10	6.35	6.99	SS001-U-IN-01250-01500-0275
31.75	41.28	7.49	8.26	SS001-U-IN-01250-01625-0325
38.10	47.63	4.67	5.13	SS001-U-IN-01500-01875-0202
38.10	47.63	8.66	9.53	SS001-U-IN-01500-01875-0375
38.10	50.80	8.66	9.53	SS001-U-IN-01500-02000-0375
38.10	53.98	10.08	11.10	SS001-U-IN-01500-02125-0437
44.45	57.15	8.66	9.53	SS001-U-IN-01750-02250-0375
44.45	60.33	11.56	12.70	SS001-U-IN-01750-02375-0500
50.80	63.50	8.66	9.53	SS001-U-IN-02000-02500-0375
50.80	66.68	8.66	9.53	SS001-U-IN-02000-02625-0375

ROYAL STANDARD SIZE LIST				
dN (f8)	DN (H9)	h	H (+0.20)	ORDER CODE
57.15	69.85	8.66	9.53	SS001-U-IN-02250-02750-0375
57.15	69.85	9.53	10.49	SS001-U-IN-02250-02750-0413
63.50	76.20	5.77	6.35	SS001-U-IN-02500-03000-0250
63.50	76.20	8.66	9.53	SS001-U-IN-02500-03000-0375
63.50	79.38	11.56	12.70	SS001-U-IN-02500-03125-0500
73.03	82.55	6.48	7.11	SS001-U-IN-02875-03250-0280
76.20	95.25	14.43	15.88	SS001-U-IN-03000-03750-0625
82.55	101.60	8.66	9.53	SS001-U-IN-03250-04000-0375
88.90	101.60	8.66	9.53	SS001-U-IN-03500-04000-0375
92.08	101.60	9.47	10.41	SS001-U-IN-03625-04000-0410
95.25	104.78	5.21	5.72	SS001-U-IN-03750-04125-0225
101.60	114.30	5.77	6.35	SS001-U-IN-04000-04500-0250
127.00	139.70	8.66	9.53	SS001-U-IN-05000-05500-0375
127.00	146.05	17.32	19.05	SS001-U-IN-05000-05750-0750
136.53	152.40	11.56	12.70	SS001-U-IN-05375-06000-0500
139.70	155.58	8.89	9.78	SS001-U-IN-05500-06125-0385
152.40	171.45	15.88	17.48	SS001-U-IN-06000-06750-0688
171.45	190.50	8.66	9.53	SS001-U-IN-06750-07500-0375
177.80	196.85	15.88	17.48	SS001-U-IN-07000-07750-0688
190.50	209.55	17.32	19.05	SS001-U-IN-07500-08250-0750
368.30	349.25	8.66	9.53	SS001-U-IN-14500-13750-0375
393.70	374.65	8.66	9.53	SS001-U-IN-15500-14750-0375
444.50	425.45	8.66	9.53	SS001-U-IN-17500-16750-0375

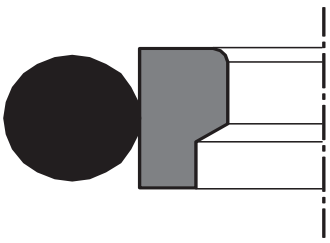
SS001 FOR ROD AND PISTON APPLICATION



* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

dN (f8)	DN (H9)	h	H (+0.20)	ORDER CODE

ROD SEAL	RS018	

Features:

Rod seal profile RS018 consists of two piece seal set for sealing rod with one seal ring and one energizer O-ring.

Composition:

SEAL COMBINATION		RS018-N	RS018-F
Seal ring	Material	PTFE+ Bronze +MoS2	PTFE+ Bronze +MoS2
	Royal Code	TB40M	TB40M
O Ring	Material	Nitrile Butadiene rubber (NBR)	Fluoro Carbon Rubber (FPM)
	Royal Code	NBR70	FPM80

For compound detail please refer Royal seal materials.

Properties:

- Low friction, free of stick slip
- High abrasion resistance.
- Wide temperature Range.
- Good thermal conductivity.
- Low wear and high extrusion resistance.
- Wide fluid application range.
- Resistance against cold flow.

Application:

- Mobile Hydraulics
- Injection moulding machines
- Machine tools
- Material Handling Equipments.

General Technical Data:

Seal Profile	RS018-N	RS018-F
Temperature range	-20°C to 100°C	-20°C to +200°C
Working Pressure(max.)	400 bar	400 bar
Surface Speed(max.)	4 m/s	4 m/s.

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Extrusion Gap dimension:

The gap dimension on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the maximum clearance Ø and min rod Ø.

Profile dimension	Maximum permissible gap (S)			
H	160 bar	260 bar	320 bar	400 bar
2.20	0.35	0.30	-	-
3.20	0.40	0.35	-	-
4.20	0.50	0.40	0.30	-
6.30	0.60	0.50	0.45	0.40
8.10	0.70	0.60	0.55	0.50

Installation:

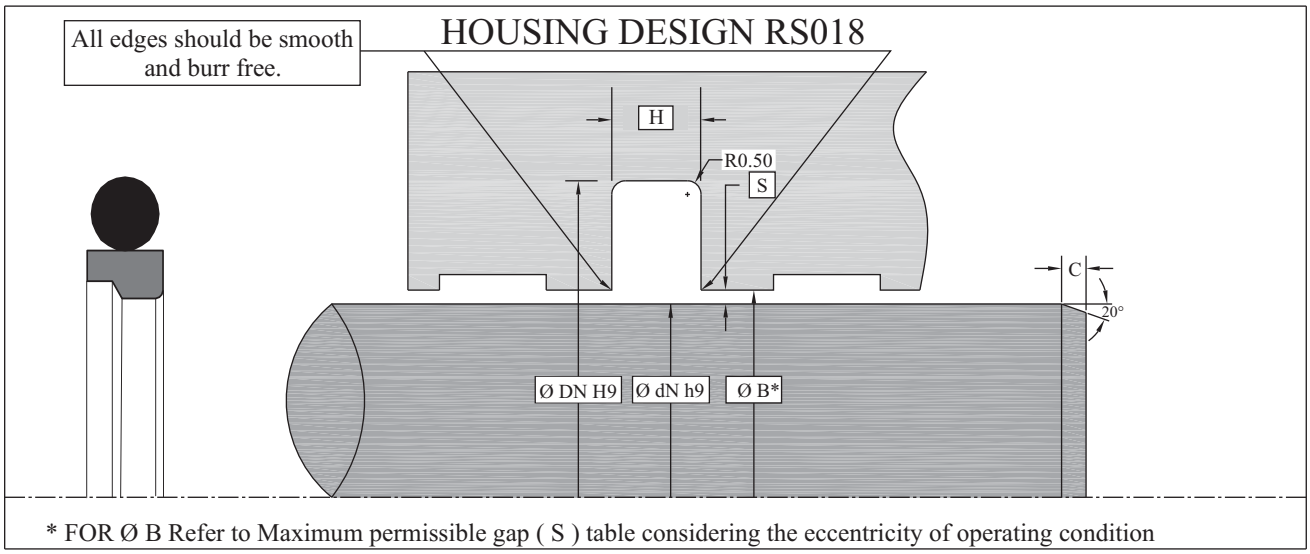
RS018 is very easy to fit. Insert the O ring first and then turn the PTFE part in kidney shape and insert in the groove. Gauge the seal by a taper rod type fixture. Careful fitting of the seal is a pre-requisite for it's perfect functioning.

Ordering format:

Please mention ROYAL order code in your order.
Example: Size 6 X 10.9 X 2.2
For Temp range: -20°C to +100°C. (O ring material is NBR70) Order code is RS018-N-00600-01090-0220
For Temp range: -20°C to +200°C. (O ring material is FPM80) Order code is RS018-F-00600-01090-0220

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN RS018

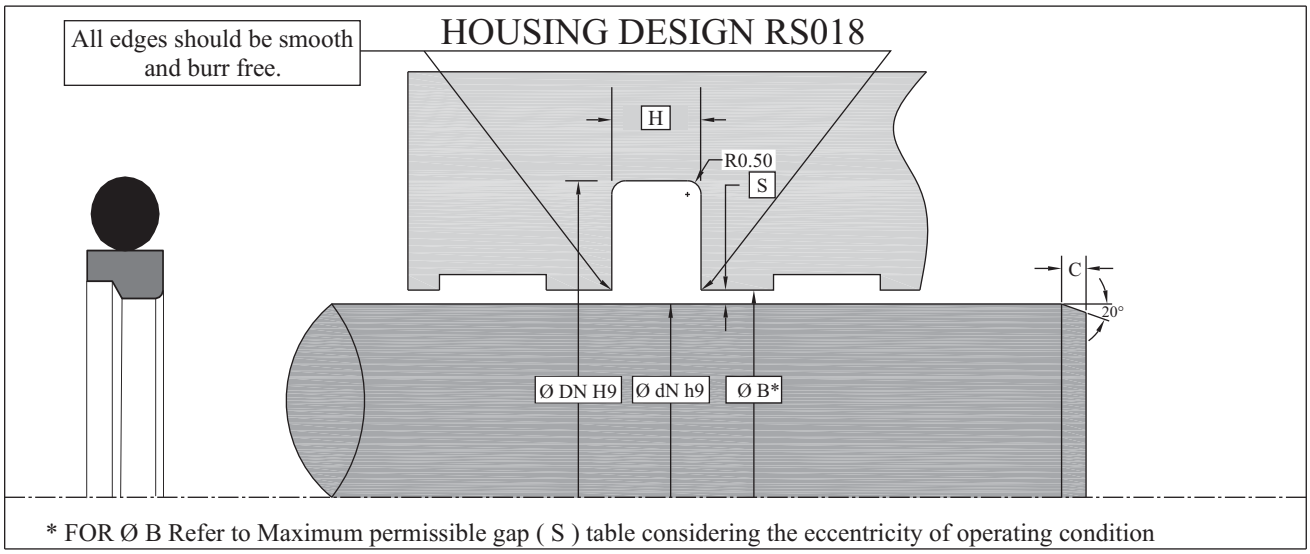


* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H9)	H (+0.20)	C	O-RING SIZE	ORDER CODE For Temp range: -20°C to +100°C use RS018-N -20°C to +200°C use RS018-F
6.00	10.90	2.20	2.50	7.65 X 1.78	RS018-N/F-00600-01090-0220
8.00	12.90	2.20	2.50	9.50 X 1.80	RS018-N/F-00800-01290-0220
10.00	14.90	2.20	2.50	12.42 X 1.78	RS018-N/F-01000-01490-0220
10.00	17.30	3.20	2.50	10.77 X 2.62	RS018-N/F-01000-01730-0320
12.00	16.90	2.20	2.50	12.42 X 1.78	RS018-N/F-01200-01690-0220
12.00	19.30	3.20	2.50	13.94 X 2.62	RS018-N/F-01200-01930-0320
14.00	18.90	2.20	2.50	15.60 X 1.78	RS018-N/F-01400-01890-0220
14.00	21.30	3.20	2.50	17.12 X 2.62	RS018-N/F-01400-02130-0320
15.00	22.30	3.20	2.50	17.12 X 2.62	RS018-N/F-01500-02230-0320
15.00	30.10	6.30	6.00	19.99 X 5.33	RS018-N/F-01500-03010-0630
16.00	20.90	2.20	2.50	17.17 X 1.78	RS018-N/F-01600-02090-0220
16.00	23.30	3.20	2.50	18.72 X 2.62	RS018-N/F-01600-02330-0320
18.00	22.90	2.20	2.50	18.77 X 1.78	RS018-N/F-01800-02290-0220
18.00	25.30	3.20	2.50	20.29 X 2.62	RS018-N/F-01800-02530-0320
18.00	28.70	4.20	3.50	21.89 X 3.53	RS018-N/F-01800-02870-0420
20.00	27.30	3.20	2.50	21.89 X 2.62	RS018-N/F-02000-02730-0320
20.00	30.70	4.20	3.50	23.40 X 3.53	RS018-N/F-02000-03070-0420
20.00	35.10	6.30	6.00	24.77 X 5.33	RS018-N/F-02000-03510-0630
22.00	29.30	3.20	2.50	25.07 X 2.62	RS018-N/F-02200-02930-0320
22.00	32.70	4.20	3.50	26.58 X 3.53	RS018-N/F-02200-03270-0420
25.00	32.30	3.20	2.50	28.24 X 2.62	RS018-N/F-02500-03230-0320
25.00	35.70	4.20	3.50	29.75 X 3.53	RS018-N/F-02500-03570-0420

HOUSING DESIGN RS018



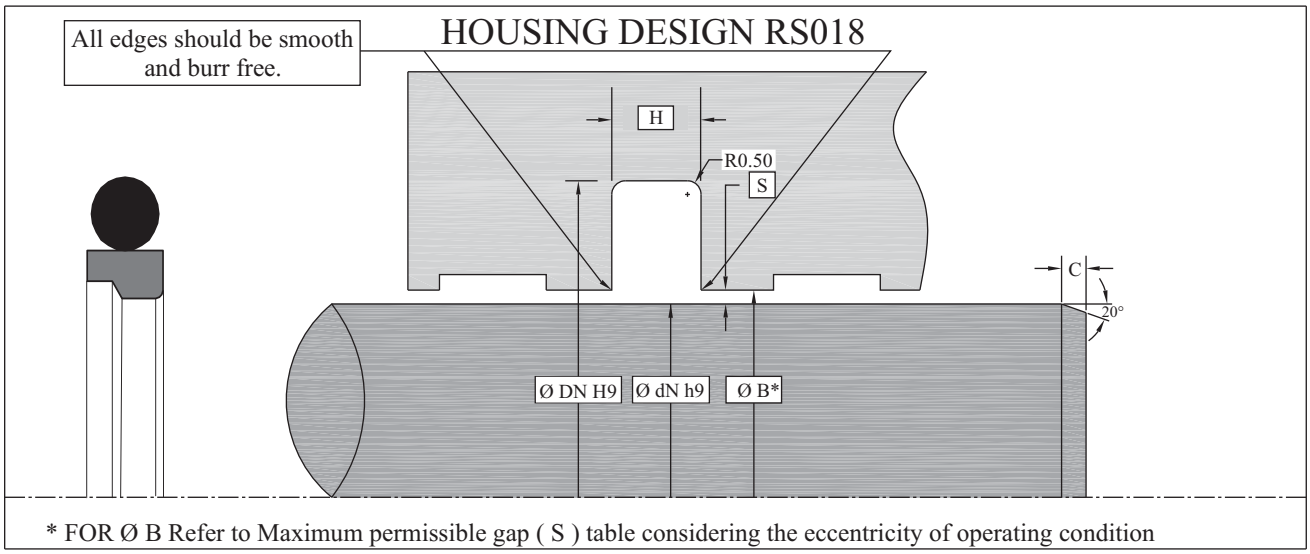
* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H9)	H (+0.20)	C	O-RING SIZE	ORDER CODE For Temp range: -20°C to +100°C use RS018-N -20°C to +200°C use RS018-F
25.00	40.10	6.30	6.00	29.51 X 5.33	RS018-N/F-02500-04010-0630
28.00	35.30	3.20	2.50	29.82 X 2.62	RS018-N/F-02800-03530-0320
28.00	38.70	4.20	3.50	32.92 X 3.53	RS018-N/F-02800-03870-0420
30.00	37.30	3.20	2.50	32.99 X 2.62	RS018-N/F-03000-03730-0320
30.00	40.70	4.20	3.50	34.52 X 3.53	RS018-N/F-03000-04070-0420
30.00	45.10	6.30	6.00	34.29 X 5.33	RS018-N/F-03000-04510-0630
32.00	39.30	3.20	2.50	34.59 X 2.62	RS018-N/F-03200-03930-0320
32.00	42.70	4.20	3.50	36.09 X 3.53	RS018-N/F-03200-04270-0420
35.00	42.30	3.20	2.50	37.77 X 2.62	RS018-N/F-03500-04230-0320
35.00	45.70	4.20	3.50	37.69 X 3.53	RS018-N/F-03500-04570-0420
35.00	50.10	6.30	6.00	40.64 X 5.33	RS018-N/F-03500-05010-0630
36.00	43.30	3.20	2.50	39.34 X 2.62	RS018-N/F-03600-04330-0320
36.00	46.70	4.20	3.50	40.87 X 3.53	RS018-N/F-03600-04670-0420
38.00	48.70	4.20	3.50	42.08 X 3.53	RS018-N/F-03800-04870-0420
38.00	53.10	6.30	5.50	43.82 X 5.33	RS018-N/F-03800-05310-0630
40.00	50.70	4.20	3.50	44.04 X 3.53	RS018-N/F-04000-05070-0420
40.00	55.10	6.30	5.50	45.70 X 5.33	RS018-N/F-04000-05510-0630
42.00	52.70	4.20	3.50	44.04 X 3.53	RS018-N/F-04200-05270-0420
42.00	57.10	6.30	5.50	46.99 X 5.33	RS018-N/F-04200-05710-0630
45.00	55.70	4.20	3.50	48.70 X 3.53	RS018-N/F-04500-05570-0420
45.00	60.10	6.30	5.50	50.17 X 5.33	RS018-N/F-04500-06010-0630
48.00	63.10	6.30	6.00	53.34 X 5.33	RS018-N/F-04800-06310-0630

ROYAL STANDARD SIZE LIST					
ØdN (f8)	**ØDN (H9)**	**H (+0.20)**	**C**	**O-RING SIZE**	**ORDER CODE** For Temp range: -20°C to +100°C use RS018-N -20°C to +200°C use RS018-F
50.00	60.70	4.20	3.50	53.57 X 3.53	RS018-N/F-05000-06070-0420
50.00	65.10	6.30	5.50	56.52 X 5.33	RS018-N/F-05000-06510-0630
55.00	65.70	4.20	3.50	59.92 X 3.53	RS018-N/F-05500-06570-0420
55.00	70.10	6.30	5.50	59.69 X 5.33	RS018-N/F-05500-07010-0630
56.00	66.70	4.20	3.50	59.92 X 3.53	RS018-N/F-05600-06670-0420
56.00	71.10	6.30	5.50	62.87 X 5.33	RS018-N/F-05600-07110-0630
60.00	70.70	4.20	3.50	63.09 X 3.53	RS018-N/F-06000-07070-0420
60.00	75.10	6.30	5.50	66.04 X 5.33	RS018-N/F-06000-07510-0630
63.00	73.70	4.20	3.50	66.27 X 3.53	RS018-N/F-06300-07370-0420
63.00	78.10	6.30	5.50	69.22 X 5.33	RS018-N/F-06300-07810-0630
65.00	75.70	4.20	3.50	69.44 X 3.53	RS018-N/F-06500-07570-0420
65.00	80.10	6.30	5.50	69.22 X 5.33	RS018-N/F-06500-08010-0630
68.00	83.10	6.30	5.50	72.39 X 5.33	RS018-N/F-06800-08310-0630
70.00	80.70	4.20	3.50	72.62 X 3.53	RS018-N/F-07000-08070-0420
70.00	85.10	6.30	5.50	75.57 X 5.33	RS018-N/F-07000-08510-0630
75.00	85.70	4.20	3.50	78.97 X 3.53	RS018-N/F-07500-08570-0420
75.00	90.10	6.30	5.50	81.92 X 5.33	RS018-N/F-07500-09010-0630
78.00	93.10	6.30	5.50	85.09 X 5.33	RS018-N/F-07800-09310-0630
80.00	90.70	4.20	3.50	85.32 X 3.53	RS018-N/F-08000-09070-0420
80.00	95.10	6.30	5.50	85.09 X 5.33	RS018-N/F-08000-09510-0630
80.00	100.50	8.10	8.00	88.00 X 7.00	RS018-N/F-08000-10050-0810
83.00	98.10	6.30	6.00	88.27 X 5.33	RS018-N/F-08300-09810-0630
85.00	95.70	4.20	3.50	88.49 X 3.53	RS018-N/F-08500-09570-0420
85.00	100.10	6.30	6.00	91.44 X 5.33	RS018-N/F-08500-10010-0630
88.00	103.10	6.30	6.00	94.62 X 5.33	RS018-N/F-08800-10310-0630
90.00	100.70	4.20	3.50	94.84 X 3.53	RS018-N/F-09000-10070-0420
90.00	105.10	6.30	6.00	94.62 X 5.33	RS018-N/F-09000-10510-0630
90.00	110.50	8.10	8.00	98.00 X 7.00	RS018-N/F-09000-11050-0810
95.00	110.10	6.30	6.00	100.97 X 5.33	RS018-N/F-09500-11010-0630
98.00	113.10	6.30	6.00	104.17 X 5.33	RS018-N/F-09800-11310-0630
100.00	115.10	6.30	6.00	104.14 X 5.33	RS018-N/F-10000-11510-0630
100.00	120.50	8.10	8.00	108.00 X 7.00	RS018-N/F-10000-12050-0810
105.00	115.70	4.20	3.50	110.72 x 3.53	RS018-N/F-10500-11570-0420
105.00	120.10	6.30	6.00	110.49 X 5.33	RS018-N/F-10500-12010-0630

HOUSING DESIGN RS018



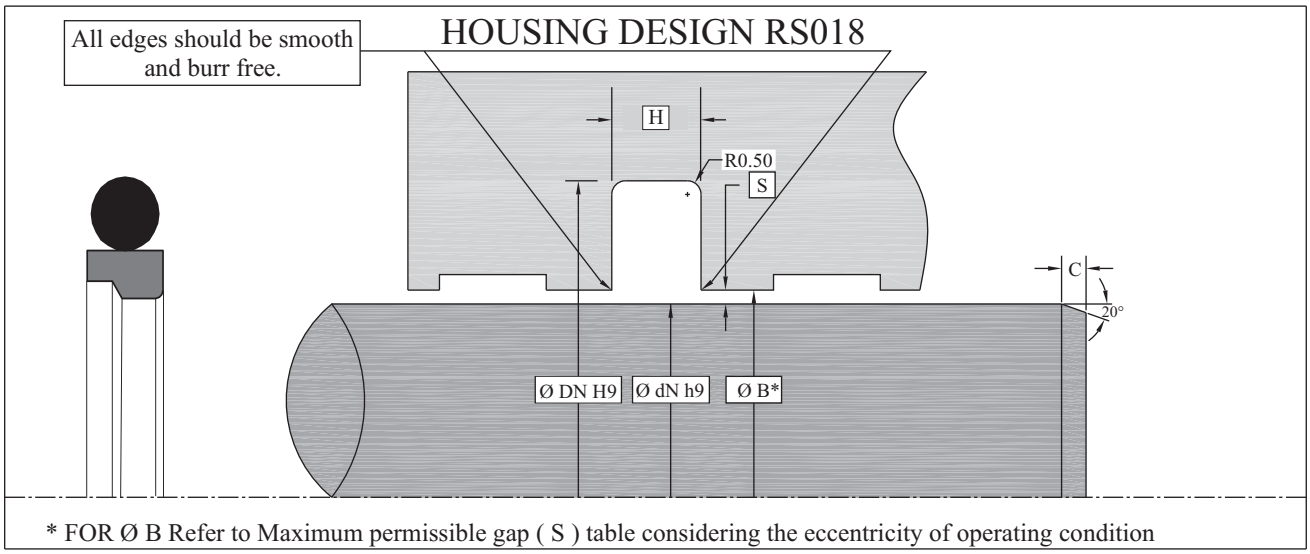
* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H9)	H (+0.20)	C	O-RING SIZE	ORDER CODE For Temp range: -20°C to +100°C use RS018-N -20°C to +200°C use RS018-F
108.00	123.10	6.30	6.00	113.67 X 5.33	RS018-N/F-10800-12310-0630
110.00	125.10	6.30	6.00	116.84 X 5.33	RS018-N/F-11000-12510-0630
110.00	130.50	8.10	8.00	116.84 X 6.99	RS018-N/F-11000-13050-0810
115.00	130.10	6.30	6.00	120.02 X 5.33	RS018-N/F-11500-13010-0630
115.00	135.50	8.10	8.00	120.02 X 6.99	RS018-N/F-11500-13550-0810
118.00	133.10	6.30	6.00	123.19 X 5.33	RS018-N/F-11800-13310-0630
120.00	135.10	6.30	6.00	126.37 X 5.33	RS018-N/F-12000-13510-0630
120.00	140.50	8.10	8.00	126.37 X 6.99	RS018-N/F-12000-14050-0810
125.00	140.10	6.30	6.00	129.54 X 5.33	RS018-N/F-12500-14010-0630
125.00	145.50	8.10	8.00	132.72 X 6.99	RS018-N/F-12500-14550-0810
127.00	137.70	4.20	3.50	129.77 x 3.53	RS018-N/F-12700-13770-0420
127.00	142.10	6.30	6.00	132.72 X 5.33	RS018-N/F-12700-14210-0630
128.00	143.10	6.30	6.00	132.72 X 5.33	RS018-N/F-12800-14310-0630
130.00	145.10	6.30	6.00	135.89 X 5.33	RS018-N/F-13000-14510-0630
130.00	150.50	8.10	8.00	135.89 X 6.99	RS018-N/F-13000-15050-0810
135.00	150.10	6.30	6.00	139.07 X 5.33	RS018-N/F-13500-15010-0630
135.00	155.50	8.10	8.00	142.24 X 6.99	RS018-N/F-13500-15550-0810
136.00	151.10	6.30	6.00	142.24 X 5.33	RS018-N/F-13600-15110-0630
138.00	153.10	6.30	6.00	142.24 X 5.33	RS018-N/F-13800-15310-0630
140.00	150.70	4.20	3.50	145.64 x 3.53	RS018-N/F-14000-15070-0420
140.00	155.10	6.30	6.00	145.42 X 5.33	RS018-N/F-14000-15510-0630
140.00	160.50	8.10	8.00	145.42 X 6.99	RS018-N/F-14000-16050-0810

ROYAL STANDARD SIZE LIST					
ØdN (f8)	**ØDN (H9)**	**H (+0.20)**	**C**	**O-RING SIZE**	**ORDER CODE** For Temp range: -20°C to +100°C use RS018-N -20°C to +200°C use RS018-F
145.00	155.70	4.20	3.50	148.82 x 3.53	RS018-N/F-14500-15570-0420
145.00	160.10	6.30	6.00	151.77 X 5.33	RS018-N/F-14500-16010-0630
150.00	165.10	6.30	6.00	154.50 X 5.33	RS018-N/F-15000-16510-0630
150.00	170.50	8.10	8.00	158.12 X 6.99	RS018-N/F-15000-17050-0810
155.00	170.10	6.30	8.00	158.12 X 5.33	RS018-N/F-15500-17010-0630
160.00	175.10	6.30	6.00	164.47 X 5.33	RS018-N/F-16000-17510-0630
160.00	180.50	8.10	8.00	170.82 X 6.99	RS018-N/F-16000-18050-0810
165.00	180.10	6.30	6.00	170.82 X 5.33	RS018-N/F-16500-18010-0630
170.00	185.10	6.30	6.00	175.02 X 5.33	RS018-N/F-17000-18510-0630
170.00	190.50	8.10	8.00	177.17 X 6.99	RS018-N/F-17000-19050-0810
175.00	185.70	4.20	3.50	177.39 x 3.53	RS018-N/F-17500-18570-0420
175.00	190.10	6.30	6.00	179.30 X 5.33	RS018-N/F-17500-19010-0630
175.00	195.50	8.10	8.00	183.52 X 6.99	RS018-N/F-17500-19550-0810
180.00	195.10	6.30	6.00	183.52 X 5.33	RS018-N/F-18000-19510-0630
180.00	200.50	8.10	8.00	189.87 X 6.99	RS018-N/F-18000-20050-0810
185.00	200.10	6.30	6.00	189.87 X 5.33	RS018-N/F-18500-20010-0630
190.00	205.50	6.30	6.00	196.22 X 5.33	RS018-N/F-19000-20550-0630
190.00	210.50	8.10	8.00	196.22 X 6.99	RS018-N/F-19000-21050-0810
195.00	210.10	6.30	6.00	202.57 X 5.33	RS018-N/F-19500-21010-0630
200.00	215.10	6.30	6.00	202.57 X 5.33	RS018-N/F-20000-21510-0630
200.00	220.50	8.10	8.00	208.50 X 6.99	RS018-N/F-20000-22050-0810
210.00	230.50	8.10	8.00	215.27 X 6.99	RS018-N/F-21000-23050-0810
215.00	235.50	8.10	8.00	221.86 X 6.99	RS018-N/F-21500-23550-0810
220.00	240.50	8.10	8.00	227.97 X 6.99	RS018-N/F-22000-24050-0810
225.00	245.50	8.10	8.00	234.32 X 6.99	RS018-N/F-22500-24550-0810
230.00	250.50	8.10	8.00	236.00 X 6.99	RS018-N/F-23000-25050-0810
235.00	255.50	8.10	8.00	240.67 X 6.99	RS018-N/F-23500-25550-0810
240.00	260.50	8.10	8.00	247.00 X 6.99	RS018-N/F-24000-26050-0810
245.00	265.50	8.10	8.00	253.37 X 6.99	RS018-N/F-24500-26550-0810
250.00	270.50	8.10	8.00	253.37 X 6.99	RS018-N/F-25000-27050-0810
260.00	284.00	8.10	8.00	266.07 X 6.99	RS018-N/F-26000-28400-0810
270.00	294.00	8.10	8.00	278.77 X 6.99	RS018-N/F-27000-29400-0810
275.00	299.00	8.10	8.00	291.47 X 6.99	RS018-N/F-27500-29900-0810
280.00	304.00	8.10	8.00	291.47 X 6.99	RS018-N/F-28000-30400-0810

HOUSING DESIGN RS018

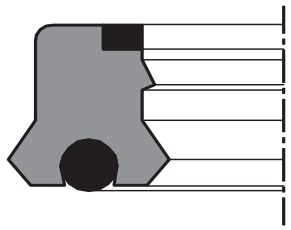


* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H9)	H (+0.20)	C	O-RING SIZE	ORDER CODE For Temp range: -20°C to +100°C use RS018-N -20°C to +200°C use RS018-F
290.00	314.00	8.10	8.00	304.17 X 6.99	RS018-N/F-29000-31400-0810
300.00	324.00	8.10	8.00	316.87 X 6.99	RS018-N/F-30000-32400-0810
303.80	327.80	8.10	8.00	316.87 X 6.99	RS018-N/F-30380-32780-0810
310.00	334.00	8.10	8.00	316.87 X 6.99	RS018-N/F-31000-33400-0810
320.00	344.00	8.10	8.00	329.57 X 6.99	RS018-N/F-32000-34400-0810
330.00	354.00	8.10	8.00	342.27 X 6.99	RS018-N/F-33000-35400-0810
340.00	364.00	8.10	8.00	354.97 X 6.99	RS018-N/F-34000-36400-0810
350.00	374.00	8.10	8.00	367.17 X 6.99	RS018-N/F-35000-37400-0810
360.00	384.00	8.10	8.00	367.17 X 6.99	RS018-N/F-36000-38400-0810
375.00	399.00	8.10	8.00	393.07 X 6.99	RS018-N/F-37500-39900-0810
380.00	404.00	8.10	8.00	393.07 X 6.99	RS018-N/F-38000-40400-0810
390.00	414.00	8.10	8.00	405.26 X 6.99	RS018-N/F-39000-41400-0810
400.00	424.00	8.10	8.00	417.96 X 6.99	RS018-N/F-40000-42400-0810
410.00	434.00	8.10	8.00	417.96 X 6.99	RS018-N/F-41000-43400-0810
420.00	444.00	8.10	8.00	430.66 X 6.99	RS018-N/F-42000-44400-0810
425.00	449.00	8.10	8.00	430.66 X 6.99	RS018-N/F-42500-44900-0810
430.00	454.00	8.10	8.00	443.36 X 6.99	RS018-N/F-43000-45400-0810
440.00	464.00	8.10	8.00	456.06 X 6.99	RS018-N/F-44000-46400-0810
450.00	474.00	8.10	8.00	468.76 X 6.99	RS018-N/F-45000-47400-0810
470.00	494.00	8.10	8.00	481.46 X 6.99	RS018-N/F-47000-49400-0810
480.00	504.00	8.10	8.00	494.16 X 6.99	RS018-N/F-48000-50400-0810
485.00	509.00	8.10	8.00	494.16 X 6.99	RS018-N/F-48500-50900-0810

ROYAL STANDARD SIZE LIST					
ØdN (f8)	**ØDN (H9)**	**H (+0.20)**	**C**	**O-RING SIZE**	**ORDER CODE** For Temp range: -20°C to +100°C use RS018-N -20°C to +200°C use RS018-F
500.00	524.00	8.10	8.00	506.86 X 6.99	RS018-N/F-50000-52400-0810
530.00	554.00	8.10	8.00	557.66 X 6.99	RS018-N/F-53000-55400-0810
550.00	574.00	8.10	8.00	557.66 X 6.99	RS018-N/F-55000-57400-0810
575.00	599.00	8.10	8.00	582.68 X 6.99	RS018-N/F-57500-59900-0810

ROD SEAL	RS027	

Features:

Rod seal profile RS027 is a single acting U cup type seal with energizer and anti extrusion polyamide ring for very high pressure and heavy duty application.

Composition:

Components	Material	Royal Code
Main seal	Polyurethane	PUHP94
Energizer O ring	NBR	NBR70
Anti extrusion ring	Polyamide	PA6000

For compound detail please refer Royal seal materials.(Section III)

Properties:

- High resistance to abrasion
- Easy installation
- Anti-extrusion ring prevent extrusion in high pressure conditions
-

Application:

- Mobile Hydraulics
- Injection molding machines
- Machine tools
- Agricultural equipment
- Construction Machineries.

General Technical Data:

Working Pressure (max.)	500 Bar
Temperature range	-20°C to 100°C
Surface Speed (max.)	0.5 m/s.

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Extrusion Gap dimension:

The gap dimension on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the maximum clearance Ø and min rod Ø.

	Maximum permissible gap (S)			
160 bar	260 bar	320 bar	400 bar	500 bar
0.5	0.4	0.35	0.3	0.25

Minimum chamfer C:

DN - dN	Up to 8	8.1 to10	10.1 to 15	15.1 to 20	20.1 to 25	25.1 and above
C	4.0	4.5	6.0	7.5	8.0	10.0

Fitting:

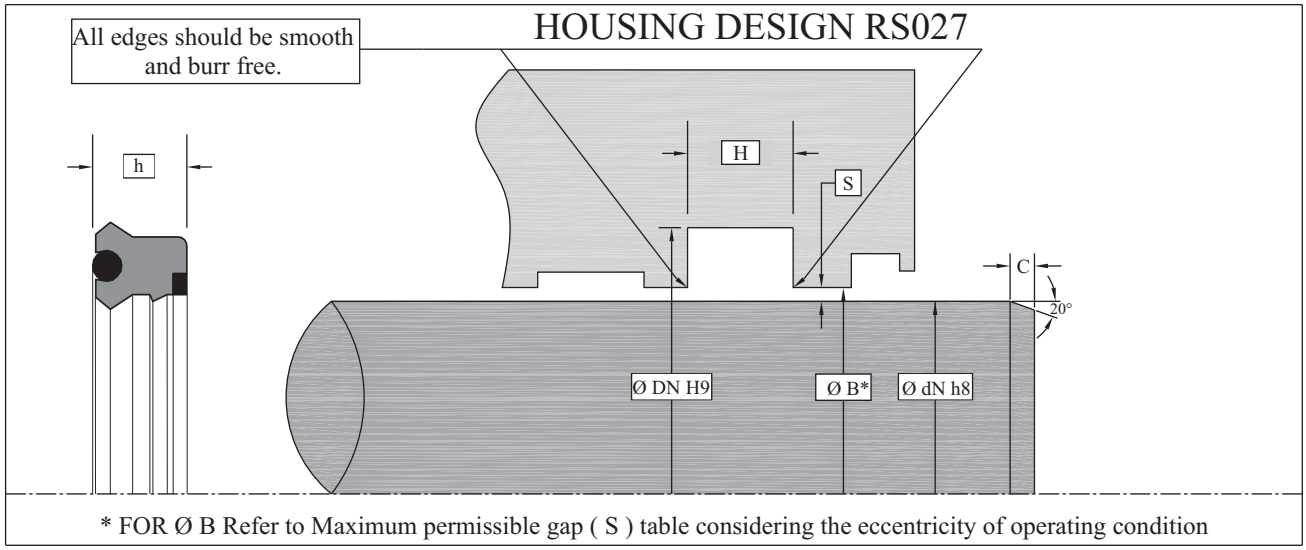
RS027 is very easy to fit. Careful fitting of the seal is a prerequisite for it's perfect functioning.

Ordering format:

Please mention ROYAL order code in your order.
Example: Size 40 X 55 X 12.5
Order code is RS027-U-04000-05500-1250

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

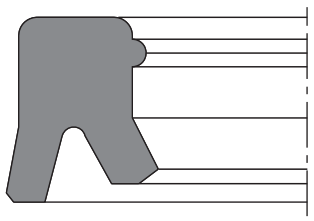
HOUSING DESIGN RS027



* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	h	H (+0.2)	ORDER CODE
30.00	42.00	8.50	9.50	RS027-U-03000-04200-0950
40.00	52.00	9.50	10.50	RS027-U-04000-05200-1050
40.00	53.00	10.00	11.00	RS027-U-04000-05300-1100
40.00	55.00	11.40	12.50	RS027-U-04000-05500-1250
45.00	60.00	11.40	12.50	RS027-U-04500-06000-1250
50.00	65.00	11.40	12.50	RS027-U-05000-06500-1250
60.00	75.00	11.40	12.50	RS027-U-06000-07500-1250
63.00	78.00	11.40	12.50	RS027-U-06300-07800-1250
65.00	80.00	11.40	12.50	RS027-U-06500-08000-1250
70.00	85.00	11.40	12.50	RS027-U-07000-08500-1250
105.00	120.00	12.00	13.00	RS027-U-10500-12000-1300
150.00	170.00	14.60	16.00	RS027-U-15000-17000-1600
IMPERIAL SIZES				
38.10	50.80	9.52	10.50	RS027-IN-U-01500-02000-0413
44.45	57.15	9.52	10.50	RS027-IN-U-01750-02250-0413
50.80	63.50	9.52	10.50	RS027-IN-U-02000-02500-0413
57.15	69.85	9.52	10.50	RS027-IN-U-02250-02750-0413

ROD SEAL	RS036	

Features:

Rod seal profile RS036 is a single acting U cup type seal with additional rib commonly in use.

Composition:

Material:	Polyurethane
Royal Code	PUET93

For compound detail please refer Royal seal materials.

Properties:

- High resistance to abrasion
- Easy installation
- Additional rib prevents ingress of dirt to a large degree

Application:

- Mobile Hydraulics
- Injection molding machines
- Machine tools
- Agricultural equipment
- Standard cylinders

General Technical Data:

Working Pressure (max.)	400 Bar*
Temperature range	-30°C to 110°C
Surface Speed (max.)	0.5 m/s.

* Maximum working pressure can be extended up to 400 bar with the use of a back up ring. If this option is required ask ROYAL.

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Extrusion Gap dimension:

The gap dimension on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the maximum clearance Ø and min rod Ø.

	Maximum permissible gap (S)		
160 bar	260 bar	320 bar	400 bar
0.5	0.4	0.35	(ask ROYAL)

Minimum chamfer C:

DN - dN	Up to 8	8.1 to10	10.1 to 15	15.1 to 20	20.1 to 25	25.1 and above
C	4.0	4.5	6.0	7.5	8.0	10.0

Fitting:

RS036 is very easy to fit. Careful fitting of the seal is a prerequisite for it's perfect functioning.

Ordering format:

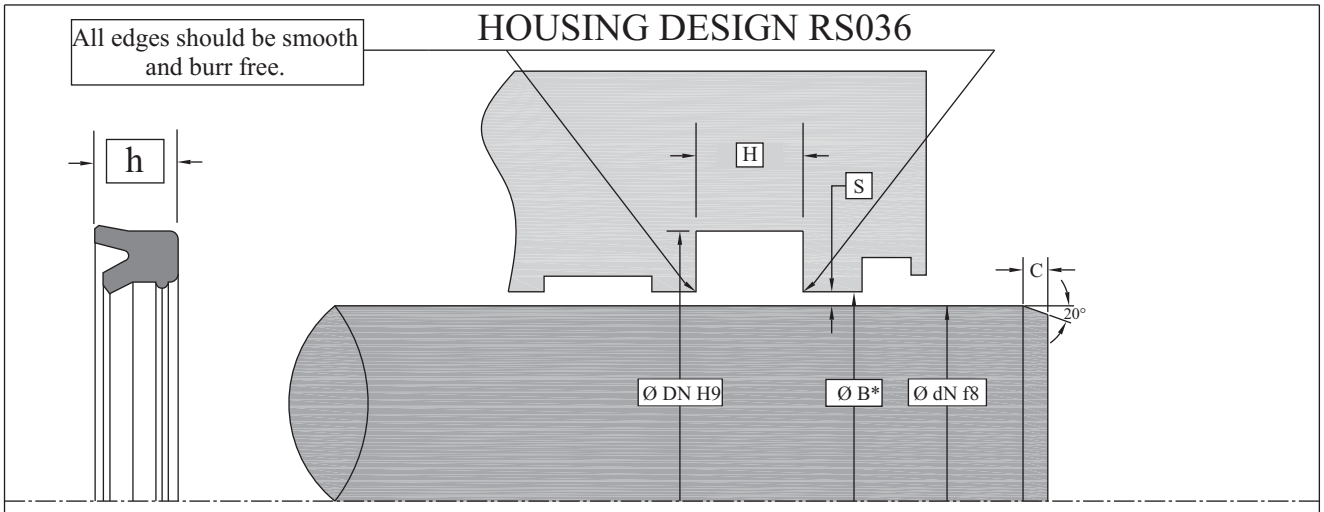
Please mention ROYAL order code in your order.

Example: Size 6 X 14 X 6.3

Order code is RS036-U-00600-01400-0630

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN RS036

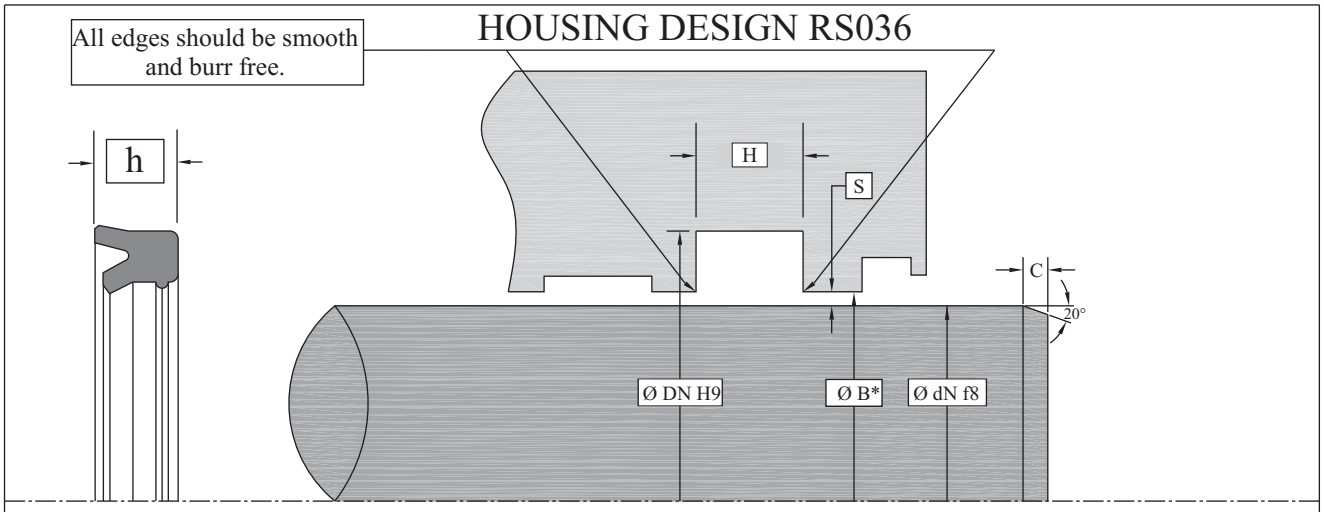


* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

dN (f8)	DN (H9)	h	H (+0.20)	ORDER CODE
6.00	14.00	5.70	6.30	RS036-00600-01400-0630
8.00	16.00	5.70	6.30	RS036-00800-01600-0630
10.00	18.00	5.70	6.30	RS036-01000-01800-0630
10.00	20.00	5.70	6.30	RS036-01000-02000-0630
12.00	19.00	5.10	5.60	RS036-01200-01900-0560
12.00	20.00	5.70	6.30	RS036-01200-02000-0630
12.00	22.00	7.30	8.00	RS036-01200-02200-0800
14.00	21.00	5.10	5.60	RS036-01400-02100-0560
14.00	24.00	7.30	8.00	RS036-01400-02400-0800
15.00	25.00	7.30	8.00	RS036-01500-02500-0800
16.00	23.50	5.10	5.60	RS036-01600-02350-0560
16.00	24.00	5.70	6.30	RS036-01600-02400-0630
16.00	24.00	5.70	6.30	RS036-01600-02400-0630
16.00	25.00	5.70	6.30	RS036-01600-02500-0630
16.00	26.00	7.30	8.00	RS036-01600-02600-0800
18.00	23.00	5.10	5.60	RS036-01800-02300-0560
18.00	25.00	5.10	5.60	RS036-01800-02500-0560
18.00	25.00	5.10	5.60	RS036-01800-02500-0560
18.00	28.00	7.30	8.00	RS036-01800-02800-0800
20.00	26.00	5.30	5.80	RS036-02000-02600-0580
20.00	26.00	5.70	6.30	RS036-02000-02600-0630
20.00	28.00	5.70	6.30	RS036-02000-02800-0630
20.00	30.00	5.70	6.30	RS036-02000-03000-0630

HOUSING DESIGN RS036



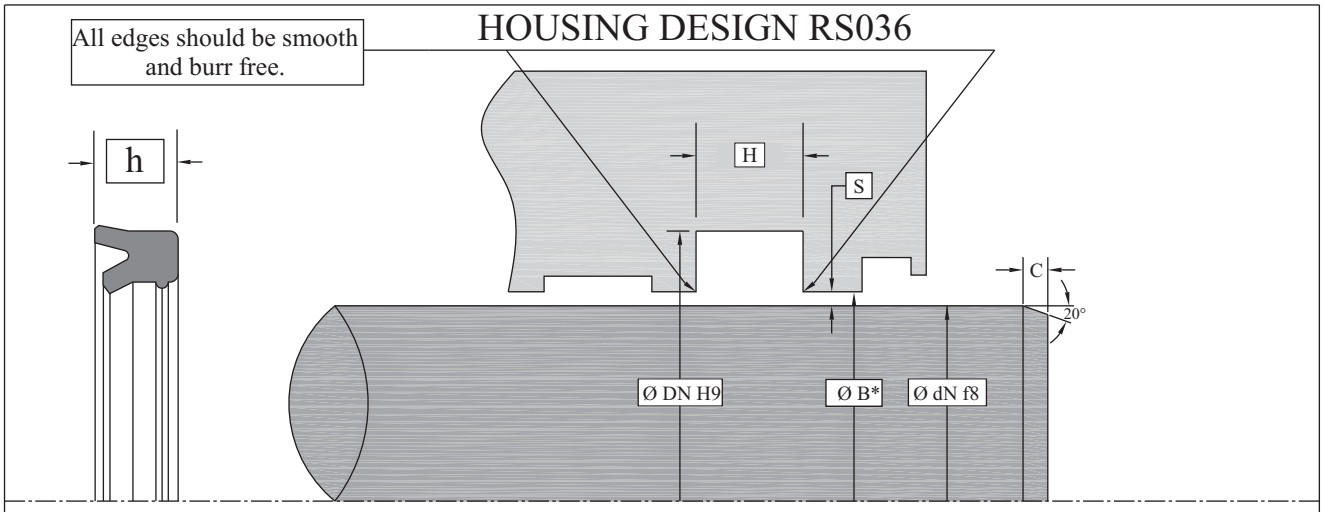
* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

dN (f8)	DN (H9)	h	H (+0.20)	ORDER CODE
20.00	30.00	7.30	8.00	RS036-02000-03000-0800
22.00	29.00	5.10	5.60	RS036-02200-02900-0560
22.00	30.00	5.70	6.30	RS036-02200-03000-0630
22.00	32.00	7.30	8.00	RS036-02200-03200-0800
25.00	33.00	5.70	6.30	RS036-02500-03300-0630
25.00	35.00	7.30	8.00	RS036-02500-03500-0800
25.00	38.00	6.80	7.50	RS036-02500-03800-0750
28.00	26.00	7.30	8.00	RS036-02800-02600-0800
28.00	36.00	5.70	6.30	RS036-02800-03600-0630
28.00	38.00	7.30	8.00	RS036-02800-03800-0800
30.00	37.00	5.30	5.80	RS036-03000-03700-0580
30.00	37.00	8.20	9.00	RS036-03000-03700-0900
30.00	40.00	5.70	6.30	RS036-03000-04000-0630
30.00	40.00	7.30	8.00	RS036-03000-04000-0800
30.00	50.00	10.00	11.00	RS036-03000-05000-1100
31.50	41.50	6.40	7.00	RS036-03150-04150-0700
32.00	40.00	5.70	6.30	RS036-03200-04000-0630
32.00	40.00	10.00	11.00	RS036-03200-04000-1100
32.00	42.00	7.30	8.00	RS036-03200-04200-0800
32.00	42.00	9.10	10.00	RS036-03200-04200-1000
33.00	43.00	7.30	8.00	RS036-03300-04300-0800
34.00	43.00	5.70	6.30	RS036-03400-04300-0630
35.00	42.00	6.90	7.60	RS036-03500-04200-0760

ROYAL STANDARD SIZE LIST				
dN (f8)	DN (H9)	h	H (+0.20)	ORDER CODE
35.00	43.00	5.70	6.30	RS036-03500-04300-0630
35.00	43.00	10.00	11.00	RS036-03500-04300-1100
35.00	45.00	5.70	6.30	RS036-03500-04500-0630
35.00	45.00	7.30	8.00	RS036-03500-04500-0800
35.00	45.00	10.00	11.00	RS036-03500-04500-1100
35.00	46.00	6.80	7.50	RS036-03500-04600-0750
36.00	43.00	6.40	7.00	RS036-03600-04300-0700
36.00	43.00	8.20	9.00	RS036-03600-04300-0900
36.00	44.00	5.70	6.30	RS036-03600-04400-0630
36.00	46.00	7.30	8.00	RS036-03600-04600-0800
36.00	46.00	10.00	11.00	RS036-03600-04600-1100
36.00	50.00	11.40	12.50	RS036-03600-05000-1250
36.00	51.00	11.40	12.50	RS036-03600-05100-1250
38.00	48.00	7.30	8.00	RS036-03800-04800-0800
40.00	50.00	7.30	8.00	RS036-04000-05000-0800
40.00	50.00	8.20	9.00	RS036-04000-05000-0900
40.00	50.00	10.00	11.00	RS036-04000-05000-1100
40.00	55.00	9.10	10.00	RS036-04000-05500-1000
40.00	55.00	11.40	12.50	RS036-04000-05500-1250
40.00	55.50	5.70	6.30	RS036-04000-05550-0630
45.00	53.00	5.70	6.30	RS036-04500-05300-0630
45.00	53.00	5.70	6.30	RS036-04500-05300-0630
45.00	53.00	8.20	9.00	RS036-04500-05300-0900
45.00	55.00	5.70	6.30	RS036-04500-05500-0630
45.00	55.00	7.30	8.00	RS036-04500-05500-0800
45.00	55.00	10.00	11.00	RS036-04500-05500-1100
45.00	60.00	5.70	6.30	RS036-04500-06000-0630
45.00	60.00	11.40	12.50	RS036-04500-06000-1250
50.00	57.00	9.10	10.00	RS036-05000-05700-1000
50.00	60.00	7.30	8.00	RS036-05000-06000-0800
50.00	60.00	10.00	11.00	RS036-05000-06000-1100
50.00	63.00	10.00	11.00	RS036-05000-06300-1100
50.00	65.00	11.40	12.50	RS036-05000-06500-1250
55.00	68.00	10.00	11.00	RS036-05500-06800-1100
55.00	70.00	10.00	11.00	RS036-05500-07000-1100

HOUSING DESIGN RS036



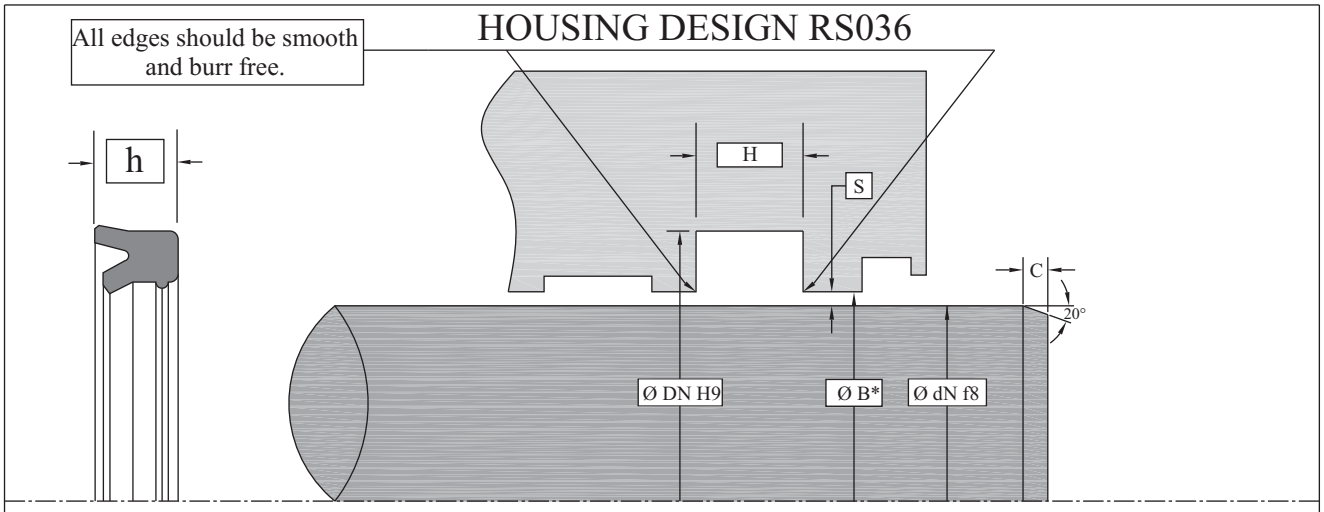
* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

dN (f8)	DN (H9)	h	H (+0.20)	ORDER CODE
55.00	75.00	11.80	13.00	RS036-05500-07500-1300
56.00	66.00	7.30	8.00	RS036-05600-06600-0800
56.00	66.00	10.00	11.00	RS036-05600-06600-1100
56.00	71.00	11.40	12.50	RS036-05600-07100-1250
60.00	70.00	7.30	8.00	RS036-06000-07000-0800
60.00	70.00	10.00	11.00	RS036-06000-07000-1100
60.00	70.00	11.80	13.00	RS036-06000-07000-1300
60.00	70.00	13.60	15.00	RS036-06000-07000-1500
60.00	73.00	10.00	11.00	RS036-06000-07300-1100
60.00	75.00	11.40	12.50	RS036-06000-07500-1250
63.00	73.00	11.80	13.00	RS036-06300-07300-1300
63.00	75.00	9.10	10.00	RS036-06300-07500-1000
63.00	75.00	11.80	13.00	RS036-06300-07500-1300
63.00	78.00	11.40	12.50	RS036-06300-07800-1250
65.00	75.00	6.40	7.00	RS036-06500-07500-0700
65.00	78.00	10.00	11.00	RS036-06500-07800-1100
65.00	80.00	10.00	11.00	RS036-06500-08000-1100
65.00	80.00	11.80	13.00	RS036-06500-08000-1300
70.00	80.00	7.30	8.00	RS036-07000-08000-0800
70.00	80.00	11.80	13.00	RS036-07000-08000-1300
70.00	83.00	10.00	11.00	RS036-07000-08300-1100
70.00	85.00	11.40	12.50	RS036-07000-08500-1250
70.00	90.00	11.80	13.00	RS036-07000-09000-1300

ROYAL STANDARD SIZE LIST				
dN (f8)	DN (H9)	h	H (+0.20)	ORDER CODE
74.00	86.00	8.60	9.50	RS036-07400-08600-0950
75.00	83.00	11.40	12.50	RS036-07500-08300-1250
75.00	85.00	11.80	13.00	RS036-07500-08500-1300
75.00	88.00	10.00	11.00	RS036-07500-08800-1100
75.00	90.00	11.80	13.00	RS036-07500-09000-1300
75.00	95.00	11.80	13.00	RS036-07500-09500-1300
78.00	88.00	13.60	15.00	RS036-07800-08800-1500
80.00	90.00	9.10	10.00	RS036-08000-09000-1000
80.00	90.00	11.80	13.00	RS036-08000-09000-1300
80.00	93.00	10.00	11.00	RS036-08000-09300-1100
80.00	95.00	11.40	12.50	RS036-08000-09500-1250
80.00	100.00	11.80	13.00	RS036-08000-10000-1300
85.00	100.00	11.80	13.00	RS036-08500-10000-1300
85.00	105.00	11.80	13.00	RS036-08500-10500-1300
90.00	100.00	11.80	13.00	RS036-09000-10000-1300
90.00	105.00	9.10	10.00	RS036-09000-10500-1000
90.00	105.00	11.40	12.50	RS036-09000-10500-1250
90.00	110.00	11.80	13.00	RS036-09000-11000-1300
95.00	110.00	9.10	10.00	RS036-09500-11000-1000
95.00	115.00	11.80	13.00	RS036-09500-11500-1300
99.00	109.00	13.60	15.00	RS036-09900-10900-1500
100.00	108.00	7.30	8.00	RS036-10000-10800-0800
100.00	115.00	9.10	10.00	RS036-10000-11500-1000
100.00	120.00	11.80	13.00	RS036-10000-12000-1300
100.00	120.00	14.50	16.00	RS036-10000-12000-1600
100.00	125.00	14.50	16.00	RS036-10000-12500-1600
105.00	120.00	10.00	11.00	RS036-10500-12000-1100
105.00	120.00	14.50	16.00	RS036-10500-12000-1600
105.00	125.00	11.80	13.00	RS036-10500-12500-1300
110.00	125.00	9.60	10.60	RS036-11000-12500-1060
110.00	125.00	11.40	12.50	RS036-11000-12500-1250
110.00	130.00	14.50	16.00	RS036-11000-13000-1600
115.00	123.00	11.40	12.50	RS036-11500-12300-1250
115.00	130.00	10.00	11.00	RS036-11500-13000-1100
115.00	135.00	14.50	16.00	RS036-11500-13500-1600

HOUSING DESIGN RS036



* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

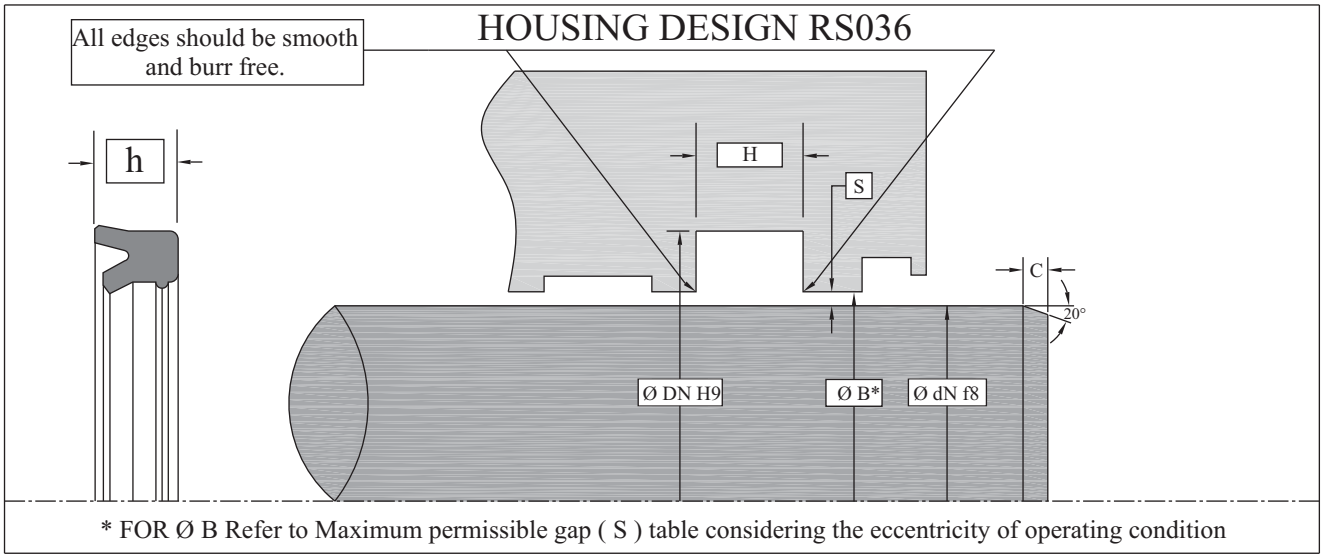
dN (f8)	DN (H9)	h	H (+0.20)	ORDER CODE
115.00	140.00	14.50	16.00	RS036-11500-14000-1600
120.00	130.00	13.60	15.00	RS036-12000-13000-1500
120.00	140.00	14.50	16.00	RS036-12000-14000-1600
125.00	135.00	11.80	13.00	RS036-12500-13500-1300
125.00	145.00	14.50	16.00	RS036-12500-14500-1600
130.00	150.00	14.50	16.00	RS036-13000-15000-1600
135.00	143.00	11.40	12.50	RS036-13500-14300-1250
135.00	150.00	9.10	10.00	RS036-13500-15000-1000
135.00	150.00	11.40	12.50	RS036-13500-15000-1250
138.00	160.00	15.50	17.00	RS036-13800-16000-1700
140.00	155.00	12.70	14.00	RS036-14000-15500-1400
140.00	160.00	14.50	16.00	RS036-14000-16000-1600
141.00	151.00	13.60	15.00	RS036-14100-15100-1500
145.00	160.00	9.10	10.00	RS036-14500-16000-1000
145.00	165.00	14.50	16.00	RS036-14500-16500-1600
150.00	170.00	14.50	16.00	RS036-15000-17000-1600
155.00	170.00	10.00	11.00	RS036-15500-17000-1100
160.00	180.00	14.50	16.00	RS036-16000-18000-1600
162.00	172.00	13.60	15.00	RS036-16200-17200-1500
165.00	180.00	10.00	11.00	RS036-16500-18000-1100
165.00	180.00	17.70	19.50	RS036-16500-18000-1950
170.00	190.00	14.50	16.00	RS036-17000-19000-1600
180.00	200.00	14.50	16.00	RS036-18000-20000-1600

ROYAL STANDARD SIZE LIST				
dN (f8)	**DN (H9)**	**h**	**H (+0.20)**	**ORDER CODE**
180.00	205.00	15.50	17.00	RS036-18000-20500-1700
183.00	193.00	13.60	15.00	RS036-18300-19300-1500
191.00	201.00	10.00	11.00	RS036-19100-20100-1100
200.00	225.00	14.50	16.00	RS036-20000-22500-1600
207.00	217.00	13.60	15.00	RS036-20700-21700-1500
210.00	226.00	17.70	19.50	RS036-21000-22600-1950
220.00	250.00	15.50	17.00	RS036-22000-25000-1700
250.00	270.00	14.50	16.00	RS036-25000-27000-1600

IMPERIAL SIZES

dN (f8)	**DN (H9)**	**h**	**H (+0.20)**	**ORDER CODE**
19.05	25.40	6.40	6.99	RS036-IN-U-00750-01000-0275
19.05	28.58	7.20	7.92	RS036-IN-U-00750-01125-0312
25.40	31.75	6.40	6.99	RS036-IN-U-01000-01250-0275
25.40	34.92	7.20	7.92	RS036-IN-U-01000-01375-0312
28.58	34.92	6.40	6.99	RS036-IN-U-01125-01375-0275
28.58	41.28	8.70	9.52	RS036-IN-U-01125-01625-0375
31.75	38.10	6.40	6.99	RS036-IN-U-01250-01500-0275
31.75	41.28	7.90	8.71	RS036-IN-U-01250-01625-0343
31.75	41.28	9.50	10.49	RS036-IN-U-01250-01625-0413
34.92	41.28	6.40	6.99	RS036-IN-U-01375-01625-0275
38.10	44.45	6.40	6.99	RS036-IN-U-01500-01750-0275
38.10	47.63	7.90	8.71	RS036-IN-U-01500-01875-0343
38.10	50.80	9.50	10.49	RS036-IN-U-01500-02000-0413
44.45	53.97	6.40	6.99	RS036-IN-U-01750-02125-0275
44.45	53.97	9.50	10.49	RS036-IN-U-01750-02125-0413
44.45	57.15	9.50	10.49	RS036-IN-U-01750-02250-0413
50.80	60.32	8.70	9.52	RS036-IN-U-02000-02375-0375
50.80	63.50	8.70	9.52	RS036-IN-U-02000-02500-0375

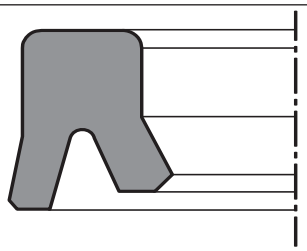
HOUSING DESIGN RS036



* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

dN (f8)	DN (H9)	h	H (+0.20)	ORDER CODE
57.15	69.85	9.50	10.49	RS036-IN-U-02250-02750-0413
63.50	74.60	8.20	8.99	RS036-IN-U-02500-02937-0354
63.50	76.20	8.70	9.52	RS036-IN-U-02500-03000-0375
76.20	88.90	8.70	9.52	RS036-IN-U-03000-03500-0375
82.55	95.25	8.70	9.52	RS036-IN-U-03250-03750-0375
88.90	101.60	8.70	9.52	RS036-IN-U-03500-04000-0375
101.60	120.65	14.50	15.88	RS036-IN-U-04000-04750-0625
114.30	133.35	14.50	15.88	RS036-IN-U-04500-05250-0625
127.00	139.70	14.30	15.72	RS036-IN-U-05000-05500-0619
152.40	171.45	14.50	15.88	RS036-IN-U-06000-06750-0625

ROD SEAL	RS045	

Features:

Rod seal profile RS045 is a single acting U cup type seal commonly in use.

Composition:

Seal Profile type	RS045-U	RS045-F*
Material	Polyurethane	Fluorocarbon Rubber
Royal Code	PUHP94	FPM80

For compound detail please refer Royal seal materials.

*** For size list of RS045-F contact ROYAL at info@royalcomposites.co.in / info@royalseals.co.in / info@royalseals.co.in**

Properties:

- High resistance to abrasion
- Easy installation

Application:

- Mobile Hydraulics
- Injection molding machines
- Machine tools
- Agricultural equipment
- Standard cylinders

General Technical Data:

Seal Profile	RS045-U	RS045-F
Temperature range	-20°C to 120°C	-20°C to +200 °C
Working Pressure(max.)	400 bar*	160 bar
Surface Speed(max.)	0.5 m/s.	0.5 m/s.

* Maximum working pressure can be extended up to 400 bar with the use of a back up ring. If this option is required ask ROYAL.

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Gap dimension:

The largest gap dimension appearing in operation on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the minimum clearance Ø and max bore Ø.

Seal Profile	Maximum permissible gap (S)			
	160 bar	260 bar	320 bar	400 bar
RS045-U	0.5	0.4	0.35	Ask ROYAL
RS045-F	0.2	Ask ROYAL	-	-

Minimum chamfer C:

DN - dN	Up to 8	8.1 to10	10.1 to 15	15.1 to 20	20.1 to 25	25.1 and above
C	4.0	4.5	6.0	7.5	8.0	10.0

Fitting:

RS045 is very easy to fit. Careful fitting of the seal is a prerequisite for it's perfect functioning.

Ordering format:

Please mention ROYAL order code in your order.

Example: Size 4 X 10 X 4.5

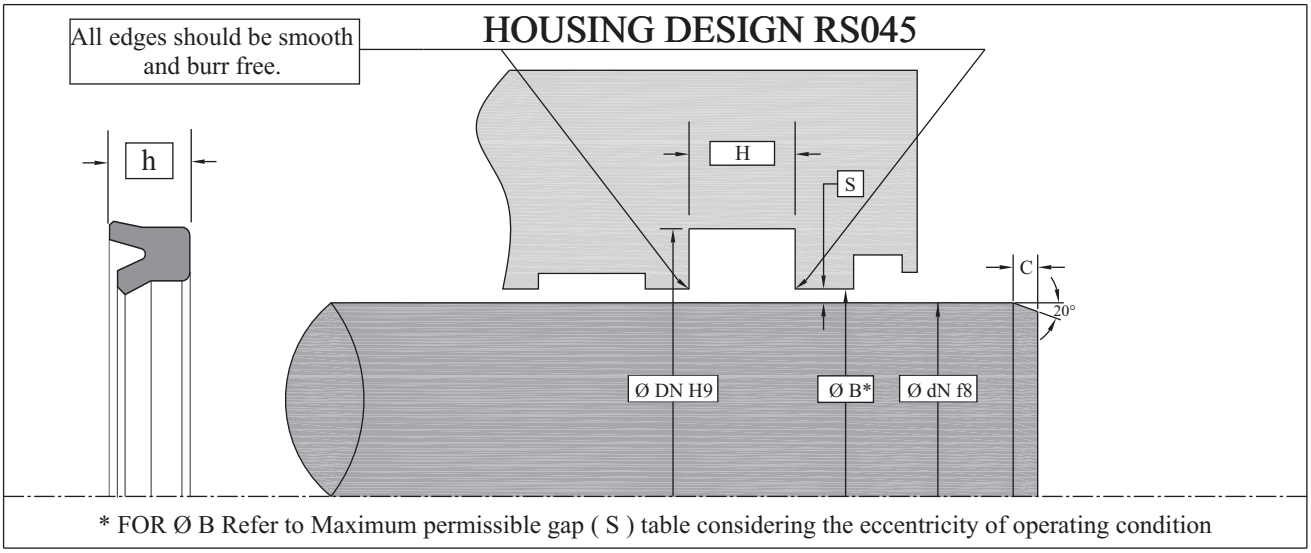
For Temp range: -20°C to +110°C Order code is RS045-U-00400-01000-0450

For Temp range: -20°C to +200°C Order code is RS045-F-00400-01000-0450

*** For size list of RS045-F contact ROYAL at info@royalcomposites.co.in / info@royalseals.co.in / info@royalseals.co.in**

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN RS045

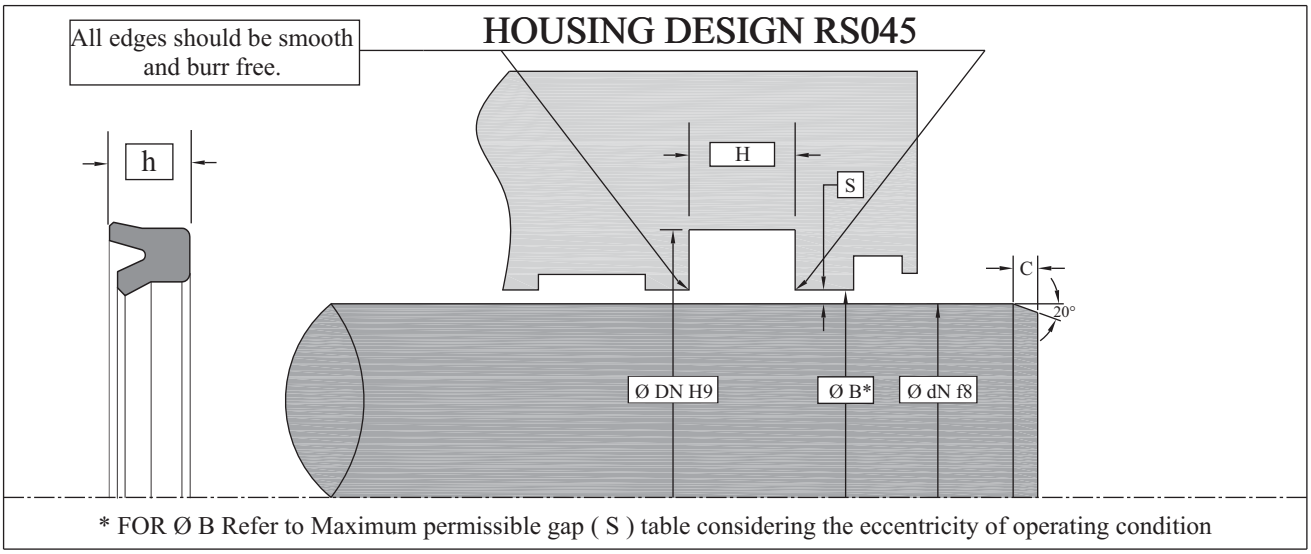


* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

dN (f8)	DN (H9)	h	H (+0.20)	ORDER CODE
4.00	10.00	4.10	4.50	RS045-U-00400-01000-0450
12.00	20.00	5.20	5.70	RS045-U-01200-02000-0570
16.00	22.00	4.10	4.50	RS045-U-01600-02200-0450
16.00	22.30	4.40	4.80	RS045-U-01600-02230-0480
14.00	24.00	10.00	11.00	RS045-U-01400-02400-1100
16.00	24.00	4.50	5.00	RS045-U-01600-02400-0500
20.00	28.00	5.20	5.70	RS045-U-02000-02800-0570
25.00	33.00	5.20	5.70	RS045-U-02500-03300-0570
25.00	33.00	5.70	6.30	RS045-U-02500-03300-0630
28.00	36.00	5.20	5.70	RS045-U-02800-03600-0570
30.00	40.00	6.40	7.00	RS045-U-03000-04000-0700
30.00	40.00	8.20	9.00	RS045-U-03000-04000-0900
30.00	40.00	5.70	6.30	RS045-U-03000-04000-0630
30.00	40.00	7.30	8.00	RS045-U-03000-04000-0800
30.00	45.00	9.10	10.00	RS045-U-03000-04500-1000
32.00	42.00	10.00	11.00	RS045-U-03200-04200-1100
35.00	45.00	8.20	9.00	RS045-U-03500-04500-0900
35.00	47.00	6.80	7.50	RS045-U-03500-04700-0750
35.00	47.70	11.80	13.00	RS045-U-03500-04770-1300
35.00	50.00	9.10	10.00	RS045-U-03500-05000-1000
40.00	50.00	8.20	9.00	RS045-U-04000-05000-0900
43.00	53.00	6.40	7.00	RS045-U-04300-05300-0700
40.00	53.00	10.00	11.00	RS045-U-04000-05300-1100

HOUSING DESIGN RS045



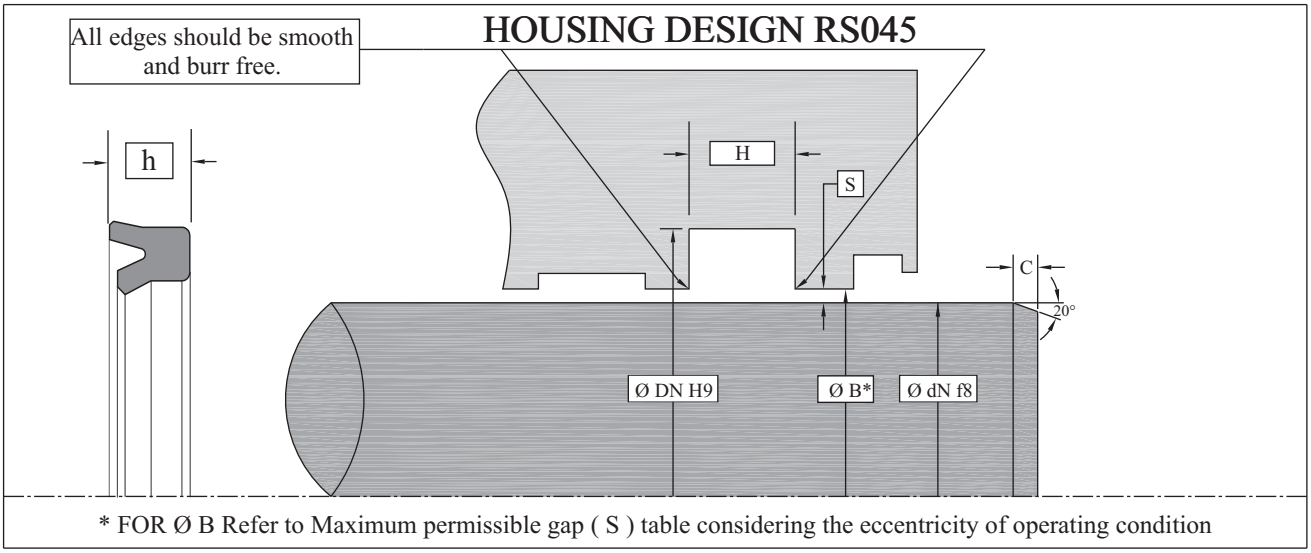
* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

dN (f8)	DN (H9)	h	H (+0.20)	ORDER CODE
40.00	55.00	9.10	10.00	RS045-U-04000-05500-1000
45.00	55.00	8.20	9.00	RS045-U-04500-05500-0900
45.00	58.00	10.00	11.00	RS045-U-04500-05800-1100
45.00	60.00	9.10	10.00	RS045-U-04500-06000-1000
40.00	60.00	11.80	13.00	RS045-U-04000-06000-1300
50.00	63.00	10.00	11.00	RS045-U-05000-06300-1100
50.00	65.00	9.10	10.00	RS045-U-05000-06500-1000
50.00	65.00	11.40	12.50	RS045-U-05000-06500-1250
55.00	68.00	10.00	11.00	RS045-U-05500-06800-1100
55.00	70.00	9.10	10.00	RS045-U-05500-07000-1000
50.00	70.00	11.80	13.00	RS045-U-05000-07000-1300
55.00	71.00	11.80	13.00	RS045-U-05500-07100-1300
60.00	73.00	10.00	11.00	RS045-U-06000-07300-1100
60.00	75.00	9.10	10.00	RS045-U-06000-07500-1000
65.00	78.00	10.00	11.00	RS045-U-06500-07800-1100
65.00	80.00	9.10	10.00	RS045-U-06500-08000-1000
65.00	80.00	11.80	13.00	RS045-U-06500-08000-1300
70.00	80.00	5.50	6.00	RS045-U-07000-08000-0600
70.00	83.00	10.00	11.00	RS045-U-07000-08300-1100
70.00	85.00	9.10	10.00	RS045-U-07000-08500-1000
75.00	85.00	11.80	13.00	RS045-U-07500-08500-1300
75.00	85.00	9.10	10.00	RS045-U-07500-08500-1000
75.00	88.00	10.00	11.00	RS045-U-07500-08800-1100

ROYAL STANDARD SIZE LIST				
dN (f8)	DN (H9)	h	H (+0.20)	ORDER CODE
75.00	88.00	10.00	11.00	RS045-U-07500-08800-1100
70.00	90.00	11.80	13.00	RS045-U-07000-09000-1300
75.00	90.00	9.10	10.00	RS045-U-07500-09000-1000
70.00	90.00	11.80	13.00	RS045-U-07000-09000-1300
80.00	90.00	6.40	7.00	RS045-U-08000-09000-0700
80.00	92.00	10.00	11.00	RS045-U-08000-09200-1100
80.00	93.00	10.00	11.00	RS045-U-08000-09300-1100
75.00	95.00	11.80	13.00	RS045-U-07500-09500-1300
80.00	95.00	9.10	10.00	RS045-U-08000-09500-1000
75.00	95.00	11.80	13.00	RS045-U-07500-09500-1300
75.00	95.00	14.50	16.00	RS045-U-07500-09500-1600
85.00	98.00	10.00	11.00	RS045-U-08500-09800-1100
80.00	100.00	11.80	13.00	RS045-U-08000-10000-1300
85.00	100.00	9.10	10.00	RS045-U-08500-10000-1000
85.00	100.00	10.00	11.00	RS045-U-08500-10000-1100
85.00	100.00	11.80	13.00	RS045-U-08500-10000-1300
90.00	103.00	10.00	11.00	RS045-U-09000-10300-1100
90.00	104.00	11.40	12.50	RS045-U-09000-10400-1250
90.00	105.00	10.00	11.00	RS045-U-09000-10500-1100
95.00	108.00	10.00	11.00	RS045-U-09500-10800-1100
90.00	110.00	11.80	13.00	RS045-U-09000-11000-1300
95.00	110.00	10.00	11.00	RS045-U-09500-11000-1100
98.00	112.00	8.60	9.50	RS045-U-09800-11200-0950
95.00	115.00	11.80	13.00	RS045-U-09500-11500-1300
100.00	115.00	10.00	11.00	RS045-U-10000-11500-1100
100.00	115.00	9.10	10.00	RS045-U-10000-11500-1000
108.00	118.00	5.70	6.30	RS045-U-10800-11800-0630
108.00	118.00	6.40	7.00	RS045-U-10800-11800-0700
100.00	120.00	11.80	13.00	RS045-U-10000-12000-1300
105.00	120.00	10.00	11.00	RS045-U-10500-12000-1100
110.00	125.00	10.00	11.00	RS045-U-11000-12500-1100
110.00	130.00	10.00	11.00	RS045-U-11000-13000-1100
115.00	130.00	5.70	6.30	RS045-U-11500-13000-0630
110.00	130.00	12.70	14.00	RS045-U-11000-13000-1400
115.00	135.00	11.80	13.00	RS045-U-11500-13500-1300

HOUSING DESIGN RS045



* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

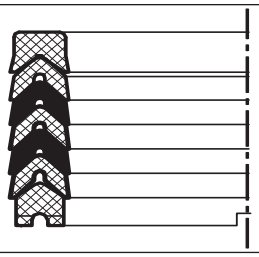
ROYAL STANDARD SIZE LIST

dN (f8)	DN (H9)	h	H (+0.20)	ORDER CODE
120.00	140.00	12.70	14.00	RS045-U-12000-14000-1400
140.00	150.00	6.40	7.00	RS045-U-14000-15000-0700
130.00	150.00	12.70	14.00	RS045-U-13000-15000-1400
140.00	155.00	10.00	11.00	RS045-U-14000-15500-1100
145.00	155.00	6.40	7.00	RS045-U-14500-15500-0700
140.00	155.00	9.10	10.00	RS045-U-14000-15500-1000
150.00	160.00	6.40	7.00	RS045-U-15000-16000-0700
140.00	160.00	14.50	16.00	RS045-U-14000-16000-1600
155.00	165.00	6.40	7.00	RS045-U-15500-16500-0700
160.00	185.00	11.80	13.00	RS045-U-16000-18500-1300

IMPERIAL SIZE

dN (f8)	DN (H9)	h	H (+0.20)	ORDER CODE
12.70	19.05	3.90	4.32	RS045-IN-U-00500-00750-0170
19.05	31.75	11.60	12.70	RS045-IN-U-00750-01250-0500
44.45	53.97	6.70	7.37	RS045-IN-U-01750-02125-0290
50.80	63.50	9.50	10.49	RS045-IN-U-02000-02500-0413
53.97	66.68	9.50	10.49	RS045-IN-U-02125-02625-0413
57.15	66.68	9.50	10.49	RS045-IN-U-02250-02625-0413
57.15	69.85	9.50	10.49	RS045-IN-U-02250-02750-0413
60.32	73.02	9.50	10.49	RS045-IN-U-02375-02875-0413

ROYAL STANDARD SIZE LIST				
dN (f8)	**DN (H9)**	**h**	**H (+0.20)**	**ORDER CODE**
63.50	76.20	9.50	10.49	RS045-IN-U-02500-03000-0413
69.85	82.55	9.50	10.49	RS045-IN-U-02750-03250-0413
76.20	88.90	9.50	10.49	RS045-IN-U-03000-03500-0413
82.55	95.25	9.50	10.49	RS045-IN-U-03250-03750-0413
88.90	98.43	9.50	10.49	RS045-IN-U-03500-03875-0413
85.60	98.55	9.50	10.49	RS045-IN-U-03370-03880-0413
88.90	101.60	9.50	10.49	RS045-IN-U-03500-04000-0413
95.25	107.95	9.50	10.49	RS045-IN-U-03750-04250-0413
101.60	114.30	9.50	10.49	RS045-IN-U-04000-04500-0413
107.95	120.65	9.50	10.49	RS045-IN-U-04250-04750-0413
152.40	165.10	17.30	19.05	RS045-IN-U-06000-06500-0750
165.10	190.50	18.10	19.84	RS045-IN-U-06500-07500-0781
177.80	203.20	18.10	19.84	RS045-IN-U-07000-08000-0781

ROD SEAL	RS054	

Features:

Rod seal profile RS054 is a seal set consist of five chevron ring (3 fabric reinforced and 2 rubber rings) or three chevron ring (2 fabric reinforced and 1 rubber ring) and a header and a pressure ring.

Composition:

Seal Profile	RS054-N		RS054-F	
	Material	Royal Code	Material	Royal Code
Header ring	Fabric + NBR	NBRCF	Fabric + FPM	FPMCF
Chevron ring(fabric)	Fabric + NBR	NBRCF	Fabric + FPM	FPCF
ring(rubber)	NBR	NBR90	FPM	FPM80
Pressure ring type-1	Fabric + NBR	NBRCF	Fabric + FPM	FPMCF
Pressure ring type-2	Polyamide	PA6000	-	-

For compound detail please refer Royal seal materials.(Section III)

Properties:

- High resistance to extrusion due to fabric reinforced header ring
- Easy installation
- Excellent sealing function
  Long working life in extreme conditions

Application:

- Mobile Hydraulics
- Machine tools
- Agricultural equipment
- standard cylinders
- hydraulic presses
- iron and steel industry
- marine hydraulics

General Technical Data :

Seal Profile	RS054-N	RS054-F
Working Pressure(max.)	400 bar	400 bar
Temperature range	-20°C to +100°C	-15°C to +140°C
Linear velocity(max.)	0.5 m/s.	0.5 m/s.

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Gap dimension

The gap dimension on the non-pressurised side of the seal is decisive for sealing function. The gap is dependent on the metal guide and its installation fit. Figures show the maximum permissible gap all on one side using the maximum clearance Ø and minimum rod Ø.

Maximum permissible gap dimension (s)			
160 bar	260 bar	320 bar	400 bar
0.65	0.45	0.40	0.35

Minimum chamfer C:

DN - dN	Up to 12	12.1 to15	15.1 to 20	20.1 to 25	25.1 to 30	30.1 and above
C	4.0	5.0	6.0	7.5	8.0	10.0

Fitting:

RS054 is very easy to fit. Careful fitting of the seal is a prerequisite for it's perfect functioning.

Ordering format:

Please mention ROYAL order code in your order.

Example: Housing Size 25 X 37 X 22.5

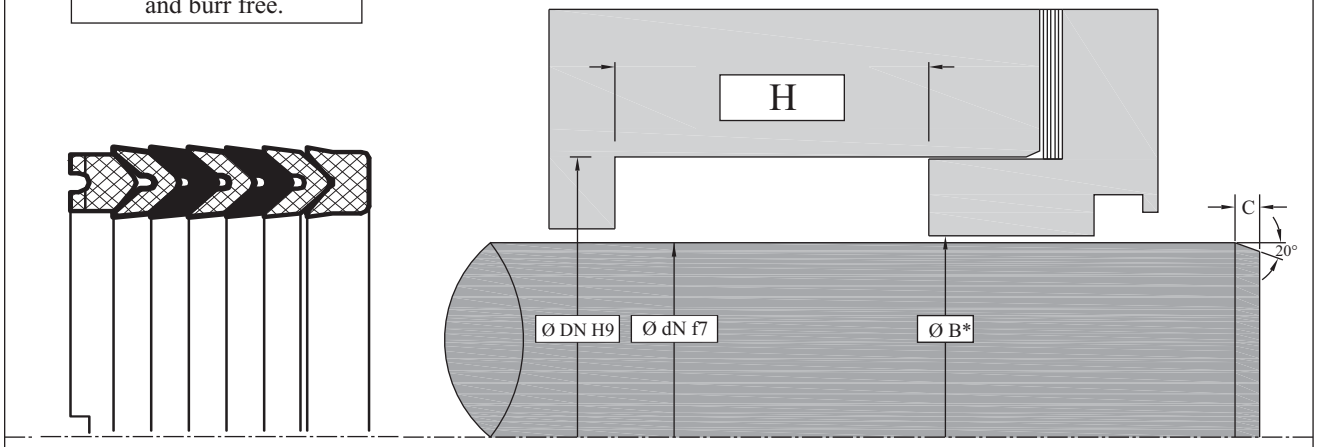
For Temp range: -20°C to +100°C Order code is RS054-N-02500-03700-2250

For Temp range: -15°C to +140°C Order code is RS054-F-02500-03700-2250

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

All edges should be smooth and burr free.

HOUSING DESIGN RS054



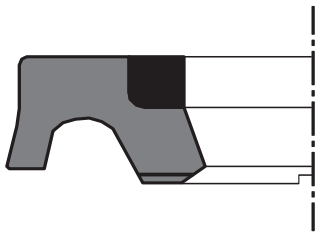
* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

ØDN (H9)	ØdN (h9)	H (+0.30)	ORDER CODE For Temp range: -20°C to +100°C use RS054-N -15°C to +140°C use RS054-F
25.00	37.00	22.50	RS054-N/F-02500-03700-2250
28.00	40.00	22.50	RS054-N/F-02800-04000-2250
32.00	47.00	22.50	RS054-N/F-03200-04700-2250
35.00	50.00	22.50	RS054-N/F-03500-05000-2250
36.00	48.00	22.50	RS054-N/F-03600-04800-2250
36.00	51.00	22.50	RS054-N/F-03600-05100-2250
40.00	55.00	22.50	RS054-N/F-04000-05500-2250
45.00	60.00	22.50	RS054-N/F-04500-06000-2250
45.00	63.00	30.00	RS054-N/F-04500-06300-3000
50.00	70.00	30.00	RS054-N/F-05000-07000-3000
56.00	76.00	37.00	RS054-N/F-05600-07600-3700
60.00	80.00	37.00	RS054-N/F-06000-08000-3700
63.00	83.00	37.00	RS054-N/F-06300-08300-3700
63.00	88.00	37.00	RS054-N/F-06300-08800-3700
70.00	90.00	40.00	RS054-N/F-07000-09000-4000
75.00	90.00	22.50	RS054-N/F-07500-09000-2250
75.00	95.00	40.00	RS054-N/F-07500-09500-4000
80.00	95.00	22.50	RS054-N/F-08000-09500-2250
80.00	100.00	40.00	RS054-N/F-08000-10000-4000
85.00	105.00	40.00	RS054-N/F-08500-10500-4000
90.00	105.00	25.00	RS054-N/F-09000-10500-2500
90.00	110.00	40.00	RS054-N/F-09000-11000-4000

ROYAL STANDARD SIZE LIST			
ØDN (H9)	**ØdN (h9)**	**H (+0.30)**	**ORDER CODE** For Temp range: -20°C to +100°C use RS054-N -15°C to +140°C use RS054-F
95.00	115.00	40.00	RS054-N/F-09500-11500-4000
100.00	120.00	40.00	RS054-N/F-10000-12000-4000
110.00	130.00	40.00	RS054-N/F-11000-13000-4000
110.00	140.00	50.00	RS054-N/F-11000-14000-5000
115.00	140.00	46.00	RS054-N/F-11500-14000-4600
120.00	145.00	46.00	RS054-N/F-12000-14500-4600
125.00	150.00	46.00	RS054-N/F-12500-15000-4600
140.00	165.00	46.00	RS054-N/F-14000-16500-4600
150.00	180.00	60.00	RS054-N/F-15000-18000-6000
160.00	190.00	60.00	RS054-N/F-16000-19000-6000
170.00	195.00	46.00	RS054-N/F-17000-19500-4600
180.00	210.00	60.00	RS054-N/F-18000-21000-6000
190.00	220.00	60.00	RS054-N/F-19000-22000-6000
200.00	220.00	50.00	RS054-N/F-20000-22000-5000
200.00	230.00	60.00	RS054-N/F-20000-23000-6000
220.00	250.00	60.00	RS054-N/F-22000-25000-6000
225.00	245.00	50.00	RS054-N/F-22500-24500-5000
225.00	255.00	60.00	RS054-N/F-22500-25500-6000
240.00	280.00	60.00	RS054-N/F-24000-28000-6000
250.00	280.00	60.00	RS054-N/F-25000-28000-6000
250.00	290.00	60.00	RS054-N/F-25000-29000-6000
260.00	290.00	60.00	RS054-N/F-26000-29000-6000
270.00	300.00	60.00	RS054-N/F-27000-30000-6000
280.00	310.00	50.00	RS054-N/F-28000-31000-5000
300.00	330.00	60.00	RS054-N/F-30000-33000-6000
320.00	360.00	63.00	RS054-N/F-32000-36000-6300

Please contact ROYAL at info@royalcomposites.co.in if your required size is not mentioned in Royal standard size list.

ROD SEAL	RS063	

Features:

Rod seal profile RS063 consists of two piece seal set with one seal ring and one backup ring used as buffer seal before Royal RS027, RS036, RS045.

Composition:

Seal Componant	Material	Royal Code
Seal Ring	Polyurethane	PUHP94
Backup Ring	Polyamide	PA6600

For compound detail please refer Royal seal materials.

Properties:

- Easy installation primary seal
- No pressure built up between primary and secondary seal
- Low friction , free of stick slip
- High abrasion resistance.
- Wide temperature Range.
- Good thermal conductivity.
- Low wear and high extrusion resistance.

Application:

- Mobile Hydraulics
- Injection molding machines
- Machine tools
- Material Handling Equipments.

General Technical Data:

Temperature range	-20°C to +110°C
Working Pressure(max.)	400 bar
Surface Speed(max.)	0.5 m/s

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Extrusion Gap dimension:

The gap dimension on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the maximum clearance Ø and min rod Ø.

Maximum permissible gap dimension (s)		
160 bar	260 bar	400 bar
0,4	0,35	0,3

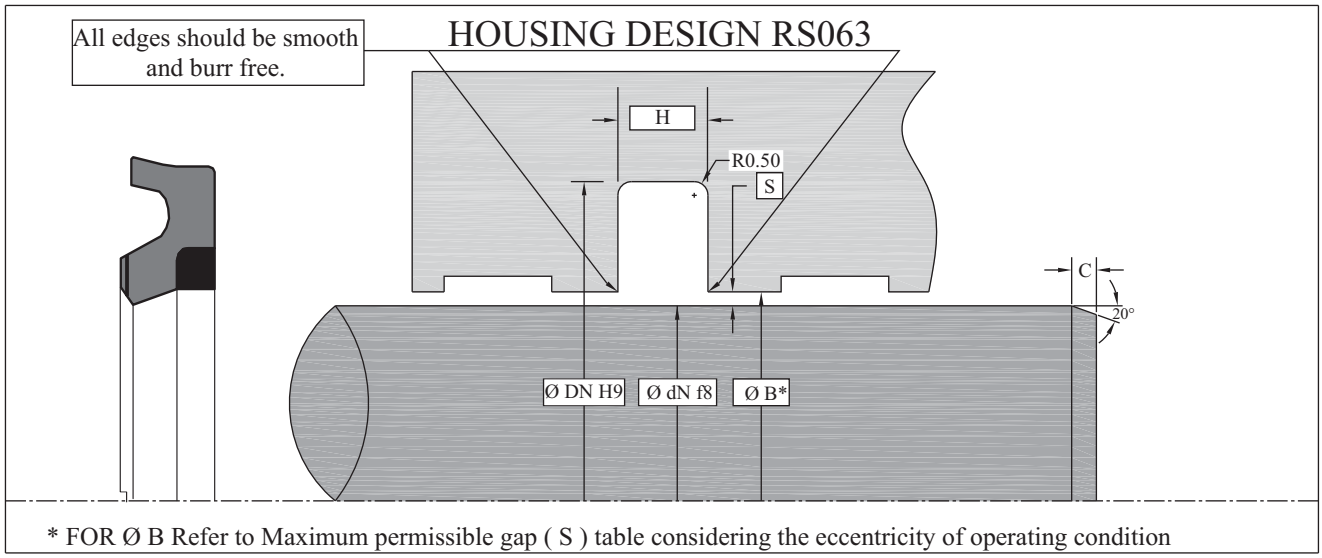
Installation:

RS063 is very easy to fit. Careful fitting of the seal is a prerequisite for it's perfect functioning.

Ordering format:

Please mention ROYAL order code in your order.
Example: Housing size 40 X 55.5 X 6.3
Order code is RS063-U-04000-05550-0630

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

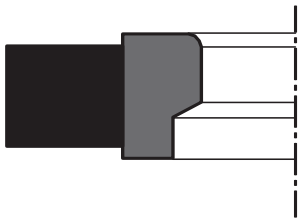


* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

ØDN (H9)	ØdN (h9)	H (+0.20)	ORDER CODE
40.00	55.50	6.30	RS063-U-04000-05550-0630
45.00	60.50	6.30	RS063-U-04500-06050-0630
50.00	65.50	6.30	RS063-U-05000-06550-0630
55.00	70.50	6.30	RS063-U-05500-07050-0630
56.00	71.50	6.30	RS063-U-05600-07150-0630
60.00	75.50	6.30	RS063-U-06000-07550-0630
63.00	78.50	6.30	RS063-U-06300-07850-0630
65.00	80.50	6.30	RS063-U-06500-08050-0630
70.00	85.50	6.30	RS063-U-07000-08550-0630
75.00	90.50	6.30	RS063-U-07500-09050-0630
80.00	95.50	6.30	RS063-U-08000-09550-0630
85.00	100.50	6.30	RS063-U-08500-10050-0630
90.00	105.50	6.30	RS063-U-09000-10550-0630
95.00	110.50	6.30	RS063-U-09500-11050-0630
100.00	115.50	6.30	RS063-U-10000-11550-0630
105.00	120.50	6.30	RS063-U-10500-12050-0630
108.00	123.50	6.30	RS063-U-10800-12350-0630
110.00	125.50	6.30	RS063-U-11000-12550-0630
115.00	130.50	6.30	RS063-U-11500-13050-0630
120.00	135.50	6.30	RS063-U-12000-13550-0630
125.00	140.50	6.30	RS063-U-12500-14050-0630
130.00	145.50	6.30	RS063-U-13000-14550-0630

ROYAL STANDARD SIZE LIST			
ØDN (H9)	**ØdN (h9)**	**H (+0.20)**	**ORDER CODE**
135.00	150.50	6.30	RS063-U-13500-15050-0630
140.00	155.50	6.30	RS063-U-14000-15550-0630
145.00	160.50	6.30	RS063-U-14500-16050-0630
150.00	165.50	6.30	RS063-U-15000-16550-0630
150.00	170.10	6.30	RS063-U-15000-17010-0630
155.00	170.10	6.30	RS063-U-15500-17010-0630
160.00	175.50	6.30	RS063-U-16000-17550-0630
170.00	185.50	6.30	RS063-U-17000-18550-0630
180.00	195.50	6.30	RS063-U-18000-19550-0630
190.00	205.50	6.30	RS063-U-19000-20550-0630
200.00	221.00	8.10	RS063-U-20000-22100-0810
IMPERIAL SIZES			
57.15	74.63	5.94	RS063-IN-U-02250-02938-0234
63.50	80.98	5.94	RS063-IN-U-02500-03188-0234

ROD SEAL	RS099	

Features:

Rod seal profile RS099 consists of two piece seal set for sealing rod with one seal ring and one energizer Square / Rectangular ring.

Composition:

SEAL COMBINATION		RS099-N
Seal ring	Material	PTFE+ Bronze +MoS2
	Royal Code	TB40M
O Ring	Material	Nitrile Butadiene rubber (NBR)
	Royal Code	NBR75

For compound detail please refer Royal seal materials.

Properties:

- Low friction, free of stick slip
- High abrasion resistance.
- Wide temperature Range.
- Good thermal conductivity.
- Low wear and high extrusion resistance.
- Wide fluid application range.
- Resistance against cold flow.

Application:

- Mobile Hydraulics
- Injection moulding machines
- Machine tools
- Material Handling Equipments.

General Technical Data:

Seal Profile	RS099
Temperature range	-20°C to 100°C
Working Pressure(max.)	400 bar
Surface Speed(max.)	4 m/s

Extrusion gap dimension (S) :

The gap dimension on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the minimum clearance Ø and max bore Ø.

Profile dimension	Maximum permissible gap (S)			
H	160 bar	260 bar	320 bar	400 bar
3.56	0.40	0.35	0.30	ASK ROYAL
5.59 / 6.35	0.50	0.40	0.30	ASK ROYAL
8.13	0.60	0.50	0.40	ASK ROYAL

Minimum chamfer C:

DN - dN	Up to 12	12.1 to15	15.1 to 20	20.1 to 25	25.1 to 30	30.1 and above
C	4.0	5.0	6.0	7.5	8.0	10.0

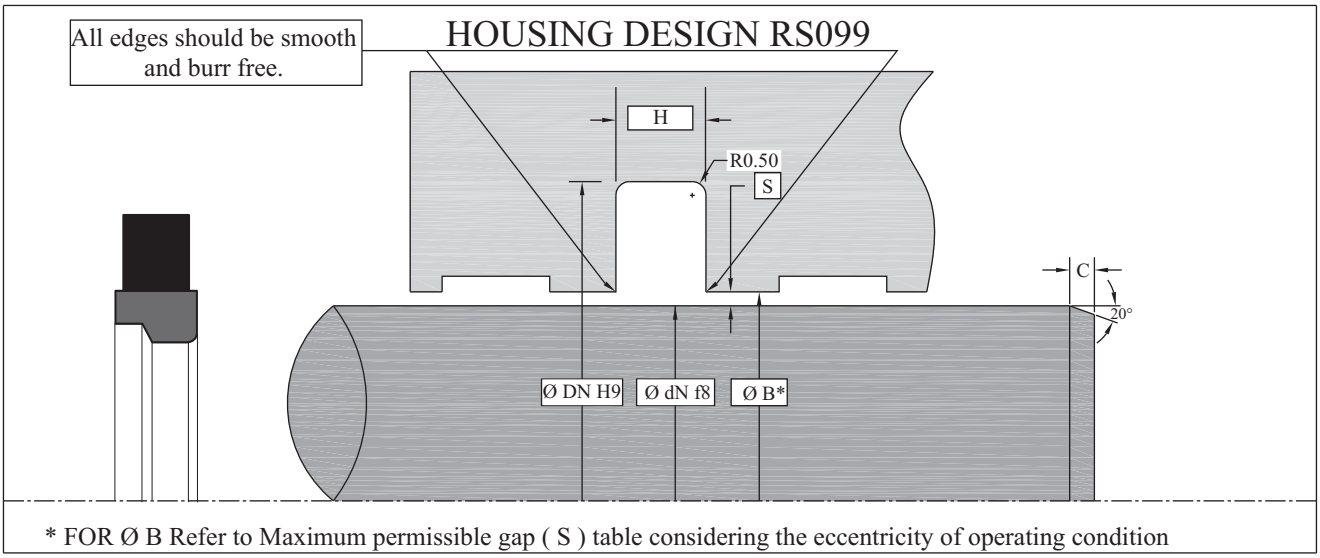
Installation :

Rs099 is very easy to fit. Insert the Square / Rectangular ring. first and then turn the PTFE part in kidney shape and insert in the groove. Gauge the seal by a taper rod type fixture. Careful fitting of the seal is a pre-requisite for it's perfect functioning.

Ordering format :

Please Follow ROYAL order code to place your order.
Like - for 31.75 X 45.29 X 3.56 - order cord is RS099-IN-01250-01783-0140

HOUSING DESIGN RS099

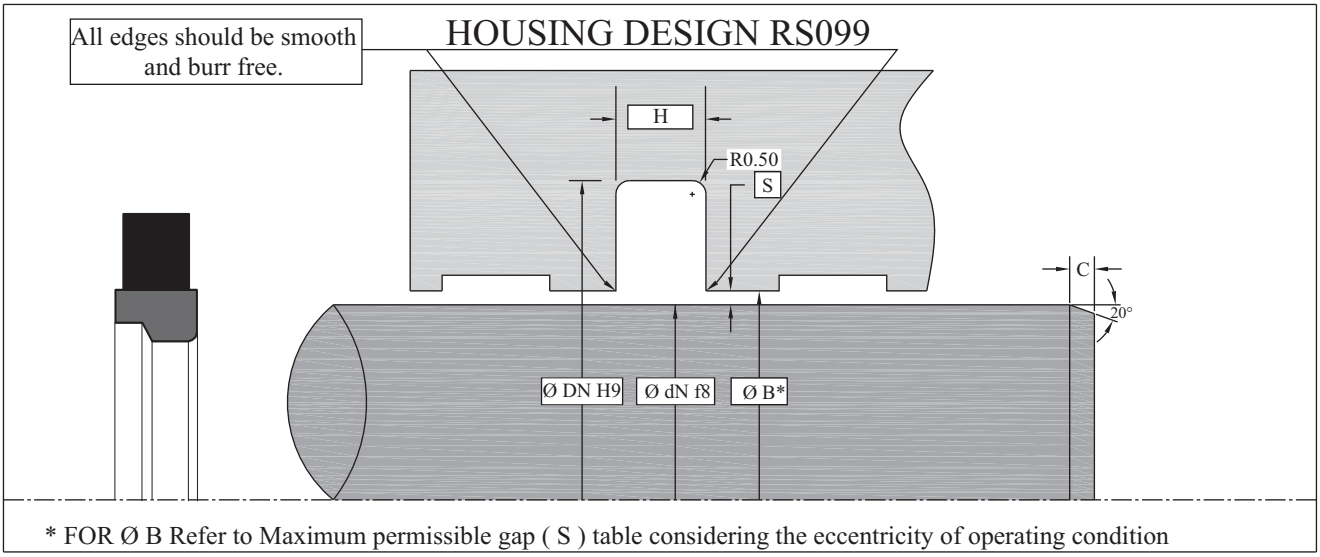


* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

ØDN (H9)	ØdN (h9)	H (+0.20)	ORDER CODE
IMPERIAL SIZE			
31.75	45.29	3.56	RS099-IN-01250-01783-0140
38.10	51.64	3.56	RS099-IN-01500-02033-0140
41.28	54.81	3.56	RS099-IN-01625-02158-0140
44.45	57.99	3.56	RS099-IN-01750-02283-0140
47.63	57.20	3.56	RS099-IN-01875-02252-0140
50.80	64.34	3.56	RS099-IN-02000-02533-0140
53.97	66.73	3.56	RS099-IN-02125-02627-0140
57.15	73.43	5.59	RS099-IN-02250-02891-0220
60.32	76.61	5.59	RS099-IN-02375-03016-0220
63.50	79.78	5.59	RS099-IN-02500-03141-0220
66.68	82.96	5.59	RS099-IN-02625-03266-0220
69.85	86.13	5.59	RS099-IN-02750-03391-0220
76.20	92.48	5.59	RS099-IN-03000-03641-0220
82.55	98.83	5.59	RS099-IN-03250-03891-0220
88.90	105.18	5.59	RS099-IN-03500-04141-0220
95.25	111.53	5.59	RS099-IN-03750-04391-0220
101.60	117.88	5.59	RS099-IN-04000-04641-0220
107.95	132.71	6.35	RS099-IN-04250-05225-0250
114.30	139.06	6.35	RS099-IN-04500-05475-0250
120.65	145.41	6.35	RS099-IN-04750-05725-0250

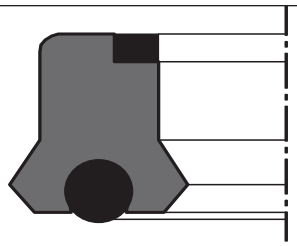
HOUSING DESIGN RS099



* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

ØDN (H9)	ØdN (h9)	H (+0.20)	ORDER CODE
127.00	151.38	6.35	RS099-IN-05000-05960-0250
133.35	157.73	6.35	RS099-IN-05250-06210-0250
139.70	155.35	6.35	RS099-IN-05500-06116-0250
146.05	170.81	6.35	RS099-IN-05750-06725-0250
152.40	168.12	6.35	RS099-IN-06000-06619-0250
165.10	190.55	6.35	RS099-IN-06500-07502-0250
171.45	195.45	6.35	RS099-IN-06750-07695-0250
177.80	201.80	6.35	RS099-IN-07000-07945-0250
184.15	205.23	8.13	RS099-IN-07250-08080-0320
190.50	211.58	8.13	RS099-IN-07500-08330-0320
203.20	224.36	8.13	RS099-IN-08000-08833-0320
209.55	230.63	8.13	RS099-IN-08250-09080-0320
222.25	243.33	8.13	RS099-IN-08750-09580-0320

ROD SEAL	RS108	

Features:

Rod seal profile RS108 is a single acting U cup type seal with energizer and anti extrusion polyamide ring for very high pressure and heavy duty application.

Composition:

Components	Material	Royal Code
Main seal	Polyurethane	PUHP94
Energizer O ring	NBR	NBR70
Anti extrusion ring	Polyamide	PA6000

For compound detail please refer Royal seal materials.(Section III)

Properties:

- High resistance to abrasion
- Easy installation
- Anti-extrusion ring prevent extrusion in
- high pressure conditions

Application:

- Mobile Hydraulics
- Injection molding machines
- Machine tools
- Agricultural equipment
- Construction Machineries.

General Technical Data:

Working Pressure (max.)	500 Bar
Temperature range	-20°C to 100°C
Surface Speed (max.)	0.5 m/s.

Extrusion Gap dimension:

The gap dimension on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the maximum clearance Ø and min rod Ø.

	Maximum permissible gap (S)			
160 bar	260 bar	320 bar	400 bar	500 bar
0.50	0.40	0.35	0.30	0.25

Installation :

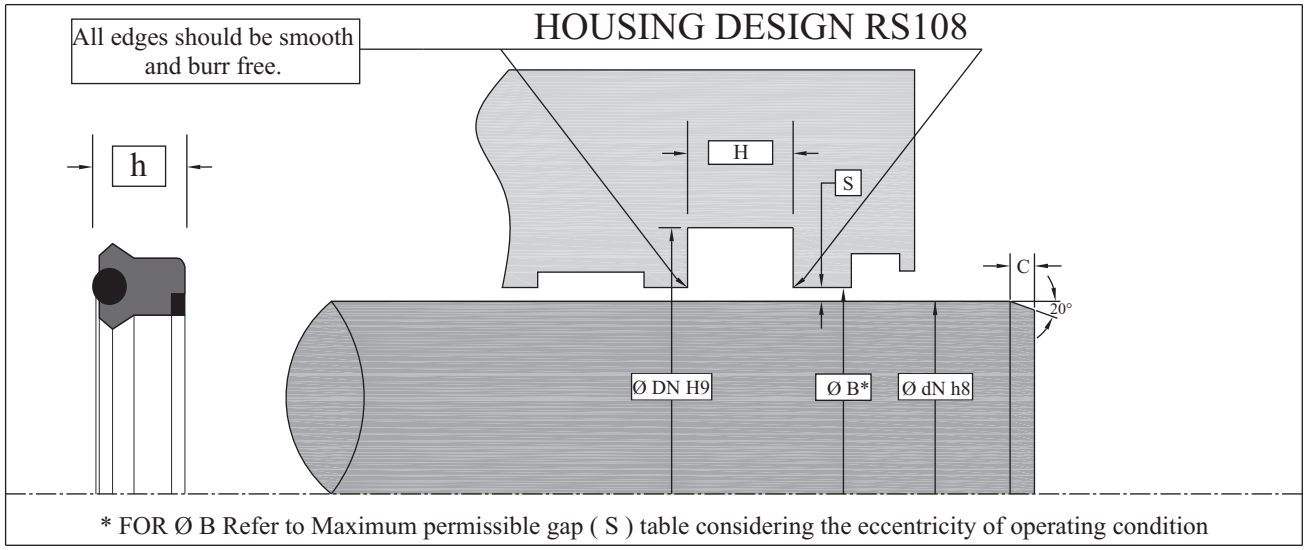
RS108 is very easy to fit. Insert the PU seal along with O ring to the housing by making kidney shape. there after insert the polyamide ring on top side of the seal. Careful fitting of the seal is a pre-requisite for it's perfect functioning.

Ordering format :

Please Follow ROYAL order code to place your order.
Like - for 40 X 55 X 12.50 - order cord is RS108-U-04000-05500-1250

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

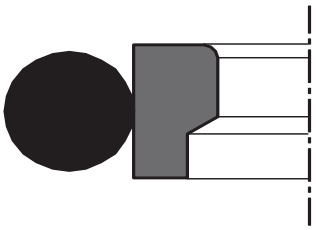
HOUSING DESIGN RS108



* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

DN (H9)	dN (h9)	h	H (+0.20)	ORDER CODE
40.00	55.00	11.40	12.50	RS108-U-04000-05500-1250
50.00	65.00	11.40	12.50	RS108-U-05000-06500-1250
60.00	75.00	11.40	12.50	RS108-U-06000-07500-1250
60.00	75.00	11.40	12.50	RS108-U-06000-07500-1250
65.00	80.00	11.40	12.50	RS108-U-06500-08000-1250
70.00	85.00	11.40	12.50	RS108-U-07000-08500-1250
125.00	140.00	11.80	13.00	RS108-U-12500-14000-1300

ROD SEAL	RS117	

Features:

Rod seal profile RS117 consists of two piece seal set for sealing rod with one polyurethane seal ring and one energizer O-ring.

Composition:

SEAL COMBINATION		RS117-U
Seal ring	Material	Polyurethane
	Royal Code	PUAB98
O Ring	Material	Nitrile Butadiene rubber (NBR)
	Royal Code	NBR70

For compound detail please refer Royal seal materials.

Properties:

- High abrasion resistance.
- Low wear
- Least oil film release
- Very Easy Installation

Application:

- Mobile Hydraulics
- Injection moulding machines
- Machine tools
- Material Handling Equipments.

General Technical Data:

Seal Profile	RS117-U
Temperature range	-20°C to 100°C
Working Pressure(max.)	320 bar
Surface Speed(max.)	1 m/s

Extrusion Gap dimension:

The gap dimension on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the maximum clearance Ø and min rod Ø.

Profile dimension	Maximum permissible gap (S)		
H	160 bar	260 bar	320 bar
3.20	0.35	0.30	-
4.20	0.40	0.35	0.30
6.30	0.50	0.40	0.30
8.10	0.60	0.50	0.40

Installation:

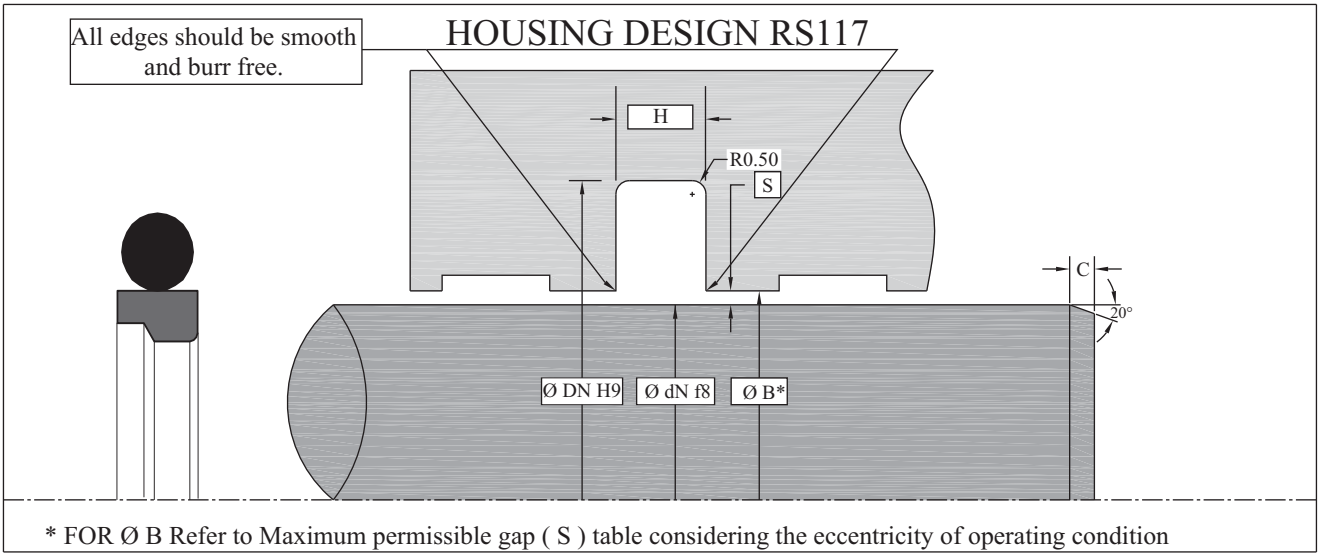
Rs117 is very easy to fit. Insert the O ring first and then turn the PU part in kidney shape and insert in the groove. Careful fitting of the seal is a pre-requisite for it's perfect functioning.

Ordering format:

Please Follow ROYAL order code to place your order.
Like - for 18 X 25.30 X 3.20 - order cord is RS117-U-01800-02530-3200

We have given all metric(mm) sizes here. For Imperial (inch) sizes please contact ROYAL at
info@royalcomposites.co.in / info@royalseals.co.in / info@royalseals.co.in

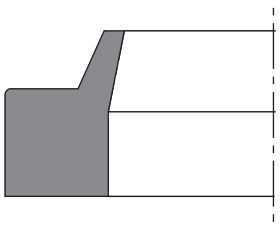
HOUSING DESIGN RS117



* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H9)	H (+0.20)	C	O-RING SIZE	ORDER CODE
18.00	25.30	3.20	2.50	20.29 X 2.62	RS117-U-01800-02530-3200
30.00	40.70	4.20	3.50	34.52 X 3.53	RS117-U-03000-04070-4200
45.00	60.10	6.30	5.50	50.17 X 5.33	RS117-U-04500-06010-6300
50.00	65.10	6.30	5.50	56.52 X 5.33	RS117-U-05000-06510-6300
60.00	75.10	6.30	5.50	66.04 X 5.33	RS117-U-06000-07510-6300
85.00	100.10	6.30	6.00	91.44 X 5.33	RS117-U-08500-10010-6300
90.00	105.10	6.30	6.00	94.62 X 5.33	RS117-U-09000-10510-6300
95.00	110.10	6.30	6.00	100.97 X 5.33	RS117-U-09500-11010-6300
110.00	125.10	6.30	6.00	116.84 X 5.33	RS117-U-11000-12510-6300
120.00	135.10	6.30	6.00	126.37 X 5.33	RS117-U-12000-13510-6300
125.00	140.10	6.30	6.00	129.54 X 5.33	RS117-U-12500-14010-6300
170.00	185.10	6.30	6.00	175.02 X 5.33	RS117-U-17000-18510-6300
225.00	245.50	8.10	8.00	234.32 X 6.99	RS117-U-22500-24550-8100

DUST SEAL	DS011	

Features:

Wiper seal DS-011 is a single acting rod wiper for integral housing.

Composition:

Seal type	DS011-U	DS011-N	DS011-F
Material:	Polyurethane	Nitrile Butadiene Rubber (NBR)	Fluorocarbon Rubber (FPM)
Royal Code	PUAB94	NBR90	FPM80

For compound detail please refer Royal seal materials. (Section III)

Properties:

- Easy snap in installation
- Very good wiping action
- Wiper can be used for a broad temperature range
- No pressure build-up between seal and wiper

Application:

- Presses, fork lift & material handling equipments
- Injection molding machines
- Standard cylinders

General Technical Data:

Seal profile	DS011-U	DS011-N	DS011-F
Temperature range	-20°C to +100°C	-20°C to +100°C	-20°C to +200°C
Linear velocity	0.5 m/s.	0.5 m/s.	0.5 m/s.

Fitting:

Snap in fitting through a kidney-shaped distortion. Careful fitting is prerequisite for perfect functioning of the wiper.

Ordering format:

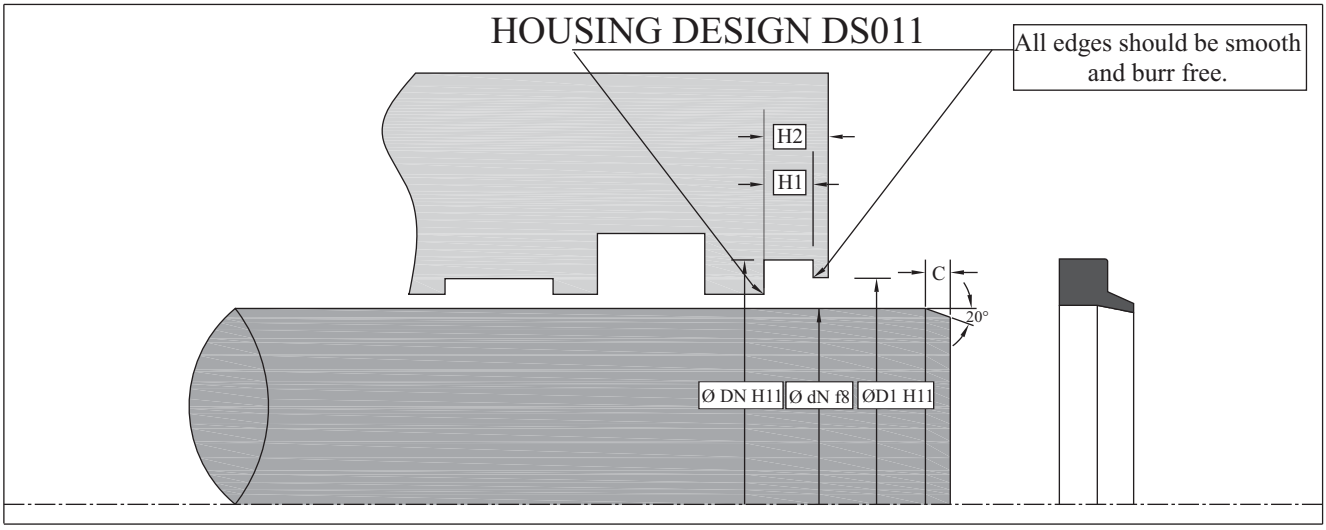
Please mention ROYAL order code in your order.

Example: Size 10 X 16.6 X 3.8 /5.3

For Temp range: - 20°C to +100°C	Material Polyurethane	Order code is DS011-U-01000-01660-038053
For Temp range: - 20°C to +100°C	Material NBR	Order code is DS011-N-01000-01660-038053
For Temp range: - 20°C to +200°C	Material FPM	Order code is DS011-F-01000-01660-038053

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN DS011

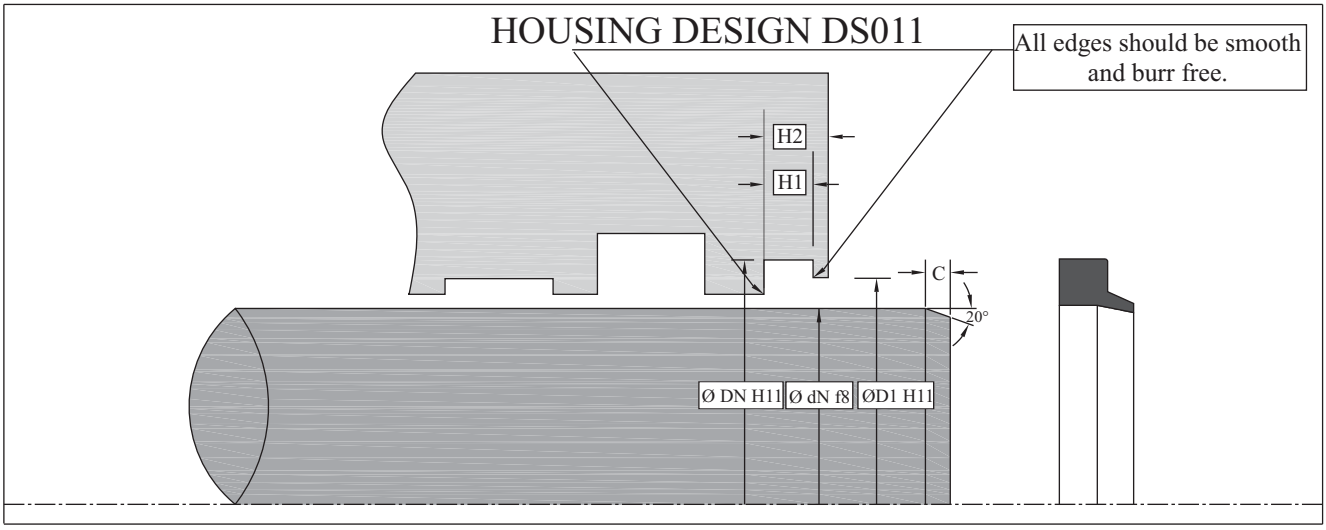


ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H11)	ØD1 (H11)	H1(+0.20)	H2	ORDER CODE
10.00	16.60	13.00	3.80	5.30	DS011-U/N/F-01000-01660-03800530
12.00	18.60	15.00	3.80	5.30	DS011-U/N/F-01200-01860-03800530
14.00	20.60	17.00	3.80	5.30	DS011-U/N/F-01400-02060-03800530
16.00	22.60	19.00	3.80	5.30	DS011-U/N/F-01600-02260-03800530
18.00	24.60	21.00	3.80	5.30	DS011-U/N/F-01800-02460-03800530
20.00	28.60	23.00	5.30	7.00	DS011-U/N/F-02000-02860-05300700
22.00	30.60	25.00	5.30	7.00	DS011-U/N/F-02200-03060-05300700
25.00	33.60	28.00	5.30	7.00	DS011-U/N/F-02500-03360-05300700
28.00	36.60	31.00	5.30	7.00	DS011-U/N/F-02800-03660-05300700
30.00	38.60	33.00	5.30	7.00	DS011-U/N/F-03000-03860-05300700
32.00	40.60	35.00	5.30	7.00	DS011-U/N/F-03200-04060-05300700
35.00	43.60	38.00	5.30	7.00	DS011-U/N/F-03500-04360-05300700
36.00	44.60	39.00	5.30	7.00	DS011-U/N/F-03600-04460-05300700
40.00	48.60	43.00	5.30	7.00	DS011-U/N/F-04000-04860-05300700
42.00	50.60	45.00	5.30	7.00	DS011-U/N/F-04200-05060-05300700
45.00	55.60	48.00	5.30	7.00	DS011-U/N/F-04500-05560-05300700
50.00	58.60	53.00	5.30	7.00	DS011-U/N/F-05000-05860-05300700
50.00	60.60	53.00	5.30	7.00	DS011-U/N/F-05000-06060-05300700
55.00	65.60	58.00	5.30	7.00	DS011-U/N/F-05500-06560-05300700
56.00	66.60	59.00	5.30	7.00	DS011-U/N/F-05600-06660-05300700
60.00	70.60	63.00	5.30	7.00	DS011-U/N/F-06000-07060-05300700
63.00	73.60	66.00	5.30	7.00	DS011-U/N/F-06300-07360-05300700

ROYAL STANDARD SIZE LIST					
ØdN (f8)	ØDN (H11)	ØD1 (H11)	H1(+0.20)	H2	ORDER CODE
65.00	75.60	68.00	5.30	7.00	DS011-U/N/F-06500-07560-05300700
70.00	80.60	73.00	5.30	7.00	DS011-U/N/F-07000-08060-05300700
75.00	87.20	81.00	7.10	12.00	DS011-U/N/F-07500-08720-07101200
80.00	92.20	86.00	7.10	12.00	DS011-U/N/F-08000-09220-07101200
85.00	97.20	91.00	7.10	12.00	DS011-U/N/F-08500-09720-07101200
90.00	102.20	96.00	7.10	12.00	DS011-U/N/F-09000-10220-07101200
100.00	112.20	106.00	7.10	12.00	DS011-U/N/F-10000-11220-07101200
110.00	122.20	116.00	7.10	12.00	DS011-U/N/F-11000-12220-07101200
115.00	127.20	121.00	7.10	12.00	DS011-U/N/F-11500-12720-07101200
120.00	132.20	126.00	7.10	12.00	DS011-U/N/F-12000-13220-07101200
125.00	137.20	131.00	7.10	12.00	DS011-U/N/F-12500-13720-07101200
130.00	142.20	136.00	7.10	12.00	DS011-U/N/F-13000-14220-07101200
140.00	152.20	146.00	7.70	12.00	DS011-U/N/F-14000-15220-07701200
140.00	155.00	147.60	10.20	16.00	DS011-U/N/F-14000-15500-10201600
150.00	162.20	156.00	7.70	12.00	DS011-U/N/F-15000-16220-07701200
150.00	165.00	157.60	10.20	16.00	DS011-U/N/F-15000-16500-10201600
155.00	167.20	161.00	7.20	12.00	DS011-U/N/F-15500-16720-07201200
160.00	172.20	166.00	7.70	12.00	DS011-U/N/F-16000-17220-07701200
160.00	175.00	167.60	10.20	16.00	DS011-U/N/F-16000-17500-10201600
170.00	182.20	176.00	7.70	12.00	DS011-U/N/F-17000-18220-07701200
180.00	200.00	190.00	10.20	18.00	DS011-U/N/F-18000-20000-10201800
190.00	210.00	200.00	10.20	18.00	DS011-U/N/F-19000-21000-10201800
200.00	220.00	210.00	10.20	18.00	DS011-U/N/F-20000-22000-10201800
210.00	230.00	220.00	10.20	18.00	DS011-U/N/F-21000-23000-10201800
220.00	240.00	230.00	10.20	18.00	DS011-U/N/F-22000-24000-10201800
230.00	250.00	240.00	10.20	18.00	DS011-U/N/F-23000-25000-10201800
240.00	260.00	250.00	10.20	18.00	DS011-U/N/F-24000-26000-10201800
250.00	270.00	260.00	10.20	18.00	DS011-U/N/F-25000-27000-10201800
260.00	280.00	270.00	10.20	18.00	DS011-U/N/F-26000-28000-10201800
270.00	290.00	280.00	10.20	18.00	DS011-U/N/F-27000-29000-10201800
280.00	300.00	290.00	10.20	18.00	DS011-U/N/F-28000-30000-10201800

HOUSING DESIGN DS011

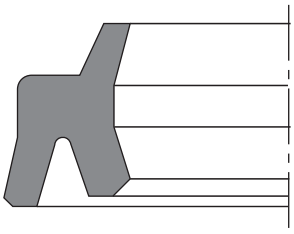


ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H11)	ØD1 (H11)	H1(+0.20)	H2	ORDER CODE
300.00	320.00	310.00	10.20	18.00	DS011-U/N/F-30000-32000-10201800
IMPERIAL SIZE					
7.92	14.27	12.07	3.15		DS011-IN-U-00312-00562-0124
12.70	19.05	16.76	3.15		DS011-IN-U-00500-00750-0124
19.05	28.58	25.27	4.75		DS011-IN-U-00750-01125-0187
25.40	34.92	31.62	4.75		DS011-IN-U-01000-01375-0187
28.58	38.10	34.80	4.75		DS011-IN-U-01125-01500-0187
31.75	41.28	37.97	4.75		DS011-IN-U-01250-01625-0187
34.92	44.45	41.15	4.75		DS011-IN-U-01375-01750-0187
38.10	47.63	44.32	6.35		DS011-IN-U-01500-01875-0250
44.45	53.97	50.67	4.75		DS011-IN-U-01750-02125-0187

ROYAL STANDARD SIZE LIST					
ØdN (f8)	ØDN (H11)	ØD1 (H11)	H1(+0.20)	H2	ORDER CODE
50.80	63.50	59.11	6.35		DS011-IN-U-02000-02500-0250
57.15	69.85	65.46	6.35		DS011-IN-U-02250-02750-0250
63.50	76.20	71.80	6.35		DS011-IN-U-02500-03000-0250
69.85	82.55	78.16	6.35		DS011-IN-U-02750-03250-0250
76.20	88.90	84.51	6.35		DS011-IN-U-03000-03500-0250
82.55	95.25	90.86	6.35		DS011-IN-U-03250-03750-0250
88.90	101.60	97.21	6.35		DS011-IN-U-03500-04000-0250
95.25	107.95	103.56	6.35		DS011-IN-U-03750-04250-0250
127.00	146.05	139.52	9.52		DS011-IN-U-05000-05750-0375
133.35	152.40	145.87	9.52		DS011-IN-U-05250-06000-0375
158.75	171.45	167.06	6.35		DS011-IN-U-06250-06750-0250
165.10	190.50	180.34	19.84		DS011-IN-U-06500-07500-0781
171.45	190.50	183.97	9.52		DS011-IN-U-06750-07500-0000
177.80	190.25	186.36	3.81		DS011-IN-U-07000-07490-0150
177.80	203.20	193.04	19.84		DS011-IN-U-07000-08000-0781
190.50	215.90	205.74	9.52		DS011-IN-U-07500-08500-0375
206.38	219.08	214.93	6.35		DS011-IN-U-08125-08625-0250
215.90	234.95	228.42	9.52		DS011-IN-U-08500-09250-0375
228.60	241.30	237.16	6.35		DS011-IN-U-09000-09500-0250
228.60	247.65	241.12	9.52		DS011-IN-U-09000-09750-0375
273.05	298.45	289.79	12.70		DS011-IN-U-10750-11750-0500

Please contact ROYAL at **info@royalcomposites.co.in** if your required size is not mentioned in size list.

DUST SEAL	DS022	

Features:
Wiper seal DS-022 is a double acting rod wiper for integral housing.

Composition:

Material:	Polyurethane
Royal Code	PUET93

For compound detail please refer Royal seal materials.(Section III)

Properties:
- Easy snap in installation
- Very good wiping action
- Highly abrasion resistance

Application:
- Presses, fork lift & material handling equipments
- Injection molding machines
- Standard cylinders

General Technical Data:

Temperature range	-20°C to +100°C
Surface Speed(max.)	0.5 m/s

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Fitting:

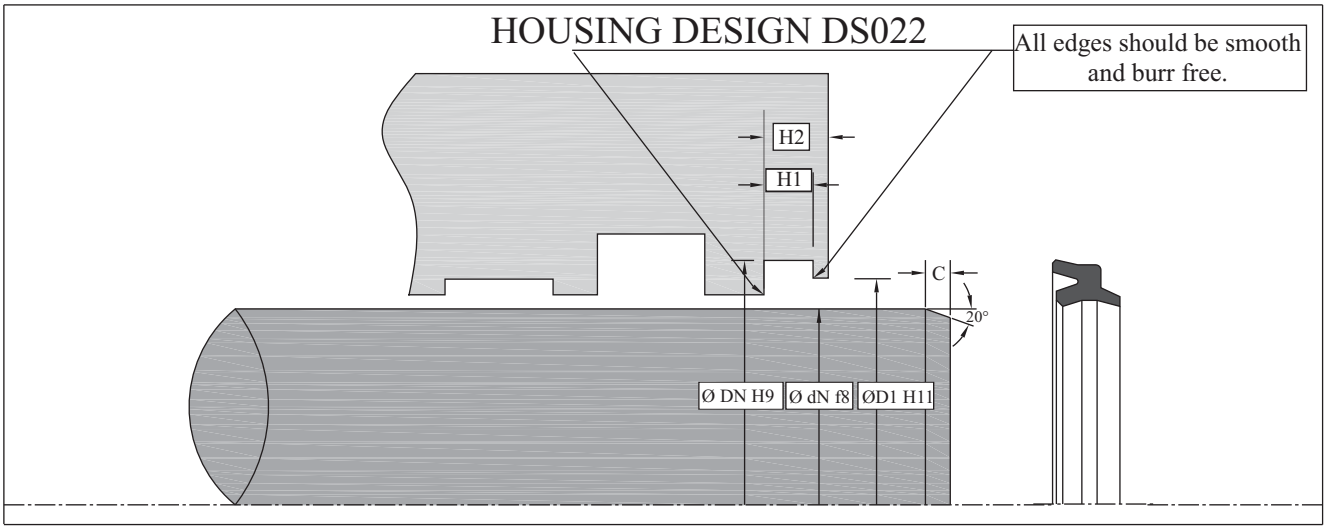
Snap in fitting through a kidney-shaped distortion. Careful fitting is prerequisite for perfect functioning of the wiper.

Ordering format:

Please mention ROYAL order code in your order.
Example: Size 12 X 18 X 4 / 7
The Order code is DS022-U-01200-01800-04000700

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

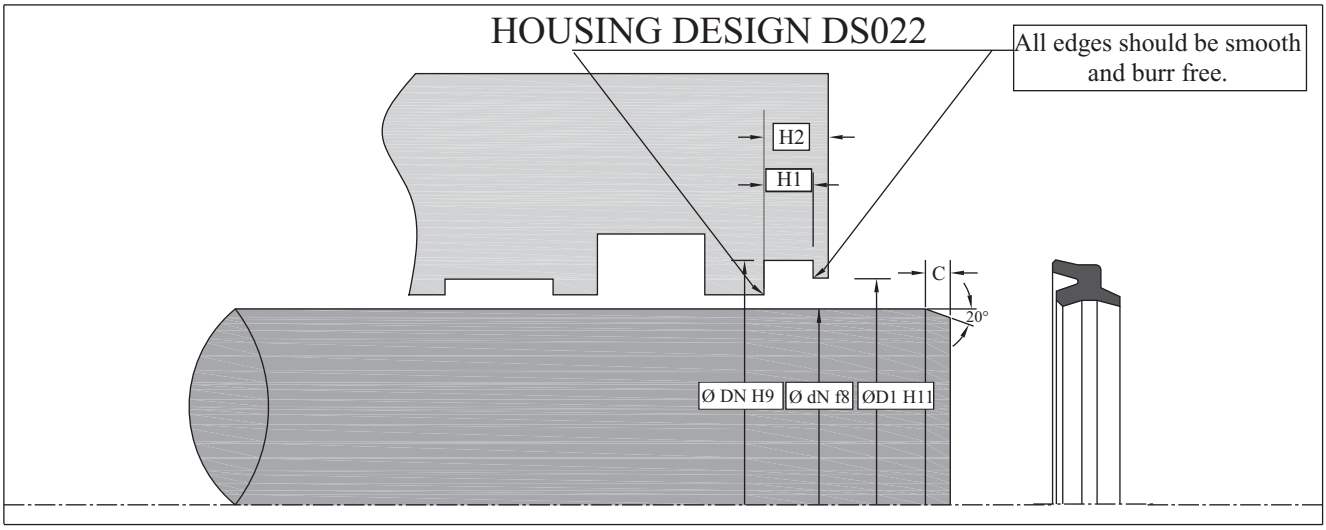
HOUSING DESIGN DS022



ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H11)	ØD1 (H11)	H1(+0.20)	H2	ORDER CODE
7.00	12.00	9.50	3.50	5.00	DS022-U-00700-01200-03500500
10.00	16.00	12.50	4.00	7.00	DS022-U-01000-01600-04000700
12.00	18.00	14.50	4.00	7.00	DS022-U-01200-01800-04000700
14.00	20.00	16.50	4.00	7.00	DS022-U-01400-02000-04000700
16.00	22.00	18.50	4.00	7.00	DS022-U-01600-02200-04000700
16.00	24.00	20.30	5.00	7.00	DS022-U-01600-02400-05000700
16.00	26.00	19.00	4.50	7.50	DS022-U-01600-02600-04500750
16.00	26.00	19.00	4.00	7.50	DS022-U-01600-02600-04000750
18.00	24.00	20.50	4.00	7.00	DS022-U-01800-02400-04000700
18.00	26.00	22.30	5.00	7.00	DS022-U-01800-02600-05000700
20.00	26.00	22.50	4.00	7.00	DS022-U-02000-02600-04000700
22.00	28.00	24.50	4.00	7.00	DS022-U-02200-02800-04000700
25.00	31.00	27.50	4.00	7.00	DS022-U-02500-03100-04000700
25.00	33.00	28.00	4.00	7.00	DS022-U-02500-03300-04000700
25.00	33.00	28.00	5.00	7.00	DS022-U-02500-03300-05000700
28.00	36.00	31.00	5.00	8.00	DS022-U-02800-03600-05000800
28.00	36.00	31.00	5.00	7.00	DS022-U-02800-03600-05000700
30.00	38.00	33.00	5.00	8.00	DS022-U-03000-03800-05000800
30.00	38.00	33.00	5.00	7.00	DS022-U-03000-03800-05000700
31.50	40.00	34.50	5.00	7.00	DS022-U-03150-04000-05000700
32.00	40.00	35.00	5.00	8.00	DS022-U-03200-04000-05000800
35.00	43.00	38.00	5.00	8.00	DS022-U-03500-04300-05000800

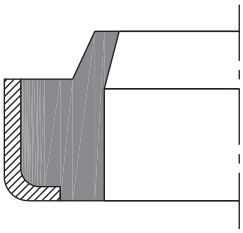
HOUSING DESIGN DS022



ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H11)	ØD1 (H11)	H1(+0.20)	H2	ORDER CODE
36.00	44.00	39.00	5.00	8.00	DS022-U-03600-04400-05000800
40.00	48.00	43.00	5.00	8.00	DS022-U-04000-04800-05000800
40.00	50.00	43.00	5.00	7.00	DS022-U-04000-05000-05000700
45.00	53.00	48.00	5.00	8.00	DS022-U-04500-05300-05000800
50.00	58.00	53.00	5.00	8.00	DS022-U-05000-05800-05000800
50.00	58.00	53.00	5.30	7.00	DS022-U-05000-05800-05300700
54.00	62.00	57.00	6.00	9.00	DS022-U-05400-06200-06000900
55.00	65.00	58.00	6.00	9.00	DS022-U-05500-06500-06000900
56.00	66.00	59.00	6.00	9.00	DS022-U-05600-06600-06000900
60.00	70.00	63.00	6.00	9.00	DS022-U-06000-07000-06000900
60.00	70.00	63.00	4.00	7.00	DS022-U-06000-07000-04000700
63.00	71.00	66.00	5.00	7.00	DS022-U-06300-07100-05000700
63.00	73.00	66.00	6.00	9.00	DS022-U-06300-07300-06000900
65.00	75.00	68.00	6.00	9.00	DS022-U-06500-07500-06000900
70.00	80.00	73.00	6.00	9.00	DS022-U-07000-08000-06000900
75.00	83.60	78.00	5.00	7.00	DS022-U-07500-08360-05000700
75.00	85.00	78.00	6.00	9.00	DS022-U-07500-08500-06000900
80.00	90.00	83.00	6.00	9.00	DS022-U-08000-09000-06000900
85.00	95.00	88.00	6.00	9.00	DS022-U-08500-09500-06000900
90.00	100.00	93.00	6.00	9.00	DS022-U-09000-10000-06000900
90.00	104.00	94.00	6.00	11.00	DS022-U-09000-10400-06001100
94.00	104.00	97.00	6.00	8.00	DS022-U-09400-10400-06000800

ROYAL STANDARD SIZE LIST					
ØdN (f8)	ØDN (H11)	ØD1 (H11)	H1(+0.20)	H2	ORDER CODE
95.00	103.60	98.00	5.00	7.00	DS022-U-09500-10360-05000700
100.00	110.00	103.00	6.00	9.00	DS022-U-10000-11000-06000900
110.00	125.00	114.00	8.50	12.00	DS022-U-11000-12500-08501200
114.00	124.00	117.00	6.00	8.00	DS022-U-11400-12400-06000800
115.00	123.60	118.00	5.00	7.00	DS022-U-11500-12360-05000700
120.00	135.00	124.00	8.50	12.00	DS022-U-12000-13500-08501200
125.00	137.00	129.00	8.00	11.00	DS022-U-12500-13700-08001100
125.00	137.00	129.00	7.00	12.00	DS022-U-12500-13700-07001200
125.00	140.00	129.00	8.50	12.00	DS022-U-12500-14000-08501200
135.00	143.60	138.00	5.00	7.00	DS022-U-13500-14360-05000700
140.00	155.00	144.00	8.50	12.00	DS022-U-14000-15500-08501200
150.00	162.00	154.00	8.00	10.00	DS022-U-15000-16200-08001000
160.00	175.00	164.00	8.50	12.00	DS022-U-16000-17500-08501200
170.00	185.00	174.00	8.50	12.00	DS022-U-17000-18500-08501200
180.00	195.00	184.00	8.50	12.00	DS022-U-18000-19500-08501200
200.00	215.00	204.00	8.50	12.00	DS022-U-20000-21500-08501200
220.00	240.00	230.00	10.20	16.00	DS022-U-22000-24000-10201600

DUST SEAL	DS033	

Features:
Wiper seal DS-033 is a single acting rod wiper with metal casing for open housing.

Composition:

Seal profile type	DS033-U	DS033-N	DS033-F
Material	Polyurethane	Nitrile Butadiene Rubber (NBR)	Fluorocarbon Rubber (FPM)
Royal Code	PUAB85	NBR90	FPM 80
Metal casing	Mild steel	Mild steel	Mild steel

For compound detail please refer Royal seal materials. (Section III)

Properties:
- Easy push fit installation
- Very good wiping action
- Wiper can be used for a broad temperature range
- No pressure build-up between seal and wiper

Application:
- Presses, fork lift & material handling equipments
- Injection molding machines
- Standard cylinders

General Technical Data:

Seal profile	DS033-U	DS033-N	DS033-F
Temperature range	-20°C to +100°C	-20°C to +100°C	-20°C to +200°C
Linear velocity	0.5 m/s.	0.5 m/s.	0.5 m/s.

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

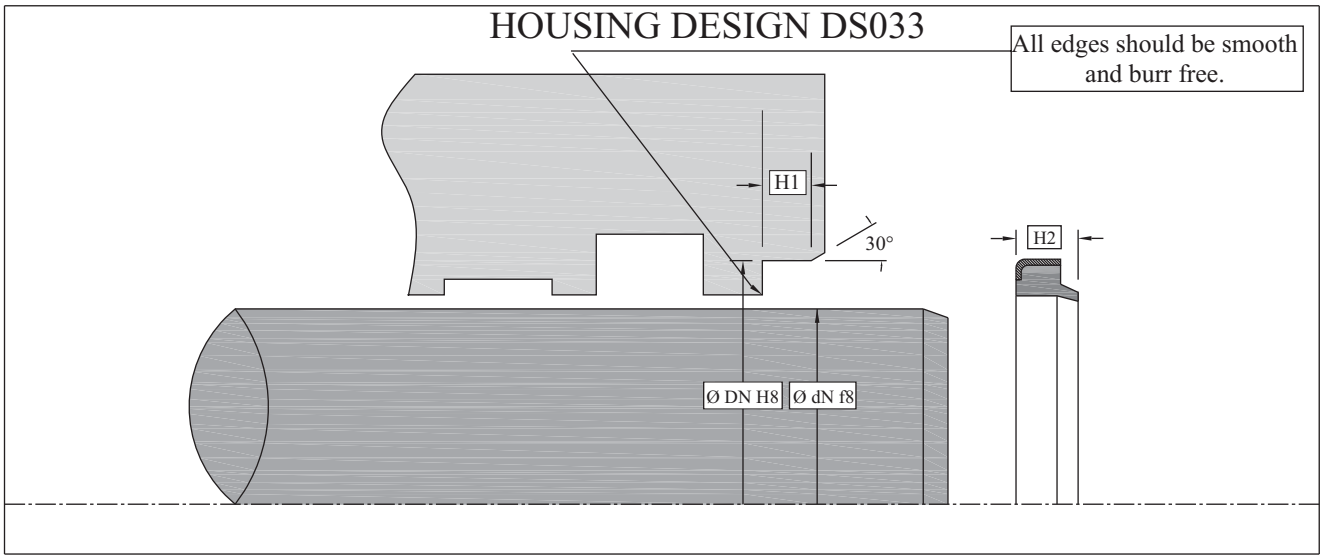
Fitting:
Wiper seal DS-033 are pressed in to open housings. Careful fitting is prerequisite for perfect functioning of the wiper seal.

Ordering format:
Please mention ROYAL order code in your order.
Example: Size 16 X 26 X 5 / 8

For Temp range: -20°C to +100°C	Material Polyurethane,	The Order code is DS033-U-02800-03800-05000700
For Temp range: -20°C to +100°C	Material NBR,	The Order code is DS033-N-02800-03800-05000700
For Temp range: -20°C to +200°C	Material FPM,	The Order code is DS033-F-02800-03800-05000700

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

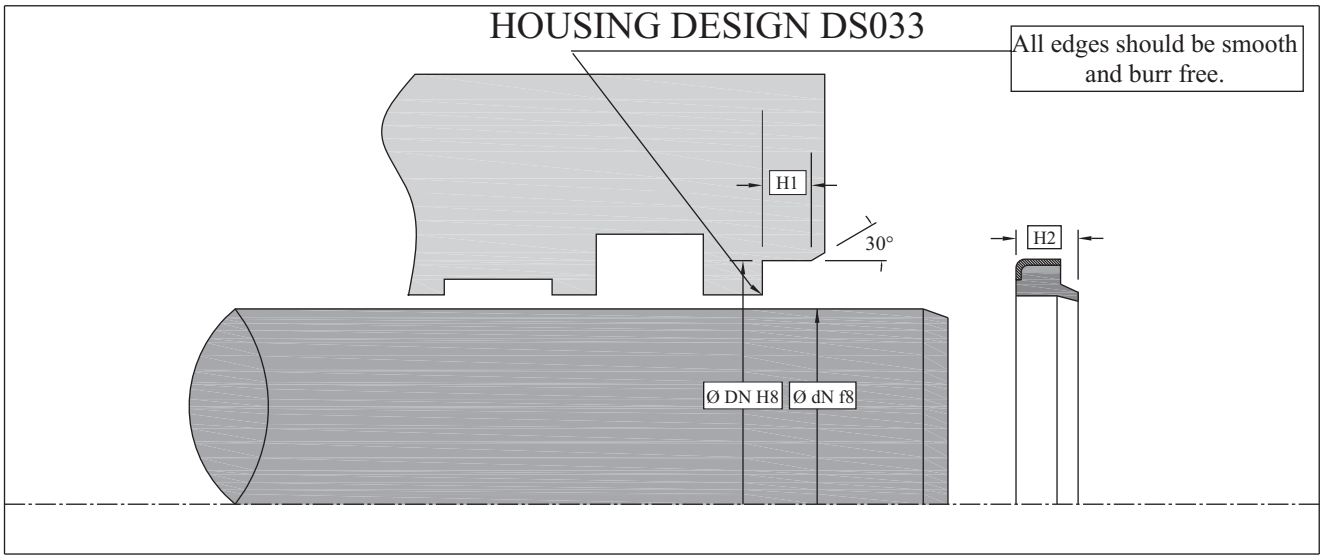
HOUSING DESIGN DS033



ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H11)	H1 (+0.20)	H2	ORDER CODE
16.00	26.00	5.00	8.00	DS033-U/N/F-01600-02600-05000800
20.00	28.00	5.00	8.00	DS033-U/N/F-02000-02800-05000800
25.00	35.00	5.00	8.00	DS033-U/N/F-02500-03500-05000800
25.00	35.00	7.00	10.00	DS033-U/N/F-02500-03500-07001000
25.00	37.00	7.00	10.00	DS033-U/N/F-02500-03700-07001000
28.00	38.00	5.00	7.00	DS033-U/N/F-02800-03800-05000700
30.00	40.00	5.00	7.00	DS033-U/N/F-03000-04000-05000700
30.00	40.00	7.00	10.00	DS033-U/N/F-03000-04000-07001000
32.00	42.00	5.00	7.00	DS033-U/N/F-03200-04200-05000700
32.00	45.00	7.00	10.00	DS033-U/N/F-03200-04500-07001000
35.00	45.00	7.00	10.00	DS033-U/N/F-03500-04500-07001000
36.00	45.00	7.00	10.00	DS033-U/N/F-03600-04500-07001000
36.00	46.00	7.00	10.00	DS033-U/N/F-03600-04600-07001000
36.00	48.00	5.00	8.00	DS033-U/N/F-03600-04800-05000800
40.00	50.00	7.00	10.00	DS033-U/N/F-04000-05000-07001000
45.00	55.00	7.00	10.00	DS033-U/N/F-04500-05500-07001000
50.00	60.00	7.00	10.00	DS033-U/N/F-05000-06000-07001000
55.00	65.00	7.00	10.00	DS033-U/N/F-05500-06500-07001000
55.00	70.00	7.00	10.00	DS033-U/N/F-05500-07000-07001000
56.00	66.00	7.00	10.00	DS033-U/N/F-05600-06600-07001000
60.00	70.00	7.00	10.00	DS033-U/N/F-06000-07000-07001000

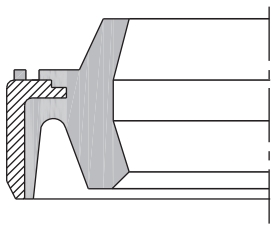
HOUSING DESIGN DS033



ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H11)	H1 (+0.20)	H2	ORDER CODE
60.00	72.00	5.00	8.00	DS033-U/N/F-06000-07200-05000800
63.00	73.00	7.00	10.00	DS033-U/N/F-06300-07300-07001000
63.00	75.00	7.00	10.00	DS033-U/N/F-06300-07500-07001000
70.00	80.00	7.00	10.00	DS033-U/N/F-07000-08000-07001000
72.00	84.00	5.00	8.00	DS033-U/N/F-07200-08400-05000800
75.00	85.00	7.00	10.00	DS033-U/N/F-07500-08500-07001000
80.00	90.00	7.00	10.00	DS033-U/N/F-08000-09000-07001000
85.00	95.00	7.00	10.00	DS033-U/N/F-08500-09500-07001000
90.00	100.00	7.00	10.00	DS033-U/N/F-09000-10000-07001000
90.00	102.00	7.00	10.00	DS033-U/N/F-09000-10200-07001000
93.00	105.00	5.00	8.00	DS033-U/N/F-09300-10500-05000800
95.00	105.00	7.00	10.00	DS033-U/N/F-09500-10500-07001000
100.00	110.00	7.00	10.00	DS033-U/N/F-10000-11000-07001000
105.00	115.00	7.00	10.00	DS033-U/N/F-10500-11500-07001000
105.00	117.00	5.00	8.00	DS033-U/N/F-10500-11700-05000800
105.00	121.00	8.00	11.00	DS033-U/N/F-10500-12100-08001100
110.00	120.00	7.00	10.00	DS033-U/N/F-11000-12000-07001000
120.00	130.00	7.00	10.00	DS033-U/N/F-12000-13000-07001000
125.00	140.00	9.00	12.00	DS033-U/N/F-12500-14000-09001200
130.00	145.00	9.00	12.00	DS033-U/N/F-13000-14500-09001200
140.00	155.00	9.00	12.00	DS033-U/N/F-14000-15500-09001200

ROYAL STANDARD SIZE LIST				
ØdN (f8)	**ØDN (H11)**	**H1 (+0.20)**	**H2**	**ORDER CODE**
150.00	165.00	9.00	12.00	DS033-U/N/F-15000-16500-09001200
160.00	175.00	9.00	12.00	DS033-U/N/F-16000-17500-09001200
170.00	185.00	10.00	14.00	DS033-U/N/F-17000-18500-10001400
180.00	195.00	10.00	14.00	DS033-U/N/F-18000-19500-10001400
200.00	220.00	12.00	16.00	DS033-U/N/F-20000-22000-12001600
200.00	225.00	12.00	18.00	DS033-U/N/F-20000-22500-12001800
240.00	260.00	12.00	16.00	DS033-U/N/F-24000-26000-12001600

DUST SEAL	DS044	

Features:
Wiper seal DS-044 is a double acting rod wiper with metal casing for open housing.

Composition:

Seal profile type	DS044-U
Material	Polyurethane
Royal Code	PUAB94
Metal casing	Mild steel

For compound detail please refer Royal seal materials. (Section III)

Properties:
- Easy press fit installation in open housing with locking circlip
- Very good wiping action
- Wiper can be used for a broad temperature range with different materials
- Sealing lips directed inwards gives both side wiping effect

Application:
- presses, fork lift & material handling equipments
- injection moulding machines
- Standard cylinders
- mining & construction machineries

General Technical Data:

Seal profile	DS044-U
Temperature range	-20°C to +100°C
Linear velocity(max.)	0.5 m/s.

Surface finish:

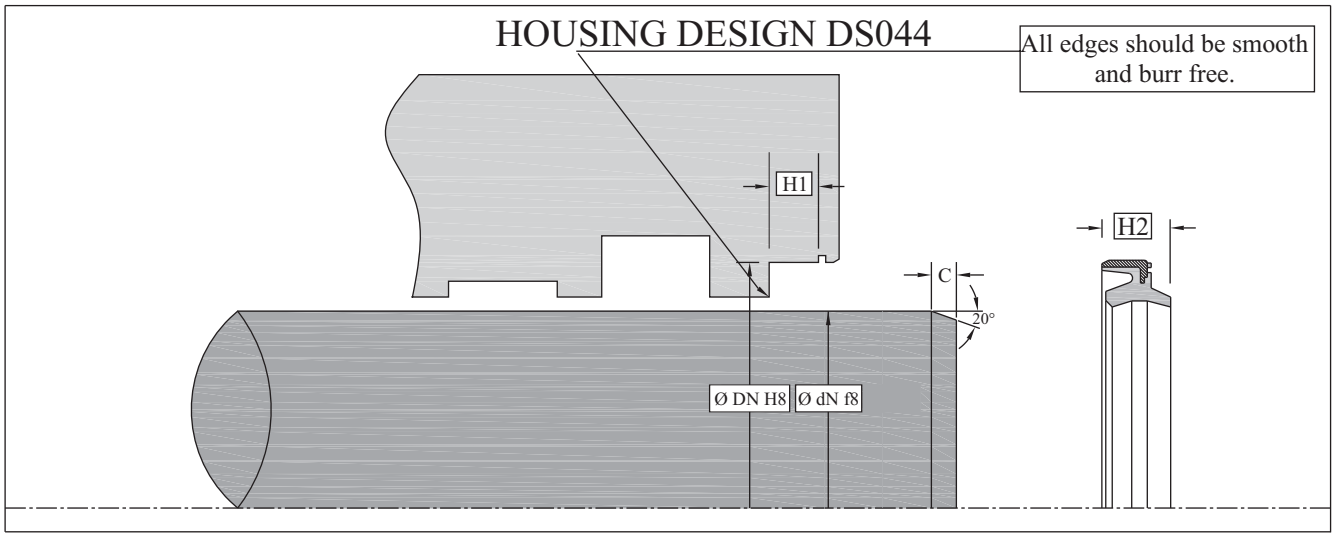
Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Fitting:
Wiper seal DS-044 are pressed in to open housings with locking circlip. Careful fitting is prerequisite for perfect functioning of the wiper seal.

Ordering format:
Please mention ROYAL order code in your order.
Example: Size 20 X 32 X 6 / 9
For Temp range: -20°C to +100°C Material-Polyurethane The Order code is DS044-U-02000-03200-06000900

We have given all metric(mm) sizes here. For Imperial (inch) sizes please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in**

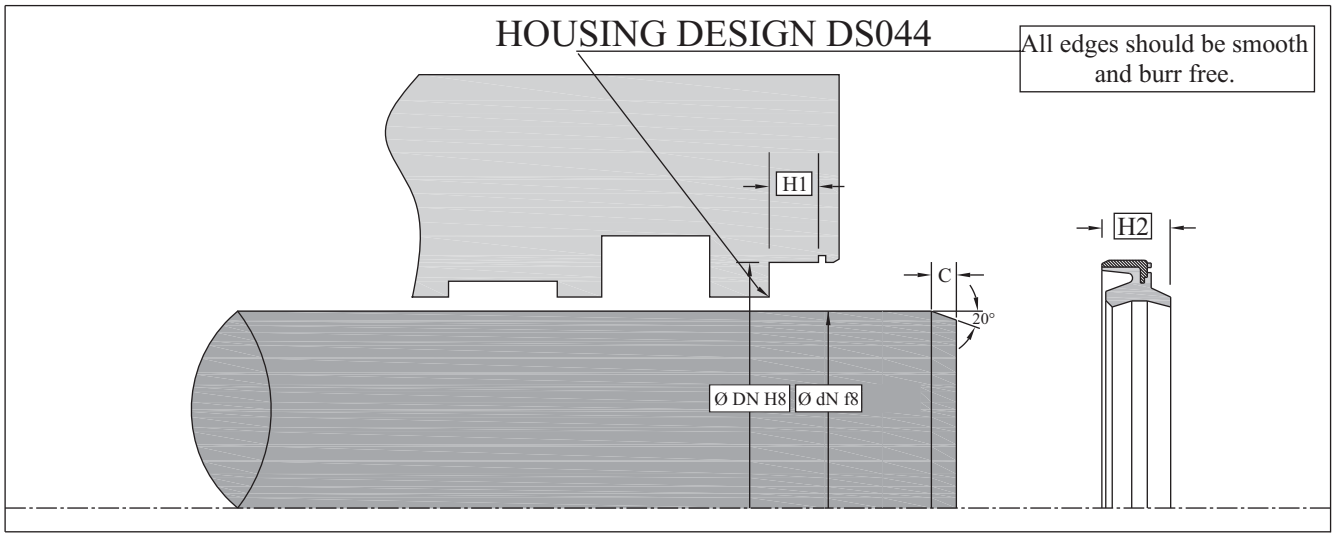
HOUSING DESIGN DS044



ROYAL STANDARD SIZE LIST

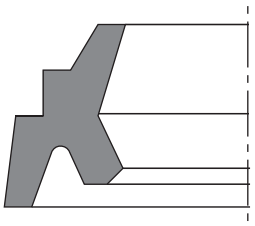
ØdN (f8)	ØDN (H8)	H1 (+0.20)	H2	ORDER CODE
20.00	32.00	6.00	9.00	DS044-U-02000-03200-06000900
22.00	34.00	6.00	9.00	DS044-U-02200-03400-06000900
25.00	37.00	6.00	9.00	DS044-U-02500-03700-06000900
28.00	40.00	6.00	9.00	DS044-U-02800-04000-06000900
30.00	42.00	6.00	9.00	DS044-U-03000-04200-06000900
32.00	44.00	7.00	10.00	DS044-U-03200-04400-07001000
35.00	47.00	7.00	10.00	DS044-U-03500-04700-07001000
36.00	48.00	7.00	10.00	DS044-U-03600-04800-07001000
38.00	50.00	7.00	10.00	DS044-U-03800-05000-07001000
40.00	52.00	7.00	10.00	DS044-U-04000-05200-07001000
45.00	57.00	7.00	10.00	DS044-U-04500-05700-07001000
50.00	62.00	7.00	10.00	DS044-U-05000-06200-07001000
55.00	69.00	8.00	11.00	DS044-U-05500-06900-08001100
56.00	70.00	8.00	11.00	DS044-U-05600-07000-08001100
60.00	74.00	8.00	11.00	DS044-U-06000-07400-08001100
63.00	77.00	8.00	11.00	DS044-U-06300-07700-08001100
65.00	79.00	8.00	11.00	DS044-U-06500-07900-08001100
70.00	84.00	8.00	11.00	DS044-U-07000-08400-08001100
75.00	89.00	8.00	11.00	DS044-U-07500-08900-08001100
80.00	94.00	8.00	11.00	DS044-U-08000-09400-08001100
85.00	99.00	8.00	11.00	DS044-U-08500-09900-08001100

HOUSING DESIGN DS044



ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H8)	H1 (+0.20)	H2	ORDER CODE
90.00	104.00	8.00	11.00	DS044-U-09000-10400-08001100
95.00	109.00	8.00	11.00	DS044-U-09500-10900-08001100
100.00	114.00	8.00	11.00	DS044-U-10000-11400-08001100
105.00	121.00	9.00	12.00	DS044-U-10500-12100-09001200
110.00	126.00	9.00	12.00	DS044-U-11000-12600-09001200
115.00	131.00	9.00	12.00	DS044-U-11500-13100-09001200
120.00	136.00	9.00	12.00	DS044-U-12000-13600-09001200
125.00	141.00	9.00	12.00	DS044-U-12500-14100-09001200
130.00	146.00	9.00	12.00	DS044-U-13000-14600-09001200
140.00	160.00	10.00	14.00	DS044-U-14000-16000-10001400
150.00	170.00	10.00	14.00	DS044-U-15000-17000-10001400
160.00	180.00	10.00	14.00	DS044-U-16000-18000-10001400
165.00	185.00	10.00	14.00	DS044-U-16500-18500-10001400
170.00	190.00	10.00	14.00	DS044-U-17000-19000-10001400
180.00	200.00	10.00	14.00	DS044-U-18000-20000-10001400
180.00	205.00	12.00	17.00	DS044-U-18000-20500-12001700
200.00	216.00	9.00	12.00	DS044-U-20000-21600-09001200
200.00	225.00	12.00	17.00	DS044-U-20000-22500-12001700

DUST SEAL	DS055	

Features:
Wiper seal DS-055 is a double acting rod wiper for integral housing.

Composition:

Seal type	DS055-U	DS055-N	DS055-F
Material:	Polyurethane	Nitrile Butadiene Rubber (NBR)	Fluorocarbon Rubber (FPM)
Royal Code	PUAB94	NBR90	FPM80

For compound detail please refer Royal seal materials. (Section III)

Properties:
- Easy snap in installation
- Very good wiping action
- Wiper can be used for a broad temperature range

Application:
- Presses, fork lift & material handling equipments
- Injection molding machines
- Standard cylinders

The double lip wiper DS055 is preferably fitted in combination with rod seal RS018. We recommend in any case a pressure release hole be provided in front of the double lip wiper in order to avoid pressure build-up between seal and wiper.

General Technical Data:

Seal profile	DS055-U	DS055-N	DS055-F
Temperature range	-20°C to +100°C	-20°C to +100°C	-20°C to +200°C
Linear velocity	0.5 m/s.	0.5 m/s.	0.5 m/s.

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

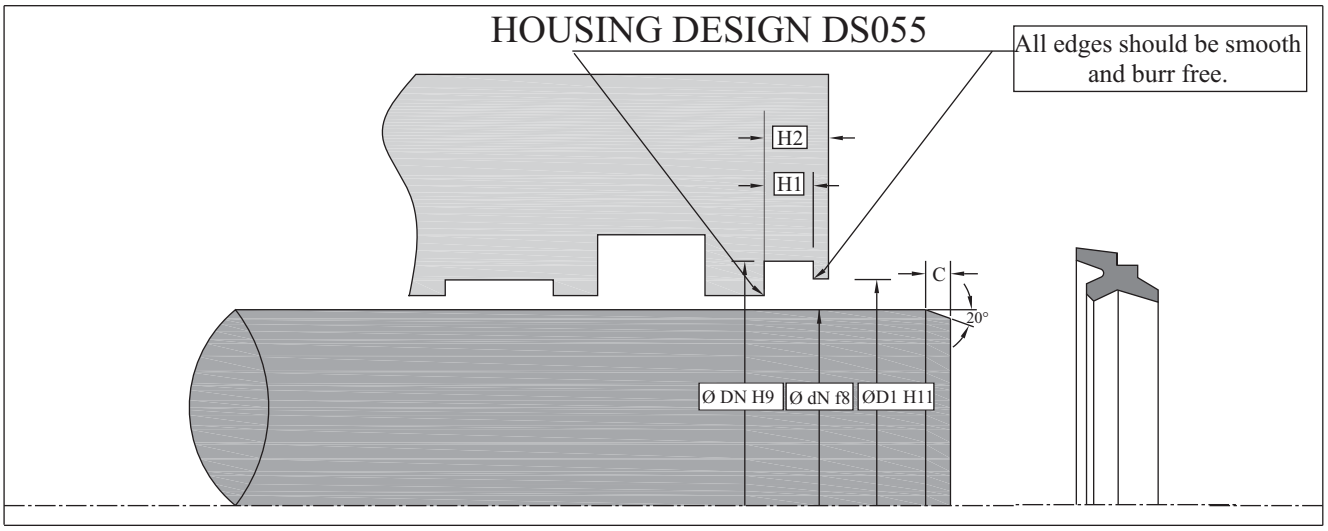
Fitting:
Snap in fitting through a kidney-shaped distortion. Careful fitting is prerequisite for perfect functioning of the wiper.

Ordering format:
Please mention ROYAL order code in your order.
Example: Size 10 X 18 X 4 / 6 / 8

For Temp range: -20°C to +100°C	Material Polyurethanes:	The Order code is DS055-U-01000-01800-06000800
For Temp range: -20°C to +100°C	Material NBR:	The Order code is DS055-N-01000-01800-06000800
For Temp range: -20°C to +200°C	Material FPM:	The Order code is DS055-F-01000-01800-06000800

We have given all metric(mm) sizes here. For Imperial (inch) sizes please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in**

HOUSING DESIGN DS055

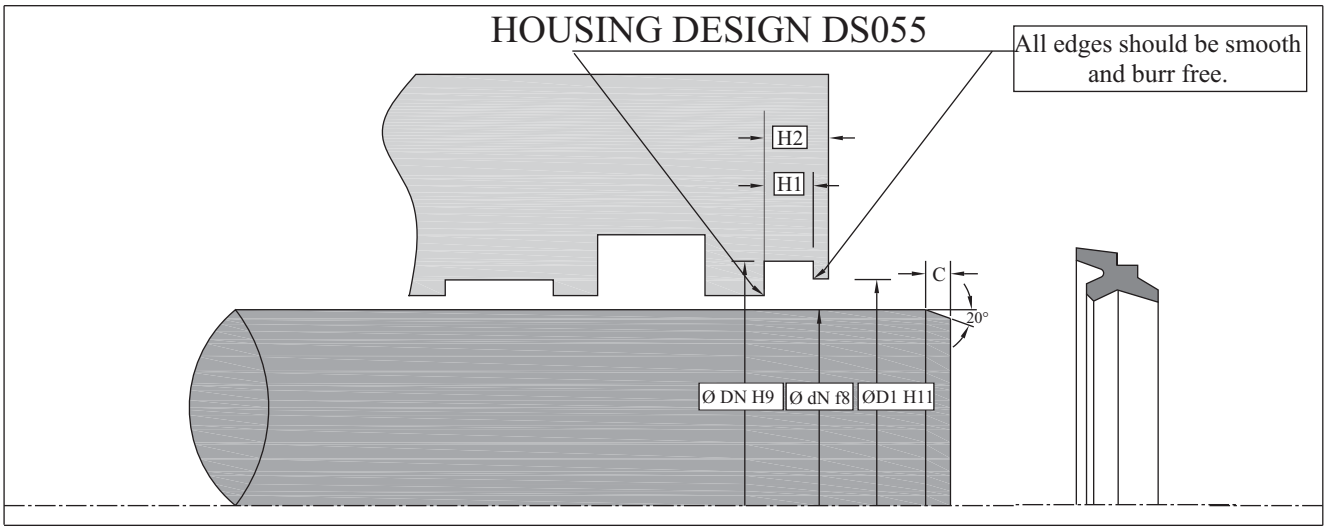


ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H11)	ØD1 (H11)	H1 (+0.20)	H2	ORDER CODE
10.00	18.00	13.50	6.00	8.00	DS055-U/N/F-01000-01800-06000800
12.00	20.00	15.50	6.00	8.00	DS055-U/N/F-01200-02000-06000800
14.00	22.00	17.50	6.00	8.00	DS055-U/N/F-01400-02200-06000800
15.00	23.00	18.50	6.00	8.00	DS055-U/N/F-01500-02300-06000800
16.00	24.00	19.50	6.00	8.00	DS055-U/N/F-01600-02400-06000800
18.00	26.00	21.50	6.00	8.00	DS055-U/N/F-01800-02600-06000800
20.00	28.00	23.50	6.00	8.00	DS055-U/N/F-02000-02800-06000800
22.00	30.00	25.50	6.00	8.00	DS055-U/N/F-02200-03000-06000800
25.00	33.00	28.50	6.00	8.00	DS055-U/N/F-02500-03300-06000800
28.00	36.00	31.50	6.00	8.00	DS055-U/N/F-02800-03600-06000800
30.00	38.00	33.50	6.00	8.00	DS055-U/N/F-03000-03800-06000800
32.00	40.00	35.50	6.00	8.00	DS055-U/N/F-03200-04000-06000800
35.00	43.00	38.50	6.00	8.00	DS055-U/N/F-03500-04300-06000800
36.00	44.00	39.50	6.00	8.00	DS055-U/N/F-03600-04400-06000800
38.00	46.00	41.50	6.00	8.00	DS055-U/N/F-03800-04600-06000800
40.00	48.00	43.50	6.00	8.00	DS055-U/N/F-04000-04800-06000800
42.00	50.00	45.50	6.00	8.00	DS055-U/N/F-04200-05000-06000800
45.00	53.00	48.50	6.00	8.00	DS055-U/N/F-04500-05300-06000800
50.00	58.00	53.50	6.00	8.00	DS055-U/N/F-05000-05800-06000800
55.00	63.00	58.50	6.00	8.00	DS055-U/N/F-05500-06300-06000800
56.00	64.00	59.50	6.00	8.00	DS055-U/N/F-05600-06400-06000800

ROYAL STANDARD SIZE LIST					
ØdN (f8)	ØDN (H11)	ØD1 (H11)	H1 (+0.20)	H2	ORDER CODE
60.00	68.00	63.50	6.00	8.00	DS055-U/N/F-06000-06800-06000800
63.00	71.00	66.50	6.00	8.00	DS055-U/N/F-06300-07100-06000800
65.00	73.00	68.50	6.00	8.00	DS055-U/N/F-06500-07300-06000800
68.00	76.00	71.50	6.00	8.00	DS055-U/N/F-06800-07600-06000800
70.00	78.00	73.50	6.00	8.00	DS055-U/N/F-07000-07800-06000800
75.00	83.00	78.50	6.00	8.00	DS055-U/N/F-07500-08300-06000800
80.00	88.00	83.50	6.00	8.00	DS055-U/N/F-08000-08800-06000800
85.00	93.00	88.50	6.00	8.00	DS055-U/N/F-08500-09300-06000800
90.00	98.00	93.50	6.00	8.00	DS055-U/N/F-09000-09800-06000800
95.00	103.00	98.50	6.00	8.00	DS055-U/N/F-09500-10300-06000800
100.00	108.00	103.50	6.00	8.00	DS055-U/N/F-10000-10800-06000800
105.00	117.00	110.00	8.20	11.10	DS055-U/N/F-10500-11700-08201110
110.00	122.00	115.00	8.20	11.10	DS055-U/N/F-11000-12200-08201110
115.00	127.00	120.00	8.20	11.10	DS055-U/N/F-11500-12700-08201110
120.00	132.00	125.00	8.20	11.10	DS055-U/N/F-12000-13200-08201110
125.00	137.00	130.00	8.20	11.10	DS055-U/N/F-12500-13700-08201110
130.00	142.00	135.00	8.20	11.10	DS055-U/N/F-13000-14200-08201110
135.00	147.00	140.00	8.20	11.10	DS055-U/N/F-13500-14700-08201110
140.00	152.00	145.00	8.20	11.10	DS055-U/N/F-14000-15200-08201110
145.00	157.00	150.00	8.20	11.10	DS055-U/N/F-14500-15700-08201110
150.00	162.00	155.00	8.20	11.10	DS055-U/N/F-15000-16200-08201110
160.00	172.00	165.00	8.20	11.10	DS055-U/N/F-16000-17200-08201110
170.00	182.00	175.00	8.20	11.10	DS055-U/N/F-17000-18200-08201110
180.00	192.00	185.00	8.20	11.10	DS055-U/N/F-18000-19200-08201110
190.00	202.00	195.00	8.20	11.10	DS055-U/N/F-19000-20200-08201110
200.00	212.00	205.00	8.20	11.10	DS055-U/N/F-20000-21200-08201110
210.00	225.00	217.00	9.50	12.50	DS055-N-21000-22500-09501250
220.00	235.00	227.00	9.50	12.50	DS055-N-22000-23500-09501250
225.00	240.00	233.00	9.50	12.50	DS055-N-22500-24000-09501250
230.00	245.00	237.00	9.50	12.50	DS055-N-23000-24500-09501250
240.00	255.00	247.00	9.50	12.50	DS055-N-24000-25500-09501250

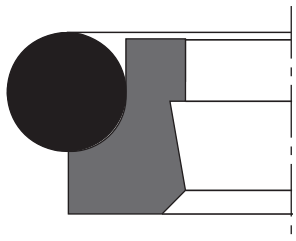
HOUSING DESIGN DS055



ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H11)	ØD1 (H11)	H1 (+0.20)	H2	ORDER CODE
250.00	265.00	257.00	9.50	12.50	DS055-N-25000-26500-09501250
255.00	270.00	262.00	9.50	12.50	DS055-N-25500-27000-09501250
260.00	275.00	267.00	9.50	12.50	DS055-N-26000-27500-09501250
265.00	280.00	272.00	9.50	12.50	DS055-N-26500-28000-09501250
270.00	285.00	277.00	9.50	12.50	DS055-N-27000-28500-09501250
280.00	295.00	287.00	9.50	12.50	DS055-N-28000-29500-09501250
300.00	315.00	307.00	9.50	12.50	DS055-N-30000-31500-09501250
320.00	335.00	327.00	9.50	12.50	DS055-N-32000-33500-09501250
350.00	365.00	357.00	9.50	12.50	DS055-N-35000-36500-09501250
360.00	375.00	367.00	9.50	12.50	DS055-N-36000-37500-09501250
380.00	395.00	387.00	9.50	12.50	DS055-N-38000-39500-09501250
400.00	415.00	407.00	9.50	12.50	DS055-N-40000-41500-09501250
410.00	425.00	417.00	9.50	12.50	DS055-N-41000-42500-09501250
450.00	465.00	457.00	9.50	12.50	DS055-N-45000-46500-09501250
520.00	535.00	527.00	9.50	12.50	DS055-N-52000-53500-09501250

DUST SEAL — DS066



Features:

Wiper seal profile DS066 consists of two piece seal set for wiping rod with one PTFE ring and one Energizer O-ring.

Composition:

SEAL PROFILE		DS066-N	DS066-F
Seal ring	Material	PTFE+ Bronze +MoS2	PTFE+ Bronze +MoS2
	Royal Code	TB40M	TB40M
O Ring	Material	Nitrile Butadiene rubber (NBR)	Fluoro Carbon Rubber (FPM)
	Royal Code	NBR70	FPM80

For compound detail please refer Royal seal materials. (Section III)

Properties:

- Low friction, free of stick slip
- High abrasion resistance
- Wide temperature Range
- Good thermal conductivity
- Low wear and high extrusion resistance
- Wide fluid application range
- Resistance against cold flow
- For short stroke and high frequency

Application:

- Mobile Hydraulics
- Injection molding machines
- Machine tools
- Material Handling Equipments.
- Standard cylinders

General Technical Data:

Seal profile	DS066-N	DS066-F
Temperature range	-20°C to +100°C	-20°C to + 200°C
Surface Speed(max.)	4 m/s.	4 m/s.

Minimum chamfer C:

DN - dN	Up to 12	12.1 to15	15.1 to 20	20.1 to 25	25.1 to 30	30.1 and above
C	4.0	5.0	6.0	7.5	8.0	10.0

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Fitting:

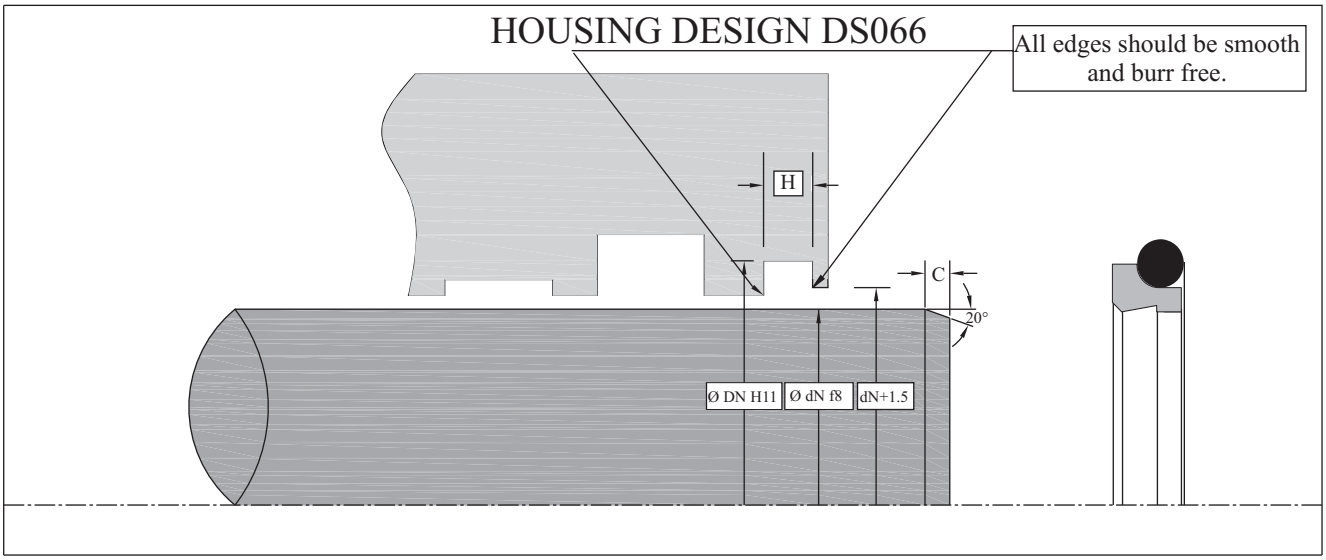
Snap in fitting through a kidney-shaped distortion. Careful fitting is prerequisite for perfect functioning of the wiper.

Ordering format:

Please mention ROYAL order code in your order.
Example: Size 12 X 18.8 X 5
For Temp range: -20°C to +100°C. (O ring material is NBR70) Order code is DS066-N-01200-01880-0500
For Temp range: -20°C to +200°C. (O ring material is FPM80) Order code is DS066-F-01200-01880-0500

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

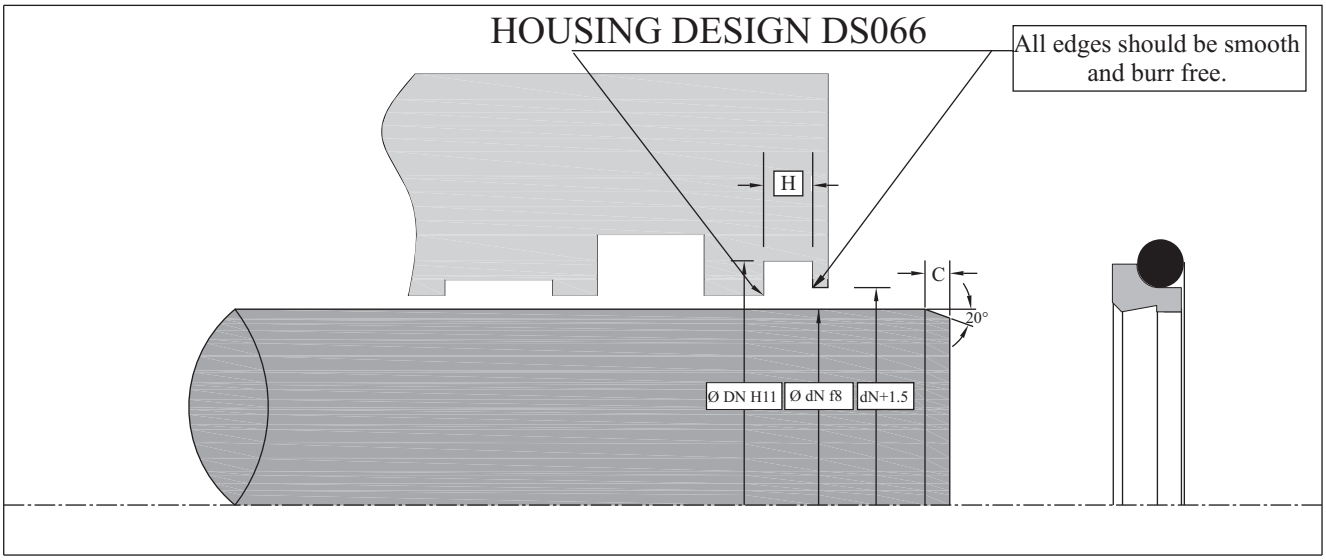
HOUSING DESIGN DS066



ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H11)	H (+0.20)	O ring size	ORDER CODE For temp. range -20°C to +100°C use DS066-N -20°C to +200°C use DS066-F
12.00	18.80	5.00	13.94 X 2.62	DS066-N/F-01200-01880-0500
14.00	20.80	5.00	15.54 X 2.62	DS066-N/F-01400-02080-0500
15.00	21.80	5.00	17.12 X 2.62	DS066-N/F-01500-02180-0500
16.00	22.80	5.00	17.12 X 2.62	DS066-N/F-01600-02280-0500
18.00	24.80	5.00	20.29 X 2.62	DS066-N/F-01800-02480-0500
20.00	26.80	5.00	21.89 X 2.62	DS066-N/F-02000-02680-0500
22.00	28.80	5.00	23.47 X 2.62	DS066-N/F-02200-02880-0500
25.00	31.80	5.00	26.64 X 2.62	DS066-N/F-02500-03180-0500
28.00	34.80	5.00	29.82 X 2.62	DS066-N/F-02800-03480-0500
30.00	36.80	5.00	31.42 X 2.62	DS066-N/F-03000-03680-0500
32.00	38.80	5.00	34.59 X 2.62	DS066-N/F-03200-03880-0500
35.00	41.80	5.00	36.17 X 2.62	DS066-N/F-03500-04180-0500
36.00	42.80	5.00	37.77 X 2.62	DS066-N/F-03600-04280-0500
38.00	44.80	5.00	39.34 X 2.62	DS066-N/F-03800-04480-0500
40.00	46.80	5.00	42.42 X 2.62	DS066-N/F-04000-04680-0500
45.00	51.80	5.00	47.29 X 2.62	DS066-N/F-04500-05180-0500
50.00	56.80	5.00	52.07 X 2.62	DS066-N/F-05000-05680-0500
55.00	61.80	5.00	56.82 X 2.62	DS066-N/F-05500-06180-0500
56.00	62.80	5.00	58.42 X 2.62	DS066-N/F-05600-06280-0500

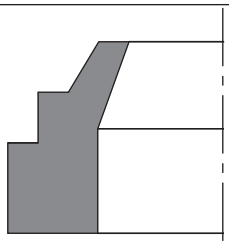
HOUSING DESIGN DS066



ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H11)	H1 (+0.20)	O ring size	ORDER CODE For temp. range -20°C to +100°C use DS066-N -20°C to +200°C use DS066-F
60.00	66.80	5.00	61.60 X 2.62	DS066-N/F-06000-06680-0500
63.00	69.80	5.00	64.77 X 2.62	DS066-N/F-06300-06980-0500
65.00	73.80	6.00	66.27 X 3.53	DS066-N/F-06500-07380-0600
70.00	78.80	6.00	72.62 X 3.53	DS066-N/F-07000-07880-0600
75.00	83.80	6.00	75.79 X 3.53	DS066-N/F-07500-08380-0600
80.00	88.80	6.00	82.14 X 3.53	DS066-N/F-08000-08880-0600
85.00	93.80	6.00	88.49 X 3.53	DS066-N/F-08500-09380-0600
90.00	98.80	6.00	91.67 X 3.53	DS066-N/F-09000-09880-0600
95.00	103.80	6.00	98.02 X 3.53	DS066-N/F-09500-10380-0600
100.00	108.80	6.00	101.19 X 3.53	DS066-N/F-10000-10880-0600
105.00	113.80	6.00	107.54 X 3.53	DS066-N/F-10500-11380-0600
110.00	118.80	6.00	110.72 X 3.53	DS066-N/F-11000-11880-0600
115.00	123.80	6.00	117.07 X 3.53	DS066-N/F-11500-12380-0600
120.00	128.80	6.00	120.24 X 3.53	DS066-N/F-12000-12880-0600
125.00	133.80	6.00	126.59 X 3.53	DS066-N/F-12500-13380-0600
130.00	138.80	6.00	132.94 X 3.53	DS066-N/F-13000-13880-0600
135.00	143.80	6.00	136.12 X 3.53	DS066-N/F-13500-14380-0600
140.00	148.80	6.00	142.47 X 3.53	DS066-N/F-14000-14880-0600
150.00	158.80	6.00	151.99 X 3.53	DS066-N/F-15000-15880-0600

ØdN (f8)	ØDN (H11)	H1 (+0.20)	O ring size	ORDER CODE For temp. range -20°C to +100°C use DS066-N -20°C to +200°C use DS066-F
160.00	168.80	6.00	158.34 X 3.53	DS066-N/F-16000-16880-0600
170.00	178.80	6.00	171.04 X 3.53	DS066-N/F-17000-17880-0600
180.00	188.80	6.00	183.74 X 3.53	DS066-N/F-18000-18880-0600
190.00	198.80	6.00	190.09 X 3.53	DS066-N/F-19000-19880-0600
200.00	208.80	6.00	202.79 X 3.53	DS066-N/F-20000-20880-0600
210.00	218.80	6.00	209.14 X 3.53	DS066-N/F-21000-21880-0600
220.00	228.80	6.00	221.84 X 3.53	DS066-N/F-22000-22880-0600
225.00	233.80	6.00	228.19 X 3.53	DS066-N/F-22500-23380-0600
230.00	238.80	6.00	228.19 X 3.53	DS066-N/F-23000-23880-0600
240.00	248.80	6.00	240.89 X 3.53	DS066-N/F-24000-24880-0600
250.00	258.80	6.00	253.59 X 3.53	DS066-N/F-25000-25880-0600
260.00	272.20	8.40	266.07 X 5.33	DS066-N/F-26000-27220-0840
270.00	282.20	8.40	278.77 X 5.33	DS066-N/F-27000-28220-0840
280.00	292.20	8.40	278.77 X 5.33	DS066-N/F-28000-29220-0840
300.00	312.20	8.40	304.17 X 5.33	DS066-N/F-30000-31220-0840
320.00	332.20	8.40	329.57 X 5.33	DS066-N/F-32000-33220-0840
350.00	362.20	8.40	354.97 X 5.33	DS066-N/F-35000-36220-0840
360.00	372.20	8.40	354.97 X 5.33	DS066-N/F-36000-37220-0840
400.00	412.20	8.40	405.26 X 5.33	DS066-N/F-40000-41220-0840

DUST SEAL	DS077	

Features:
Wiper seal DS-077 is a single acting rod wiper for integral housing.

Composition:

Seal type	DS077-U
Material	Polyurethane
Royal Code	PUAB94

For compound detail please refer Royal seal materials. (Section III)

Properties:
- Easy snap in installation
- Very good wiping action
- Wiper can be used for a broad temperature range
- No pressure build-up between seal and wiper

Application:
- Presses, fork lift & material handling equipments
- Injection molding machines
- Standard cylinders

General Technical Data:

Seal profile	DS077-U
Temperature range	-20°C to +100°C
Linear velocity(max.)	1 m/s.

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Minimum chamfer C:

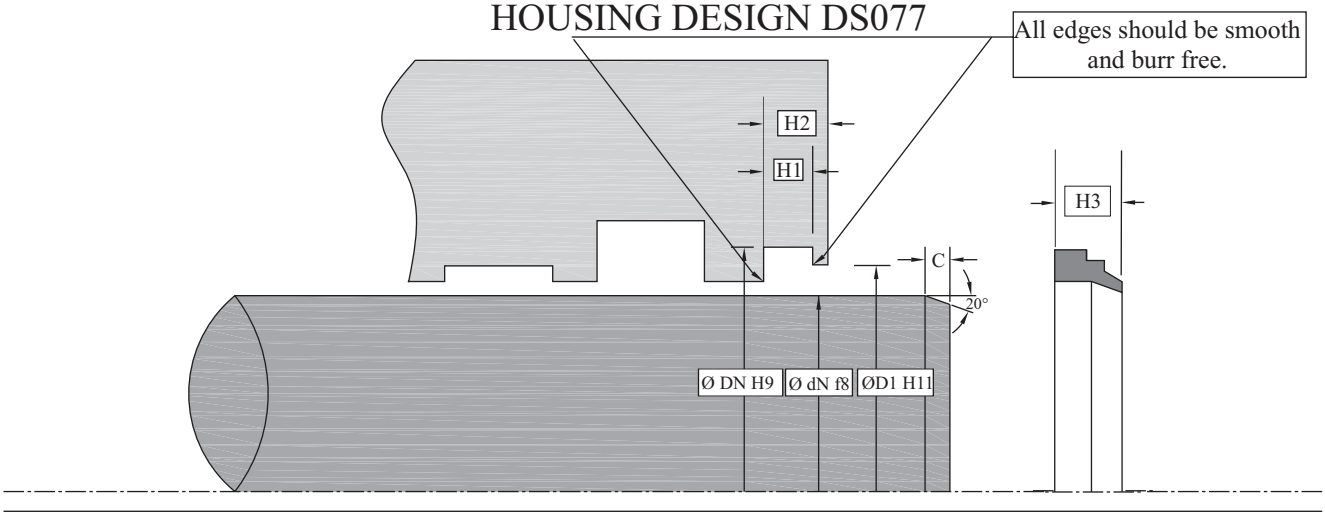
DN - dN	Up to 12	12.1 to15	15.1 to 20	20.1 to 25	25.1 to 30	30.1 and above
C	4.0	5.0	6.0	7.5	8.0	10.0

Fitting:
Snap in fitting through a kidney-shaped distortion. Careful fitting is prerequisite for perfect functioning of the wiper.

Ordering format:
Please mention ROYAL order code in your order.
Example: Size 12 X 20 X 4 / 5 / 7
The Order code is DS077-U-01200-02000-4005000700

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

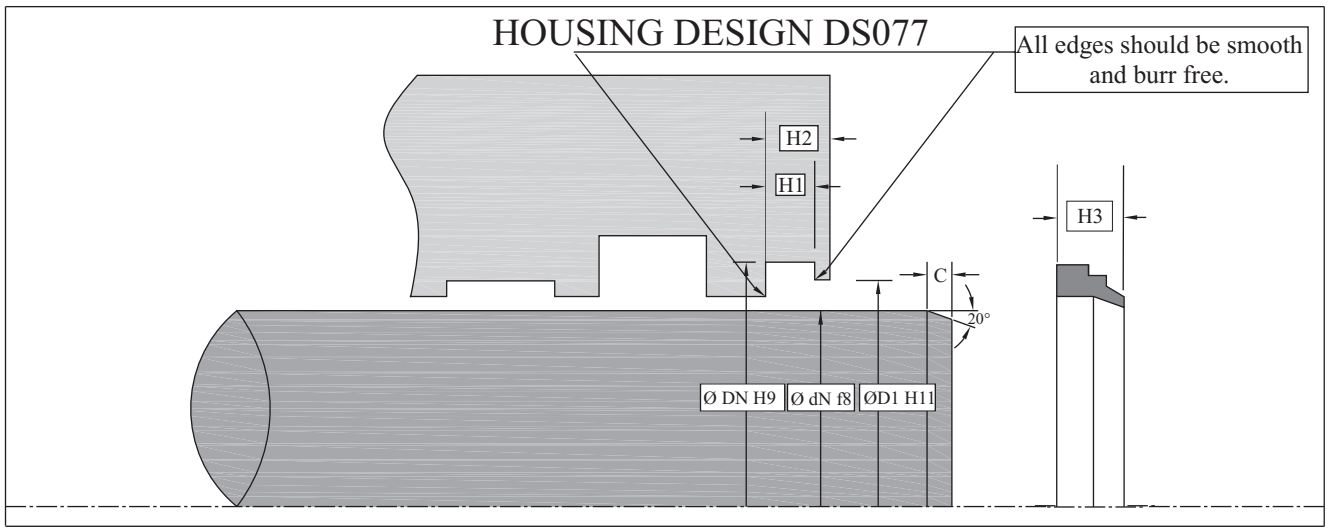
HOUSING DESIGN DS077



ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H11)	ØD1 (H11)	H1 (+0.20)	H2	H3	ORDER CODE
12.00	20.00	18.00	4.00	5.00	7.00	DS077-U-01200-02000-4005000700
14.00	22.00	20.00	4.00	5.00	7.00	DS077-U-01400-02200-4005000700
15.00	23.00	21.00	4.00	5.00	7.00	DS077-U-01500-02300-4005000700
16.00	24.00	22.00	4.00	5.00	7.00	DS077-U-01600-02400-4005000700
18.00	26.00	24.00	4.00	5.00	7.00	DS077-U-01800-02600-4005000700
20.00	28.00	26.00	4.00	5.00	7.00	DS077-U-02000-02800-4005000700
22.00	30.00	28.00	4.00	5.00	7.00	DS077-U-02200-03000-4005000700
25.00	33.00	31.00	4.00	5.00	7.00	DS077-U-02500-03300-4005000700
28.00	36.00	34.00	4.00	5.00	7.00	DS077-U-02800-03600-4005000700
30.00	38.00	36.00	4.00	5.00	7.00	DS077-U-03000-03800-4005000700
32.00	40.00	38.00	4.00	5.00	7.00	DS077-U-03200-04000-4005000700
33.00	41.00	39.00	4.00	5.00	7.00	DS077-U-03300-04100-4005000700
35.00	43.00	41.00	4.00	5.00	7.00	DS077-U-03500-04300-4005000700
36.00	44.00	42.00	4.00	5.00	7.00	DS077-U-03600-04400-4005000700
38.00	46.00	44.00	4.00	5.00	7.00	DS077-U-03800-04600-4005000700
40.00	48.00	46.00	4.00	5.00	7.00	DS077-U-04000-04800-4005000700
42.00	50.00	48.00	4.00	5.00	7.00	DS077-U-04200-05000-4005000700
45.00	53.00	51.00	4.00	5.00	7.00	DS077-U-04500-05300-4005000700
50.00	58.00	56.00	4.00	5.00	7.00	DS077-U-05000-05800-4005000700
55.00	63.00	61.00	4.00	5.00	7.00	DS077-U-05500-06300-4005000700
56.00	64.00	62.00	4.00	5.00	7.00	DS077-U-05600-06400-4005000700

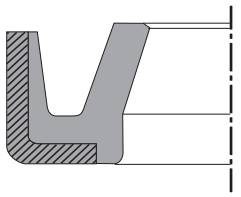
HOUSING DESIGN DS077



ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H11)	ØD1 (H11)	H1 (+0.20)	H2	H3	ORDER CODE
60.00	68.00	66.00	4.00	5.00	7.00	DS077-U-06000-06800-4005000700
63.00	71.00	69.00	4.00	5.00	7.00	DS077-U-06300-07100-4005000700
65.00	73.00	71.00	4.00	5.00	7.00	DS077-U-06500-07300-4005000700
70.00	78.00	76.00	4.00	5.00	7.00	DS077-U-07000-07800-4005000700
72.00	80.00	78.00	4.00	5.00	7.00	DS077-U-07200-08000-4005000700
75.00	83.00	81.00	4.00	5.00	7.00	DS077-U-07500-08300-4005000700
80.00	88.00	86.00	4.00	5.00	7.00	DS077-U-08000-08800-4005000700
82.00	90.00	88.00	4.00	5.00	7.00	DS077-U-08200-09000-4005000700
85.00	93.00	91.00	4.00	5.00	7.00	DS077-U-08500-09300-4005000700
90.00	98.00	96.00	4.00	5.00	7.00	DS077-U-09000-09800-4005000700
95.00	103.00	101.00	4.00	5.00	7.00	DS077-U-09500-10300-4005000700
100.00	108.00	106.00	4.00	5.00	7.00	DS077-U-10000-10800-4005000700
105.00	117.00	114.00	5.50	7.00	10.00	DS077-U-10500-11700-5507001000
110.00	122.00	119.00	5.50	7.00	10.00	DS077-U-11000-12200-5507001000
115.00	127.00	124.00	5.50	7.00	10.00	DS077-U-11500-12700-5507001000
120.00	132.00	129.00	5.50	7.00	10.00	DS077-U-12000-13200-5507001000
125.00	137.00	134.00	5.50	7.00	10.00	DS077-U-12500-13700-5507001000
130.00	142.00	139.00	5.50	7.00	10.00	DS077-U-13000-14200-5507001000
135.00	147.00	144.00	5.50	7.00	10.00	DS077-U-13500-14700-5507001000
140.00	152.00	149.00	5.50	7.00	10.00	DS077-U-14000-15200-5507001000
150.00	162.00	159.00	5.50	7.00	10.00	DS077-U-15000-16200-5507001000

ROYAL STANDARD SIZE LIST						
ØdN (f8)	ØDN (H11)	ØD1 (H11)	H1 (+0.20)	H2	H3	ORDER CODE
160.00	172.00	169.00	5.50	7.00	10.00	DS077-U-16000-17200-5507001000
170.00	182.00	179.00	5.50	7.00	10.00	DS077-U-17000-18200-5507001000
180.00	192.00	189.00	5.50	7.00	10.00	DS077-U-18000-19200-5507001000
190.00	202.00	199.00	5.50	7.00	10.00	DS077-U-19000-20200-5507001000
200.00	212.00	209.00	5.50	7.00	10.00	DS077-U-20000-21200-5507001000

DUST SEAL	DS088	

Features:
Wiper seal DS-088 is a single acting rod wiper with metal casing for open housing.

Composition:

Seal profile type	DS088-U
Material	Polyurethane
Royal Code	PUAB94
Metal casing	Mild steel

For compound detail please refer Royal seal materials. (Section III)

Properties:
- Easy press fit installation in open housing
- Very good wiping action
- Wiper can be used for a broad temperature range with different materials

Application:
- presses, fork lift & material handling equipments
- injection moulding machines
- Standard cylinders
- mining & construction machineries

General Technical Data:

Seal profile	DS088-U
Temperature range	-20°C to +100°C
Linear velocity(max.)	0.5 m/s.

Minimum chamfer C:

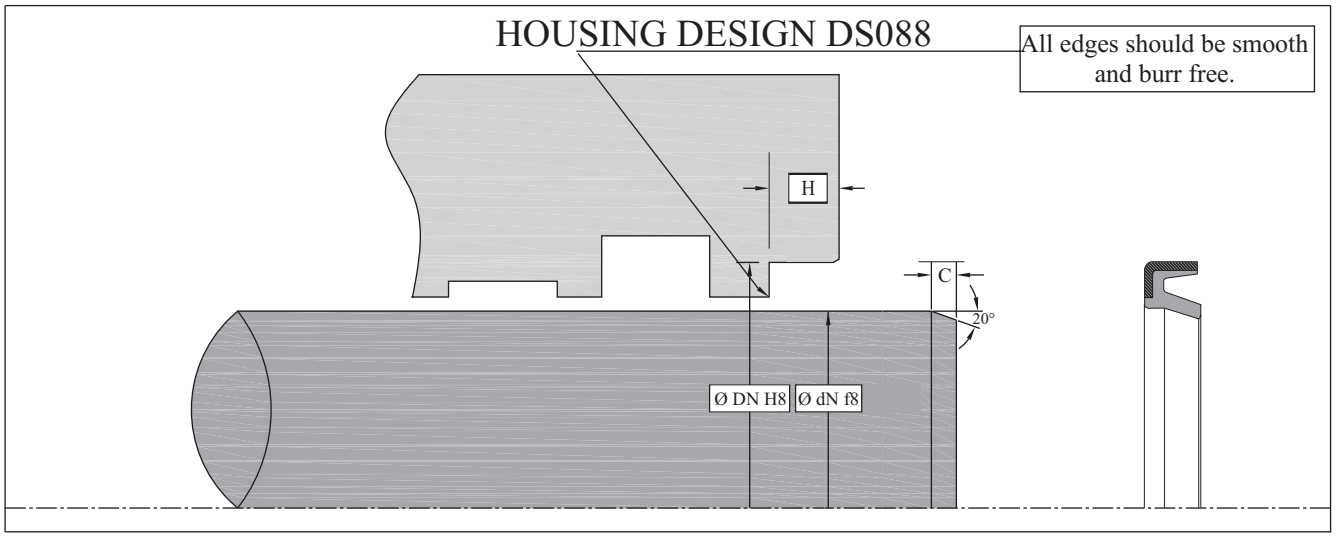
DN - dN	Up to 12	12.1 to15	15.1 to 20	20.1 to 25	25.1 to 30	30.1 and above
C	4.0	5.0	6.0	7.5	8.0	10.0

Fitting:
Wiper seal DS-088 are pressed in to open housings with locking circlip. Careful fitting is prerequisite for perfect functioning of the wiper seal.

Ordering format:
Please mention ROYAL order code in your order.
Example: Size 25 X 35 X 4
For Temp range: -20°C to +100°C Material-Polyurethane The Order code is DS088-U-02500-03500-0400

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

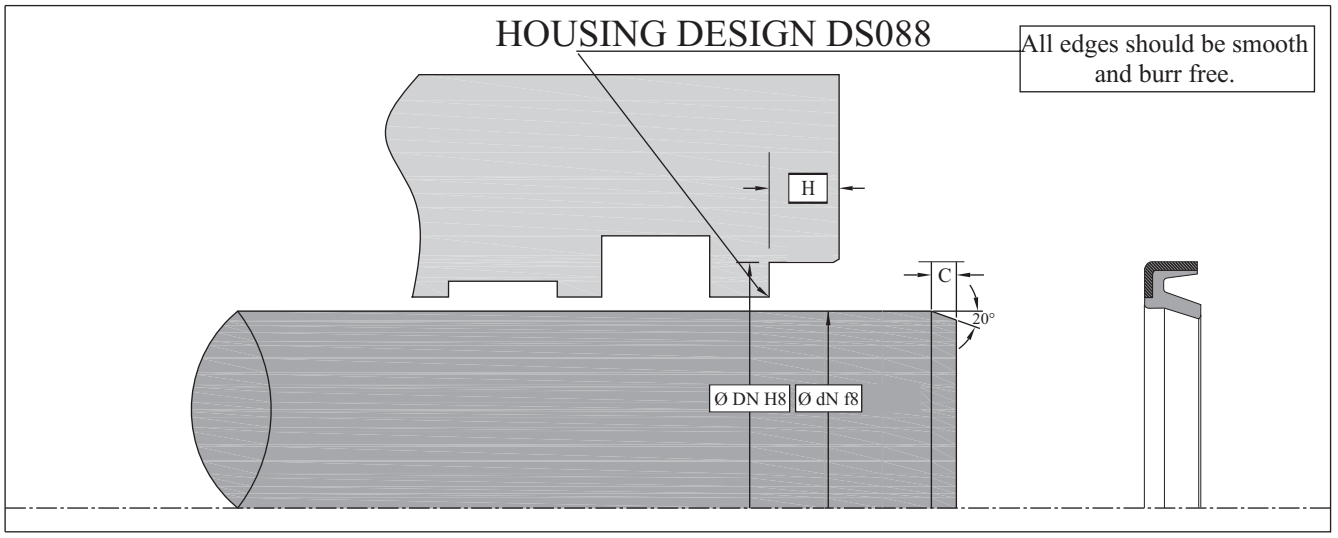
HOUSING DESIGN DS088



ROYAL STANDARD SIZE LIST

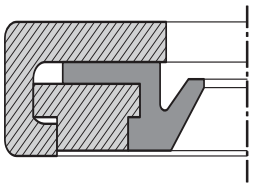
ØdN (f8)	ØDN (H11)	H (+0.20)	ORDER CODE
25.00	35.00	4.00	DS088-U-02500-03500-0400
35.00	45.00	4.00	DS088-U-03500-04500-0400
40.00	50.00	5.00	DS088-U-04000-05000-0500
45.00	55.00	4.00	DS088-U-04500-05500-0400
45.00	55.00	5.00	DS088-U-04500-05500-0500
45.00	55.00	6.00	DS088-U-04500-05500-0600
50.00	60.00	5.00	DS088-U-05000-06000-0500
60.00	70.00	5.00	DS088-U-06000-07000-0500
60.00	75.00	8.00	DS088-U-06000-07500-0800
65.00	80.00	8.00	DS088-U-06500-08000-0800
70.00	80.00	7.00	DS088-U-07000-08000-0700
70.00	83.50	6.00	DS088-U-07000-08350-0600
70.00	85.00	8.00	DS088-U-07000-08500-0800
75.00	90.00	8.00	DS088-U-07500-09000-0800
85.00	100.00	8.00	DS088-U-08500-10000-0800
95.00	110.00	8.00	DS088-U-09500-11000-0800
110.00	125.00	8.00	DS088-U-11000-12500-0800
115.00	135.00	6.00	DS088-U-11500-13500-0600
120.00	135.00	8.00	DS088-U-12000-13500-0800
125.00	140.00	7.00	DS088-U-12500-14000-0700
140.00	155.00	8.00	DS088-U-14000-15500-0800

HOUSING DESIGN DS088



ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H11)	H1+0.2	ORDER CODE
145.00	160.00	8.00	DS088-U-14500-16000-0800
IMPERIAL SIZE			
1.375	1.875	0.315	DS088-U-IN-1375-1875-0315
2.000	2.500	0.312	DS088-U-IN-2000-2500-0312
2.000	2.500	0.315	DS088-U-IN-2000-2500-0315
2.250	2.750	0.132	DS088-U-IN-2250-2750-0132
2.500	3.000	0.312	DS088-U-IN-2500-3000-0312
2.500	3.000	0.375	DS088-U-IN-2500-3000-0375

DUST SEAL	DS099	

Features:
Wiper seal DS-099 is a single acting pin dust wiper with metal casing for open housing.

Composition:

Seal profile type	DS099-U
Material	Polyurethane
Royal Code	PUAB92
Metal casing	Mild steel

For compound detail please refer Royal seal materials. (Section III)

Properties:
- Easy press fit installation in open housing
- Very good Dust protection

Application:
- Outer pin
- Gear box

General Technical Data:

Seal profile	DS099-U
Temperature range	-20°C to +100°C
Linear velocity(max.)	0.5 m/s.

Minimum chamfer C:

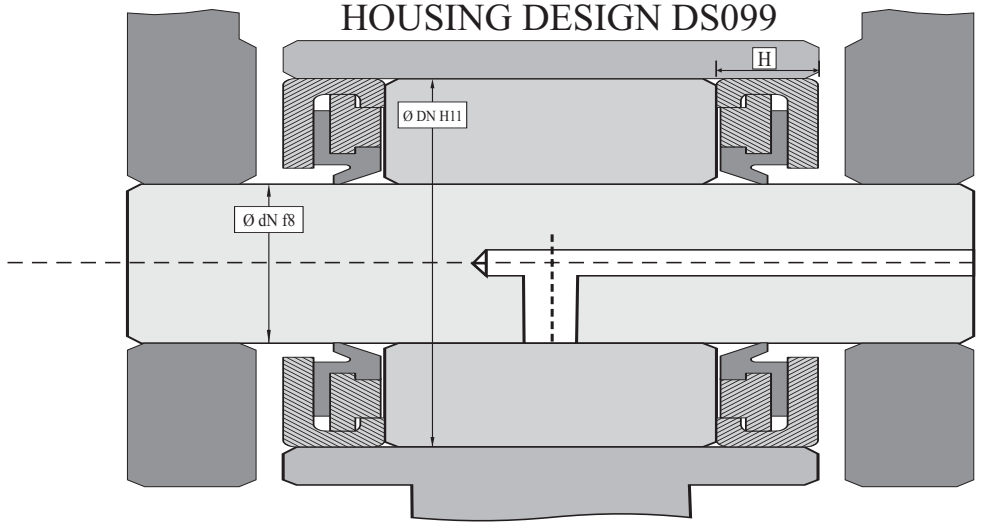
DN - dN	Up to 12	12.1 to15	15.1 to 20	20.1 to 25	25.1 to 30	30.1 and above
C	4.0	5.0	6.0	7.5	8.0	10.0

Fitting:
Wiper seal DS-099 are pressed in to axially-open housings. Careful fitting is prerequisite for perfect functioning of the wiper seal.

Ordering format:
Please mention ROYAL order code in your order.
Example: Size 25 X 35 X 4
For Temp range: -20°C to +100°C Material - Polyurethane - The Order code is DS099-U-02500-03500-0400

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

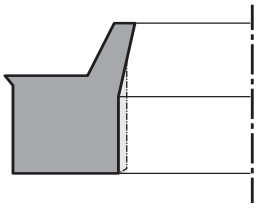
HOUSING DESIGN DS099



All edges should be smooth and burr free.

ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H11)	H (+0.20)	ORDER CODE
25.00	35.00	4.00	DS099-U-02500-03500-0400
30.00	40.00	4.00	DS099-U-03000-04000-0400
35.00	45.00	4.00	DS099-U-03500-04500-0400
40.00	50.00	4.00	DS099-U-04000-05000-0400
50.00	60.00	4.00	DS099-U-05000-06000-0400
55.00	70.00	6.00	DS099-U-05500-07000-0600
60.00	75.00	4.00	DS099-U-06000-07500-0400
60.00	75.00	6.00	DS099-U-06000-07500-0600
65.00	80.00	4.00	DS099-U-06500-08000-0400
70.00	85.00	4.00	DS099-U-07000-08500-0400
75.00	90.00	4.00	DS099-U-07500-09000-0400
80.00	95.00	4.00	DS099-U-08000-09500-0400
85.00	100.00	4.00	DS099-U-08500-10000-0400
90.00	105.00	4.00	DS099-U-09000-10500-0400
95.00	110.00	4.00	DS099-U-09500-11000-0400
100.00	115.00	4.00	DS099-U-10000-11500-0400

DUST SEAL	DS 110	

Features:
Wiper seal DS-110 is a single acting rod wiper for integral housing with extermal dust protecting lip

Composition:

Seal type	DS110-U
Material:	Polyurethane
Royal Code	PUAB98

For compound detail please refer Royal seal materials. (Section III)

Properties:
- Easy snap in installation
- Very good wiping action
- Wiper can be used for a broad temperature range
- No pressure build-up between seal and wiper

Application:
- Presses, fork lift & material handling equipments
- Injection molding machines
- Standard cylinders

General Technical Data:

Seal profile	DS110-U
Temperature range	-20°C to +110°C
Linear velocity	0.5 m/s.

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Minimum chamfer C:

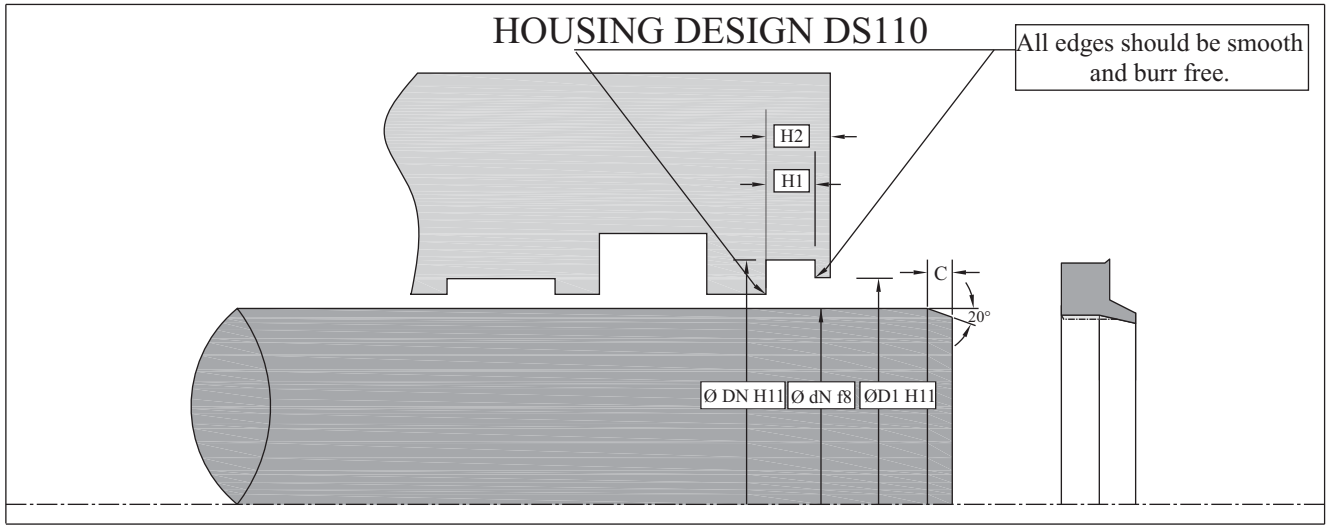
DN - dN	Up to 12	12.1 to15	15.1 to 20	20.1 to 25	25.1 to 30	30.1 and above
C	4.0	5.0	6.0	7.5	8.0	10.0

Fitting:
Snap in fitting through a kidney-shaped distortion. Careful fitting is prerequisite for perfect functioning of the wiper.

Ordering format:
Please mention ROYAL order code in your order.
Example: Size 50 X 58 X 5 / 8
The order code is DS110-U-05000-05800-05000800

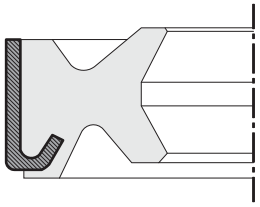
Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN DS110



ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H11)	ØD1 (H11)	H1 (+0.20)	H2	ORDER CODE
50.00	58.00	55.50	5.00	8.00	DS110-U-05000-05800-05000800
56.00	66.00	63.00	6.30	10.00	DS110-U-05600-06600-06301000
80.00	90.00	87.00	6.30	10.00	DS110-U-08000-09000-06301000
110.00	125.00	120.00	9.50	14.00	DS110-U-11000-12500-09501400
125.00	140.00	135.00	9.50	14.00	DS110-U-12500-14000-09501400

DUST SEAL	DS121	

Features:
Wiper seal DS-121 is a double acting rod wiper with metal casing for open housing.

Composition:

Seal profile type	DS121-U
Material	Polyurethane
Royal Code	PUAB94
Metal casing	Mild steel

For compound detail please refer Royal seal materials. (Section III)

Properties:
- Easy press fit installation in open housing with locking circlip
- Very good wiping action
- Wiper can be used for a broad temperature range with different materials
- Sealing lips directed inwards gives both side wiping effect

Application:
- presses, fork lift & material handling equipments
- injection moulding machines
- Standard cylinders
- mining & construction machineries

General Technical Data:

Seal profile	DS121-U
Temperature range	-20°C to +100°C
Linear velocity(max.)	0.5 m/s.

Minimum chamfer C:

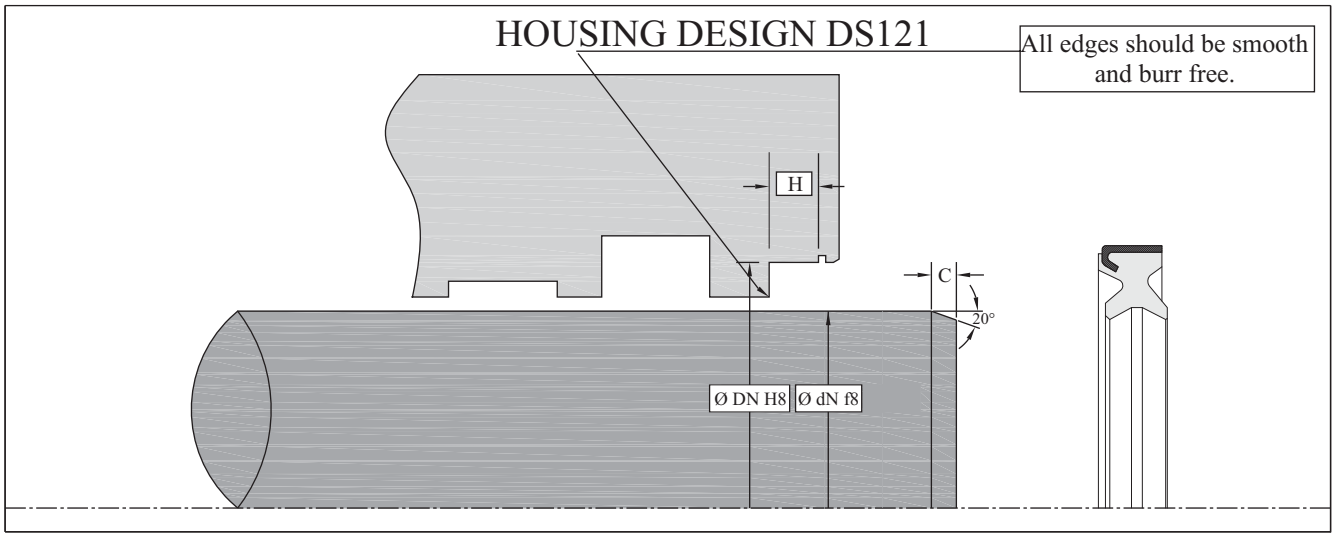
DN - dN	Up to 12	12.1 to15	15.1 to 20	20.1 to 25	25.1 to 30	30.1 and above
C	4.0	5.0	6.0	7.5	8.0	10.0

Fitting:
Wiper seal DS-121 are pressed in to open housings with locking circlip. Careful fitting is prerequisite for perfect functioning of the wiper seal.

Ordering format:
Please mention ROYAL order code in your order.
Example: Size 38.10 X 57.15 X 10.41
The Order code is DS121-U-IN01500-02250-0410

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

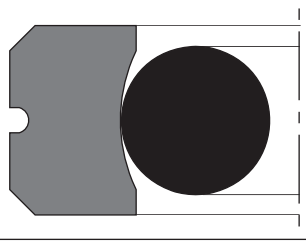
HOUSING DESIGN DS121



ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H11)	H (+0.20)	ORDER CODE
38.10	57.15	10.41	DS121-U-IN-01500-02250-0410
41.28	60.33	10.41	DS121-U-IN-01625-02375-0410
44.20	61.98	10.41	DS121-U-IN-01740-02440-0410
44.45	63.50	10.41	DS121-U-IN-01750-02500-0410
47.63	65.07	10.41	DS121-U-IN-01875-02562-0410
47.63	66.68	10.41	DS121-U-IN-01875-02625-0410
50.80	68.25	10.41	DS121-U-IN-02000-02687-0410
50.80	69.85	10.41	DS121-U-IN-02000-02750-0410
53.98	73.03	10.41	DS121-U-IN-02125-02875-0410
57.15	76.20	10.41	DS121-U-IN-02250-03000-0410
60.33	79.38	10.41	DS121-U-IN-02375-03125-0410
63.50	82.55	10.41	DS121-U-IN-02500-03250-0410
66.68	85.73	10.41	DS121-U-IN-02625-03375-0410
69.85	95.25	13.72	DS121-U-IN-02750-03750-0540
76.20	101.60	13.72	DS121-U-IN-03250-04250-0540
88.90	114.30	13.72	DS121-U-IN-03500-04500-0540
95.25	120.65	13.72	DS121-U-IN-03750-04750-0540
101.60	127.00	13.72	DS121-U-IN-04000-05000-0540
107.95	133.35	13.72	DS121-U-IN-04250-05250-0540
114.30	139.70	13.72	DS121-U-IN-04500-05500-0540

ROYAL STANDARD SIZE LIST			
ØdN (f8)	**ØDN (H11)**	**H (+0.20)**	**ORDER CODE**
120.65	146.05	13.72	DS121-U-IN-04750-5750-0540
127.00	152.40	13.72	DS121-U-IN-05000-6000-0540
133.35	158.75	13.72	DS121-U-IN-05250-6250-0540
139.70	165.10	13.72	DS121-U-IN-05500-6500-0540
146.05	171.45	13.72	DS121-U-IN-05750-6750-0540
152.40	177.80	13.72	DS121-U-IN-06000-7000-0540
158.75	184.15	13.72	DS121-U-IN-06250-7250-0540
222.25	247.65	13.72	DS121-U-IN-08750-9750-0540

ROTARY SEAL	ROP001	

Features:

Rotary piston seal profile ROP001 consists of two piece seal set for sealing bore with one seal ring and one energizer O-ring.

Composition:

SEAL PROFILE		ROP001-N	ROP001-F
Seal ring	Material	PTFE+ Bronze +MoS2	PTFE+ Bronze +MoS2
	Royal Code	TB40M	TB40M
O Ring	Material	Nitrile Butadiene rubber (NBR)	Fluoro Carbon Rubber (FPM)
	Royal Code	NBR70	FPM80

For compound detail please refer Royal seal materials. (Section III)

Properties:

- Low friction, free of stick slip
- High abrasion resistance
- Wide temperature Range
- Good thermal conductivity
- Low wear and high extrusion resistance
- Wide fluid application range
- Resistance against cold flow
- Minimum installation space required

Application:

- Rotary joints
- Mobile Hydraulics
- Machine tools
- Material Handling Equipments
- Control and regulating apparatus

General Technical Data :

Seal Profile	ROP001-N	ROP001-F
Temperature range	-20°C to +100°C	-20 °C to +200 °C
Working Pressure(max.)	400 bar	400 bar
Surface Speed	4 m/s	4 m/s

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Extrusion Gap dimension:

The gap dimension on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the minimum clearance Ø and max bore Ø.

Profile dimension	Maximum permissible gap (S)			
H	160 bar	260 bar	320 bar	400 bar
2,2	0,35	0,3	-	-
3,2	0,4	0,35	-	-
4,2	0,5	0,4	0,3	-
6,3	0,6	0,5	0,45	0,45
8,1	0,7	0,6	0,55	0,5

Installation:

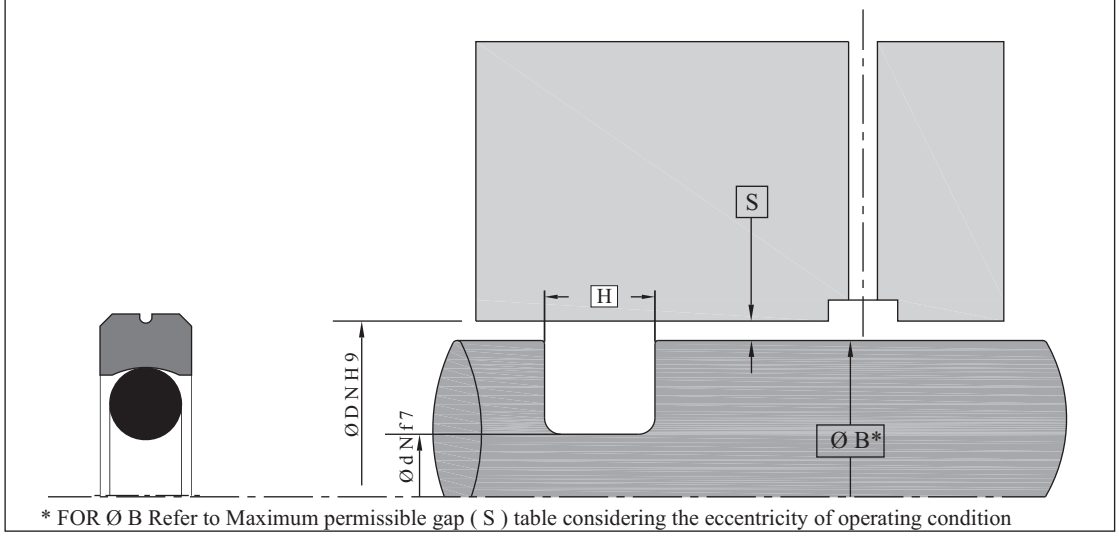
ROP001 rotary piston seals can be fitted on single-piece pistons. First insert the rubber 'O' Ring into the housing carefully to avoid twisting of O Ring Chord. The PTFE ring is then carefully pushed over the shaft into place above the O Ring. An assembly tool eases installation of small, PTFE rings. The PTFE seal normally returns to its original size after installation. It is advisable to calibrate the PTFE ring by mounting it on a steel tube (duly machined and polished), having inner diameter equal to housing inner diameter. Immediately after calibration, the shaft should be installed to the housing body

Ordering format:

Please mention ROYAL order code in your order.
Example: Size 16 X 11.1 X 2.2
For Temp range: -20°C to +100°C. (O ring material is NBR70) Order code is ROP001-N-01600-01110-0220
For Temp range: -20°C to +200°C. (O ring material is FPM80) Order code is ROP001-F-01600-01110-0220

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

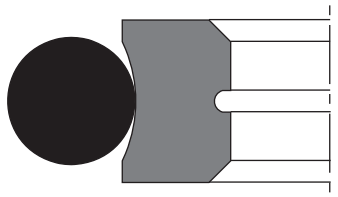
HOUSING DESIGN ROP001



* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

ØDN (H9)	ØdN (f7)	H1 (+0.20)	O ring size	ORDER CODE For temp. range -20°C to +100°C use ROP001-N -20°C to +200°C use ROP001-F
16.00	11.10	2.20	10.82 X 1.78	ROP001-N/F-01600-01110-0220
20.00	15.10	2.20	14.00 X 1.78	ROP001-N/F-02000-01510-0220
25.00	20.10	2.20	18.77 X 1.78	ROP001-N/F-02500-02010-0220
32.00	27.10	2.20	26.70 X 1.78	ROP001-N/F-03200-02710-0220
40.00	32.50	3.20	31.42 X 2.62	ROP001-N/F-04000-03250-0320
50.00	42.50	3.20	40.94 X 2.62	ROP001-N/F-05000-04250-0320
63.00	55.50	3.20	53.64 X 2.62	ROP001-N/F-06300-05550-0320
80.00	69.00	4.20	66.27 X 3.53	ROP001-N/F-08000-06900-0420
100.00	89.00	4.20	85.32 X 3.53	ROP001-N/F-10000-08900-0420
125.00	114.00	4.20	110.72 X 3.53	ROP001-N/F-12500-11400-0420
160.00	144.50	6.30	142.24 X 5.33	ROP001-N/F-16000-14450-0630
200.00	184.50	6.30	183.52 X 5.33	ROP001-N/F-20000-18450-0630
250.00	234.50	6.30	234.32 X 5.33	ROP001-N/F-25000-23450-0630

ROTARY SEAL	ROR002	

Features:

Rotary seal profile ROR002 consists of two piece seal set for sealing shaft with one seal ring and one energizer O-ring.

Composition:

SEAL PROFILE		ROR002-N	ROR002-F
Seal ring	Material	PTFE+ Bronze +MoS2	PTFE+ Bronze +MoS2
	Royal Code	TB40M	TB40M
O Ring	Material	Nitrile Butadiene rubber (NBR)	Fluoro Carbon Rubber (FPM)
	Royal Code	NBR70	FPM80

For compound detail please refer Royal seal materials.(Section III)

Properties:

- Low friction , free of stick slip
- High abrasion resistance.
- Wide temperature Range.
- Good thermal conductivity.
- Low wear and high extrusion resistance.
- Wide fluid application range.
- Resistance against cold flow.

Application:

- Mobile Hydraulics
- Injection molding machines
- Machine tools
- Material Handling Equipments.

General Technical Data:

Seal Profile	ROR002-N	ROR002-F
Temperature range	-20°C to +100°C	-20°C to +200 °C
Working Pressure(max.)	400 bar	400 bar
Surface Speed(max.)	4 m/s	4 m/s

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Extrusion gap dimension:

The gap dimension on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the maximum clearance Ø and min rod Ø.

Profile dimension	Maximum permissible gap (S)			
H	160 bar	260 bar	320 bar	400 bar
2,2	0,35	0,3	-	-
3,2	0,4	0,35	-	-
4,2	0,5	0,4	0,3	-
6,3	0,6	0,5	0,4	0,3
8,1	0,7	0,6	0,5	0,4

Installation:

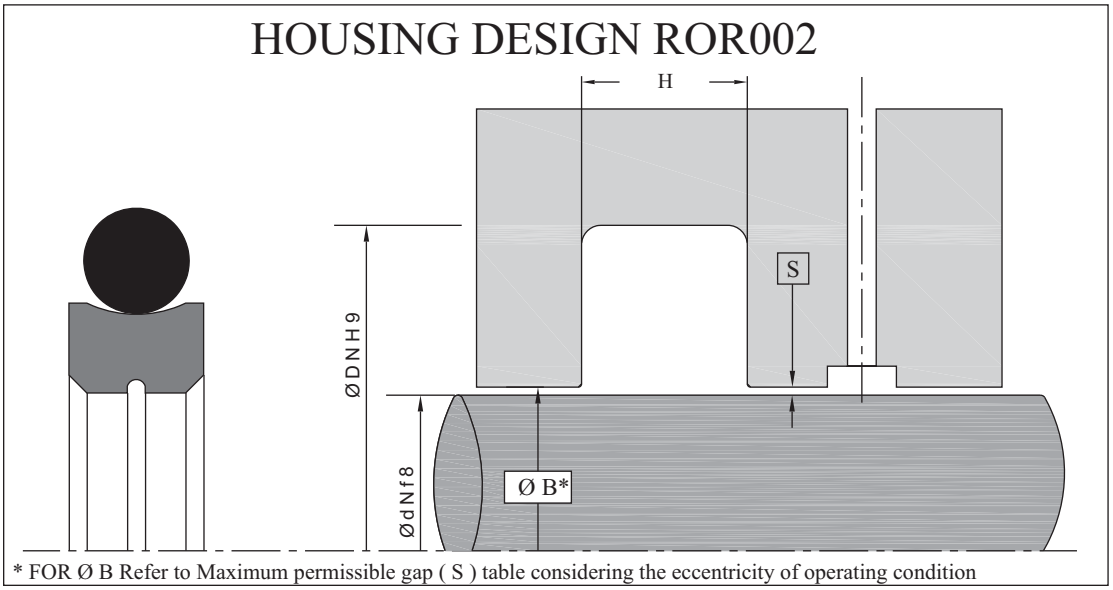
ROR002 is very easy to fit. Careful fitting of the seal is a prerequisite for it's perfect functioning.

Ordering format:

Please mention ROYAL order code in your order.
Example: Size 12 X 16.9 X 2.2
For Temp range: -20°C to +100°C. (O ring material is NBR70) Order code is ROR002-N-01200-01690-0220
For Temp range: -20°C to +200°C. (O ring material is FPM80) Order code is ROR002-F-01200-01690-0220

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN ROR002

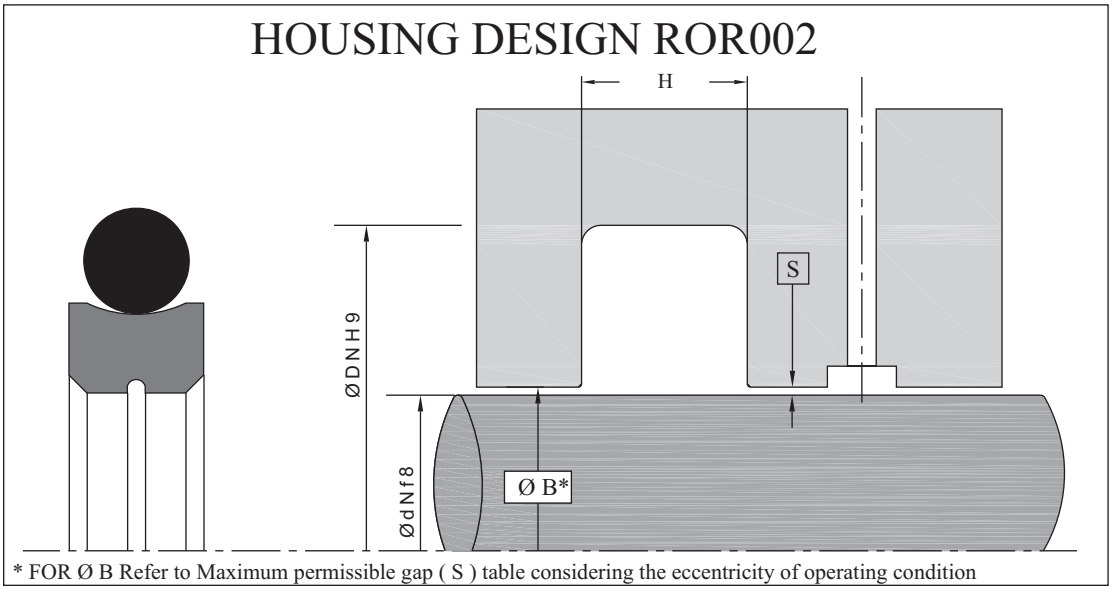


* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H9)	H (+0.20)	O ring size	ORDER CODE For temp. range -20°C to +100°C use ROR002-N -20°C to +200°C use ROR002-F
12.00	16.90	2.20	14.00 X 1.78	ROR002-N/F-01200-01690-0220
14.00	18.90	2.20	15.60 X1.78	ROR002-N/F-01400-01890-0220
16.00	20.90	2.20	17.17 X 1.78	ROR002-N/F-01600-02090-0220
18.00	22.90	2.20	18.77 X 1.78	ROR002-N/F-01800-02290-0220
20.00	27.50	3.20	21.89 X 2.62	ROR002-N/F-02000-02750-0320
22.00	29.50	3.20	25.07 X 2.62	ROR002-N/F-02200-02950-0320
25.00	32.50	3.20	28.24 X 2.62	ROR002-N/F-02500-03250-0320
28.00	35.50	3.20	31.42 X 2.62	ROR002-N/F-02800-03550-0320
32.00	39.50	3.20	34.59 X 2.62	ROR002-N/F-03200-03950-0320
36.00	43.50	3.20	39.34 X 2.62	ROR002-N/F-03600-04350-0320
40.00	51.00	4.20	44.04 X 3.53	ROR002-N/F-04000-05100-0420
45.00	56.00	4.20	50.39 X 3.53	ROR002-N/F-04500-05600-0420
50.00	61.00	4.20	53.57 X 3.53	ROR002-N/F-05000-06100-0420
56.00	67.00	4.20	59.92 X 3.53	ROR002-N/F-05600-06700-0420
63.00	74.00	4.20	66.27 X 3.53	ROR002-N/F-06300-07400-0420
70.00	81.00	4.20	75.79 X 3.53	ROR002-N/F-07000-08100-0420
80.00	91.00	4.20	85.32 X 3.53	ROR002-N/F-08000-09100-0420
90.00	101.00	4.20	94.84 X 3.53	ROR002-N/F-09000-10100-0420
100.00	111.00	4.20	104.37 X 3.53	ROR002-N/F-10000-11100-0420

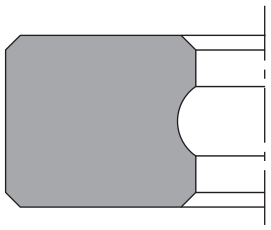
HOUSING DESIGN ROR002



* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H9)	H (+0.20)	O ring size	ORDER CODE For temp. range -20°C to +100°C use ROR002-N -20°C to +200°C use ROR002-F
110.00	121.00	4.20	113.89 X 3.53	ROR002-N/F-11000-12100-0420
125.00	136.00	4.20	129.77 X 3.53	ROR002-N/F-12500-13600-0420
140.00	151.00	4.20	145.64 X 3.53	ROR002-N/F-14000-15100-0420
160.00	171.00	4.20	164.69 X 3.53	ROR002-N/F-16000-17100-0420
180.00	191.00	4.20	183.74 X 3.53	ROR002-N/F-18000-19100-0420
200.00	215.50	6.30	208.97 X 5.33	ROR002-N/F-20000-21550-0630
220.00	235.50	6.30	227.97 X 5.33	ROR002-N/F-22000-23550-0630
250.00	265.50	6.30	253.37 X 5.33	ROR002-N/F-25000-26550-0630
280.00	301.00	8.10	291.47 X 6.99	ROR002-N/F-28000-30100-0810

ROTARY SEAL	ROS004	

Features:

Rotary seal profile ROS004 is a single piece seal commonly in use.

Composition:

Material:	Polyurethane
Royal Code	PUHP94

For compound detail please refer Royal seal materials.(Section III)

Properties:

- High resistance to abrasion
- Easy installation

Application:

- Mobile Hydraulics
- Rotary joints

General Technical Data:

Temperature range	-20°C to +110°C
Working Pressure(max.)	400 bar
Surface Speed(max.)	0.5 m/s

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Extrusion Gap dimension:

The gap dimension on the non-pressurised side of the seal is decisive for sealing function. Figures show the maximum permissible gap all on one side using the maximum clearance Ø and min rod Ø.

Maximum permissible gap dimension (s)			
160 bar	260 bar	320 bar	400 bar
0.5	0.4	0.35	(ask ROYAL)

Fitting:

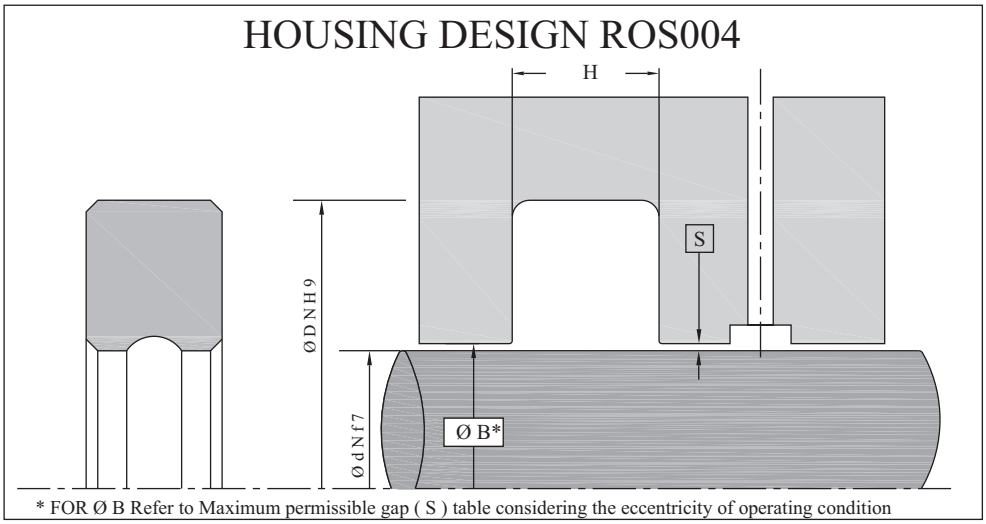
ROS004 is very easy to fit. Careful fitting of the seal is a prerequisite for it's perfect functioning.

Ordering format:

Please mention ROYAL order code in your order.
Example: Size 60 X 70 X 5.3
Order code is ROS004-U-06000-07000-0530

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

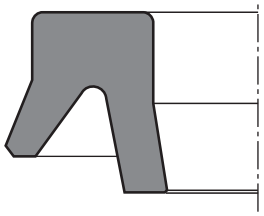
HOUSING DESIGN ROS004



* FOR Ø B Refer to Maximum permissible gap (S) table considering the eccentricity of operating condition

ROYAL STANDARD SIZE LIST

ØdN (f8)	ØDN (H11)	H (+0.20)	ORDER CODE
60.00	70.00	5.30	ROS004-U-06000-07000-0530
70.00	80.00	5.30	ROS004-U-07000-08000-0530
75.00	85.00	5.30	ROS004-U-07500-08500-0530
80.00	90.00	5.30	ROS004-U-08000-09000-0530
85.00	95.00	5.30	ROS004-U-08500-09500-0530
90.00	100.00	5.30	ROS004-U-09000-10000-0530
95.00	105.00	5.30	ROS004-U-09500-10500-0530
100.00	110.00	5.30	ROS004-U-10000-11000-0530
105.00	115.00	5.30	ROS004-U-10500-11500-0530
110.00	120.00	5.30	ROS004-U-11000-12000-0530
115.00	125.00	5.30	ROS004-U-11500-12500-0530
120.00	130.00	5.30	ROS004-U-12000-13000-0530
125.00	135.00	5.30	ROS004-U-12500-13500-0530
130.00	140.00	5.30	ROS004-U-13000-14000-0530
135.00	145.00	5.30	ROS004-U-13500-14500-0530
140.00	150.00	5.30	ROS004-U-14000-15000-0530
145.00	155.00	5.30	ROS004-U-14500-15500-0530
150.00	160.00	5.30	ROS004-U-15000-16000-0530
155.00	165.00	5.30	ROS004-U-15500-16500-0530
160.00	170.00	5.30	ROS004-U-16000-17000-0530

PISTON / SEAL	PNPS007	

Features:

Piston seal PNPS007 is a U cup type piston seal made of highly abrasion resistant polyurethane having very thin outer contact lip suitable for low pressure pneumatic application.

Composition:

Material:	Polyurethane
Royal Code	PUAB92

For compound detail please refer Royal seal materials.(Section III)

Properties:

- Low friction due to thin outer lip
- High resistance to abrasion
- Easy installation

Application:

- Pneumatics tools & equipments
- Standard Cylinder

General Technical Data:

Temperature range	-20°C to +100°C
Working Pressure	16 bar
Surface Speed	1 m/s.

Fitting:

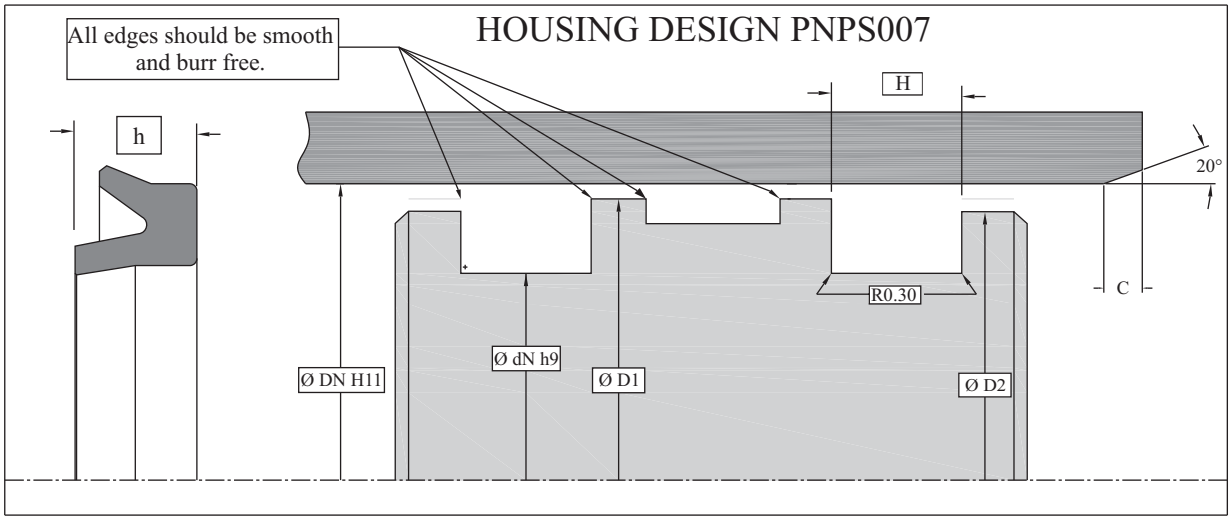
PNPS007 is very easy to fit. Careful fitting of the seal is a prerequisite for it's perfect functioning.

Ordering format:

Please mention ROYAL order code in your order.
Example: Size 25 X 17 X 6
Order code is PNPS007-02500-01700-0600

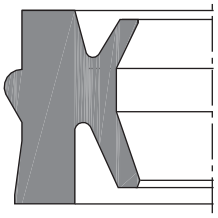
Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN PNPS007



ROYAL STANDARD SIZE LIST

DN	dN	D1	D2	h	H (+0.20)	ITEM CODE
25.00	17.00	24.80	24.00	5.50	6.00	PNPS007-02500-01700-0600
32.00	24.00	31.70	30.50	5.50	6.00	PNPS007-03200-02400-0600
40.00	30.00	39.70	38.50	7.00	7.50	PNPS007-04000-03000-0750
45.00	35.00	44.60	43.50	7.00	7.50	PNPS007-04500-03500-0750
50.00	40.00	49.60	48.50	7.00	7.50	PNPS007-05000-04000-0750
60.00	50.00	59.60	58.50	7.00	7.50	PNPS007-06000-05000-0750
63.00	53.00	62.60	61.50	7.00	7.50	PNPS007-06300-05300-0750
70.00	58.00	69.60	68.50	8.50	9.50	PNPS007-07000-05800-0950
80.00	68.00	79.60	78.50	8.50	9.50	PNPS007-08000-06800-0950
90.00	78.00	89.60	88.50	8.50	9.50	PNPS007-09000-07800-0950
100.00	88.00	99.50	98.00	8.50	9.50	PNPS007-10000-08800-0950
125.00	110.00	124.30	123.00	10.00	11.00	PNPS007-12500-11000-1100
150.00	135.00	149.30	148.00	10.00	11.00	PNPS007-15000-13500-1100
160.00	145.00	159.30	158.00	10.00	11.00	PNPS007-16000-14500-1100
200.00	180.00	199.00	198.00	11.00	12.00	PNPS007-20000-18000-1200

ROD SEAL	PNRS010	

Features:

Rod seal PNRS010 is a U cup type rod seal made of highly abrasion resistant polyurethane having two very thin inner contact lip suitable for low pressure pneumatic application. This design is often used as rod cum wiper seal for many pneumatic applications.

Composition:

Material:	Polyurethane
Royal Code	PUAB92

For compound detail please refer Royal seal materials.(Section III)

Properties:

- Low friction due to thin outer lip
- High resistance to abrasion
- Easy installation

Application:

- Pneumatics tools & equipments
- Standard cylinders

General Technical Data:

Temperature range	-20°C to +100°C
Working Pressure	Up to 16 bar
Surface Speed	1 m/s.

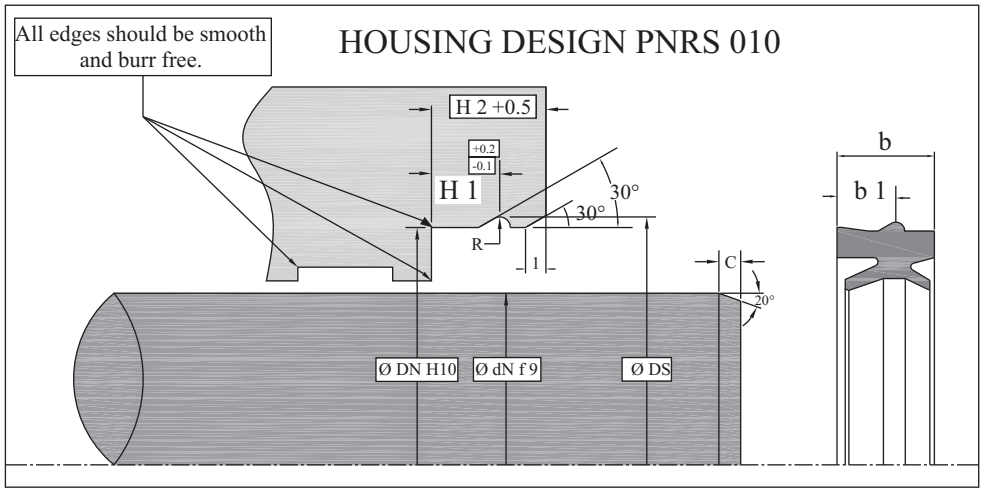
Fitting:

PNRS-010 is very easy to fit. Careful fitting of the seal is a prerequisite for it's perfect functioning.

Ordering format:

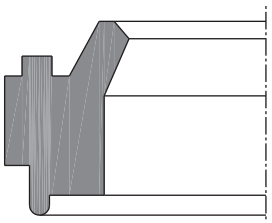
Please mention ROYAL order code in your order.
Example: Size 12 X 22 X 7.7
Order code is PNRS010-01200-02200-0770

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not available in Royal standard size list



ROYAL STANDARD SIZE LIST

dN (f9)	DN (H10)	b	b1	DS	H2	H1	R	ORDER CODE
12.00	22.00	10.50	7.00	24.00	13.00	7.70	1.00	PNRS010-01200-02200-0770
14.00	24.00	10.50	7.00	26.00	13.00	7.70	1.00	PNRS010-01400-02400-0770
16.00	26.00	10.50	7.00	28.00	13.00	7.70	1.00	PNRS010-01600-02600-0770
18.00	28.00	10.50	7.00	30.00	13.00	7.70	1.00	PNRS010-01800-02800-0770
20.00	30.00	10.50	7.00	32.00	13.00	7.70	1.00	PNRS010-02000-03000-0770
22.00	32.00	10.50	7.30	35.00	13.00	8.00	1.00	PNRS010-02200-03200-0800
25.00	35.00	10.50	7.30	38.00	13.00	8.00	1.00	PNRS010-02500-03500-0800
30.00	40.00	10.50	7.30	43.00	13.00	8.00	1.00	PNRS010-03000-04000-0800
32.00	42.00	10.50	7.30	45.00	13.00	8.00	1.00	PNRS010-03200-04200-0800
35.00	45.00	10.50	7.30	48.00	13.00	8.00	1.00	PNRS010-03500-04500-0800
40.00	50.00	10.50	7.30	53.00	13.00	8.00	1.00	PNRS010-04000-05000-0800
50.00	60.00	10.50	7.70	64.00	13.00	8.60	1.80	PNRS010-05000-06000-0860

CUSHION SEAL	PNCS012	

Features:

Cushion seal PNCS-012 is specially designed seal for cushioning in pneumatic cylinders made of very high abrasion resistant polyurethane.

Composition:

Material:	Polyurethane
Royal Code	PUAB92

For compound detail please refer Royal seal materials.(Section III)

Properties:

- Seal acts as a check valve
- High resistance to abrasion
- Easy installation

Application:

- Pneumatic tools & equipments

General Technical Data:

Temperature range	-20 to 100°C
Working Pressure	Up to 16 bar
Surface Speed	1 m/s.

Fitting:

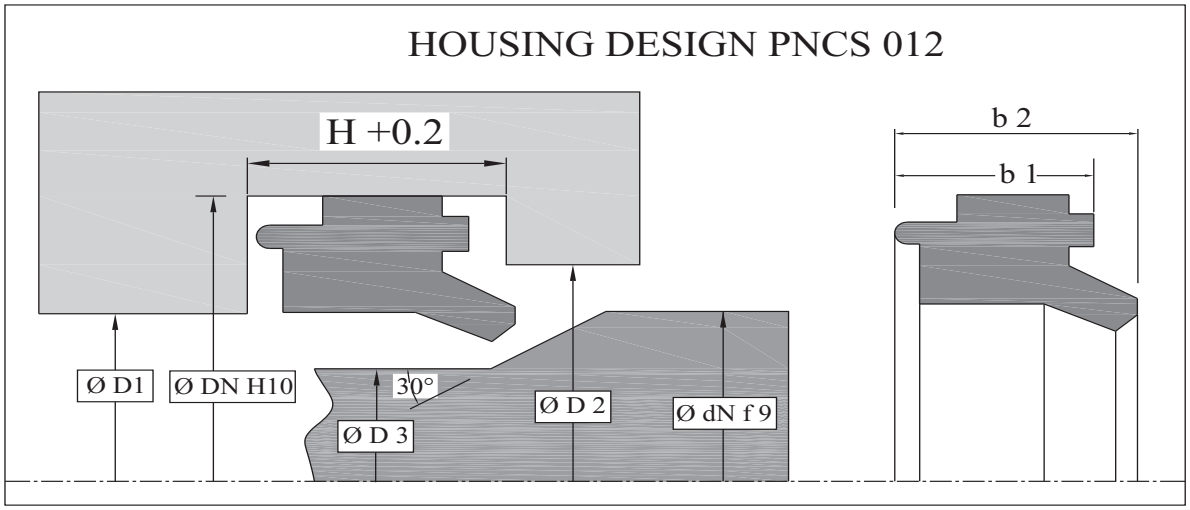
PNCS-012 is very easy to fit. Careful fitting of the seal is a prerequisite for it's perfect functioning.

Ordering format:

Please mention ROYAL order code in your order.
Example: Size 14 X 22 X 7.8
Order code is PNCS012-01400-02200-0780

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN PNCS 012



ROYAL STANDARD SIZE LIST

dN (f9)	DN (H10)	D1	D2	D3	b1	b2	H	ORDER CODE
14.00	22.00	15.00	19.00	10.00	6.30	7.80	7.00	PNCS012-01400-02200-0780
16.00	24.00	17.00	21.00	12.00	6.30	7.80	7.00	PNCS012-01600-02400-0780
18.00	26.00	19.00	23.00	14.00	6.30	7.80	7.00	PNCS012-01800-02600-0780
20.00	28.00	21.00	24.00	16.00	6.30	7.80	7.00	PNCS012-02000-02800-0780
22.00	30.00	23.00	26.00	18.00	6.30	7.80	7.00	PNCS012-02200-03000-0780
25.00	33.00	26.00	29.00	21.00	6.30	7.80	7.00	PNCS012-02500-03300-0780
28.00	36.00	29.00	32.00	24.00	6.30	7.80	7.00	PNCS012-02800-03600-0780
30.00	40.00	31.50	35.00	26.00	6.30	7.80	7.00	PNCS012-03000-04000-0780
32.00	42.00	33.50	37.00	28.00	6.30	7.80	7.00	PNCS012-03200-04200-0780
36.00	46.00	37.50	41.00	30.00	6.30	7.80	7.00	PNCS012-03600-04600-0780
40.00	50.00	41.50	45.00	36.00	6.30	7.80	7.00	PNCS012-04000-05000-0780
45.00	55.00	46.50	50.00	40.00	6.30	7.80	7.00	PNCS012-04500-05500-0780
50.00	60.00	51.50	55.00	45.00	6.30	7.80	7.00	PNCS012-05000-06000-0780

WEAR / GUIDE RING	WRPBRG	

Features:

Guide ring WRPBRG is a PTFE+ Bronze strip available in long length and cut to the size.

Composition:

Material	PTFE+ Bronze compound	PTFE+ Bronze compound
Royal Code	TB40(Green Colour)	TB40M(Dark Brown)

For compound detail please refer Royal seal materials. (Section III)

Properties:

- Low friction, free of stick slip
- High abrasion resistance
- Wide temperature Range
- Good thermal conductivity
- Low wear and high extrusion resistance
- Wide fluid application range
- Resistance against cold flow
- chamfered profile edges prevent pressing of edge in the corner radii of the housing groove

Application:

- Mobile Hydraulics
- Injection molding machines
- Machine tools
- Material Handling Equipments
- Control and regulating apparatus

General Technical Data:

Maximum speed	5 m/s
Load bearing capacity	15 N/mm² at 20 °C
	10 N/mm² at 60 °C
	5 N/mm² at 120 °C
	(permitted specific surface pressing*)

* The permissible load of the guide strip is calculated from the projected area multiplied by the permissible specific surface pressing. The non linear contact pressure distribution, produces by the load with relation to temperature and tolerances, is taken into account in the value of the permissible load.

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Gap dimension:

The radial clearance S min controls the minimum metal to metal clearance between the gland and rod or bore and piston. S max control the maximum extrusion gap seen by a seal associated with the guide ring.
The radial clearance S and the given tolerances are guidelines. Use of the guide and tolerances are to be designed in accordance with the used seal. The radial clearance S given in the table of dimensions is to be seen exclusively as reference to the guide ring. The corresponding diameter of the seal housing is to be adjusted to the sealing component. The cut gap ' Z' arising after fitting of guide ring is necessary to compensate the thermal expansion of polymer

Determination of the stretched length L2 of guide strip:

The stretched length is the circumferential length of the centre line of the strip when installed.

Calculation of stretched length L2 = π X (ØDN-T) -Z

Guide strip tolerances:

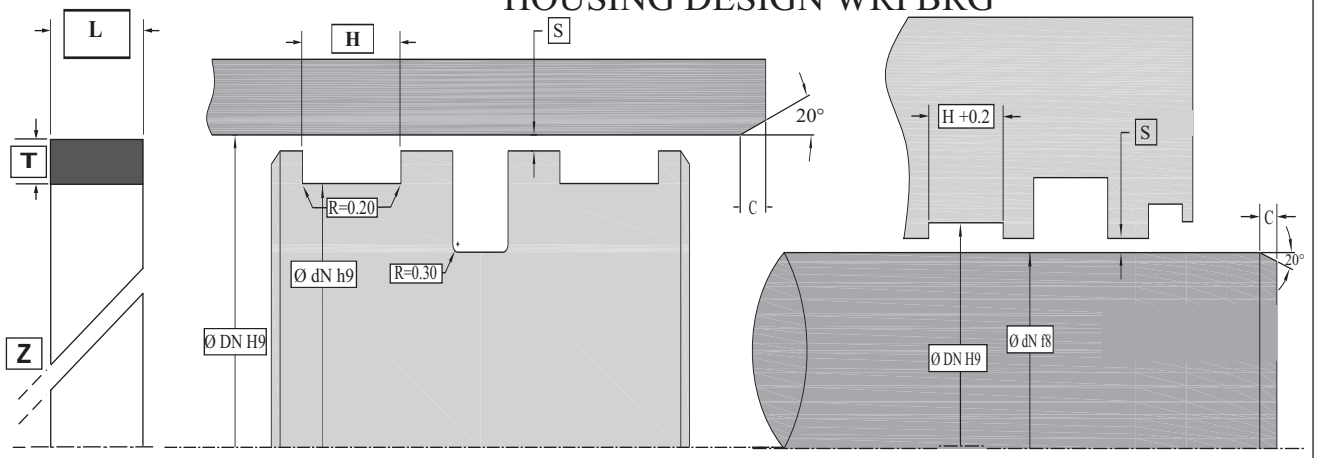
Width L (mm)	Thickness T (mm)
- 0.1 to - 0.6	- 0.03 to – 0.06

Installation:

Fitting is prerequisite for perfect functioning of the guide ring.

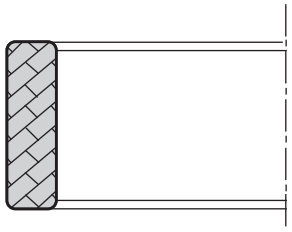
Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN WRPBRG



ROYAL STANDARD SIZE LIST

CROSS SECTION (H X T)(MM)	USE UP TO DIA RANGE (MM)ØDN H9	GROOVE DIA(MM) ØdN h8	GROOVE WIDTH(MM) H +0.2	S (Max.)	CUT GAP Z(min)	STD.ROLL LENGTH (MTR.)	ORDER CODE
5.00 X 1.60	10-40	ØDN-3.20	5.00	0.25	0.8-2.0	15	PB040-0500-160
5.60 X 2.50	20-140	ØDN-5.00	5.60	0.40	1.0-3.0	10	PB040-0560-250
6.00 X 2.00	20-60	ØDN-4.00	6.00	0.30	1.0-2.0	12	PB040-0600-200
6.00 X 2.50	25-140	ØDN-5.00	6.00	0.40	1.0-3.0	10	PB040-0600-250
6.00 X 4.00	70-200	ØDN-8.00	6.00	0.50	2.0-3.5	6	PB040-0600-400
6.00 X 4.50	100-200	ØDN-9.00	6.00	0.50	2.5-4.0	5	PB040-0600-450
8.00 X 2.00	25-80	ØDN-4.00	8.00	0.30	1.0-3.0	12	PB040-0800-200
8.00 X 2.50	30-160	ØDN-5.00	8.00	0.40	1.0-3.0	10	PB040-0800-250
8.00 X 4.00	100-400	ØDN-8.00	8.00	0.50	2.0-4.0	6	PB040-0800-400
9.50 X 3.17	60-160	ØDN-6.35	9.50	0.40	2.0-3.5	8	PB040-0950-317
9.50 X 4.00	120-500	ØDN-8.00	9.50	0.50	2.0-4.0	6	PB040-0950-400
9.70 X 2.50	40-160	ØDN-5.00	9.70	0.40	2.0-3.5	10	PB040-0970-250
10.00 X 2.00	40-100	ØDN-4.00	10.00	0.30	2.0-3.5	12	PB040-1000-200
12.50 X 2.50	50-180	ØDN-5.00	12.50	0.40	2.0-3.5	10	PB040-1250-250
12.70 X 3.17	50-200	ØDN-6.35	12.70	0.50	2.0-3.5	8	PB040-1270-317
14.80 X 2.50	60-200	ØDN-5.00	14.80	0.40	2.0-3.5	10	PB040-1480-250
15.00 X 2.50	60-200	ØDN-5.00	15.00	0.40	2.0-3.5	10	PB040-1500-250
15.00 X 4.00	160-900	ØDN-8.00	15.00	0.50	3.0-5.0	6	PB040-1500-400
15.85 X 3.17	60-250	ØDN-6.35	15.85	0.50	2.0-4.0	8	PB040-1585-317
19.05 X 3.17	80-380	ØDN-6.35	19.05	0.50	2.0-4.0	8	PB040-1905-317
20.00 X 2.00	50-180	ØDN-4.00	20.00	0.30	2.0-3.5	12	PB040-2000-200
20.00 X 2.50	80-300	ØDN-5.00	20.00	0.40	2.0-3.5	10	PB040-2000-250
20.00 X 3.20	100-400	ØDN-6.40	20.00	0.50	2.0-4.0	8	PB040-2000-320
25.00 X 2.50	100-400	ØDN-5.00	25.00	0.40	3.0-5.0	10	PB040-2500-250
25.40 X 3.17	100-400	ØDN-6.35	25.40	0.50	3.0-5.0	8	PB040-2540-317
30.00 X 2.00	100-225	ØDN-4.00	30.00	0.30	3.0-5.0	12	PB040-3000-200
30.00 X 2.50	100-400	ØDN-5.00	30.00	0.40	3.0-6.0	10	PB040-3000-250

WEAR / GUIDE RING	WRFRPR	

Features:
Guide ring WRFRPR is a pre-molded ring ready to the size.

Composition:

Material	Polyester fabric with polyester resin + PTFE
Royal Code	FRPR

For compound detail please refer Royal seal materials.

Properties:
- High load bearing capacity.
- Low friction,free of stick slip
- High abrasion resistance.
- Low wear and high extrusion resistance.
- Resistance against cold flow .

Application:
- Mobile Hydraulics
- Injection moulding machines
- Machine tools
- Material Handling Equipments
- Standard cylinders

General Technical Data:

Temp. range - 20°C to +100°C

Guide tolerances:

Width L (mm)	Thickness T (mm)
- 0.1 to - 0.6	- 0.03 to – 0.06

Operating limit:

Ultimate compressive strength	360 N/mm²
Load bearing capacity	120 N/mm² at 20 °C
	80 N/mm² at 60 °C
	63 N/mm² at 120 °C
	(permitted specific surface pressing*)

* The permissible load of the guide strip is calculated from the projected area multiplied by the permissible specific surface pressing. The non linear contact pressure distribution, produces by the load with relation to temperature and tolerances, is taken into account in the value of the permissible load.

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Gap dimension:

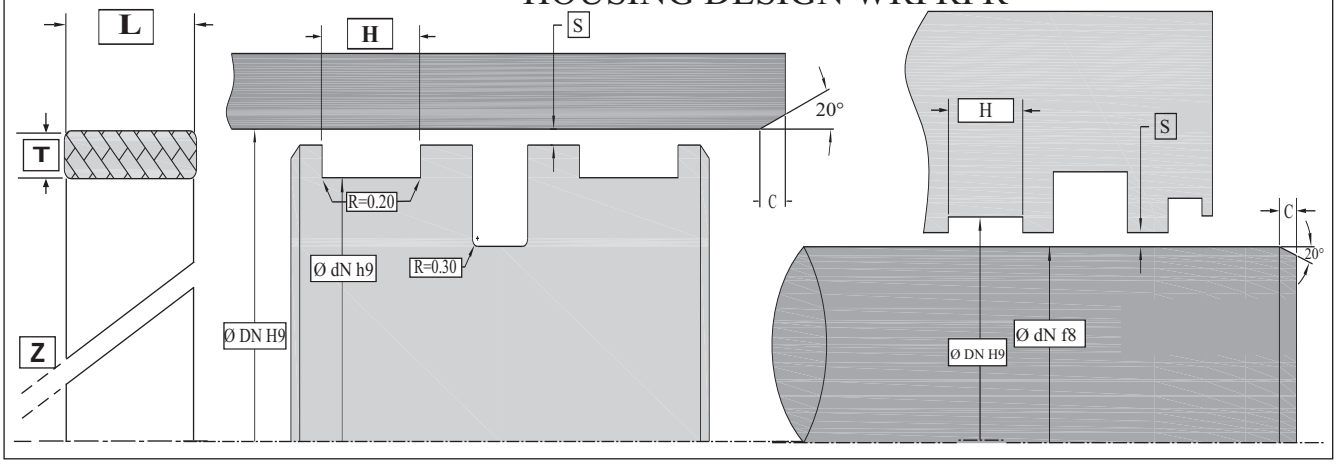
The radial clearance S min controls the minimum metal to metal clearance between the gland and rod or bore and piston. S max control the maximum extrusion gap seen by a seal associated with the guide ring.
The radial clearance S and the given tolerances are guidelines. Use of the guide and tolerances are to be designed in accordance with the used seal. The radial clearance S given in the table of dimensions is to be seen exclusively as reference to the guide ring. The corresponding diameter of the seal housing is to be adjusted to the sealing component. The cut gap ' Z' arising after fitting of guide ring is necessary to compensate the thermal expansion of polymer

Installation:

Careful fitting of the Guide ring is a prerequisite for it's perfect functioning.

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

HOUSING DESIGN WRFRPR

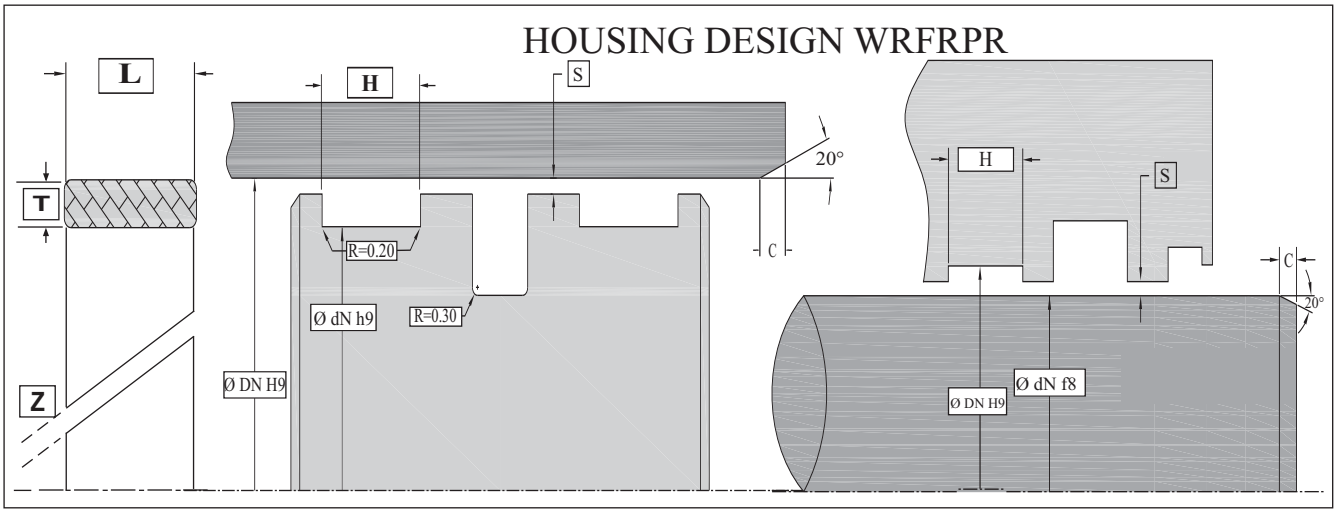


ROYAL STANDARD SIZE LIST

CROSS SECTION (H X T)	RECOMMENDED UP TO DIA (MM) ØDN H9	GROOVE DIA(MM) ØdN h8	GROOVE WIDTH (MM) H (+0.20)	S(MAX.)	CUT GAP Z	ORDER CODE Example for ØDN=090 and cross section 05.60 X 2.50 WRFRPR-09000-0560-250
05.60 X 2.50	20-140	ØDN-5.00	5.60	0.30	1.00-2.00	WRFRPR-ØDN-0560-250
06.00 X 2.00	20-60	ØDN-4.00	6.00	0.30	1.00-2.00	WRFRPR-ØDN-0600-200
06.00 X 2.50	25-140	ØDN-5.00	6.00	0.40	1.00-3.00	WRFRPR-ØDN-0600-250
06.00 X 4.00	70-200	ØDN-8.00	6.00	0.50	2.00-3.50	WRFRPR-ØDN-0600-400
06.00 X 4.50	100-200	ØDN-9.00	6.00	0.50	2.50-4.00	WRFRPR-ØDN-0600-450
08.00 X 2.00	25-80	ØDN-4.00	8.00	0.30	1.00-3.00	WRFRPR-ØDN-0800-200
08.00 X 2.50	30-160	ØDN-5.00	8.00	0.40	1.00-3.00	WRFRPR-ØDN-0800-250
08.00 X 4.00	100-300	ØDN-8.00	8.00	0.50	2.00-4.00	WRFRPR-ØDN-0800-400
09.50 X 4.00	120-300	ØDN-8.00	9.50	0.50	2.00-4.00	WRFRPR-ØDN-0950-400
09.70 X 2.50	40-160	ØDN-5.00	9.70	0.40	2.00-3.50	WRFRPR-ØDN-0970-250
10.00 X 2.00	40-100	ØDN-4.00	10.00	0.30	2.00-3.50	WRFRPR-ØDN-1000-200
12.50 X 2.50	50-180	ØDN-5.00	12.50	0.40	2.00-3.50	WRFRPR-ØDN-1250-250
15.00 X 2.50	60-200	ØDN-5.00	15.00	0.40	2.00-3.50	WRFRPR-ØDN-1500-250
15.00 X 4.00	160-300	ØDN-8.00	15.00	0.50	3.00-5.00	WRFRPR-ØDN-1500-400
20.00 X 2.00	50-180	ØDN-4.00	20.00	0.30	2.00-3.50	WRFRPR-ØDN-2000-200
20.00 X 2.50	80-300	ØDN-5.00	20.00	0.40	2.00-3.50	WRFRPR-ØDN-2000-250
20.00 X 3.20	100-300	ØDN-6.40	20.00	0.50	2.00-4.00	WRFRPR-ØDN-2000-320
25.00 X 2.50	100-300	ØDN-5.00	25.00	0.40	3.00-5.00	WRFRPR-ØDN-2500-250
30.00 X 2.00	100-225	ØDN-4.00	30.00	0.30	3.00-5.00	WRFRPR-ØDN-3000-200
30.00 X 2.50	100-300	ØDN-5.00	30.00	0.40	3.00-6.00	WRFRPR-ØDN-3000-250

Guide rings in pre-molded form can be provided up to Ø350 mm thickness (T) up to 10mm & height (L) up to 180mm

HOUSING DESIGN WRFRPR



ROYAL STANDARD SIZE LIST

CROSS SECTION (H X T)	RECOMMENDED UP TO DIA (MM) ØDN H9	GROOVE DIA(MM) ØdN h8	GROOVE WIDTH (MM) H (+0.20)	S(MAX.)	CUT GAP Z	ORDER CODE If ØDN=50.80 mm (2.000") and cross sec. 12.70X3.17(0.500"Ht.X0.125" Thick Order code is -WRFRPR-IN-02000-0500-125
2.92 X 1.58	30-50	ØDN-3.17	2.92	0.30	1.00-1.50	WRFRPR-IN-ØDN-0115-062
6.35 X 1.58	20-50	ØDN-3.17	6.35	0.30	1.00-1.50	WRFRPR-IN-ØDN-0250-062
9.53 X 1.58	20-65	ØDN-3.17	9.53	0.30	1.00-1.50	WRFRPR-IN-ØDN-0375-062
12.70 X 1.58	20-85	ØDN-3.17	12.70	0.30	1.00-2.00	WRFRPR-IN-ØDN-0500-062
19.05 X 1.58	35-90	ØDN-3.17	19.05	0.30	1.00-2.00	WRFRPR-IN-ØDN-0750-062
25.40 X 1.58	35-90	ØDN-3.17	25.40	0.40	2.00-3.50	WRFRPR-IN-ØDN-1000-062
4.19 X 3.17	25-90	ØDN-6.35	4.19	0.40	1.00-2.00	WRFRPR-IN-ØDN-0165-125
4.75 X 3.17	25-90	ØDN-6.35	4.75	0.40	1.00-2.00	WRFRPR-IN-ØDN-0187-125
6.35 X 3.17	25-150	ØDN-6.35	6.35	0.40	1.00-3.00	WRFRPR-IN-ØDN-0250-125
7.92 X 3.17	25-200	ØDN-6.35	7.92	0.40	1.00-3.10	WRFRPR-IN-ØDN-0312-125
9.53 X 3.17	60-300	ØDN-6.35	9.53	0.40	2.00-3.50	WRFRPR-IN-ØDN-0375-125
12.70 X 3.17	50-200	ØDN-6.35	12.70	0.50	2.00-3.50	WRFRPR-IN-ØDN-0500-125
15.88 X 3.17	60-250	ØDN-6.35	15.88	0.50	2.00-4.00	WRFRPR-IN-ØDN-0625-125
19.05 X 3.17	80-300	ØDN-6.35	19.05	0.50	2.00-4.00	WRFRPR-IN-ØDN-0750-125
22.23 X 3.17	80-300	ØDN-6.35	22.23	0.50	2.00-4.10	WRFRPR-IN-ØDN-0875-125
25.40 X 3.17	100-300	ØDN-6.35	25.40	0.50	3.00-5.00	WRFRPR-IN-ØDN-1000-125
31.75 X 3.17	100-325	ØDN-6.35	31.75	0.50	3.00-5.10	WRFRPR-IN-ØDN-1250-125
38.10 X 3.17	100-325	ØDN-6.35	38.10	0.50	3.00-5.20	WRFRPR-IN-ØDN-1500-125

Guide rings in pre-molded form can be provided up to Ø350 mm thickness (T) up to 10mm & height (L) up to 180mm

WEAR / GUIDE RING	WRPAGF	

Features:
Guide ring WRPAGF is a pre-molded ring ready to the size.

Composition:

Material	40% Glass Fibre Filled Polyamide 6
Royal Code	PG6040

For compound detail please refer Royal seal materials.(Section III)

Properties:
- High load bearing capacity.
- Low friction, free of stick slip
- High abrasion resistance.
- Low wear and high extrusion resistance.
- Resistance against cold flow.

Application:
- Mobile Hydraulics
- Injection moulding machines
- Machine tools
- Material Handling Equipments
- Standard cylinders

General Technical Data:

Temp. range - 55°C to 140°C

Operating limit:

Ultimate compressive strength	180 N/mm²
Load bearing capacity	40 N/mm² at 20 °C
	30 N/mm² at 100 °C
	(Permitted specific surface pressing*)

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Guide tolerances:

Width L (mm)	Thickness T (mm)
- 0.1 to - 0.6	- 0.03 to – 0.06

Gap dimension:

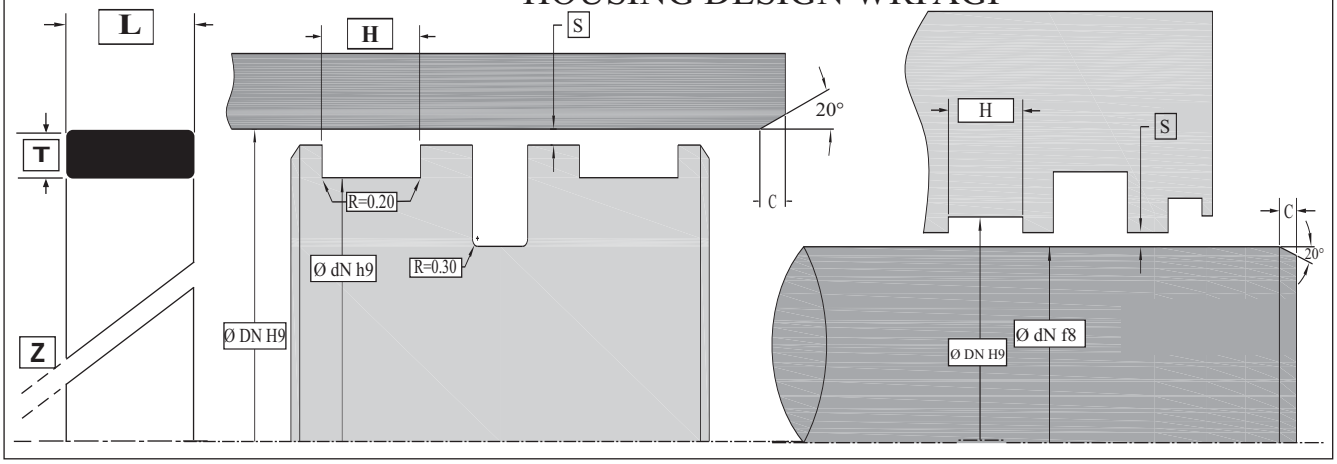
The radial clearance S min controls the minimum metal to metal clearance between the gland and rod or bore and piston. S max control the maximum extrusion gap seen by a seal associated with the guide ring.
The radial clearance S and the given tolerances are guidelines. Use of the guide and tolerances are to be designed in accordance with the used seal. The radial clearance S given in the table of dimensions is to be seen exclusively as reference to the guide ring. The corresponding diameter of the seal housing is to be adjusted to the sealing component. The cut gap ' Z' arising after fitting of guide ring is necessary to compensate the thermal expansion of polymer

Installation:

Careful fitting of the Guide ring is a prerequisite for it's perfect functioning.

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not mentioned in Royal standard size list

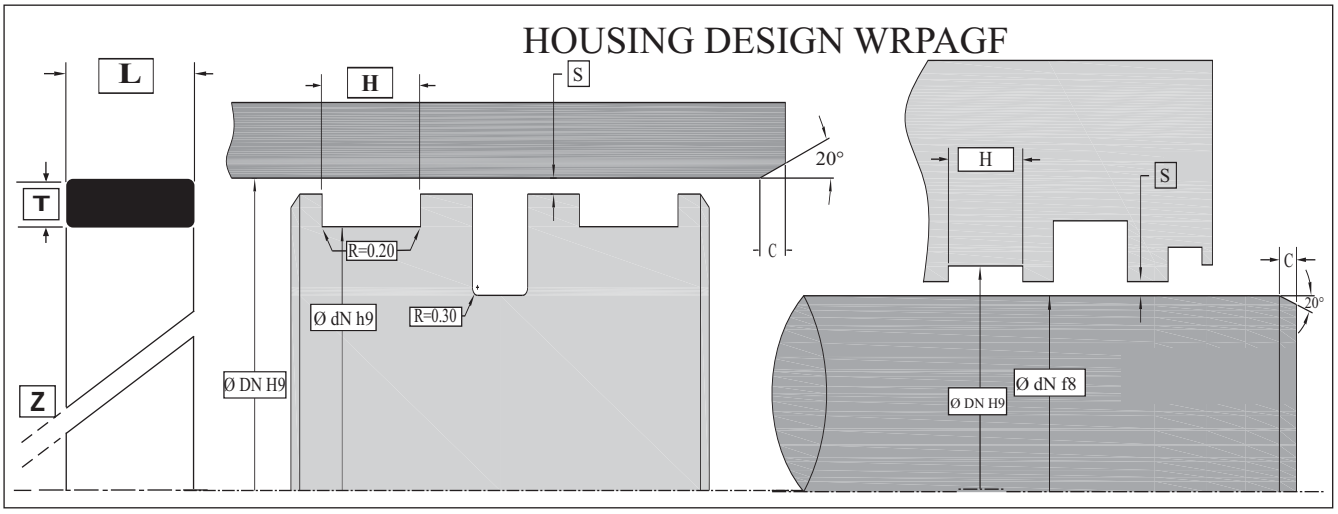
HOUSING DESIGN WRPAGF



ROYAL STANDARD SIZE LIST

CROSS SECTION (H X T)	RECOMMENDED UP TO DIA(MM) ØDN H9	GROOVE DIA(MM) ØdN h8	GROOVE WIDTH (MM) H +0.2	S(MAX.)	CUT GAP Z	ORDER CODE Example for ØDN=090 and cross section 05.60 X 2.50 WRPAGF-09000-0560-250
05.60 X 2.50	20-140	ØDN-5.00	5.60	0.30	1.00-2.00	WRPAGF-ØDN-0560-250
06.00 X 2.00	20-60	ØDN-4.00	6.00	0.30	1.00-2.00	WRPAGF-ØDN-0600-200
06.00 X 2.50	25-140	ØDN-5.00	6.00	0.40	1.00-3.00	WRPAGF-ØDN-0600-250
06.00 X 4.00	70-200	ØDN-8.00	6.00	0.50	2.00-3.50	WRPAGF-ØDN-0600-400
06.00 X 4.50	100-200	ØDN-9.00	6.00	0.50	2.50-4.00	WRPAGF-ØDN-0600-450
08.00 X 2.00	25-80	ØDN-4.00	8.00	0.30	1.00-3.00	WRPAGF-ØDN-0800-200
08.00 X 2.50	30-160	ØDN-5.00	8.00	0.40	1.00-3.00	WRPAGF-ØDN-0800-250
08.00 X 4.00	100-300	ØDN-8.00	8.00	0.50	2.00-4.00	WRPAGF-ØDN-0800-400
09.50 X 4.00	120-300	ØDN-8.00	9.50	0.50	2.00-4.00	WRPAGF-ØDN-0950-400
09.70 X 2.50	40-160	ØDN-5.00	9.70	0.40	2.00-3.50	WRPAGF-ØDN-0970-250
10.00 X 2.00	40-100	ØDN-4.00	10.00	0.30	2.00-3.50	WRPAGF-ØDN-1000-200
12.50 X 2.50	50-180	ØDN-5.00	12.50	0.40	2.00-3.50	WRPAGF-ØDN-1250-250
15.00 X 2.50	60-200	ØDN-5.00	15.00	0.40	2.00-3.50	WRPAGF-ØDN-1500-250
15.00 X 4.00	160-300	ØDN-8.00	15.00	0.50	3.00-5.00	WRPAGF-ØDN-1500-400
20.00 X 2.00	50-180	ØDN-4.00	20.00	0.30	2.00-3.50	WRPAGF-ØDN-2000-200
20.00 X 2.50	80-300	ØDN-5.00	20.00	0.40	2.00-3.50	WRPAGF-ØDN-2000-250
20.00 X 3.20	100-300	ØDN-6.40	20.00	0.50	2.00-4.00	WRPAGF-ØDN-2000-320
25.00 X 2.50	100-300	ØDN-5.00	25.00	0.40	3.00-5.00	WRPAGF-ØDN-2500-250
30.00 X 2.00	100-225	ØDN-4.00	30.00	0.30	3.00-5.00	WRPAGF-ØDN-3000-200
30.00 X 2.50	100-300	ØDN-5.00	30.00	0.40	3.00-6.00	WRPAGF-ØDN-3000-250

HOUSING DESIGN WRPAGF

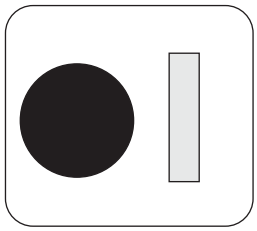


ROYAL STANDARD SIZE LIST

CROSS SECTION (H X T)	RECOMMEDED UP TO DIA(MM) ØDN H9	GROOVE DIA(MM) ØdN h8	GROOVE WIDTH (MM) H +0.2	S(MAX.)	CUT GAP Z	ORDER CODE If ØDN=50.80 mm (2.000") and cross sec. 12.70X3.17(0.500"Ht.X0.125" Thick Order code is -WRPAGF-IN-02000-0500-125
02.92 X 1.58	30-50	ØDN-3.17	2.92	0.30	1.00-1.50	WRPAGF-IN-ØDN-0115-062
06.35 X 1.58	20-50	ØDN-3.17	6.35	0.30	1.00-1.50	WRPAGF-IN-ØDN-0250-062
09.53 X 1.58	20-65	ØDN-3.17	9.53	0.30	1.00-1.50	WRPAGF-IN-ØDN-0375-062
12.70 X 1.58	20-85	ØDN-3.17	12.70	0.30	1.00-2.00	WRPAGF-IN-ØDN-0500-062
19.05 X 1.58	35-90	ØDN-3.17	19.05	0.30	1.00-2.00	WRPAGF-IN-ØDN-0750-062
25.40 X 1.58	35-90	ØDN-3.17	25.40	0.40	2.00-3.50	WRPAGF-IN-ØDN-1000-062
4.19 X 3.17	25-90	ØDN-6.35	4.19	0.40	1.00-2.00	WRPAGF-IN-ØDN-0165-125
4.75 X 3.17	25-90	ØDN-6.35	4.75	0.40	1.00-2.00	WRPAGF-IN-ØDN-0187-125
6.35 X 3.17	25-150	ØDN-6.35	6.35	0.40	1.00-3.00	WRPAGF-IN-ØDN-0250-125
7.92 X 3.17	25-200	ØDN-6.35	7.92	0.40	1.00-3.10	WRPAGF-IN-ØDN-0312-125
9.53 X 3.17	60-300	ØDN-6.35	9.53	0.40	2.00-3.50	WRPAGF-IN-ØDN-0375-125
12.70 X 3.17	50-200	ØDN-6.35	12.70	0.50	2.00-3.50	WRPAGF-IN-ØDN-0500-125
15.88 X 3.17	60-250	ØDN-6.35	15.88	0.50	2.00-4.00	WRPAGF-IN-ØDN-0625-125
19.05 X 3.17	80-300	ØDN-6.35	19.05	0.50	2.00-4.00	WRPAGF-IN-ØDN-0750-125
22.23 X 3.17	80-300	ØDN-6.35	22.23	0.50	2.00-4.10	WRPAGF-IN-ØDN-0875-125
25.40 X 3.17	100-300	ØDN-6.35	25.40	0.50	3.00-5.00	WRPAGF-IN-ØDN-1000-125
31.75 X 3.17	100-325	ØDN-6.35	31.75	0.50	3.00-5.10	WRPAGF-IN-ØDN-1250-125
38.10 X 3.17	100-325	ØDN-6.35	38.10	0.50	3.00-5.20	WRPAGF-IN-ØDN-1500-125

O RING AND BACK UP RING

OR & BRT



O-RING

Features:

O-Rings are ring shaped seals of circular cross section. They are mainly used to seal machined components-in static situations - against fluid and gaseous media. In certain conditions, they can be used as a dynamic sealing element for axial, rotating and oscillating movement.

Composition:

O ring type	OR7N	OR9N	OR8F	ORPU*
Material:	NBR	NBR	FPM	Polyurethane
Royal Code	NBR70	NBR90	FPM80	PUAB92

For compound detail please refer Royal seal materials.(Section III)

*** For O ring size list in Polyurethane ask ROYAL at info@royalcomposites.co.in / info@royalseals.co.in**

Properties:

- O ring can be used for both static and dynamic sealing application
- Wide temperature range
- Easy installation

Application:

- All short of hydraulic and pneumatic sealing system

General Technical Data:

O ring type	OR7N	OR9N	OR8F	ORPU
Material	NBR	NBR	FPM	Polyurethane
Working Pressure(without back up ring)	160 bar	240 bar	160 bar	400 bar
Working Pressure(with back up ring)	250 bar	400 bar	400 bar	-
Temperature range	-20°C to +100°C	-20°C to +100°C	-20°C to +200°C	-20°C to +100°C

Surface finish:

Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Fitting:

Careful fitting of the seal is a prerequisite for it's perfect functioning.

BACKUP RING

Features:
Standard backup element with rectangular cross section.

Composition:

	BRT	BRH
Material:	PTFE	Polyester
Royal Code	TFPR	HY55

For compound detail please refer Royal seal materials.

Application:
Backup element to prevent gap extrusion of radially sealing O-Rings on the outside in dynamic sealing.

General Technical Data:

Type	BRT	BRH
Material:	PTFE	Polyester
Temperature	-50°C to +200°C	-35°C to +120°C

Surface finish:

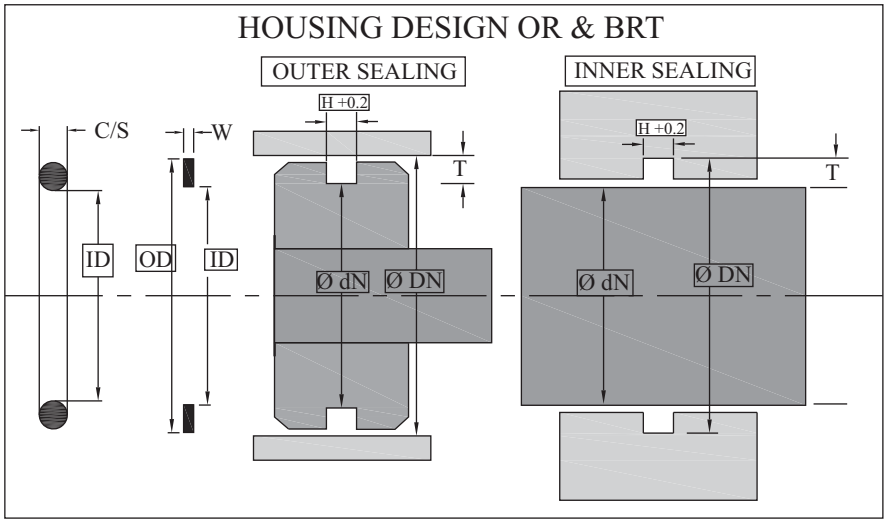
Roughness depth	R(max)	Ra
Sliding surface	≤ 2.5 µm	0.05-0.3 µm
Bottom of groove	≤ 6.3 µm	≤ 1.6 µm
Side of groove	≤ 15 µm	≤ 3µm

Fitting
Careful fitting is prerequisite for perfect functioning of the backup ring.

Ordering format:
Please mention ROYAL order code in your order.
Example: Size 25 X 30X 1.3
Order code for PTFE is BRT-G-025
Order code for polyester is BRH-G-025

Please contact ROYAL at **info@royalcomposites.co.in / info@royalseals.co.in** if your required size is not available in Royal standard size list

HOUSING DESIGN OR & BRT



ROYAL STANDARD SIZE LIST

CROSS SECTION		GLAND DEPTH(MM) T				H+0.2(MM)			BACKUP RING W (-0.10)
C/S	TOL.	HYD.	TOL.	PNEU.	TOL.	NO BACK UP	ONE BACK UP	TWO BACK UP	THICKNESS
1.78	±0.08	1.45	±0.02	1.55	±0.02	2.40	3.80	5.20	1.40
1.90	±0.08	1.50	±0.02	1.65	±0.02	2.60	4.00	5.40	1.40
2.40	±0.09	2.00	±0.02	2.10	±0.02	3.20	4.60	6.00	1.40
2.62	±0.09	2.25	±0.02	2.35	±0.02	3.60	5.00	6.40	1.40
3.10	±0.10	2.50	±0.02	2.80	±0.02	4.10	5.50	6.90	1.40
3.50	±0.10	3.00	±0.05	3.20	±0.05	4.60	6.00	7.40	1.40
3.53	±0.10	3.10	±0.05	3.20	±0.05	4.80	6.20	7.60	1.40
5.33	±0.13	4.70	±0.05	4.90	±0.05	7.10	9.10	11.10	2.00
5.70	±0.13	5.00	±0.05	5.30	±0.05	7.20	9.20	11.20	2.00
6.99	±0.15	6.00	±0.05	6.55	±0.05	9.50	12.00	14.50	2.50
8.40	±0.18	7.50	±0.05	8.00	±0.05	10.00	12.50	15.00	2.50

O RING INSIDE DIAMETER TOLERANCES

I.D. (mm)	2.80 to 6.30	6.31 to 11.20	11.21 to 21.20	21.21 to 40.00	41.20 to 80.00	80.01 to 160	165 to 300	300 to 650
TOL.(mm)	±0.13 mm	±0.16 mm	±0.19 mm	±0.95%	±0.86%	±0.78%	±0.74%	±0.67%
				of corresponding inside diameter				

ROYAL STANDARD SIZE LIST

I.D.	C/S	O RING (N=NBR, F=FPM) ORDER CODE	BACKUP RING ORDER CODE
		CROSS SECTION 1.9 mm	
2.8	1.9	OR-N/F-P-003	BRT-P-003
3.8	1.9	OR-N/F-P-004	BRT-P-004
4.8	1.9	OR-N/F-P-005	BRT-P-005
5.8	1.9	OR-N/F-P-006	BRT-P-006
6.8	1.9	OR-N/F-P-007	BRT-P-007
7.8	1.9	OR-N/F-P-008	BRT-P-008
8.8	1.9	OR-N/F-P-009	BRT-P-009
9.8	1.9	OR-N/F-P-010	BRT-P-010
		CROSS SECTION 2.4 mm	
9.8	2.4	OR-N/F-P-010A	BRT-P-010A
10.8	2.4	OR-N/F-P-011	BRT-P-011
11.0	2.4	OR-N/F-P-011.2	BRT-P-011.2
11.8	2.4	OR-N/F-P-012	BRT-P-012
12.3	2.4	OR-N/F-P-012.5	BRT-P-012.5
13.8	2.4	OR-N/F-P-014	BRT-P-014
14.8	2.4	OR-N/F-P-015	BRT-P-015
15.8	2.4	OR-N/F-P-016	BRT-P-016
17.8	2.4	OR-N/F-P-018	BRT-P-018
19.8	2.4	OR-N/F-P-020	BRT-P-020
20.8	2.4	OR-N/F-P-021	BRT-P-021
21.8	2.4	OR-N/F-P-022	BRT-P-022
		CROSS SECTION 3.1 mm	
24.4	3.1	OR-N/F-G-025	BRT-G-025
29.4	3.1	OR-N/F-G-030	BRT-G-030
34.4	3.1	OR-N/F-G-035	BRT-G-035
39.4	3.1	OR-N/F-G-040	BRT-G-040
44.4	3.1	OR-N/F-G-045	BRT-G-045
49.4	3.1	OR-N/F-G-050	BRT-G-050
54.4	3.1	OR-N/F-G-055	BRT-G-055
59.4	3.1	OR-N/F-G-060	BRT-G-060
64.4	3.1	OR-N/F-G-065	BRT-G-065
69.4	3.1	OR-N/F-G-070	BRT-G-070
74.4	3.1	OR-N/F-G-075	BRT-G-075
79.4	3.1	OR-N/F-G-080	BRT-G-080
84.4	3.1	OR-N/F-G-085	BRT-G-085
89.4	3.1	OR-N/F-G-090	BRT-G-090
94.4	3.1	OR-N/F-G-095	BRT-G-095
99.4	3.1	OR-N/F-G-100	BRT-G-100
104.4	3.1	OR-N/F-G-105	BRT-G-105
109.4	3.1	OR-N/F-G-110	BRT-G-110
114.4	3.1	OR-N/F-G-115	BRT-G-115
119.4	3.1	OR-N/F-G-120	BRT-G-120

ROYAL STANDARD SIZE LIST

I.D.	C/S	O RING (N=NBR, F=FPM) ORDER CODE	BACKUP RING ORDER CODE
124.4	3.1	OR-N/F-G-125	BRT-G-125
129.4	3.1	OR-N/F-G-130	BRT-G-130
134.4	3.1	OR-N/F-G-135	BRT-G-135
139.4	3.1	OR-N/F-G-140	BRT-G-140
144.4	3.1	OR-N/F-G-145	BRT-G-145
CROSS SECTION 3.50 mm			
21.7	3.5	OR-N/F-P-022A	BRT-P-022A
22.1	3.5	OR-N/F-P-022.4	BRT-P-022.4
23.7	3.5	OR-N/F-P-024	BRT-P-024
24.7	3.5	OR-N/F-P-025	BRT-P-025
25.2	3.5	OR-N/F-P-025.5	BRT-P-025.5
25.7	3.5	OR-N/F-P-026	BRT-P-026
27.7	3.5	OR-N/F-P-028	BRT-P-028
28.7	3.5	OR-N/F-P-029	BRT-P-029
29.2	3.5	OR-N/F-P-029.5	BRT-P-029.5
29.7	3.5	OR-N/F-P-030	BRT-P-030
30.7	3.5	OR-N/F-P-031	BRT-P-031
31.2	3.5	OR-N/F-P-031.5	BRT-P-031.5
31.7	3.5	OR-N/F-P-032	BRT-P-032
33.7	3.5	OR-N/F-P-034	BRT-P-034
34.7	3.5	OR-N/F-P-035	BRT-P-035
35.2	3.5	OR-N/F-P-035.5	BRT-P-035.5
35.7	3.5	OR-N/F-P-036	BRT-P-036
37.7	3.5	OR-N/F-P-038	BRT-P-038
38.7	3.5	OR-N/F-P-039	BRT-P-039
39.7	3.5	OR-N/F-P-040	BRT-P-040
40.7	3.5	OR-N/F-P-041	BRT-P-041
41.7	3.5	OR-N/F-P-042	BRT-P-042
43.7	3.5	OR-N/F-P-044	BRT-P-044
44.7	3.5	OR-N/F-P-045	BRT-P-045
45.7	3.5	OR-N/F-P-046	BRT-P-046
47.7	3.5	OR-N/F-P-048	BRT-P-048
48.7	3.5	OR-N/F-P-049	BRT-P-049
49.7	3.5	OR-N/F-P-050	BRT-P-050
CROSS SECTION 5.70 mm			
47.6	5.7	OR-N/F-P-048A	BRT-P-048A
49.6	5.7	OR-N/F-P-050A	BRT-P-050A
51.6	5.7	OR-N/F-P-052	BRT-P-052
52.6	5.7	OR-N/F-P-053	BRT-P-053
54.6	5.7	OR-N/F-P-055	BRT-P-055
55.6	5.7	OR-N/F-P-056	BRT-P-056
57.6	5.7	OR-N/F-P-058	BRT-P-058
59.6	5.7	OR-N/F-P-060	BRT-P-060
61.6	5.7	OR-N/F-P-062	BRT-P-062

ROYAL STANDARD SIZE LIST

I.D.	C/S	O RING (N=NBR, F=FPM) ORDER CODE	BACKUP RING ORDER CODE
62.6	5.7	OR-N/F-P-063	BRT-P-063
64.6	5.7	OR-N/F-P-065	BRT-P-065
66.6	5.7	OR-N/F-P-067	BRT-P-067
69.6	5.7	OR-N/F-P-070	BRT-P-070
70.6	5.7	OR-N/F-P-071	BRT-P-071
74.6	5.7	OR-N/F-P-075	BRT-P-075
79.6	5.7	OR-N/F-P-080	BRT-P-080
84.6	5.7	OR-N/F-P-085	BRT-P-085
89.6	5.7	OR-N/F-P-090	BRT-P-090
94.6	5.7	OR-N/F-P-095	BRT-P-095
99.6	5.7	OR-N/F-P-100	BRT-P-100
101.6	5.7	OR-N/F-P-102	BRT-P-102
104.6	5.7	OR-N/F-P-105	BRT-P-105
109.6	5.7	OR-N/F-P-110	BRT-P-110
111.6	5.7	OR-N/F-P-112	BRT-P-112
114.6	5.7	OR-N/F-P-115	BRT-P-115
119.6	5.7	OR-N/F-P-120	BRT-P-120
124.6	5.7	OR-N/F-P-125	BRT-P-125
129.6	5.7	OR-N/F-P-130	BRT-P-130
131.6	5.7	OR-N/F-P-132	BRT-P-132
134.6	5.7	OR-N/F-P-135	BRT-P-135
139.6	5.7	OR-N/F-P-140	BRT-P-140
144.6	5.7	OR-N/F-P-145	BRT-P-145
149.6	5.7	OR-N/F-P-150	BRT-P-150
149.3	5.7	OR-N/F-G-150	BRT-G-150
154.3	5.7	OR-N/F-G-155	BRT-G-155
159.3	5.7	OR-N/F-G-160	BRT-G-160
164.3	5.7	OR-N/F-G-165	BRT-G-165
169.3	5.7	OR-N/F-G-170	BRT-G-170
174.3	5.7	OR-N/F-G-175	BRT-G-175
179.3	5.7	OR-N/F-G-180	BRT-G-180
184.3	5.7	OR-N/F-G-185	BRT-G-185
189.3	5.7	OR-N/F-G-190	BRT-G-190
194.3	5.7	OR-N/F-G-195	BRT-G-195
199.3	5.7	OR-N/F-G-200	BRT-G-200
204.3	5.7	OR-N/F-G-205	BRT-G-205
209.3	5.7	OR-N/F-G-210	BRT-G-210
214.3	5.7	OR-N/F-G-215	BRT-G-215
219.3	5.7	OR-N/F-G-220	BRT-G-220
224.3	5.7	OR-N/F-G-225	BRT-G-225
229.3	5.7	OR-N/F-G-230	BRT-G-230
234.3	5.7	OR-N/F-G-235	BRT-G-235
239.3	5.7	OR-N/F-G-240	BRT-G-240
244.3	5.7	OR-N/F-G-245	BRT-G-245

ROYAL STANDARD SIZE LIST

I.D.	C/S	O RING (N=NBR, F=FPM) ORDER CODE	BACKUP RING ORDER CODE
249.3	5.7	OR-N/F-G-250	BRT-G-250
254.3	5.7	OR-N/F-G-255	BRT-G-255
259.3	5.7	OR-N/F-G-260	BRT-G-260
264.3	5.7	OR-N/F-G-265	BRT-G-265
269.3	5.7	OR-N/F-G-270	BRT-G-270
274.3	5.7	OR-N/F-G-275	BRT-G-275
279.3	5.7	OR-N/F-G-280	BRT-G-280
284.3	5.7	OR-N/F-G-285	BRT-G-285
289.3	5.7	OR-N/F-G-290	BRT-G-290
294.3	5.7	OR-N/F-G-295	BRT-G-295
299.3	5.7	OR-N/F-G-300	BRT-G-300
304.3	5.7	OR-N/F-G-305	
309.3	5.7	OR-N/F-G-310	
314.3	5.7	OR-N/F-G-315	
319.3	5.7	OR-N/F-G-320	
324.3	5.7	OR-N/F-G-325	
329.3	5.7	OR-N/F-G-330	
334.3	5.7	OR-N/F-G-335	
339.3	5.7	OR-N/F-G-340	
344.3	5.7	OR-N/F-G-345	
349.3	5.7	OR-N/F-G-350	
354.3	5.7	OR-N/F-G-355	
359.3	5.7	OR-N/F-G-360	
364.3	5.7	OR-N/F-G-365	
369.3	5.7	OR-N/F-G-370	
374.3	5.7	OR-N/F-G-375	
379.3	5.7	OR-N/F-G-380	
384.3	5.7	OR-N/F-G-385	
389.3	5.7	OR-N/F-G-390	
394.3	5.7	OR-N/F-G-395	
399.3	5.7	OR-N/F-G-400	
CROSS SECTION 8.40 mm			
149.5	8.4	OR-N/F-P-150A	BRT-P-150A
154.5	8.4	OR-N/F-P-155	BRT-P-155
159.5	8.4	OR-N/F-P-160	BRT-P-160
164.5	8.4	OR-N/F-P-165	BRT-P-165
169.5	8.4	OR-N/F-P-170	BRT-P-170
174.5	8.4	OR-N/F-P-175	BRT-P-175
179.5	8.4	OR-N/F-P-180	BRT-P-180
184.5	8.4	OR-N/F-P-185	BRT-P-185
189.5	8.4	OR-N/F-P-190	BRT-P-190
194.5	8.4	OR-N/F-P-195	BRT-P-195
199.5	8.4	OR-N/F-P-200	BRT-P-200
204.5	8.4	OR-N/F-P-205	BRT-P-205

ROYAL STANDARD SIZE LIST

I.D.	C/S	O RING (N=NBR, F=FPM) ORDER CODE	BACKUP RING ORDER CODE
208.5	8.4	OR-N/F-P-209	BRT-P-209
209.5	8.4	OR-N/F-P-210	BRT-P-210
214.5	8.4	OR-N/F-P-215	BRT-P-215
219.5	8.4	OR-N/F-P-220	BRT-P-220
224.5	8.4	OR-N/F-P-225	BRT-P-225
229.5	8.4	OR-N/F-P-230	BRT-P-230
234.5	8.4	OR-N/F-P-235	BRT-P-235
239.5	8.4	OR-N/F-P-240	BRT-P-240
244.5	8.4	OR-N/F-P-245	BRT-P-245
249.5	8.4	OR-N/F-P-250	BRT-P-250
254.5	8.4	OR-N/F-P-255	BRT-P-255
259.5	8.4	OR-N/F-P-260	BRT-P-260
264.5	8.4	OR-N/F-P-265	BRT-P-265
269.5	8.4	OR-N/F-P-270	BRT-P-270
274.5	8.4	OR-N/F-P-275	BRT-P-275
279.5	8.4	OR-N/F-P-280	BRT-P-280
284.5	8.4	OR-N/F-P-285	BRT-P-285
289.5	8.4	OR-N/F-P-290	BRT-P-290
294.5	8.4	OR-N/F-P-295	BRT-P-295
299.5	8.4	OR-N/F-P-300	
314.5	8.4	OR-N/F-P-315	
319.5	8.4	OR-N/F-P-320	
334.5	8.4	OR-N/F-P-335	
339.5	8.4	OR-N/F-P-340	
354.5	8.4	OR-N/F-P-355	
359.5	8.4	OR-N/F-P-360	
374.5	8.4	OR-N/F-P-375	
384.5	8.4	OR-N/F-P-385	
399.5	8.4	OR-N/F-P-400	
CROSS SECTION 1.78 mm			
2.90	1.78	OR-N/F-D-006	
3.68	1.78	OR-N/F-D-007	
4.47	1.78	OR-N/F-D-008	
5.28	1.78	OR-N/F-D-009	
6.07	1.78	OR-N/F-D-010	
7.65	1.78	OR-N/F-D-011	
9.25	1.78	OR-N/F-D-012	
10.82	1.78	OR-N/F-D-013	
12.42	1.78	OR-N/F-D-014	
14.00	1.78	OR-N/F-D-015	
15.60	1.78	OR-N/F-D-016	
17.17	1.78	OR-N/F-D-017	
18.77	1.78	OR-N/F-D-018	

ROYAL STANDARD SIZE LIST

I.D.	C/S	O RING (N=NBR, F=FPM) ORDER CODE	BACKUP RING ORDER CODE
20.35	1.78	OR-N/F-D-019	
21.95	1.78	OR-N/F-D-020	
23.52	1.78	OR-N/F-D-021	
25.12	1.78	OR-N/F-D-022	
26.70	1.78	OR-N/F-D-023	
28.30	1.78	OR-N/F-D-024	
29.87	1.78	OR-N/F-D-025	
31.47	1.78	OR-N/F-D-026	
33.05	1.78	OR-N/F-D-027	
34.65	1.78	OR-N/F-D-028	
37.82	1.78	OR-N/F-D-029	
41.00	1.78	OR-N/F-D-030	
44.17	1.78	OR-N/F-D-031	
47.35	1.78	OR-N/F-D-032	
50.52	1.78	OR-N/F-D-033	
53.70	1.78	OR-N/F-D-034	
56.87	1.78	OR-N/F-D-035	
60.05	1.78	OR-N/F-D-036	
63.22	1.78	OR-N/F-D-037	
66.40	1.78	OR-N/F-D-038	
69.57	1.78	OR-N/F-D-039	
72.75	1.78	OR-N/F-D-040	
75.92	1.78	OR-N/F-D-041	
82.27	1.78	OR-N/F-D-042	
88.62	1.78	OR-N/F-D-043	
94.97	1.78	OR-N/F-D-044	
101.32	1.78	OR-N/F-D-045	
107.67	1.78	OR-N/F-D-046	
114.02	1.78	OR-N/F-D-047	
120.37	1.78	OR-N/F-D-048	
126.72	1.78	OR-N/F-D-049	
133.07	1.78	OR-N/F-D-050	
CROSS SECTION 2.62 mm			
4.42	2.62	OR-N/F-D-106	
5.23	2.62	OR-N/F-D-107	
6.02	2.62	OR-N/F-D-108	
7.59	2.62	OR-N/F-D-109	
9.19	2.62	OR-N/F-D-110	
10.77	2.62	OR-N/F-D-111	
12.37	2.62	OR-N/F-D-112	
13.94	2.62	OR-N/F-D-113	
15.54	2.62	OR-N/F-D-114	
17.12	2.62	OR-N/F-D-115	

ROYAL STANDARD SIZE LIST

I.D.	C/S	O RING (N=NBR, F=FPM) ORDER CODE	BACKUP RING ORDER CODE
18.72	2.62	OR-N/F-D-116	
20.29	2.62	OR-N/F-D-117	
21.89	2.62	OR-N/F-D-118	
23.47	2.62	OR-N/F-D-119	
25.07	2.62	OR-N/F-D-120	
26.64	2.62	OR-N/F-D-121	
28.24	2.62	OR-N/F-D-122	
29.82	2.62	OR-N/F-D-123	
31.42	2.62	OR-N/F-D-124	
32.99	2.62	OR-N/F-D-125	
34.59	2.62	OR-N/F-D-126	
36.17	2.62	OR-N/F-D-127	
37.77	2.62	OR-N/F-D-128	
39.34	2.62	OR-N/F-D-129	
40.94	2.62	OR-N/F-D-130	
42.52	2.62	OR-N/F-D-131	
44.12	2.62	OR-N/F-D-132	
45.69	2.62	OR-N/F-D-133	
47.29	2.62	OR-N/F-D-134	
48.90	2.62	OR-N/F-D-135	
50.47	2.62	OR-N/F-D-136	
52.07	2.62	OR-N/F-D-137	
53.64	2.62	OR-N/F-D-138	
55.25	2.62	OR-N/F-D-139	
56.82	2.62	OR-N/F-D-140	
58.42	2.62	OR-N/F-D-141	
59.99	2.62	OR-N/F-D-142	
61.60	2.62	OR-N/F-D-143	
63.17	2.62	OR-N/F-D-144	
64.77	2.62	OR-N/F-D-145	
66.34	2.62	OR-N/F-D-146	
67.95	2.62	OR-N/F-D-147	
69.52	2.62	OR-N/F-D-148	
71.12	2.62	OR-N/F-D-149	
72.69	2.62	OR-N/F-D-150	
75.87	2.62	OR-N/F-D-151	
82.22	2.62	OR-N/F-D-152	
88.57	2.62	OR-N/F-D-153	
94.92	2.62	OR-N/F-D-154	
101.27	2.62	OR-N/F-D-155	
107.62	2.62	OR-N/F-D-156	
113.97	2.62	OR-N/F-D-157	
120.32	2.62	OR-N/F-D-158	

ROYAL STANDARD SIZE LIST

I.D.	C/S	O RING (N=NBR, F=FPM) ORDER CODE	BACKUP RING ORDER CODE
126.67	2.62	OR-N/F-D-159	
133.02	2.62	OR-N/F-D-160	
139.37	2.62	OR-N/F-D-161	
145.72	2.62	OR-N/F-D-162	
152.07	2.62	OR-N/F-D-163	
158.42	2.62	OR-N/F-D-164	
164.77	2.62	OR-N/F-D-165	
171.12	2.62	OR-N/F-D-166	
177.47	2.62	OR-N/F-D-167	
183.82	2.62	OR-N/F-D-168	
190.17	2.62	OR-N/F-D-169	
196.52	2.62	OR-N/F-D-170	
202.87	2.62	OR-N/F-D-171	
209.22	2.62	OR-N/F-D-172	
215.57	2.62	OR-N/F-D-173	
221.92	2.62	OR-N/F-D-174	
228.27	2.62	OR-N/F-D-175	
234.62	2.62	OR-N/F-D-176	
240.97	2.62	OR-N/F-D-177	
247.32	2.62	OR-N/F-D-178	
CROSS SECTION 3.53 mm			
9.12	3.53	OR-N/F-D-204	
10.69	3.53	OR-N/F-D-205	
12.29	3.53	OR-N/F-D-206	
13.87	3.53	OR-N/F-D-207	
15.47	3.53	OR-N/F-D-208	
17.04	3.53	OR-N/F-D-209	
18.64	3.53	OR-N/F-D-210	
20.22	3.53	OR-N/F-D-211	
21.82	3.53	OR-N/F-D-212	
23.39	3.53	OR-N/F-D-213	
24.99	3.53	OR-N/F-D-214	
26.57	3.53	OR-N/F-D-215	
28.17	3.53	OR-N/F-D-216	
29.74	3.53	OR-N/F-D-217	
31.34	3.53	OR-N/F-D-218	
32.92	3.53	OR-N/F-D-219	
34.52	3.53	OR-N/F-D-220	
36.09	3.53	OR-N/F-D-221	
37.69	3.53	OR-N/F-D-222	
40.87	3.53	OR-N/F-D-223	
44.04	3.53	OR-N/F-D-224	
47.22	3.53	OR-N/F-D-225	

ROYAL STANDARD SIZE LIST

I.D.	C/S	O RING (N=NBR, F=FPM) ORDER CODE	BACKUP RING ORDER CODE
50.39	3.53	OR-N/F-D-226	
53.57	3.53	OR-N/F-D-227	
56.74	3.53	OR-N/F-D-228	
59.92	3.53	OR-N/F-D-229	
63.09	3.53	OR-N/F-D-230	
66.27	3.53	OR-N/F-D-231	
69.44	3.53	OR-N/F-D-232	
72.62	3.53	OR-N/F-D-233	
75.79	3.53	OR-N/F-D-234	
78.97	3.53	OR-N/F-D-235	
82.14	3.53	OR-N/F-D-236	
85.32	3.53	OR-N/F-D-237	
88.49	3.53	OR-N/F-D-238	
91.67	3.53	OR-N/F-D-239	
94.84	3.53	OR-N/F-D-240	
98.02	3.53	OR-N/F-D-241	
101.19	3.53	OR-N/F-D-242	
104.37	3.53	OR-N/F-D-243	
107.54	3.53	OR-N/F-D-244	
110.72	3.53	OR-N/F-D-245	
113.89	3.53	OR-N/F-D-246	
117.07	3.53	OR-N/F-D-247	
120.24	3.53	OR-N/F-D-248	
123.42	3.53	OR-N/F-D-249	
126.59	3.53	OR-N/F-D-250	
129.77	3.53	OR-N/F-D-251	
132.94	3.53	OR-N/F-D-252	
136.12	3.53	OR-N/F-D-253	
139.29	3.53	OR-N/F-D-254	
142.47	3.53	OR-N/F-D-255	
145.64	3.53	OR-N/F-D-256	
148.82	3.53	OR-N/F-D-257	
151.99	3.53	OR-N/F-D-258	
158.34	3.53	OR-N/F-D-259	
164.69	3.53	OR-N/F-D-260	
171.04	3.53	OR-N/F-D-261	
177.39	3.53	OR-N/F-D-262	
183.74	3.53	OR-N/F-D-263	
190.09	3.53	OR-N/F-D-264	
196.44	3.53	OR-N/F-D-265	
202.79	3.53	OR-N/F-D-266	
209.14	3.53	OR-N/F-D-267	
215.49	3.53	OR-N/F-D-268	

ROYAL STANDARD SIZE LIST

I.D.	C/S	O RING (N=NBR, F=FPM) ORDER CODE	BACKUP RING ORDER CODE
221.84	3.53	OR-N/F-D-269	
228.19	3.53	OR-N/F-D-270	
234.54	3.53	OR-N/F-D-271	
240.89	3.53	OR-N/F-D-272	
247.24	3.53	OR-N/F-D-273	
253.59	3.53	OR-N/F-D-274	
266.29	3.53	OR-N/F-D-275	
278.99	3.53	OR-N/F-D-276	
291.69	3.53	OR-N/F-D-277	
304.39	3.53	OR-N/F-D-278	
329.79	3.53	OR-N/F-D-279	
355.19	3.53	OR-N/F-D-280	
380.59	3.53	OR-N/F-D-281	
405.26	3.53	OR-N/F-D-282	
430.66	3.53	OR-N/F-D-283	
456.06	3.53	OR-N/F-D-284	
CROSS SECTION 5.33 mm			
10.46	5.33	OR-N/F-D-309	
12.07	5.33	OR-N/F-D-310	
13.64	5.33	OR-N/F-D-311	
15.24	5.33	OR-N/F-D-312	
16.81	5.33	OR-N/F-D-313	
18.42	5.33	OR-N/F-D-314	
19.99	5.33	OR-N/F-D-315	
21.59	5.33	OR-N/F-D-316	
23.16	5.33	OR-N/F-D-317	
24.77	5.33	OR-N/F-D-318	
26.34	5.33	OR-N/F-D-319	
27.94	5.33	OR-N/F-D-320	
29.51	5.33	OR-N/F-D-321	
31.12	5.33	OR-N/F-D-323	
32.69	5.33	OR-N/F-D-323	
34.29	5.33	OR-N/F-D-324	
37.47	5.33	OR-N/F-D-325	
40.64	5.33	OR-N/F-D-326	
43.82	5.33	OR-N/F-D-327	
46.99	5.33	OR-N/F-D-328	
50.17	5.33	OR-N/F-D-329	
53.34	5.33	OR-N/F-D-330	
56.52	5.33	OR-N/F-D-331	
59.69	5.33	OR-N/F-D-332	
62.87	5.33	OR-N/F-D-333	
66.04	5.33	OR-N/F-D-334	

ROYAL STANDARD SIZE LIST

I.D.	C/S	O RING (N=NBR, F=FPM) ORDER CODE	BACKUP RING ORDER CODE
69.22	5.33	OR-N/F-D-335	
72.39	5.33	OR-N/F-D-336	
75.57	5.33	OR-N/F-D-337	
78.74	5.33	OR-N/F-D-338	
81.91	5.33	OR-N/F-D-339	
85.09	5.33	OR-N/F-D-340	
88.27	5.33	OR-N/F-D-341	
91.44	5.33	OR-N/F-D-342	
94.62	5.33	OR-N/F-D-343	
97.79	5.33	OR-N/F-D-344	
100.97	5.33	OR-N/F-D-345	
104.14	5.33	OR-N/F-D-346	
107.32	5.33	OR-N/F-D-347	
110.49	5.33	OR-N/F-D-348	
113.67	5.33	OR-N/F-D-349	
116.84	5.33	OR-N/F-D-350	
120.02	5.33	OR-N/F-D-351	
123.19	5.33	OR-N/F-D-352	
126.37	5.33	OR-N/F-D-353	
129.54	5.33	OR-N/F-D-354	
132.72	5.33	OR-N/F-D-355	
135.89	5.33	OR-N/F-D-356	
139.07	5.33	OR-N/F-D-357	
142.24	5.33	OR-N/F-D-358	
145.42	5.33	OR-N/F-D-359	
148.59	5.33	OR-N/F-D-360	
151.77	5.33	OR-N/F-D-361	
158.12	5.33	OR-N/F-D-362	
164.47	5.33	OR-N/F-D-363	
170.82	5.33	OR-N/F-D-364	
177.17	5.33	OR-N/F-D-365	
183.52	5.33	OR-N/F-D-366	
189.87	5.33	OR-N/F-D-367	
196.22	5.33	OR-N/F-D-368	
202.57	5.33	OR-N/F-D-369	
208.97	5.33	OR-N/F-D-370	
215.27	5.33	OR-N/F-D-371	
221.62	5.33	OR-N/F-D-372	
227.97	5.33	OR-N/F-D-373	
234.32	5.33	OR-N/F-D-374	
240.67	5.33	OR-N/F-D-375	
247.02	5.33	OR-N/F-D-376	
253.37	5.33	OR-N/F-D-377	

ROYAL STANDARD SIZE LIST

I.D.	C/S	O RING (N=NBR, F=FPM) ORDER CODE	BACKUP RING ORDER CODE
266.07	5.33	OR-N/F-D-378	
278.77	5.33	OR-N/F-D-379	
291.47	5.33	OR-N/F-D-380	
304.17	5.33	OR-N/F-D-381	
329.57	5.33	OR-N/F-D-382	
354.97	5.33	OR-N/F-D-383	
380.37	5.33	OR-N/F-D-384	
405.26	5.33	OR-N/F-D-385	
430.66	5.33	OR-N/F-D-386	
CROSS SECTION 6.99 mm			
113.67	6.99	OR-N/F-D-425	
116.84	6.99	OR-N/F-D-426	
120.02	6.99	OR-N/F-D-427	
123.19	6.99	OR-N/F-D-428	
126.37	6.99	OR-N/F-D-429	
129.54	6.99	OR-N/F-D-430	
132.72	6.99	OR-N/F-D-431	
135.89	6.99	OR-N/F-D-432	
139.07	6.99	OR-N/F-D-434	
142.24	6.99	OR-N/F-D-434	
145.42	6.99	OR-N/F-D-435	
148.59	6.99	OR-N/F-D-436	
151.77	6.99	OR-N/F-D-437	
158.12	6.99	OR-N/F-D-438	
164.47	6.99	OR-N/F-D-439	
170.82	6.99	OR-N/F-D-440	
177.17	6.99	OR-N/F-D-441	
183.52	6.99	OR-N/F-D-442	
189.87	6.99	OR-N/F-D-443	
196.22	6.99	OR-N/F-D-444	
202.57	6.99	OR-N/F-D-445	
215.27	6.99	OR-N/F-D-446	
227.97	6.99	OR-N/F-D-447	
240.67	6.99	OR-N/F-D-448	
253.37	6.99	OR-N/F-D-449	
266.07	6.99	OR-N/F-D-450	
278.77	6.99	OR-N/F-D-451	
291.47	6.99	OR-N/F-D-452	
304.17	6.99	OR-N/F-D-453	
316.87	6.99	OR-N/F-D-454	
329.57	6.99	OR-N/F-D-455	
342.27	6.99	OR-N/F-D-456	

ROYAL STANDARD SIZE LIST

I.D.	C/S	O RING (N=NBR, F=FPM) ORDER CODE	BACKUP RING ORDER CODE
354.97	6.99	OR-N/F-D-457	
367.67	6.99	OR-N/F-D-458	
380.37	6.99	OR-N/F-D-459	
393.07	6.99	OR-N/F-D-460	
405.26	6.99	OR-N/F-D-461	
417.96	6.99	OR-N/F-D-462	
430.66	6.99	OR-N/F-D-463	
443.36	6.99	OR-N/F-D-464	
456.06	6.99	OR-N/F-D-465	
468.76	6.99	OR-N/F-D-466	
481.46	6.99	OR-N/F-D-467	
494.16	6.99	OR-N/F-D-468	
506.86	6.99	OR-N/F-D-469	
532.26	6.99	OR-N/F-D-470	
557.66	6.99	OR-N/F-D-471	
582.68	6.99	OR-N/F-D-472	
608.08	6.99	OR-N/F-D-473	
633.48	6.99	OR-N/F-D-474	
658.88	6.99	OR-N/F-D-475	

Section - V
STORAGE & FITTING

Storage

When storing **ROYAL** seals, please check the following points:

1) Do not open the packing unnecessarily. Dust may stick to the seal or scratch it.
2) Avoid exposure to direct sunlight and store in a cool place. Ultraviolet light and humidity may sometimes promote deterioration and dimensional change of rubber and resin materials.
3) When storing products already unpacked, be careful that foreign materials do not stick or become embedded. Store them in their original condition, in tightly sealed poly to prevent dimensional change resulting from humidity.
4) Do not place seals near any heat source such as a boiler, stove, oven etc. Heat may deteriorate the material.
5) Do not place seals near electric motors or equipments generating ozone.
6) Avoid hanging seals with a nail, wire or suspending them with a string as this may cause deformation and scratches on the top end of the lip.
7) Sometimes, colour changes or white powder appears on the surface of seals (blooming phenomenon). This will not affect the function of seal.
8) PTFE rings of combination seals may be easily scratched if dropped or impacted by an external force. Be careful when handling.
9) The store room should be cool, dry and dust-free; it should have moderate ventilation.
10) The seals should never be exposed to any ozone or ionizing radiation.

Check points before installation of seals :-

ROYAL quality control methods for material and manufacturing processes ensure that all seals going into application field are in a condition capable of giving a long and reliable service life. Sealing performance of a seal is greatly influence by the method of assembling of a cylinder. On the basis of many years experience, to avoid premature seal failure the following recommendations should be considered at the design and manufacturing stage of the cylinder:

1) Specify piston and gland bearings which are adequately proportioned to support the cylinder loads. As a result of mounting misalignment and / or the working action of the cylinder, piston and gland bearings will be subjected to side loading, causing damage to the rod or the tube surface and hence the seal, if the bearings are inadequate.
2) Check that the seal housing is free from damage likely to harm the seal. Remove all sharp edges and burrs from metal parts, paying particular attention to ports, grooves and threads over or through which the seal passes during assembly.
3) Clean all seal housing areas, ensuring that all metallic particles and other contaminants have been removed. Check that other surfaces adjacent to the passage of the seal on fitting are also free of dirt or other contaminants. Check that both static and dynamic housing surface finishes meet specifications.
4) Where the difference between a thread diameter over which the seal must pass and the seal diameter is small, use some form of protection over the thread, such as a fitting sleeve made of hard plastic.
5) Lubricate all seals and metal components liberally with clean operating fluid or a compatible grease prior to assembly. Note that silicone grease should not be used in normal hydraulic applications.
6) Where seals fitted to sub-assemblies, such as pistons, are awaiting further fitting operations, ensure that the seals are not subjected to any misaligned or localized loading which will cause local deformation. Ensure that sub-assemblies remain clean.
7) The use of metal levers is not recommended but should they be used it is imperative that they are completely smooth and free from nicks and burrs. When using them ensure that the metal surfaces adjacent to the seal are not damaged.
8) Make sure that hydraulic cylinder & system is free from dust, any burr or steel particle.

NOTES

Nominal sizes		Shaft (outside diameter)						Bore (inside diameter)			
mm		Units 0.001 mm						Units 0.001mm			
over	to	f8	f9	h8	h9	h10	h11	H8	H9	H10	H11
1.6	3	-6	-6	0	0	0	0	14	25	40	60
		-20	-31	-14	-25	-40	-60	0	0	0	0
3	6	-10	-10	0	0	0	0	18	30	48	75
		-28	-40	-18	-30	-48	-75	0	0	0	0
6	10	-13	-13	0	0	0	0	22	36	58	90
		-35	-49	-22	-36	-58	-90	0	0	0	0
10	18	-16	-16	0	0	0	0	27	43	70	110
		-43	-59	-27	-43	-70	-110	0	0	0	0
18	30	-20	-20	0	0	0	0	33	52	84	130
		-53	-72	-33	-52	-84	-130	0	0	0	0
30	50	-25	-25	0	0	0	0	39	62	100	160
		-64	-87	-39	-62	-100	-160	0	0	0	0
50	80	-30	-30	0	0	0	0	46	74	120	190
		-76	-104	-46	-74	-120	-190	0	0	0	0
80	120	-36	-36	0	0	0	0	54	87	140	220
		-90	-123	-54	-87	-140	-220	0	0	0	0
120	180	-43	-43	0	0	0	0	63	100	160	250
		-106	-143	-63	-100	-160	-250	0	0	0	0
180	250	-50	-50	0	0	0	0	72	115	185	290
		-122	-165	-72	-115	-185	-290	0	0	0	0
250	315	-56	-56	0	0	0	0	81	130	210	320
		-137	-186	-81	-130	-210	-320	0	0	0	0
315	400	-62	-62	0	0	0	0	89	140	230	360
		-151	-202	-89	-140	-230	-360	0	0	0	0
400	500	-68	-68	0	0	0	0	97	155	250	400
		-165	-223	-97	-155	-250	-400	0	0	0	0
500	630	-76	-76	0	0	0	0	110	175	280	440
		-186	-251	-110	-175	-280	-440	0	0	0	0
630	800	-80	-80	0	0	0	0	125	200	320	500
		-205	-280	-125	-200	-320	-500	0	0	0	0

ROYAL











SEAL APPLICATIONS











R O Y A L

ROYAL

SEALS FOR HYDRAULICS & PNEUMATICS











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